

CODED APERTURE IMAGING OF NUCLEAR FUSION IN THE PLASMA FOCUS DEVICE

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Dedicated to:

my dear wife, Elahe

and my lovely kids, Azam, Akram and Hamed

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Contents

Abstract.....	i
Publication.....	iii
List of Figures.....	vi
List of tables	xv
1 Introduction to Plasma Focus	3
1.1 Structure and Working Principle	3
1.2 Dynamics of Plasma Focus.....	4
1.2.1 The Breakdown phase (Inverse Pinch phase).....	5
1.2.2 The Axial Acceleration (Run Down) phase in Mather geometry.....	6
1.2.3 The Radial Collapse phase	7
1.3 Plasma Instabilities	10
1.3.1 Rayleigh-Taylor Instability	10
1.4 Plasma Focus Models	12
1.4.1 Snow Plow model:.....	12
1.4.2 Quasi-Steady-State model:	12
1.4.3 Slug model:.....	12
1.4.4 Gas dynamic model:	13
1.4.5 Ion beam acceleration and neutron production models:.....	13
1.5 Plasma Focus Radiation:.....	17
1.5.1 Ion-electron production	17
1.5.2 X-ray production	18
1.6 Radiation Measurement	20
1.6.1 Neutron Yield	20
1.6.2 Neutron Energy Measurement.....	21
1.6.3 Proton and Ion Measurement.....	22
1.6.4 Hard X-ray Measurement	23
1.6.5 Soft X-ray Measurement	23
1.7 NX2 Plasma Focus.....	24
1.7.1 Chamber	25
1.7.2 Electric System.....	25
1.7.3 Electrical Diagnostics	26
1.8 Layout of Thesis:	30

2	Coded Aperture Imaging (CAI).....	32
2.1	CAI Theory	32
2.1.1	Overview	32
2.1.2	Coded Aperture Families:.....	37
2.1.3	Coded Aperture Geometry	41
2.1.4	Singer set array, a new approach for coded aperture imaging:	42
2.1.5	Signal to Noise Ratio (SNR) in Coded Aperture Imaging	45
2.1.6	Mask Design.....	50
2.1.7	Image Reconstruction.....	55
2.2	CAI Experimental Procedure.....	56
2.2.1	Mask Fabrication	56
2.2.2	Charge Particle Detection.....	58
2.2.3	Scanning	67
2.2.4	Image-Pro for Track Analysis	69
2.2.5	Data Analysis.....	70
2.3	Monte Carlo Numerical Simulation of CAI System.....	72
2.3.1	Simulated Source Distributions	75
2.3.2	Fusion Source Simulation with mask-1365 and mask-4680	77
3	Beryllium Activation Neutron Detector	80
3.1	Overview.....	80
3.2	Construction.....	85
3.3	Energy Calibration and Procedure for Setting the Energy Threshold	86
3.4	Rate count and capability.....	90
3.5	Numerical Calibration of Detector Response	94
3.5.1	General Purpose Calibration.....	94
3.5.2	Special Calibration for NX2 Plasma Focus Device	100
3.6	Experimental Setup and Results	109
3.6.1	Data Acquisition System	109
3.6.2	Interfering Activation	111
3.6.3	Comparison of Response for Two Be Activation Detectors	112
3.6.4	NX2 neutron yield and anisotropy measurements.....	114
3.7	Conclusion	119
4	Other Diagnostics	121

4.1	X-Ray Pinhole Imaging	121
4.2	Plastic Scintillator	124
4.3	Cesium Iodide (CsI)	125
4.4	Barium Fluoride	125
5	Experiments and Results	128
5.1	Alpha Source Imaging	128
5.1.1	Experimental Setup	128
5.1.2	Results and Discussion	131
5.2	Fusion Source 3-D Imaging	134
5.2.1	Experimental Setup	134
5.2.2	Experimental Results	139
5.3	Fusion Imaging with Spaced Mask	142
5.3.1	Experimental Setup	142
5.3.2	Experimental Results of operation in pure deuterium	151
5.3.3	Experimental Results of operation in admixture gas	157
5.3.4	Image registration	164
5.3.5	Fusion imaging with pure deuterium gas	169
5.3.6	Fusion imaging with admixture gas	202
6	Discussion	218
6.1	Mask design:	218
6.2	Simulation:	221
6.3	Detector:	221
6.4	Scanning and Analyzing:	222
6.5	Other diagnostics:	223
6.6	Alpha source imaging:	224
6.7	Fusion source imaging with mask-400:	224
6.8	Fusion source imaging with spaced mask:	225
6.8.1	Pure deuterium:	225
6.8.2	D ₂ +2%Kr Admixture gas:	226
6.9	Final conclusion and suggested future work:	227
	Appendices	229
	References	276

Abstract

The Coded Aperture Imaging (CAI) technique has been used to image the deuterium-deuterium (D-D) fusion source in the NX2 plasma focus (PF) device, using protons emitted from the $D(d,p)T$ reaction. CAI is a form of multiplexed pinhole imaging which uses many small pinholes arranged in specific patterns so as to obtain images with the good spatial resolution expected for a small pinhole, together with high brightness and signal-to-noise ratio (SNR) associated with a much larger open aperture. Our theoretical study of CAI led us to the conclusion that for extended sources, such as the PF pinch, coded mask patterns of relatively low open fraction (i.e. significantly less than 50%) are optimal. Therefore the mask patterns used in our experimental work are based on Singer cyclic difference sets. Monte Carlo simulations of CAI and pinhole imaging were performed in order to validate our theoretical SNR formulae and test our decoding algorithm. CR-39 polymer nuclear track detectors were used to register the ~ 3 MeV protons emitted from the PF pinch. As an additional test of the CR-39 detector scanning procedures, theoretical formulae, and CAI decoding algorithm, various source shapes were simulated experimentally using a ^{226}Ra alpha particle source. As anticipated, the obtained alpha particle CAI images were found to be considerably better than the corresponding single-pinhole images. Significantly, this work represents the first application of the CAI technique to imaging the fusion source in a plasma focus device.

The time-integrated neutron yield and anisotropy for each NX2 plasma focus shot were measured using fast-neutron detectors based on the production of radioactive ^6He via the $^9\text{Be}(n,\alpha)^6\text{He}$ reaction. The newly fabricated detectors comprise a beryllium metal sheet sandwiched between two large-area xenon-filled proportional counters. A methodology was developed for calculating the absolute neutron response function of these beryllium activation detectors, based on: the $^9\text{Be}(n,\alpha)^6\text{He}$ reaction cross-section, energy calibration of the proportional counters, and numerical simulations (using MCNP5) of neutron and beta-particle interactions.

In one series of experiments, five CAI cameras with identical masks were employed simultaneously: one placed on-axis (0°) and four at 45° to the plasma focus axis at the distances of 10.3 cm and 4.9 cm from the source respectively, to investigate the spatial distribution of D-D fusion in the NX2 PF device operated in pure D_2 gas and

with a PF bank energy of 1.6 kJ. These conditions represent the neutron-optimized regime, for which the neutron yield is typically $1\text{--}3 \times 10^8$ n/shot. In a second series of experiments, two larger coded masks: mask-1345 (91×15 array and 341 holes each 0.3 mm side) and mask-4680 (104×45 arrays and 585 holes each 0.27 mm side) were placed at 90° to the plasma focus axis to investigate the fusion source in pure deuterium and deuterium-krypton admixture working gases. The number of proton tracks registered on the CR-39 detectors per shot was typically $(1\text{--}3) \times 10^5$. The results clearly show the different size, density and shape of the fusion source in pure D_2 and $\text{D}_2\text{-Kr}$ admixture operation. For this second series of experiments, an x-ray pinhole imaging system with suitable filtering was employed simultaneously to record the associated x-ray images of the hot dense pinched plasma column for comparison with the fusion source images. A plastic scintillation detector was used to measure the time-resolved neutron signal, while the time-resolved hard x-ray pulse was measured using both cesium-iodide and barium-fluoride scintillation detectors.

The results show little apparent correlation between the shape of the D-D fusion source and the corresponding soft x-ray image of the plasma pinch. For pure D_2 discharges the proton CAI images show a reasonably symmetrical “cigar-shaped” fusion source surrounding the pinch column, without any appreciable indication of $m = 0$ or $m = 1$ plasma instabilities. Soft x-ray emission from the pinch column is much stronger for $\text{D}_2\text{-Kr}$ admixture operation than for pure D_2 operation, and micro-pinches are frequently present within the main pinch column. Despite this, there are again no discernable indications of plasma instabilities in the associated proton CAI images. Proton CAI images show that the D-D fusion source is much narrower (i.e. of smaller diameter) for $\text{D}_2\text{-Kr}$ admixture operation of the NX2 than it is for pure D_2 operation. It is concluded that a collective mechanism of deuteron acceleration occurs throughout the length of the pinch column in the NX2 device, and therefore the fusion is distributed rather evenly around the pinch column. Although $m = 0$ instabilities in the pinch column may initiate deuteron acceleration, the fast deuterons and resulting fusion are not concentrated within, or around, $m = 0$ instabilities.

Publication

Recent Journal Papers:

1. **A. Talebitaher**, S.V. Springham, P.M.E. Shutler, P. Lee, R.S. Rawat “*Imaging of Plasma Focus Fusion by Proton Coded Aperture Technique*” Journal of Fusion Energy, DOI 10.1007/s10894-011-9462-8
2. **A. Talebitaher**, S.V. Springham, R.S. Rawat, P. Lee “*Beryllium Neutron Activation Detector for Pulsed DD Fusion Sources*” Nuclear Inst. and Methods in Physics Research A, DOI:10.1016/j.nima.2011.08.063
3. **A. Talebitaher**, P.M.E. Shutler, S.V. Springham, R.S. Rawat, P. Lee “*Coded Aperture Imaging of Alpha Source Spatial Distribution*” (submitted to Radiation Measurements).
4. **A. Talebitaher**, P.M.E. Shutler, S.V. Springham, R.S. Rawat, P. Lee “*Coded Aperture Imaging of Fusion Source in a Plasma Focus Operated with pure D₂ and a D₂-Kr Gas Admixture*” (submitted to Applied Physics Letter).
1. P.M.E. Shutler, **A. Talebitaher**, S.V. Springham “*Signal-to-Noise ratio in Coded Aperture Imaging*” Nuclear Instr. and Methods in Physics Research Section A, **669**, (21 March 2012), Pages 22-31, DOI:10.1016/j.nima.2011.12.023
2. **A. Talebitaher**, S.V. Springham, R.S. Rawat, P. Lee “*Neutron Yield and Anisotropy Measurement in Plasma Focus using Beryllium Activation detector*” (under revision).
3. S. Lee, S H Saw, R.S. Rawat, P. Lee, R. Verma, **A. Talebitaher**, et al. “*Measurement and Processing of Fast Pulsed Discharge Current in Plasma Focus Machines*” Journal of Fusion Energy, DOI 10.1007/s10894-011-9456-6.
4. S. Lee, S H Saw, R.S. Rawat, P. Lee, **A. Talebitaher**, et al. “*Correlation of Measured soft x-ray pulses with modeled dynamics of the plasma focus*” IEEE Transactions on Plasma Science, **39**, 11, 3196-3202 (2011).
5. M V Roshan, S V Springham, **A Talebitaher**, R S Rawat, P Lee “*Magnetic spectroscopy of high energy deuteron beams from pulsed plasma system*” Plasma Phys. Control Fusion **52** 085007 pp17 (2010).

6. M V. Roshan, St V Springham, **A Talebitaher**, R S Rawat, P Lee “*Nuclear Activation Measurements of High Energy Deuterons from a Small Plasma Focus*” Physics Letters A, 373, 851 (2009).
7. M V. Roshan, P. Lee, S. Lee, **A. Talebitaher**, R. S. Rawat, and S. V. Springham “*backward high energy ion beams from plasma focus*” Physics of Plasma **16**, 074506 (2009).
8. M V. Roshan, S.V. Springham, **A.R. Talebitaher**, R.S. Rawat, P. Lee, “*Intense deuteron beam investigation by activation yield-ratio technique*” Physics Letters A, 373, 3771-3774 (2009).
9. S.V. Springham, F. Malik, M.V. Roshan, **A. Talebitaher**, R.S. Rawat, P. Lee, “*Ferrofluidic Masking of Solid State Nuclear Track Detectors During Etching*” Radiation Measurements, 44 173-175 (2009).
10. M.V. Roshan, **Alireza Talebitaher**, et al. “*High Energy Deuterons Emission in NX2 Plasma Focus*” Journal of Plasma and Fusion Research, Vol. 8 (2009)
11. M.V. Roshan, R.S. Rawat, **A. Talebitaher**, P. Lee, S.V. Springham “*Neutron and high energy deuteron anisotropy investigation in Plasma Focus Device*” Physics of Plasmas 16, 053301 (2009)
12. **A.R. Talebitaher**, M.V. Roshan, R. Verma, P. Lee, R.S. Rawat, S.V. Springham “*Soft x-ray emission investigation in plasma focus device (NX2) with neon filling gas*” Iranian Physical Journal 2-3, 38-41 (2008)

Recent Conference Papers:

13. **A.R. Talebitaher**, S.V. Springham, P.M.E. Shutler, R.S. Rawat, P. Lee “*Coded Aperture Imaging Technique for Investigation of Fusion Source Spatial Distribution its Correlation with X-Ray Images in Plasma Focus Device*” ICOPS conference, June 2011, Chicago-USA
14. R. Verma, T. L. Tan, P. Lee, **A.R. Talebitaher**, et al. “*NX-3 Plasma Focus Device: High Flux Pulsed Neutron Sources*” ICOPS conference, June 2011, Chicago-USA
15. **A.R. Talebitaher**, et al. “*A new approach for Investigation of Fusion Source by Coded Aperture Imaging*” FPPT5 conference, April 2011. Singapore.

16. **A.R. Talebitaher**, S.V. Springham, P.M.E. Shutler, R.S. Rawat, P. Lee “*Coded Aperture Imaging Technique for Investigation of Fusion Source Spatial Distribution in Plasma Focus Device*” IPS meeting Singapore, February 2011
17. R. Verma, T. Tan, P. Lee, **A. Talebitaher**, H. Shariff, S.V. Springham, R.S. Rawat “*20KJ Plasma Focus Device Neutron Source: Development and Recent Results*” FPPT5 conference, April 2011. Singapore.
18. A. Bisht, **A. Talebitaher**, et al. “*Synthesis of Nanoparticles Using Atmospheric Microplasma Discharge*” FPPT5 conference, April 2011. Singapore.
19. **A.R. Talebitaher**, et al. “*Preliminary Result of Coded Aperture Imaging on NX2 Plasma Focus Device*” Proceeding of International Workshop on Plasma Diagnostics and Application (IWPDA), July 2009 Singapore. ISBN: 978-983-44947-2-8
20. **A.R. Talebitaher**, et al. “*Soft X-ray emission in Plasma Focus device with Neon filling gas*” IPS March Meeting Singapore, 6,7 March 2009
21. P. Lee, R.S. Rawat, S.V. Springham, **A. Talebitaher**, S. K. Munirah, Lyna, C.C. Lim, C.M. Yau and S.K. Foong “*The need for special courses in physics for NIE student teachers*” Redesigning Pedagogies: Designing New Learning Contexts for a Globalizing World, 1-3 June 2009, National Institute of Education (NIE), Singapore
22. S.V. Springham, M.V. Roshan, **A. Talebitaher** et al. “*Neutron & Charged-Particle Diagnostics and Preliminary Studies of Coded Aperture Imaging*” ICTP 2008 Italy Conference

List of Figures

Figure 1: Schematic diagram of Mather-type and Filippov-type PF device	4
Figure 2: a) Schematic of plasma focus dynamic b) current and voltage signals	5
Figure 3: sausage ($m = 0$) and kink ($m = 1$) instability.....	11
Figure 4: Radial trajectories of shock (r_s) and piston (r_p) versus time in collapse phase (Slug model)	13
Figure 5: NX2 Plasma Focus device.	24
Figure 6: Rogowski coil and its equivalent circuit.....	27
Figure 7: current trace for NX2 short circuit test. The experiment was done with Argon gas at pressure 20 mbar	28
Figure 8: dI/dt RC signal by NX2 with single pinch (left) and double pinch (right)	28
Figure 9: High Voltage Probe using in PF	29
Figure 10: Coded Mask pattern of 95×95 elements formed by a replicated 53×53 MURA basic pattern. [from: A. Goldwurm et al.: IBIS science data analysis (2003)]..	32
Figure 11: a) pinhole imaging and b) collimator imaging systems	35
Figure 12: Pinhole camera and its limitation.....	35
Figure 13: Fresnel Zone Plate; theoretical (left) and practical (right).....	38
Figure 14: Random array and its periodic self-correction.....	38
Figure 15: non-redundant array	39
Figure 16: 61×59 Uniformly Redundant Array.....	39
Figure 17: MURA pattern, while the 79×79 is anti-symmetric about its center, the 73 $\times 73$ is symmetric.	40
Figure 18: CAI geometry to identify the magnification, object pixel size (resolution), image pixel size and field of view.	42
Figure 19: 1D Cyclic Difference Singer Set in modulo 2	44

Figure 20: schematic of reconstruction algorithm (Deconvolution)	44
Figure 21: SNR vs. open fraction (ρ) for objects (of the same total brightness) which illuminate different numbers of pixels (m) the vertical black lines represent the possible open fraction for Singer Set.....	48
Figure 22: mask-400 (20×20) pattern with $\rho = 1/7$ open fraction	50
Figure 23: mask-1365 (91×15) pattern with $\rho = 1/4$ open fraction	50
Figure 24: mask-4680 (104×45) pattern with $\rho = 1/8$ open fraction, with one pixel truncated from the complete CDS length of 4681.....	51
Figure 25: mask 1365 (91×15) spaced by factor of $k=6$	52
Figure 26: mask-4680 (104×45) spaced by factor of $k=2$	52
Figure 27: SNR of mask-1365; basic pattern (blue lines) and spaced mask (red lines) for different object size ($m=32, 64$ and 128). Bold black line represents $\rho=1/4$ (the optimum open fraction for $m=64$).....	54
Figure 28: SNR of mask-4680; basic pattern (blue lines) and spaced mask (red lines) for different object size ($m=32, 64$ and 128). Bold black line represents $\rho=1/8$	54
Figure 29: schematic diagram for reconstruction algorithm (cyclic convolution)	55
Figure 30: mask-400, Physical mask (right) made by Micro-EDM drilling and its pattern (left)	56
Figure 31: The mask-400 pattern (a), and a photograph of the fabricated mask (b) made by laser drilling.....	57
Figure 32: Stopping power graph for protons in CR-39.....	59
Figure 33: The SRIM simulation result by 3 MeV proton pass through the CR-39 with 250 μm thickness	60
Figure 34: very high dense deuteron tracks on the detector (occasionally) evidence the existence of extremely high energy deuterons in plasma focus device. The deuteron	

tracks can be seen in images, taken by different microscope object lens: $\times 50$ at the left side and $\times 5$ at the right side.	62
Figure 35: the scanned image of detector (right) shows the same pattern as the mask (left) which shows the present of a point source. The shadows of mask pattern in up-left and down parts of image also evidence that there are more than one point source which are coming like a narrow beam.....	62
Figure 36: Bragg curve for proton particle. The relative dose (y-axis) and depth (x-axis) are arbitrary unit.	63
Figure 37: Etching speed and normal track shape	64
Figure 38: NRU program output shows the 5 MeV α track size after 5 h etching.	65
Figure 39: Over etched track shape is shallow and appears like bright circle.....	66
Figure 40: normal track (a) and over etched track (b) taken by confocal microscope. (c) Distinguishing between usual proton and deuteron tracks under microscope.....	66
Figure 41: Microscope Olympus BX51 and its attachments.	68
Figure 42: overlapped tracks (left) have been separated by watershed filter (right).....	70
Figure 43: Scatter plots of measured parameters of dark features on CR-39 detector surface: (a) grey-value vs. area, (b) roundness vs. area, and (c) roundness vs. grey-value, with user-defined polygon for each.(d) Positions (x,y) of genuine particle tracks (for 1-spot source) selected by applying the logical AND of within-polygon criteria shown in a-c. This (x,y) data represents the coded image	71
Figure 44: reconstructed image of 1-spot α source with mask-400.....	72
Figure 45: isotropic particles distribution within a cone from a point source	74
Figure 46: Reconstructed images from Monte Carlo generated data for different source shapes with pinhole (top row) and 20×20 CAI mask (bottom row). All images are of	

10 cm side (in the source plane). 1-point source for pinhole (a1) and mask (a2); 2-point source for pinhole (b1) and mask (b2); ridge source for pinhole (c1) and mask (c2)	76
Figure 47: mask-1365: reconstructed image from Monte Carlo generated data for a 10 mm×2 mm rectangular source.	78
Figure 48: mask-4680: reconstructed image from Monte Carlo generated data for a 10 mm×2 mm rectangular source.	78
Figure 49: The ${}^9\text{Be}(n,\alpha){}^6\text{He}$ reaction cross-section as function of neutron energy.	83
Figure 50: (a) Assembled activation counter comprising beryllium sandwiched between two xenon-filled proportional counters, aluminium holding plate, and two PDT10A units attached. (b) One PDT10A unit providing preamplifier, shaping amplifier and discriminator functions.	87
Figure 51: Plot of channel centroid for 5.90, 6.49 and 22.16 keV x-rays for LND-PC as a function of applied bias voltage. Overlaid on this plot is the energy spectrum in the vicinity of the 5.90 and 6.49 keV x-rays from ${}^{55}\text{Fe}$ acquired with Ortec pulse processing electronics.	88
Figure 52: Plot of relative gas-gain for each of the four LND proportional counters, normalized to unity at 1500 V for each LND proportional counter. The dashed curve represents the empirical function $gg(V)$	89
Figure 53: β -particle energy spectrum for the decay of ${}^6\text{He}$	96
Figure 54: MCNP simulation results of frequency distribution of energy deposition in PC xenon gas layer (12.7 mm depth) by ${}^6\text{He}$ β particles which escape from Be sheet and enter PC. Separate curves are for ${}^6\text{He}$ β particles originating in each 1 mm thick layer of the 5 mm thick Be sheet. In this simulation a PC is located on one side only of the Be sheet, and adjacent to layer 1.	96

Figure 55: Response function $R(E_n)$ of beryllium neutron activation detector for a parallel neutron beam of fluence 1 neutron/cm ² , and contributions $R_L(E_n)$ from each 1 mm thick layer of the Be sheet. Note that the vertical axis is broken and the two parts are scaled differently. The absolute $R(E_n)$ values across the curve as a whole, have an estimated uncertainty of $\pm 10\%$	98
Figure 56: Beryllium neutron detector response function $R_{ps}(E_n = 2.45 \text{ MeV})$ for a point source at distance d.	99
Figure 57: Center of Mass (C.M.) cross-section for D-D reaction with the deuteron energy (in keV) as a parameter	100
Figure 58: Lab frame cross-section for D-D reaction with the deuteron energy (in keV) as a parameter	101
Figure 59: Lab frame (L.F.) neutron energy distribution for D-D reaction with the deuteron energy (in keV) as a parameter	102
Figure 60: Simulation of NX2 chamber and Beryllium detectors in MCNP-visual	103
Figure 61: total efficiency of neutron detector in axial and radial direction at 25.0 cm from anode tip.....	105
Figure 62: the result of neutron anisotropy versus deuteron energy by MCNP	106
Figure 63: neutron fluence vs. energy at different angles with 100keV deuteron energy 30° forward cone at $r = 100\text{cm}$ from centre of pinch, without chamber.	107
Figure 64: neutron fluence vs. energy at different angles with 100keV deuteron energy 30° forward cone at $r = 100\text{cm}$ from centre of pinch, with NX2 chamber.	108
Figure 65: Ortec MCS-pci board.....	109
Figure 66: MCS-32 software for Data Acquisition System – showing ^6He decay plot for a series of 20 NX2 shots fired at 0.125 Hz repetition rate.	110
Figure 67: Ortec 556 H.V. power supply	111

Figure 68: Experimental setup for the NX2 plasma focus neutron fluence measurements and detectors' cross comparing.	113
Figure 69: Experimental data from the two beryllium detectors placed in radial directions, for 60 plasma focus shots.....	113
Figure 70: Experimental setup for neutron anisotropy measurements using both beryllium activation detectors.	114
Figure 71: Average neutron anisotropy and beryllium count (0° and 90°) vs. pressure (top plots) neutron anisotropy vs. neutron yield in different filling gas pressure	115
Figure 72: proton tracks distribution on PM-355 detector with pinhole imaging at a) AP1 (0°) and b) AP2 (90°)	116
Figure 73: Experiment setup for anisotropy with Pyrex plate (6 mm thick and diameter slightly less than chamber)	117
Figure 74: anisotropy variation with Pyrex which prevents the deuterons reaching the upper part of the chamber.....	118
Figure 75: Circuit diagram of Synchronization Board using for x-ray imaging.	123
Figure 76: components of Scintillator-PMT combination [PhD thesis: Rishi Verma]	124
Figure 77: BaF ₂ scintillation emission spectrum [SAINT-GORBAIN CRYSTALS] .	126
Figure 78: Radium-226 alpha-particle source (left) and its Decay Chain (right).....	128
Figure 79: Experimental arrangement of CR-39 detector, CAI mask and Ra-226 α source within vacuum enclose.....	129
Figure 80: Optical micrograph of etched alpha-particle tracks on CR-39 surface. The overall dimensions of the image are 262×152 μm^2	131
Figure 81: Reconstructed image from experimental track data for different source shapes with pinhole (top row) and 20×20 CAI mask (bottom row). All images are of	

10 cm side (in the source plane). 1-point source for pinhole (a1) and mask (a2); 2-point source for pinhole (b1) and mask (b2); ridge source for pinhole (c1) and mask (c2) ..	133
Figure 82: The hollow anode using in NX2	134
Figure 83: experimental arrangement for 3-D fusion imaging in NX2	135
Figure 84 Trajectories of 3 MeV protons (which impinge on coded mask at 45 degree) in plasma pinch magnetic field.....	137
Figure 85: Magnitude of mean displacement of apparent source position δzm versus actual z position of source within the pinch for mask-400 at 45 degree.	139
Figure 86: elongation of image for 45° view.....	140
Figure 87: reconstructed images: Camera-1 to Camera-4 at 45° and Camera-0 at 0° .	141
Figure 88: insert tapered anode with 2 mm diameter hole (middle) and complete anode shape (right).....	142
Figure 89: mask-1365 (left) and the pattern (right).....	143
Figure 91: mask-4680 (left) and the pattern (right).....	144
Figure 90: both masks were housed on the base	144
Figure 92: Trajectories of 3 MeV protons (which impinge on coded mask) in plasma pinch magnetic field for mask-1365 and mask-4680 at side-on.....	145
Figure 93: Magnitude of mean displacement of apparent source position δzm versus actual z position of source within the pinch for masks at side-on.....	147
Figure 94: the position of masks in NX2 chamber.....	148
Figure 95: plastic Scintillator with lead sheet in front, Cesium-Iodide and Barium-Fluoride, with included PMTs.	149
Figure 96: YOKOGAWA 4-channels digital oscilloscope	149
Figure 97: The experimental setup on NX2: Be detector, pinhole x-ray imaging and admixture gas valves.	150

Figure 98: Beryllium filter transmission for x-ray starts from ~ 700 eV	151
Figure 99: x-ray images in pure deuterium. While the left image shows a strong pinch at middle of anode, the right image does not have any particular shape.....	152
Figure 100: recorded signals: current derivative, plastic Scintillator (neutron and hard x-ray), CsI and BaF ₂ (hard x-ray) signals.	152
Figure 101: anisotropy versus neutron yield with pure deuterium gas at 13 mbar.....	153
Figure 102: hard x-ray (integral of CsI signal) versus neutron yield with pure deuterium gas at 13 mbar.....	154
Figure 103: hard x-ray (integral of CsI signal) versus neutron anisotropy with pure deuterium as working gas at 13 mbar	155
Figure 104: recorded signals: current derivative, plastic Scintillator, CsI and BaF ₂ signals for the best shot.	156
Figure 105: x-ray image in pure deuterium for the best shot shows a dense and strong pinch.	156
Figure 106: saturation of x-ray image in krypton seeding gas admixture with beryllium filter thickness: 20 μm (left) and 50 μm (right).....	157
Figure 107: x-ray transmission of combination filter: titanium (12.5 μm) and beryllium (50 μm) for admixture gas.....	158
Figure 108: x-ray images in D ₂ +2% Kr admixture gas with combination filter: titanium (12.5 μm) and beryllium (50 μm).....	158
Figure 109: x-ray images in pure deuterium gas with combination filter: titanium (12.5 μm) and beryllium (50 μm).....	159
Figure 110: anisotropy versus neutron yield with admixture gas.....	159
Figure 111: hard x-ray (integral of CsI signal) versus neutron anisotropy with admixture gas.....	160

Figure 112: hard x-ray (integral of CsI signal) vs. neutron yield with admixture gas .	160
Figure 113: x-ray image in pure deuterium for one of the best shots shows a multi-hot spot pinch.....	162
Figure 114: recorded signals: current derivative, plastic scintillator, CsI and BaF ₂ signals for one of the best shot.	162
Figure 115: anode tip has been changed after 100s shots.....	164
Figure 116: The reconstructed image of plasma focus fusion source from mask-1365 (91×90 pixels). The actual fusion source position must be zoomed-in, calibrated and registered by referring to anode tip place (left down image).	165
Figure 117: 5.5 MeV α tracks on the detector after 6 hours etching	166
Figure 118: Am-241 α source (left) with 9 mm actual diameter and 14 mm case diameter. It is covered partially by a set of metal mesh. The reconstructed image (right) shows clearly the detail of the source and mesh.....	167
Figure 119: α tracks (upper size) and the D-D reaction proton tracks (lower size) on the detector (after 12 hours etching) can be distinguished very clearly.	168
Figure 120: Reconstructed image of mixed α and proton tracks.....	168

List of tables

Table 1: Main parameter of NX2 Plasma Focus	24
Table 2: characteristic of CR-39 elements for stopping power calculation	59
Table 3: minimum energy of proton and deuteron to pass the Kapton filter.....	61
Table 4: Track number and SNR for CAI and pinhole system from experimental results and theoretical calculation	75
Table 5: Beryllium layers (each 1 mm) contribution to ^6He production	95
Table 6: Sum of frequencies of β particle originating in each 1 mm layer of the Be cell entering a Xe cell and depositing energy $> 2.5\text{keV}$	95
Table 7: Counts from layers for a parallel neutron beam fluence of 1 neutron/cm ²	98
Table 8: Efficiency and anisotropy for different deuteron energy	104
Table 9: Changing percentage of efficiency and anisotropy by scattering effect	105
Table 10: average neutron count and anisotropy for different distances of Pyrex	118
Table 11: Track number and SNR for CAI and pinhole system from Experimental results.....	132
Table 12: Displacements of apparent proton source position versus emission angle, for trajectories which impinge on coded mask at 45 degree. Actual source position is mid- pinch: $z = 5\text{ mm}$	138
Table 13: proton count and yield for five coded aperture imaging cameras	140
Table 14: Displacements of apparent proton source position versus emission angle, for trajectories which impinge on coded mask at side-on. Actual source position is mid- pinch: $z = 5\text{ mm}$	146

Chapter 1

Introduction to Plasma Focus

1 Introduction to Plasma Focus

1.1 Structure and Working Principle

The Plasma Focus (PF) device is a pulsed electrical discharge in different gases which was discovered independently by Mather (1961) [1] and Filippov (1954) [2] in 1965. Although these two systems were different in terms of constructional geometries, they were very similar as regards the dynamics of the current sheath and in the scaling laws [3] for neutron yield, X-ray and other kinds of emission. Since then PF devices have been built with energies ranging from a few joules to a few mega-joules with a consequent variation in the physical size. The main difference between Mather-type and Filippov-type is in the electrode dimension and the aspect ratio (diameter/length) of the anode, as is shown in Figure 1. The Mather-type device has an aspect ratio less than 1 (typically 0.25), while for the Filippov-type device the typical aspect ratio is around 5.

Physicists are interested in the Plasma Focus because of the high $n\tau$ value for the pinched plasma (where n is the particle density $\sim 10^{19} \text{ cm}^{-3}$ and τ is the confinement time $\sim 50 \text{ ns}$, giving $n\tau = 5 \times 10^{11}$), and bursts of fusion neutrons when operated in deuterium. However to reach ignition in a thermonuclear reaction requires fulfilling the Lawson Criterion of $n\tau$ greater than $3 \times 10^{14} \text{ cm}^{-3}$. Also the hot ($\sim 1 \text{ keV}$) dense plasma emits abundant soft and hard x-rays, especially when operated with high-Z gases like neon or argon.

The PF has a very complex behavior and exhibits a rich variety of plasma phenomena. The PF is essentially a two dimensional Z-pinch formed on, or near, the axis at the end of a coaxial plasma accelerator. The Filippov machine was developed as a modification of the straight Z-pinch and the Mather-type device was modified from the coaxial plasma gun. The device used for the present investigation in the Plasma Radiation Source Laboratory (PRSL)–NIE–NTU is a Mather-type PF device, called the NX2. As with most other PF devices, the NX2 is operated with the inner electrode as the anode and the outer electrode as the cathode. Operation of PF devices with reverse polarity decreases the neutron and x-ray outputs substantially [4].

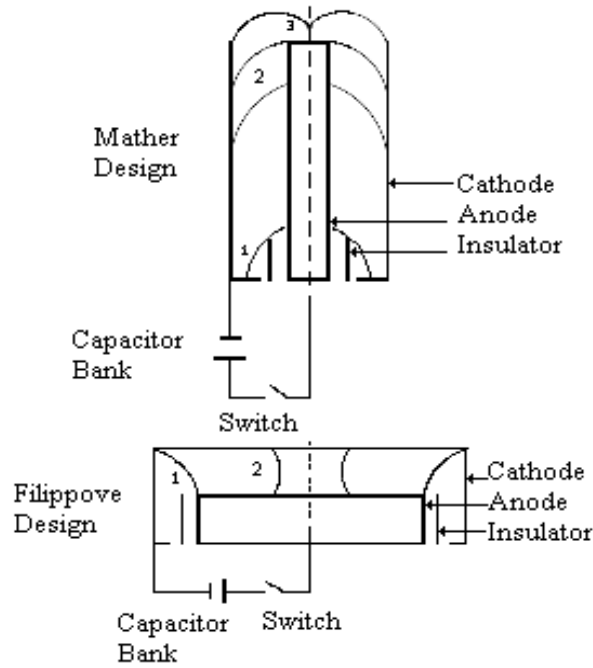


Figure 1: Schematic diagram of Mather-type and Filippov-type PF device

The PF has several sub-systems which include: the capacitor bank providing energy storage prior to each shot, the charger/power supply to charge the capacitors to high voltage, fast high-power spark-gap switches, a vacuum system for evacuating air (and any impurities) from the chamber and a gas injector system for filling the chamber with a working gas. Naturally, there are also diagnostic systems for investigation of the plasma behavior and its products.

1.2 Dynamics of Plasma Focus

In a Mather-type device, the study of Plasma Focus dynamics can be divided into three distinct phases:

- The breakdown phase
- The axial acceleration phase
- The radial phase

The three phases are illustrated in Figure 2 which shows the current and voltage signals corresponding to the different phases of the discharge.

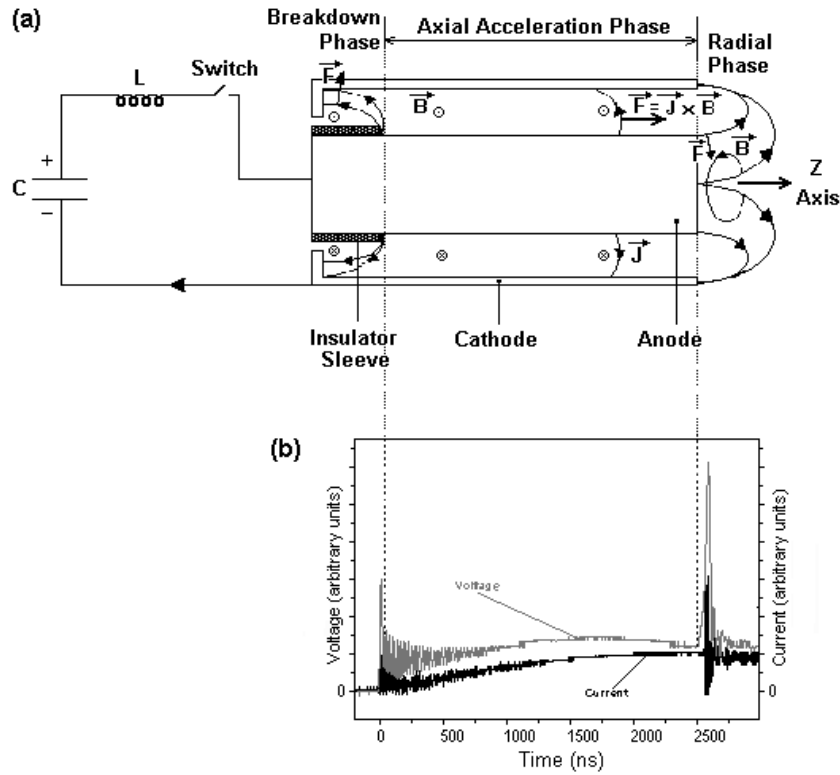


Figure 2: a) Schematic of plasma focus dynamic b) current and voltage signals

1.2.1 The Breakdown phase (Inverse Pinch phase)

When the high voltage pulse is applied between the electrodes of the PF filled with the working gas at an appropriate pressure (usually Deuterium, Neon or Argon at few mbar), an azimuthal symmetric electrical discharge will be initiated along the surface of the insulator. This discharge is preceded by an electrical breakdown phase which generates an initial plasma configuration through which the discharge current can flow. According to Paschen's law [5] the values of the static breakdown voltage in any PF device operating with 1 to 10 mbar deuterium is lower than 1 kV DC voltage, which is smaller than the usual working voltage (10 to 60 kV). Therefore the breakdown is always a highly overvoltage phenomenon.

The development of the high current discharge depends on factors such as the voltage applied across the electrodes, the initial gas conditions [6, 7] and the parameters of the electrode-insulator configuration [8].

For a given electrode-insulator configuration, an applied voltage at a low pressure (e.g. 1 mbar) will cause current flow to develop throughout the whole inter-electrode volume [9]. On the other hand, at high pressure (e.g. 50 mbar), the applied voltage will result in radial filamentary current discharged between the coaxial electrodes [10]. At the optimum initial gas pressure, the gas starts to breakdown along the insulator surface after a delay of a few nanoseconds [6].

Due to the initial high dielectric constant and high surface resistivity of the insulator, free electrons in the vicinity have a tendency to drift towards its surface and hence create a negative potential there with respect to the electrodes. To ensure a symmetrical breakdown, the polarity of the inner electrode (anode) should be positive so as to concentrate the electrons at the insulator surface. Further electrons created by field emission from metallic edges and by ionization of the ambient gas are accelerated along the radially directed electric field lines towards the insulator surface. As the discharge develops, the surface is charged gradually while the highest electric field remains at the leading-edge of the discharge. Due to this effect, an ionization front propagates along the insulator surface until it reaches the conducting anode.

Another mechanism is based on the electron avalanche principle. The initial free electrons in the vicinity of the electrodes are accelerated toward the anode due to the electric field, which strike the insulator, causing more electrons to be emitted from it. This causes a buildup or avalanche of electrons on the insulator surface, resulting in a sliding discharge across it.

When the sliding discharge reaches the end of the insulator, both electrodes are connected by weak current filaments which blend to form a uniform current sheath within 100 to 300 ns. “Double-layered structure” in DPF consists of a current sheath (CS) and a shock wave in front [11, 12]. The current sheath subsequently lifts off from the insulator surface in an inverse pinch manner, propelled by the $\vec{J} \times \vec{B}$ force (see Figure 2) which indicates the end of the breakdown phase.

1.2.2 The Axial Acceleration (Run Down) phase in Mather geometry

This phase begins with the magnetically-driven current sheath accelerating along the axial direction (z-axis) and ends when it arrives at the open end of the inner electrode (anode) (see Figure 2). The magnetic field is produced by the electric current which flows from the central electrode (anode). The radial component of the $\vec{J} \times \vec{B}$ force pushes the current sheath towards the outer electrode (cathode) while the axial component pushes it along the electrodes. As the $\vec{J} \times \vec{B}$ force has a $\frac{1}{r^2}$ dependence, the axial component is stronger near the anode, resulting in a higher acceleration close to the anode. This gives rise to a current sheath that is parabolic [13] (rather than like a flat disc) and is thinner in the region near the anode [14].

Two important factors affect the subsequent formation of the hot dense plasma pinch. Firstly, the current sheath should reach the axis at the time close to the first maximum (near 1/4 cycle time) of the discharge current. This condition represents the optimization of the energy transfer from the capacitor bank to the pinch plasma. This is important in the sense that it does not only determine the time instant when the plasma begins to collapse, but also the final current sheath profile that directly affects the focus/pinch efficiency of the plasma. The first condition is a requirement representing the optimization of the energy transfer from the power source (capacitor bank) to the pinch plasma. For DPF devices, optimization studies require the discharge current to reach its maximum value near the end of the axial phase for an optimum energy transfer through reasonable choice of central electrode dimensions, filling gas pressure, charging voltage, etc. Lee et al. [15] have lumped these various parameters into a single parameter called the “drive parameter”, $\frac{I_p}{a\sqrt{\rho}}$ (where I_p is the peak discharge current driving the PF, a is the anode radius, and ρ is the ambient gas density) to investigate the neutron optimized regime from various DPF devices.

The second condition is that the structure, i.e. the (r, z) profile, of the current sheath should have certain characteristics for a good focusing effect. This condition requires smooth and uniform axially symmetric current sheath (r, z) profiles with optimized angle between the current sheath and the inner electrode. The pattern of the driving current determines the complete structure of the sheath: the more uniform the pattern, the smoother will be the current sheath.

1.2.3 The Radial Collapse phase

At the end of the axial phase, the current sheath pinches inwardly sweeping around the end of the inner electrode and finally collapses in about 50-200 ns, depending on the DPF machine characteristics (mainly the anode radius) [15, 16]. The radial phase, as the final stage of plasma compression, plays the most important role in the plasma focus evolution due to its extremely high energy density, its transient character and for being a rich source of phenomena such as the emission of intense electromagnetic radiation, high energy particle beams, and a burst of nuclear fusion products (i.e. neutrons and protons). The radial phase can be divided into four sub-phases, compression phase, quiescent phase, unstable phase and decay phase.

Compression phase:

The compression phase begins when the current sheath reaches the top of the anode and ends when the plasma column reaches the minimum radius with maximum plasma density ($n_e \sim 10^{19} \text{ cm}^{-3}$) [17]. During this phase the current sheath sweeps around the end of the central electrode, and collapses radially with azimuthal symmetry, with a non-cylindrical, funnel-shaped profile. The main heating mechanism for the plasma before the front of the current sheath collapses along the z-axis is shock heating and the ions with temperature around 300 eV play more important role than the electrons with temperature of ~ 100 eV [18]. After the transformation of plasma structure to plasma column, at the end of the compression phase, Joule heating becomes the main heating mechanism [19] as the maximum discharge current flows through the pinched plasma column, and the kinetic energy of the collapsing current sheath is converted into the thermal energy of the plasma. The plasma column is then further compressed adiabatically [18] to form the final focus. Peacock et al [20] observed the onset of the Rayleigh-Taylor instability (instability of an interface between two fluids of different densities) about 30 ns before the end of this phase. This type of instability is damped out when the radial velocity is diminished.

Towards the end of this phase the magnetic field starts diffusing into the plasma column. The quick and complete diffusion of magnetic field into the focused plasma column leads to an anomalously high plasma resistance. Also, due to the faster speed of the collapsing current sheath in the compression phase, the rate of change of inductance of the plasma is increased [21]. This induces an electromotive force and contributes to the sharp spike and the voltage signal. The sharp voltage spike and the current dip are typical features of the plasma column's increasing impedance. The final electron temperature will be around 1 keV. The maximum values of electron density and electron (ion) temperature have been determined by laser scattering [19], spectroscopy [22], interferometry [23] and other methods. According to Bennett equilibrium the final temperature should depend only on the current I and the line density N , i.e. $T \sim I^2/N$. This shows that the lower the line density, the higher the temperature of the pinched plasma column.

Quiescent phase:

The quiescent phase marks the beginning of the expansion of the pinch plasma column. During this phase, the rate of radial expansion is limited by the confining magnetic pressure. The increasing length of the plasma column results in an increase in the plasma resistance and inductance, which in turn leads to sharp increase in voltage signal due to the changing current. The sharp change in plasma inductance induces an electric field in the plasma column. This electric field accelerates the ions and electrons in opposite directions. The relative drift velocity between the electrons and ions increases and exceeds the electron thermal velocity, which is the condition for the onset of micro-instabilities (such as the electron cyclotron and various forms of beam-plasma instability) in the focused plasma column. At the end of the quiescent phase, the $m = 0$ instability develops rapidly due to the increase of electron temperature. The plasma column is locally compressed again due to the $m = 0$ instability. The pinch lifetime, which is defined as the time between the first compression and the instant when the $m = 0$ instability occurs, affects the emission characteristics of DPF device: the shorter the pinch lifetime, the higher the neutron yield for fixed DPF discharge parameters [24].

Unstable phase:

The sub-phase during which these micro-instabilities occur and disrupt the pinch plasma column is called the unstable phase. The unstable phase is the richest stage of the plasma focus evolution in terms of associated phenomena like soft and hard x-ray emission, fast deuterons, electrons and D-D reaction products (neutrons and protons) when operated in deuterium. The induced electric field is enhanced locally due to growing $m = 0$ instabilities. Therefore, electrons are accelerated towards the inner electrode and ions in the opposite direction (when operated with a positive polarity central electrode), and a fairly large amount of impurities are injected into the plasma column due to the bombardment of the anode by accelerated runaway electrons. The “explosion” proceeds sequentially along the plasma column. This disruption of the plasma column continues until the whole plasma column has completely broken up, and the plasma density decreases.

Decay phase:

The last phase of the radial collapse and also the last stage of the plasma focus dynamics is the decay phase. The plasma density drops below $2 \times 10^{17} \text{ cm}^{-3}$. During the decay phase a large, hot and thin plasma cloud is formed due to the complete breakup of the plasma column. This plasma cloud emits a large amount of Bremsstrahlung and line radiation. The neutron pulse which started from the beginning of the unstable phase reaches its peak during this phase.

1.3 Plasma Instabilities

Over the several decades of plasma physics research, more than 100 types of plasma instability have been identified, and more than 20 of these have been observed in plasma focus devices. Plasma instabilities can be divided into two general groups Magneto-Hydro-Dynamic (MHD) instabilities and kinetic instabilities. Some of the more important instabilities in PF device are described here:

1.3.1 Rayleigh-Taylor Instability

In plasma focus device, the Rayleigh-Taylor instability is happen because the magnetic field as a light fluid supports the dense and incompressible plasma. During the radial compression, the interface between the plasma and magnetic field will be unstable [25]. The rapid compression of relatively dense plasma heats the plasma to a high temperature ($\sim 1 \text{ keV}$). Then the plasma collapses toward the axis with approximately constant acceleration and flutes on the outer surface of the plasma. Rayleigh-Taylor instability prevents the compression of the plasma column from being radially uniform. The magnetic field then begins to diffuse into the plasma column, leading to an increase in the inductance of the system due to the increasing density of the plasma within the pinch column.

Sausage ($m = 0$) Instability:

A radial perturbation of the plasma column may causes it to become constricted in a place where the magnetic pressure exceeds the kinetic pressure [26]. Then the current flowing in the smaller cross sectional area produces a stronger magnetic field around the constricted region which exerts greater inward pressure on the column, and subsequently causes it to become still more constricted and unstable. Rapid change in radial magnetic field produces a strong axial electric field ($E_z \equiv dB_\theta / dt$) which

accelerates the ion-electron in plasma column up to hundreds of keV in opposite directions. The high energy particles are the source of multiple radiation emission.

Kink ($m = 1$) Instability:

The bending of the plasma column causes the poloidal field on the inner edge of the bend to become stronger than that at the outer edge, and the associated larger magnetic pressure on the inner edge causes the column to bend further. Figure 3 shows $m = 0$ and $m = 1$ instabilities.

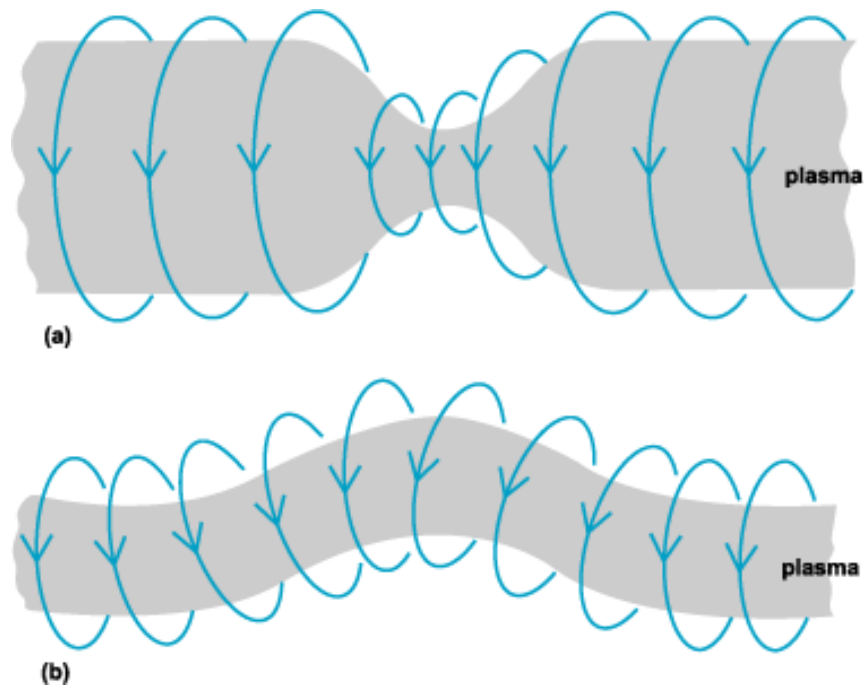


Figure 3: sausage ($m = 0$) and kink ($m = 1$) instability

1.4 Plasma Focus Models

Based on properties of the electrical system and the working gas, the dynamics of the current sheath can be understood in terms of a few different models. The main differences between these models can be identified by their approach to representing the accumulation of mass near the driving magnetic piston.

1.4.1 Snow Plow model:

It is presumed that a very thin layer swept up by the magnetic piston and the plasma balance the driving magnetic pressure. In the snow-plow model, the arbitrarily large compression of mass is assumed as it is swept up and accumulated in a thin layer which is considered to be highly ionized, resulting in an infinite electrical conductivity and thus requiring no energy to ionize the fluid. This model has been applied in studying the shock phenomena associated with current sheath propagation [27].

1.4.2 Quasi-Steady-State model:

In this model, the advancing shock front overtakes the stationary fluid, and this overtaken fluid is then uniformly distributed between the shock front and the magnetic piston. This region of shocked fluid moves at the piston velocity [28].

1.4.3 Slug model:

The simplest physical model assumes that the incompressible mass of the current sheath increases as encountered ambient gas is swept into the sheath. The thickness of the sheath increases. In this model [15], the magnetic pressure drives a shock wave ahead of it, creating a space for the magnetic piston (current sheath) to move into. Lee has used this model for the radial phase of Plasma Focus dynamics, as is shown in Figure 4 [29]. In slug model a zero length pinch starts at $r = a$ (anode radius) and elongates as it pinches inwards. The collapsing magnetic piston at position r_p pushes ahead of it a radially converging structure at position r_s . With assuming a constant speed in radial direction, the compression time (t_{comp}) is depended to anode radius, when the shock converges on the axis and piston reaches to r_c . After hitting the shock wave to the axis, a reflected shock propagates outward until reaches to the piston at position $r_p = r_{\text{min}}$ and then the plasma column pinches for a while (t_f) before explosion.

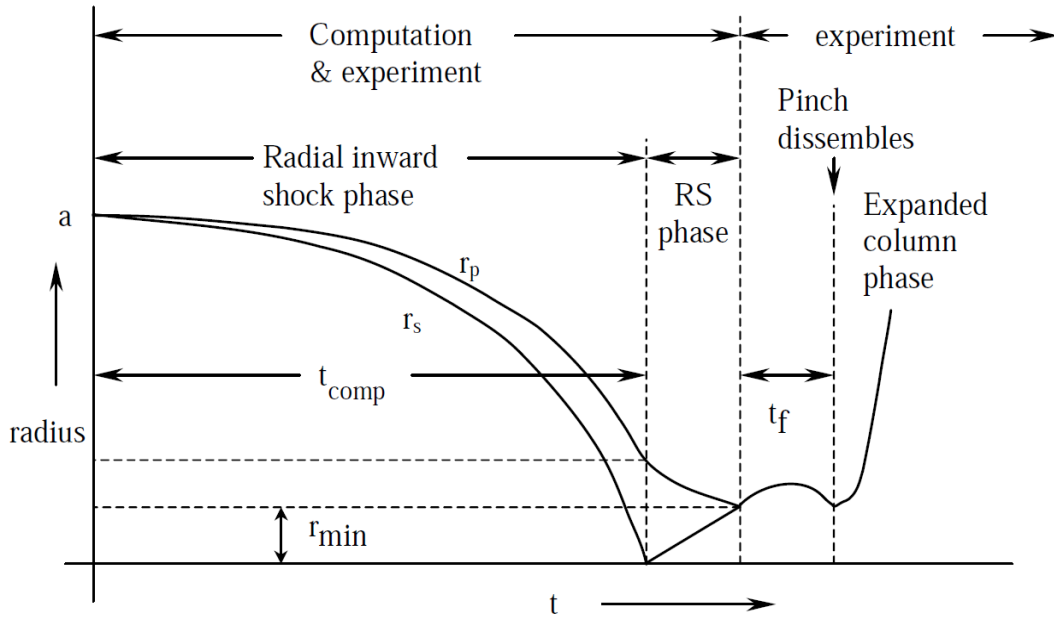


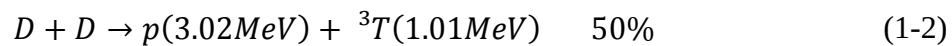
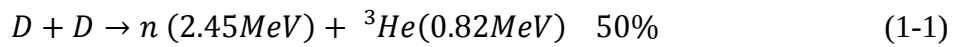
Figure 4: Radial trajectories of shock (r_s) and piston (r_p) versus time in collapse phase (Slug model)

1.4.4 Gas dynamic model:

This model tries to include the non-steady-state gas-dynamic motion between the shock front and the magnetic piston. The mass is continuously overtaken by the advancing shock and is distributed between the shock and current sheath.

1.4.5 Ion beam acceleration and neutron production models:

The ${}^2\text{H}(\text{d},\text{n}){}^3\text{He}$ and ${}^2\text{H}(\text{d},\text{p}){}^3\text{H}$ reactions are of interest for fusion energy applications when deuterium is the working gas. The D-D fusion reactions are:



Classically, the energy required to overcome the Coulomb barrier which is due to electrostatic repulsion, so that D-D fusion takes place is:

$$E = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{R_0} \quad (1-3)$$

Where Z_1 and Z_2 are the atomic number of two nucleus and R_0 is the distance between nuclei centers at which the short-range nuclear force exceeds long-range repulsive Coulomb force. However, according to quantum mechanics, there is a finite probability

that two nuclei will fuse even though they do not have sufficient energy to overcome the Coulomb barrier. The penetration probability for a given nuclear fusion reaction increases as the energy of the fusing nuclei approach the top of the barrier. Fusion reactors such as Tokamak are expected to operate at temperatures in the range $kT = 1$ to 30 keV [30].

The principal mechanisms for proton and neutron production in the plasma focus can be categorized as thermonuclear and non-thermonuclear (beam target).

Thermonuclear reaction:

For the thermonuclear mechanism the ions are approximately in thermal equilibrium with one another. If the thermal plasma is considered to have an isotropic Maxwellian ion-energy distribution, then the energy spectra are independent of proton emission angle, and FWHM of proton line will be significant. Concerning the proton fluence anisotropy, for a stationary isotropic Maxwellian energy distribution, the proton intensity is spatially isotropic. One characteristic of a thermonuclear reaction is that the reacting center of mass of any two deuterons should be statistically stationary in the laboratory frame of reference. This is in contrast to the case of an accelerated beam striking a target, in which case the reaction center of mass has a directed momentum equal to the incident particle momentum in the beam. The resulting velocity of the reacting system results in a non-isotropic velocity distribution (in the laboratory frame).

An accelerated ion beam incident upon a cold target cannot produce useful (positive net output) thermonuclear power because the irreversible energy loss of slowing down in the target is always greater than the reaction energy produced. In this case the bulk of the kinetic energy of the impinging beam is dissipated uselessly by ionization, radiation, and energy transfer to the atomic electrons of the target [31].

Approximately one-half of the thermonuclear reactions produce protons, so the number of protons (and neutrons) in thermal mechanism, is:

$$\begin{aligned}
 Y_p = Y_n &= \left(\frac{1}{2} \times \frac{1}{2}\right) n_{d-pin}^2 \langle \sigma v \rangle_{DD} \times volume \times reaction\ time \\
 &= \frac{1}{4} n_d^2 \langle \sigma v \rangle_{DD-th} (\pi r_{pin}^2 h_{pin}) \tau
 \end{aligned} \tag{1-4}$$

where $n_d \approx 3 \times 10^{18} \text{cm}^{-3}$ is the deuteron density inside the pinch, $\langle \sigma v \rangle \approx 2 \times 10^{-22} \text{cm}^3 \text{s}^{-1}$ is the cross-section velocity product of D-D fusion reaction at ~ 1 keV.

Also $r_p \approx 0.1$ cm and $h_p \approx 1$ cm are the pinch radius and height respectively for NX2 plasma focus device. By replacing the $\tau \approx 50$ ns as neutron emission time, the neutron yield will be in order of 10^6 .

This model was rejected very soon with observing the relatively high anisotropy in neutron emission. Moreover, the actual neutron yield is much higher (2 order of magnitude) than calculated yield based on plasma temperature ($\sim$ keV) in pinch plasma.

Moving Boiler model:

This model, based on thermonuclear mechanism, tries to explain the anisotropy in neutron production where it is assumed the plasma centre of mass is moving along the axis of plasma focus device [32]. The required velocity to produce anisotropy of 1.5 is about 5×10^8 cm/sec. However, it is in contrary to many experimental results such as optical observation which indicates a velocity of 5×10^7 cm/sec. Moreover, regarding to the neutron emission interval time (50-100 nsec) gives us a fusion source length around 25-50 cm which is very unrealistic (the average length of fusion source in NX2 is about 1 cm). In addition, the neutron energy spectrum shows a large range of energy (2.2-2.9 MeV) which indicates the involvement of high energy deuteron in neutron production.

Beam target mechanism:

In order to explain the neutron energy spectrum, several models for acceleration mechanisms have been considered for energetic electron and ion (in our case, deuteron) beams. However, mechanisms for ion acceleration and the formation of intense ion beams have still not been enough clarified yet. Some of the proposed mechanisms are fluid-like and others are kinetic in nature. An important test of any hypothetical theory is whether total axial momentum is conserved during the ion beam formation.

With regard to the ion beam, Bernard et al. [33] Claim that it is the main source of neutrons, as the deuterons bombard both the dense gas and the ionized but relatively cold gas bubble between the ionization-front and the shock-front.

The non-thermonuclear (beam target) model considers a high current of charged particles accelerated in an electromagnetic field giving a different proton and neutron production mechanism. In this mechanism the accelerated deuterons strike ions at rest. The beam-target yield is written in the form [29]:

$$Y_p = Y_n = \frac{1}{2} n_b n_i \sigma_{DD-b_t} v (\pi r_{pinch}^2 h_{pinch}) \tau \quad (1-5)$$

where $n_b \approx 10^{16} \text{cm}^{-3}$ is the number of beam ions per unit plasma volume, $n_i = n_d \approx 3 \times 10^{18} \text{cm}^{-3}$ is the target ion density, $\sigma \approx 0.02 \text{ barn} = 2 \times 10^{-26} \text{ cm}^2$ is the cross-section of D-D fusion reaction and $v \approx 3 \times 10^8 \text{ cms}^{-1}$ at $\sim 100 \text{ keV}$. So the total neutron yield will be in order of 10^8 which is matched with experimental result from NX2.

Gyrating Particle Model (GPM):

In a generalized beam-target model GPM, the trajectories of trapped high energy deuterons are simulated in the time varying structure pinch plasma column. It is assumed that the deuteron source is a series of local point source positioned on the axis. The main difference between conventional beam target model and GPM is the stretching of deuteron path by gyro-motion in GPM against a straight and un-scattered axial beam in origin version. Jager shows the results of this new model [34] in comparison with spectrally and spatially resolved measurements of fusion protons (pinhole imaging) and accelerated deuterons in POSEIDON plasma focus device ($W_0 = 280 \text{ kJ}$, $U_0 = 60 \text{ kV}$, Mather type).

Cross Field Acceleration:

Bernstein and Comisar [35] proposed another generalized beam-target model which is assumed: (1) high energy deuterons have angular distributions consistent with a cross field acceleration mechanism and have velocity vectors distributed anisotropically in space and energy. (2) These energetic deuterons undergo fusion collisions primarily with stationary target ions. The principle assumption is that the azimuthal magnetic field diffuses rapidly toward the axis of self-pinching discharge.

Plasma Diode:

The study of ion-induced pinch and enhancement of ion current by pinched electron flow in relativistic diodes was done by S.A. Goldstein and R. Lee [36]. In this model, the initial motion of the ions, accelerated from the anode plasma, forms an ion sheath which modifies the electric field. When an electron enters the ion sheath with grazing incidence, it will be reflected back into the diode by the unaffected magnetic field but reduced electric field. The necessary ion-sheath thickness to provide such electron reflection is reached in times which are short enough ($< 1 \text{ nsec}$) to explain the fast collapsing pinch. The electric field in the diode is of order $\sim 1 \text{ MV/cm}$.

Double-layer pulse current model:

H. Hora et al. developed a new model [37] for the mechanism of the plasma focus based on the assumption that the space charge term in Maxwell's equation (usually neglected) is dominating due to the double layer of discharge. The model assumes that the current pulse is being produced during the several nanoseconds in the focus by the charge separation in the double layer between the collapsing plasma and the remaining gas in the focus.

Surfing model:

The simulation of high energy proton production by fast magneto-sonic shock waves in pinched plasma focus was done by Y. Mizuguchi et al. [38]. It is focused on the shock formation and the shock acceleration during the pinched current. Some protons are trapped in the electrostatic potential produced near the shock front can be accelerated to a few MeV by surfatron acceleration mechanism.

Collective focusing model:

Recently the collective focusing of intense ion beam pulses model was proposed by M.A. Dorf et al. [39]. The main concept is that a weak magnetic lens (several hundred gauss) can provide a strong focusing of an intense ion beam pulse carrying a neutralizing electron background. Note that for a single-species non-neutral ion beam, a several Tesla magnetic field would be required to achieve the same focal length. Collective focusing will only occur if no background plasma or secondary electrons are present inside the lens.

1.5 Plasma Focus Radiation:

Depending on working gas, the plasma focus device can be used as a strong source of various radiations such as high energy neutron and proton (fusion products which has been already explained) different range of energetic electron-ion and intense electromagnetic wave such as ultra-violet and x-ray.

1.5.1 Ion-electron production

Some of the most interesting applications of energetic ion beams are in materials science, such as ion implantation, surface modification, thin film deposition and semiconductor doping. Ion beams are also of importance for the production of short-

lived isotopes for medical imaging. The high energy electron beam interaction with the metallic anode is the source of hard x-ray emission.

The disruption of plasma column in radial phase mostly because of $m = 0$ instability, anomalous resistivity and fast magneto-sonic shock waves have been proposed as the source high electric field which accelerates the ion-electron in opposite direction to gain their energy from tens of keV to few MeV. Different theories have been proposed for acceleration mechanisms [34, 40], however it is not well understood and none of them satisfies the experimental results. Even it has not been proved that the mechanisms for the acceleration of ions and electrons are the same. Most measurements on electrons and ion beams show the independency of the peak energy from charging voltage and total stored energy of capacitor bank.

For the NX2 plasma focus device, the deuteron energy range has been measured by different methods such as the nuclear activation of low-Z elements such as B, C, and N. Examples of such studies are the activation of graphite targets via the reaction $^{12}\text{C}(\text{d},\text{n})^{13}\text{N}$ [41], and boron-carbide targets via the previous reaction and also $^{10}\text{B}(\text{d},\text{n})^{11}\text{C}$ [42]. Magnetic spectroscopy represents another important technique for investigation of ion beams [43]. It has even been found that there can be a high energy deuteron beam in backward direction (towards the anode) [44].

1.5.2 X-ray production

Hard X-ray radiation:

Moderately high energy (several keV) electrons striking the copper anode producing bremsstrahlung and characteristic radiation is the main mechanism of hard x-ray production in plasma focus devices. Measurements of x-ray energy spectra in different devices have revealed a very broad range of energies covering the range 10 to 500 keV.

Bremsstrahlung (Free-Free) radiation:

Acceleration of the electrons in the ion coulomb field causes a continuous emitted radiation as the initial and final state is continuous.

Recombination (Free-Bound) radiation:

Capturing a free electron into ion bound state causes the excess energy which will be emitted as a photon. The chainlike emission spectrum for each bound state has a low-frequency cut-off, the minimum energy for removing the electron from the bound state.

Line (Bound-Bound transition) radiation:

Electron transition between the discrete energy levels in atoms or ions gives a quantized energy photon with the energy of level difference: $h\nu = E_f - E_i$. It is known as characteristic radiation because the x-ray spectra are associated with certain series of lines, *K, L, M, etc.*

1.6 Radiation Measurement

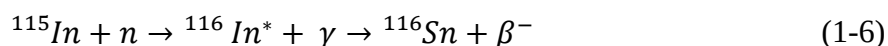
In this section, different radiation measurement techniques which are often employed in the plasma focus labs are presented.

1.6.1 Neutron Yield

As the neutron and proton yield in PF device is the same, by measuring the neutron yield, we can estimate the total protons for each shot.

Indium foil activation detector:

Natural indium consists of ^{115}In (95.8% abundance) and ^{113}In (4.2%). The activation cross-section is about 200 barns. Here detection involves three: the fast neutrons are first thermalized in steps using moderator such as paraffin or polyethylene. The thermal neutrons are captured predominantly by the ^{115}In within an indium foil, making it beta radioactive as described by the following reaction:



The activated ^{116}In has three separate half-lives: 14.1 s, 54.15 min, 2.18 s associated with the ground state and two isomeric levels. For the indium activation detector used on the NX2, only the beta decay with the half-life of 14.1 s (and with end-point energy of 3.3 MeV) are counted. The beta particles are counted by the Geiger-Muller tube detector (or plastic scintillator with photomultiplier). This detector can be calibrated by ^{252}Cf fission source (positioned temporarily at the top of the anode).

Silver activation detector:

Natural silver consists of ^{107}Ag (51.35% abundance) and ^{109}Ag (48.65%). Both isotopes have a sufficiently high activation cross section for thermal neutrons and a relatively short decay time. The activation cross-sections for thermal neutrons are 35 and 92 barns, respectively. When ^{109}Ag is activated with thermal neutrons, ^{110}Ag (half-life of 24.2 s) and $^{110\text{m}}\text{Ag}$ (half-life of 270 day) are produced which undergo beta decay to ^{110}Cd . Activation of ^{107}Ag results in the generation of ^{108}Ag with a half-life of 2.3 min. So the total activity is the sum of the activities of ^{110}Ag and ^{108}Ag . A paraffin cylinder moderates the fast neutrons, and the so thermalized neutrons activate a natural silver foil.

Bubble detector:

It is a reusable and passive dosimeter that allows instant visible detection of neutron emission. It can be used to calibrate the other neutron detectors. It consists of droplets of a superheated liquid dispersed throughout an elastic polymer. When neutrons strikes these droplets, they form small gaseous bubbles that remain fixed in the polymer and hence providing a real time and immediate visual record of the dose that is directly proportional to the number of bubbles. Neutron elastically scatters from a nucleus, e.g. ^{35}Cl , in a superheated drop; the recoil of nucleus provides energy to nucleate the boiling of the droplet. The number of neutrons per square centimeter equivalent to a dose of 1 rem for 2.5 MeV D-D neutrons from plasma focus is $\sim 3 \times 10^7 \text{ n/cm}^2$ [45].

Beryllium detector:

It works with fast neutron so it doesn't need neutron moderation. Important advantages of beryllium activation are its short decay half-life (0.807 s), maximum sensitivity at 2.8 MeV neutrons, and practically no sensitivity to the neutrons with energies less than 1 MeV (due to cross-section). It is also insensitive to hard x-ray. It employs a beryllium sheet sandwiched between two proportional counters using xenon gas. The relevant reaction $^9\text{Be} (n, \alpha) ^6\text{He}$ has a threshold energy of 670 keV and a peak cross section of about 100 mb for 2.8 MeV neutrons. The resulting beta decay of ^6He nuclei has a half-life of 0.807 s and end point energy of 3.5 MeV [46].

Helium-3 detector:

Helium-3 neutron proportional counters are sensitive to thermal neutrons and are normally used in conjunction with a moderator material. For applications where the surrounding medium acts as a moderator, the detectors can be used un-moderated. They utilize the $^3\text{He}(n, p)^3\text{H}$ reaction for the detection of thermal neutrons. The energy of reaction is carried away as kinetic energy of the products, which move in opposite direction. This detector provides an output pulse that is proportional to 764 keV (reaction Q-value) for thermal neutrons. The cross section of He-3 for thermal neutrons is 5330 barns. The cross section follows an inverse proportionality with neutron velocity up to about 200 keV. The ionization potential of helium is about 25 eV.

1.6.2 Neutron Energy Measurement

Plastic scintillator is employed for fast neutron measurements. Some of the neutrons entering the plastic scintillator will undergo elastic scattering collisions with

hydrogen nuclei (i.e. protons) which comprise part of the polymer chain in the plastic scintillator. The colliding neutron transfers some of its kinetic energy to the proton in one collision. The fraction of kinetic energy given to the proton varies between ~ 0 and ~ 1 depending on the scattering angle, and averaging one-half of the neutron energy. As a result, the scintillator material gains energy from the proton and the excited scintillating molecule will produce light photons during the de-excitation process. The light photons are then detected by a photomultiplier tube. The scintillation decay-time is approximately 2 ns for most types of plastic scintillator.

The neutron energy can be measured by its time of flight over a known distance through the use of plastic scintillator plus photomultiplier. The neutrons produced from the D-D fusion reaction are non-relativistic since they have energy of 2.45 MeV which is much less than 940 MeV (rest mass energy of neutrons). The energy resolution is given by

$$\frac{\Delta E}{E} = \frac{2\Delta t}{\tau}, \quad \tau = \frac{x}{v} \quad (1-7)$$

where Δt is the emission duration, and τ is the average time of flight of the neutron from the source to the detector. The scintillation decay-time can be neglected as it is so much shorter than the neutron emission time in the plasma focus. Then the neutron velocity can be calculated by:

$$v = \left(\frac{2E}{m}\right)^{1/2} \quad (1-8)$$

1.6.3 Proton and Ion Measurement

The ion energy and yield investigation in plasma focus device can be done by different methods. The most popular and easiest way is using a Faraday Cup to find the ion energy range. Mixing it with a magnetic system introduces the ion magnetic spectroscopy.

The most efficient (almost 100%) system for ion detection is using the nuclear track detector [47]. When a particle (in our case, proton or deuteron) crosses the nuclear track detector, it produces damages at the level of polymeric bonds within a cylindrical region extending to a few tens of nm, forming the so-called latent track. They can be made visible under an optical microscope by etching the exposed detector foils with appropriate chemical solutions. During the etching the material is removed along the

latent track at velocity V_T and isotropically at velocity V_B from the bulk of the material, if $V_T > V_B$ then conical etch-pits are formed.

1.6.4 Hard X-ray Measurement

Hard X-rays are produced by non-thermal high energy electron beams striking the anode surface. This radiation falling upon a suitably activated scintillation material produces light pulses which may be detected with photomultiplier. Of particular interest among these materials are inorganic crystal scintillators (e.g. NaI(Tl), BaF₂ and CsI) and various plastic scintillators. Inorganic scintillators, such as NaI(Tl), produce a greater light output for a given energy deposition in the crystal than does plastic and will resolve lower-energy photons from the photomultiplier noise. The hard X-ray emission rate as well as the neutron time resolved measurements can be performed using a fast plastic scintillator-photomultiplier system.

1.6.5 Soft X-ray Measurement

Soft X-rays are generated by Bremsstrahlung (free-free transition), recombination emission (free-bound transition), and line emission (bound-bound transition).

When a semiconductor detector absorbs an x-ray photon of energy $h\nu$, $h\nu/w$ electron-hole pairs are produced where w is the average energy needed to create an electron-hole pair (for Si, $W=3.62$ eV). The collected charge is proportional to $h\nu$. Silicon p-i-n diodes such as the BPX6 (with its glass window removed) are a well proven method of making time-resolved soft x-ray measurements on plasma focus devices. When used in combination with various filters (such as Ross filter pairs) this photodiode yields spectral information as well as time resolved x-ray measurements.

1.7 NX2 Plasma Focus

NX2 plasma focus [48], (Figure 5), is a high repetition rate plasma focus with the capacitance $27.0\ \mu\text{F}$, inductance $26\ \text{nH}$, and charging voltage up to $15\ \text{kV}$. The main components of the NX2 plasma focus are given in Table 1:

Table 1: Main parameter of NX2 Plasma Focus

Capacitance (C_0)	$27.0\ \mu\text{F}$ ($0.6\ \mu\text{F} \times 45$)
Charging voltage (operational)	$12\ \text{kV}$
Bank storage energy (at $12\ \text{kV}$)	$1.9\ \text{kJ}$
Inductance of circuit (L_0)	$26\ \text{nH}$
Impedance (Z_0)	$30\ \text{m}\Omega$
Anode radius (a)	$1.15\ \text{cm}$
Cathode and anode separation (b)	$2.2\ \text{cm}$
Repetition rate (maximum)	$16\ \text{Hz}$
Maximum current (at $12\ \text{kV}$)	$380\ \text{kA}$

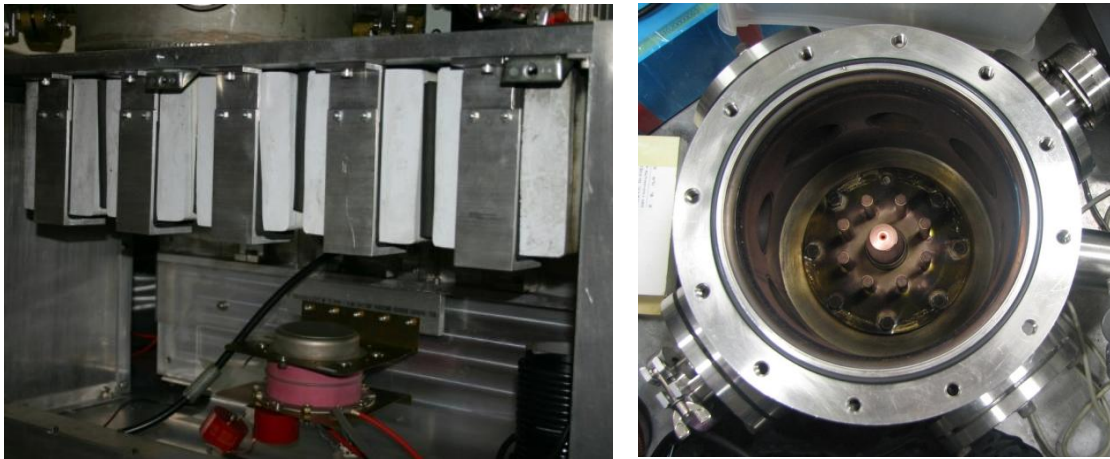


Figure 5: NX2 Plasma Focus device.

Right: main chamber. Left: capacitors and pseudo-spark-gap switch

1.7.1 Chamber

Figure 5-left shows NX2 Plasma Focus chamber which includes the discharge chamber, electrode system, vacuum nozzles, diagnostic ports, vacuum and pressure gauges, gas injection valves. The vacuum system for NX2 plasma focus consists of one single stage Leybold Rotary pump and a Varian turbo-molecular pump. The electrodes in NX2 plasma focus are made of copper. The cathode is in the form of a squirrel cage consisting of 8 rods arranged concentrically around the anode with a separation gap of 22 mm and length of 55 mm. The anode is a cylindrical tube with an outer diameter of 23 mm and inner diameter of 18 mm. The anode effective length is 40 mm, whereas the total length of anode taking into account the insulator sleeve is 100 mm. The insulator plays an important role in the symmetrical formation of the current sheath and is tightly mounted on the anode using rubber O-rings to avoid damage resulting from vibration. The back wall insulator is a nylon disc and it is essential to the structure of the focus tube.

The arrangement of the connecting plates, the switches, and the energy banks are designed to minimize the circuit inductance. The earth plate of the capacitor is directly connected to the mainframe. And its high voltage plate is directly connected to the input plate of the pseudo spark gap switch. The output plate of the pseudo spark gap is directly connected to the anode collector plate. Therefore, there are no wire connections in the discharge loop. These arrangements greatly reduced the circuit impedance (inductance and resistance).

The cylindrical section of the anode (within a 35 mm diameter and 60 mm height Pyrex insulator sleeve) has a diameter of 31 mm, which tapers down to 23 mm at the top of the anode over the last 25 mm of its length. The coaxial cathode comprises 8 copper rods arranged in a squirrel cage configuration.

1.7.2 Electric System

High Voltage Power Supply:

Four modules of ALE 802 charger are used in parallel to charge the capacitors. These units are especially designed for charging capacitors in pulsed power applications with an average power exceeding 8 kW. For intervals it can supply a power of 38 kJ/s. The maximum charging rate for the bank energy of the NX2 is therefore 16 shots per second. A high performance control module regulates the output voltage, automatically

compensating for line, load, temperature, repetition rate, and program voltage variations.

Capacitor bank:

The energy bank comprises 4 modules, each one composed of 12 high-voltage (0.6 μ F / 30kV) capacitors connected in parallel. The total capacitance is 27.0 μ F (The total capacitance was 28.8 μ F, but 3 of them have been spoiled). Each module uses a pseudo-spark gap to switch 12 capacitors connected in parallel.

Switch and Trigger:

They include low inductance (less than 20 nH), high current (150 kA max) and fast (jitter time $\approx$ 4 ns) semiconductor discharge igniter, TDI1-150/25 pseudo-spark switch with their associated heater and trigger circuit;

1.7.3 Electrical Diagnostics

Rogowski Coil:

This is a multi-turn solenoid bent into the shape of a torus which is used to measure the high current due to the capacitor bank discharge. The discharge current passes through the Rogowski coil, and the rate of the discharge current change dI/dt , produces induced emf in the coil, proportional to dI/dt . To get the current signal, the coil has to be terminated with a small resistor (not applicable in our experiments with NX2). The circuit equation can be written as (see Figure 6):

$$L \frac{di}{dt} + (R + r)i = k \frac{dI}{dt} \quad (1-9)$$

where L and R are the inductance and resistance of the coil, I is the discharge current, and i is the circuit current of the coil. Usually the Rogowski coil is operated in one of two limiting cases as described in the following. Firstly: $L \frac{di}{dt} \gg (R + r)i$ then $i = k \frac{I}{L}$ so the output voltage is:

$$V = ri = rk \frac{I}{L} = k_1 I \quad (1-10)$$

and in this case the coil acts as a current transformer. The output voltage is proportional to the transient discharge current I passing through it.

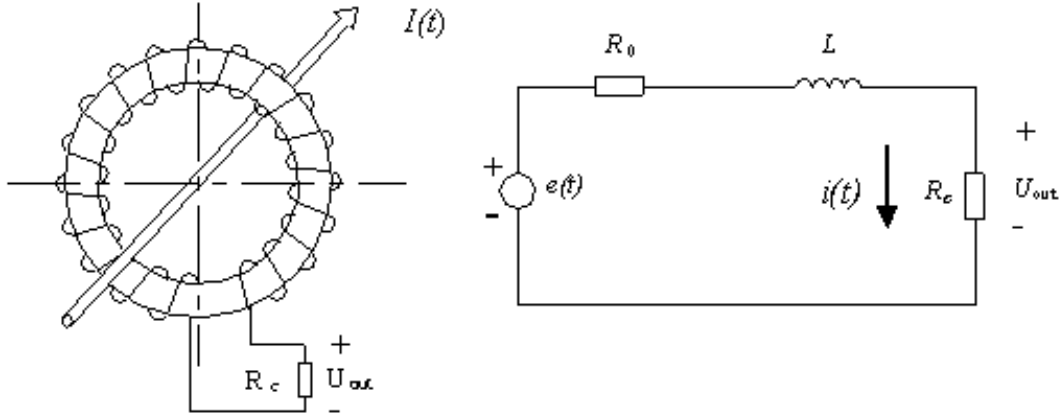


Figure 6: Rogowski coil and its equivalent circuit

In the second limiting case, $L \frac{di}{dt} \ll (R + r)i$ then

$$V = ri = \left(\frac{rk}{R + r} \right) \frac{dI}{dt} \quad (1-11)$$

and here the output voltage is proportional to the derivative of the discharge current, therefore, it can be used to measure the current derivative (dI/dt).

The Rogowski coil, when used as a current transformer, needs to be calibrated. The plasma focus, with the electrodes shorted from the end, was employed as a simple RLC circuit to perform the calibration. The calibration factor K_1 can be found using a simple formula ($k_1 = I_1/V_1$). For the actual peak current flowing:

$$I_1 = \frac{\pi C_0 V_0 (1 + f)}{T} \quad (1-12)$$

where C_0 and V_0 are known capacitance and voltage, f is reversal ratio, which is the ratio voltage peaks between successive cycles in the recorded dI/dt or *current* trace:

$$f = \frac{1}{n-1} \left(\frac{V_2}{V_1} + \frac{V_3}{V_2} + \dots + \frac{V_n}{V_{n-1}} \right) \quad (1-13)$$

where n is the number cycles in the signal and V_1 to V_n are the absolute values of the decreasing amplitude of the consecutive half cycles of the Rogowski coil signal (see Figure 7). T is the periodic time and same procedure as f applies to the determination of T , several cycles are taken to get the average of T .

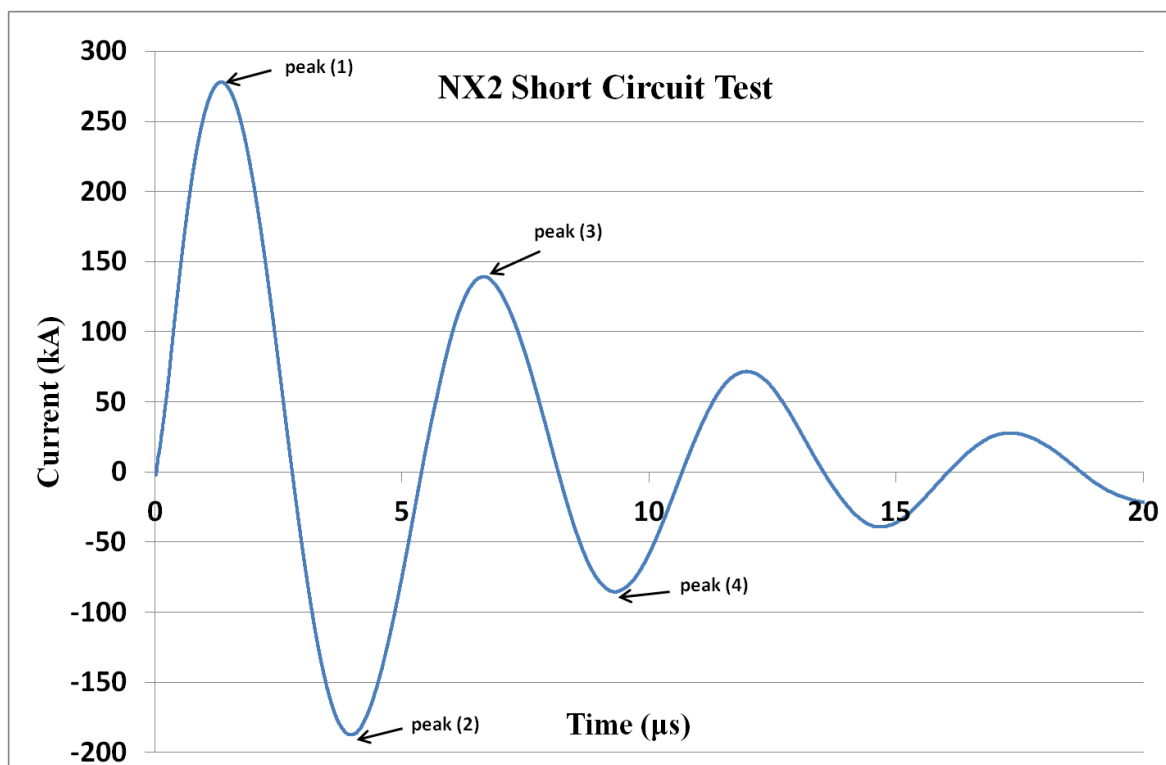


Figure 7: current trace for NX2 short circuit test. The experiment was done with Argon gas at pressure 20 mbar

Figure 8 shows two typical dI/dt (current derivative) signal traces acquired using the Rogowski Coil for the NX2 system. Sometimes we have single pinch (left) and sometimes a double pinch (right). The likelihood of each depending on many parameter such as pressure, voltage (energy), shape of anode, etc.

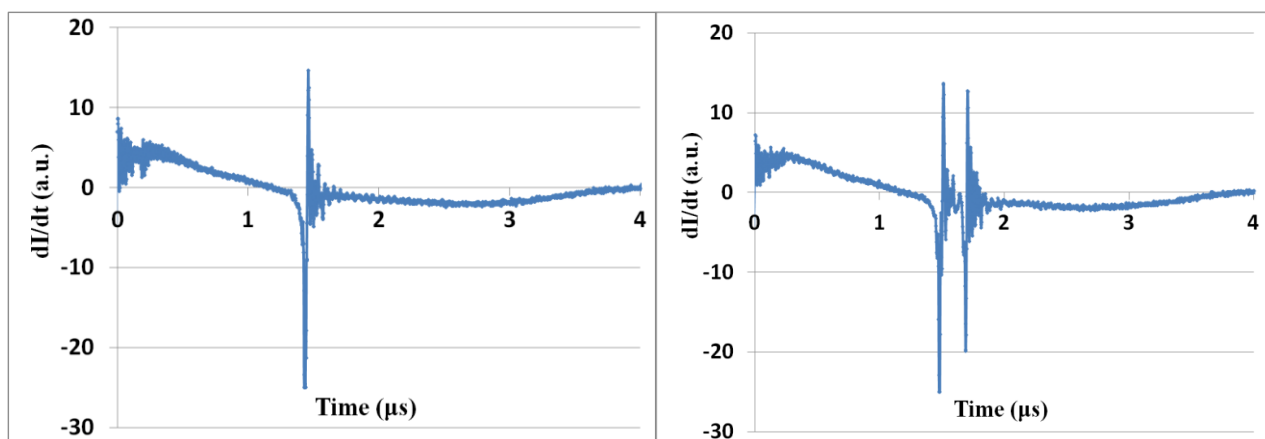


Figure 8: dI/dt RC signal by NX2 with single pinch (left) and double pinch (right)

High Voltage Probe:

The resistive dividers are commonly used as the voltage probe, because of their simple design and fabrication. The resistive divider based voltage probe is connected at the lower end of the focus tube across the electrodes. The divider consists of a series of ten resistors of $510\ \Omega$ each with a shunt resistor of $51\ \Omega$ (Figure 9); it therefore, attenuates the voltage by a factor of 100. The divider is enclosed in a copper tube of 20 mm diameter. The response time of the probe divider is about 10 ns.

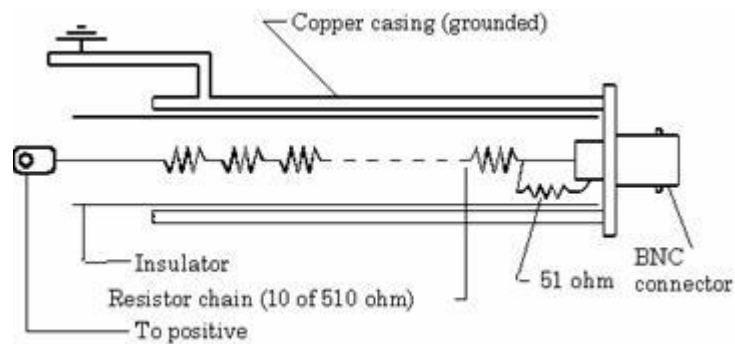


Figure 9: High Voltage Probe using in PF

1.8 Layout of Thesis:

Having presented an introduction to the Plasma Focus device, dynamics, instabilities, different models, radiation and measurement techniques, giving a brief history of employed NX2 plasma focus, the review will proceed as follow.

In section 2, the theory of Coded Aperture Imaging (CAI) will be discussed by regarding to CA families, the new Singer Set approach, Signal to Noise Ratio (SNR) concept and calculation, desired mask design and image reconstruction. Then the CAI experimental procedure will be described via mask fabrication, charge particle detection, scanning system and data analyzing. Later on, our optimized de-convolution algorithm will be introduced to correct the undesirable experiment geometry misalignment and mismeasurement. Monte Carlo numerical simulation is the last part of section 2 to show the feasibility and reliability of CAI technique for semi-extended sources.

The beryllium activation neutron detector will be dissected in section 3 which includes an overview of neutron detection, construction of Be based detector and its absolute numerical calibration with MCNP simulation. Then the experimental setup for two manufactured Be detectors will be explained followed by results of experiments on NX2.

In section 4, the other employed diagnostics such as soft x-ray pinhole imaging, Plastic Scintillator (hard x-ray and neutron detector), Cesium Iodide and Barium Fluoride (both are hard x-ray detectors) will be introduced.

In section 5, the experimental setup and results for alpha source imaging (as a more controllable source) followed by 3D plasma focus fusion source with mask-400 and side-on imaging with mask-1365 and mask-4680 will be described. Image registration technique with alpha source will be explained to achieve the actual proton source position. Then 17 selected shots with pure deuterium as working gas and 7 shots with admixture gas (Deuterium+ 2% volumetric Krypton) will be elaborated with regarding to the other diagnostics.

A brief discussion about the results and possible introduced models will be included in section 6.

Chapter 2

Coded Aperture Imaging

2 Coded Aperture Imaging (CAI)

2.1 CAI Theory

2.1.1 Overview

Coded Aperture Imaging is a technique originally developed for x-ray and γ -ray astronomy, where typical imaging problems are characterized by far-field geometry and an object comprising many point sources distributed over a mainly dark background. These conditions provide, respectively, the basis of artifact-free and high Signal to Noise Ratio (SNR) imaging. Coded aperture techniques were first proposed in 1961 by Mertz and Young. The aperture they proposed was a Fresnel Zone Plate (FZP), in theory a circularly-symmetric mask of continuously varying transmittance. The advent of more manageable apertures gave new momentum to the field. In 1968, Dicke and Ables independently pointed out that a square arrangement of randomly distributed square openings (a random array) has reasonable self-correlation properties. In 1971 the Non Redundant Arrays (NRAs) were proposed. These arrays are compact but have ideal properties only on a small field of view and contain a small number of holes, which prevents great improvements in the SNR. The difficulty was overcome in 1978, when Fenimore and Cannon introduced the rectangular Uniformly Redundant Arrays (URAs), which have an ideal Point Spread Function (PSF) and are finite. A decade later URAs were followed by the Modified URAs (MURAs), which have the additional convenience of being square. Meanwhile, a number of other apertures were discovered. A coded mask pattern of 95×95 elements formed by a replicated 53×53 MURA basic pattern (Gottesman and Fenimore 1989) is shown in Figure 10.

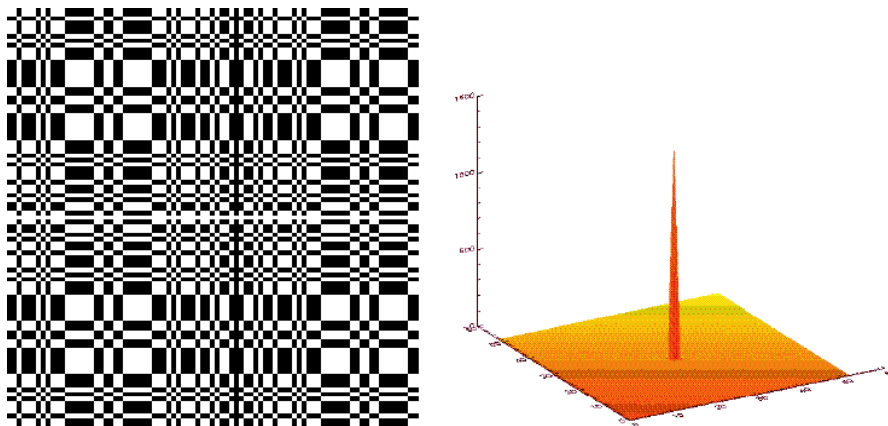


Figure 10: Coded Mask pattern of 95×95 elements formed by a replicated 53×53 MURA basic pattern. [from: A. Goldwurm et al.: IBIS science data analysis (2003)]

Since its invention, the most salient applications of the CAI technique have been in x-ray and gamma-ray astronomy [49, 50]. However in recent years the CAI technique has also been investigated for a broader range of applications. For example, in medical imaging it may be a means of obtaining high resolution images while decreasing the radiation dose for patients [51]. And its uses for particle-source imaging include Coded Aperture Fast Neutron Analysis (CNFNA) [52, 53] and Inertial Confinement Fusion (ICF) neutron and proton imaging [54-56].

The two most essential components of a CAI camera are a position-sensitive detector and a patterned mask of alternately transparent and opaque areas (or pixels). The detector and mask planes are arranged to be parallel, and the pixels are usually square or hexagonal. In essence all CAI schemes are a form of multiplexed pinhole imaging. The pinhole is of course the simplest and earliest known method of imaging, and its limitations, in terms of resolution vs. signal, are well known. However it is still important for types of radiation for which coherent imaging is not practicable (e.g. vacuum-ultraviolet, x-rays (>30 keV), gamma-rays, etc.). The basic concept of the CAI technique is to open multiple small pinholes, thereby maintaining good angular resolution, while achieving the high signal throughput of a large aperture.

It is notable that all the apertures that in far-field ensure artifact-free imaging, deviate significantly from ideality when used in near-field. The goal of devising a rational design procedure for the coded aperture was now expanding to include near-field artifact reduction. Also, among the many apertures that provide ideal far-field imaging, some types have a SNR significantly superior to others, which are preferred to use.

Coded apertures try to achieve the good resolution of small pinholes while maintaining a high signal throughput. The basic idea is to overcome photon (or, in our case proton) shortage by opening many small pinholes to give the same effective open area as a large pinhole. A large pinhole necessarily results in a loss of resolution [48, 57]. These many small pinholes are placed in specially designed arrays called patterns. The coded aperture (or mask) is the physical realization of a pattern. The detector and mask, together, form the coded aperture camera. The design of a coded aperture system for charged particles such as protons and deuterons requires work on two components: a position and energy sensitive detector and the coded aperture. The complexity of the CAI system requires theoretical investigation together with verification by numerical

and experiment simulations. In order to design a suitable CAI mask and camera it is necessary to have an understanding of the trade-offs between various imaging parameters such as field-of-view, resolution, collection efficiency, and signal-to-noise ratio. Clearly for a detector of reasonable size, and a mask of useful open fraction, there will be considerable overlap of the multiple (inverted) pinhole images. Therefore a deconvolution step is necessary to extract the final image, known as the decoded image. The unprocessed image from the detector is referred to as the coded image.

In this project we intend to apply the coded aperture for fusion source imaging in plasma focus device. In our case the irradiative particles are protons and the detector is a polymer nuclear track detector (CR-39).

It is worth mentioning that for high neutron (proton) yield device such as Frascati, 1 MJ plasma focus device with the $Y_n \approx 10^{11}$ per shot, the pinhole imaging is quite sufficient [55]. However, in our case with NX2 PF with $Y_n \approx 10^8$, the precise fusion imaging can only be done by coded aperture imaging technique.

2D Imaging:

An image is a mapping over space of some distribution, in our case that of a proton source. A pinhole is the simplest imaging device, being a sheet of material through which light or charged particles cannot pass, which is punctured by a small (ideally dimensionless) hole. As every ray (or proton) emitted from the object passes through the pinhole following a straight line trajectory (see Figure 11-a), every point of the detector plane is related to a point in the source. Another simple imaging system is the collimator: an array of narrow parallel tubes placed side by side, and aligned perpendicular to the detector plane. Then protons must arrive from directions perpendicular to the detector surface (see Figure 11-b), which effectively identifies the place of emission. The image is directly the proton distribution recorded by the detector. So a one-to-one correspondence between object and image is the property of ideal pinholes and collimators [58]. Also, in both cases, for a given source point, only protons traveling in a single direction are collected.

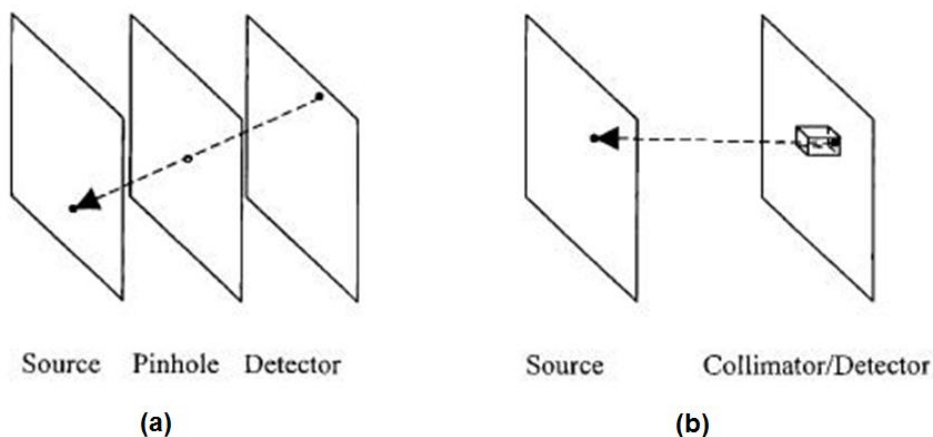


Figure 11: a) pinhole imaging and b) collimator imaging systems

In theory, the resolution obtainable with an ideal dimensionless pinhole is perfect, in the sense that two point sources arbitrarily close to one another in the object will always be separated on the detector. However, an ideal pinhole results in no counts at all on the detector, as the ideal pinhole has zero area. Real pinholes must have a finite size to allow some protons to pass, but the problem is if two point sources are close enough, their projections on the detector would not be distinct. As the black lines show, (see Figure 12-right) light rays from the same source point can hit the detectors over a small region. In the same way, the red lines show that the light rays from different points on the object can hit the same point on the film. Light from the same point on the object hits a circular area on the film, creating a so-called “circle of confusion”. The larger the hole, the larger the circle of confusion, and the blurrier the photograph becomes. In conclusion, the pinhole (or collimator) hole size cannot be increased indefinitely to increase efficiency because some resolution limit will be reached.

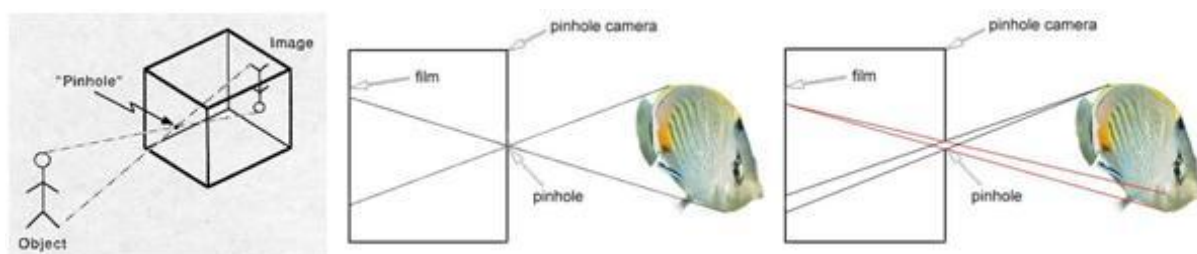


Figure 12: Pinhole camera and its limitation.

Concept of Coded Aperture Imaging

For a coded aperture arrangement, the number of protons passing through each pinhole is independent of all the others; each pinhole producing an inverted image on the detector. Therefore the tracks collected on the detector are, the superposition of many shifted copies of the object by different pinholes. In a far-field approximation, the projection process follows the equation:

$$O * A = R \quad (2-1)$$

Where O is the number of emitted protons per unit area from the object, A the transmission function of the coded aperture (mask), R the counts recorded on the detector and $*$ indicates non-periodic correlation.

As the mask can have a large number of pinholes, R does not resemble O . This is the mask itself, as follows from eq. (2-1), when O is replaced with Dirac's delta function (δ):

$$R = \delta * A = A \quad (2-2)$$

Therefore, each point of the object will project a pattern on the detector, rather than a point. Different point sources are characterized by a pattern shift. So a point source is counted once for every pinhole of the pattern, which is expected to increase the counting statistics and, consequently, improve the SNR. The second thing is that each point of detector presents information about many points of the source. In this sense information is multiplexed. The third consequence is that the recorded pattern R must be decoded to obtain an image by separating the overlapped copies through an associated decoding pattern G such that:

$$A \otimes G = \delta \quad (2-3)$$

where $\otimes$ indicates periodic correlation. Given the linearity of correlation operations and eq. (2-1) one can write:

$$\hat{O} \equiv R \otimes G = (O * A) \otimes G = O * (A \otimes G) \quad (2-4)$$

where $\hat{O}$ is the reconstructed image. This shows that the output of the imaging system is not directly the object, but as in all linear systems, it is a convolution of the object with a kernel, in this case $A \otimes G$ also called the Point Spread Function (PSF):

$$\hat{O} = O * \text{PSF} \quad (2-5)$$

From eq. (2-4) and (2-5), in coded aperture imaging:

$$\text{PSF} = A \otimes G \quad (2-6)$$

We can design and generate pairs of A and G somehow to satisfy the criteria of eq. (2-3) and have perfect imaging properties. Substitution of eq. (2-3) in eq. (2-6) and then in eq. (2-5) gives:

$$\hat{O} = O * \delta = O \quad (2-7)$$

which means that if the PSF is a δ function, the reconstruction is perfect. Naturally, the choice of aperture pattern A and decoding function G have a large influence on the quality of the final image [59, 60].

In conclusion, coded aperture imaging is able to produce a high precision copy of the object via a two-step process, which is physical projection of the source through the aperture pattern, followed by computational decoding. The motivation to go through this complication is the potential of achieving a higher SNR than for a small pinhole of equivalent angular resolution.

2.1.2 Coded Aperture Families:

Fresnel Zone Plate

The first Coded aperture technique was proposed in 1961 by Mertz and Young [61]. It was the Fresnel Zone Plate (FZP), in theory a circularly-symmetric mask having transmission:

$$\cos(r^2); \quad 0 \leq r \leq \infty \quad (2-8)$$

where r is the radius from the center of symmetry. In early experiments the projection was recorded on a film, which was developed and then exposed to coherent light of wavelength comparable to the size of the projected pattern. Since each projected zone plate is refocused in a point, the image is decoded. The perfect imaging properties hold only if the plate is infinite and has a continuously varying transmission but in real application the FZP is often approximated by a finite series of concentric circles of radius:

$$r_n = r_0 \sqrt{n}, \quad n = 1, 2, 3, \dots, n_{\max} \quad (2-9)$$

where the annuli are alternatively totally opaque and totally transparent (see Figure 13). The first actual demonstration of FZP imaging was a study of a thyroid phantom [62].

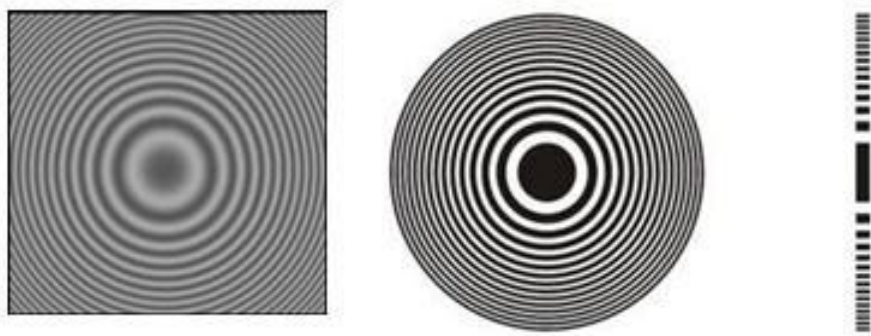


Figure 13: Fresnel Zone Plate; theoretical (left) and practical (right)

Random Arrays

In the random array [63, 64] holes are placed at random. The decoding array is the array itself, $G=A$. Since there is no correlation between the neighbouring pixels, the sidelobe is not flat and the final image does not have ideal properties. But even so, they are still attractive because they do not contribute structured artifacts to the image and can easily be made of any open fraction and size. Furthermore, they can be large enough so that the inherent noise is negligible.

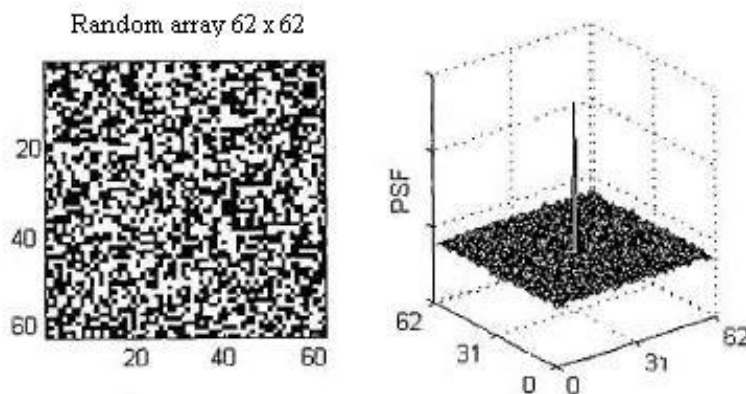


Figure 14: Random array and its periodic self-correction.

Non-Redundant Arrays

When the autocorrelation of a binary pattern is taken, the number of hole coincidences at each shift equals the number of holes of the pattern separated by a vertical distance equal to the shift. If this number is 1 for a certain shift, only one hole in the pattern is separated from any other holes by that shift. For this pattern, the self-correlation function would be “1” everywhere, except for no shift, where a peak of height equal to the number of holes in the pattern is formed. Golay reports a number of

patterns with optimal or nearly optimal imaging characteristics [65]. The major problem of the NRAs are the non-optimal imaging properties which limit the field of view [49, 66] and the small number of holes, results in a small signal throughput [67, 68]. Nevertheless, they were used in nuclear medicine studies [69].

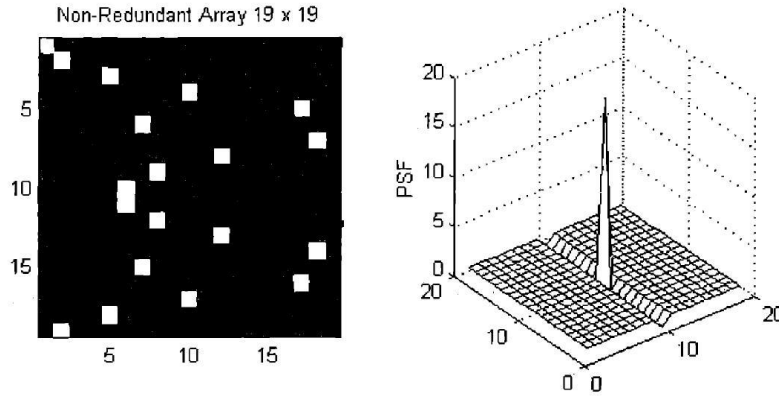


Figure 15: non-redundant array

Uniformly Redundant Arrays (URA)

The Uniformly Redundant Array (URA) was introduced by Calabro and Wolf in 1968 [70] and then were applied for imaging in 1978 by Fenimore and Cannon by giving a procedure to generate two dimensional bypassing folding [67]. In this pattern, all separations between the holes occur M times. Hence, the $M = 1$ case corresponds to an NRA. In other cases, all distances appear M times, and this is referred to as uniform redundancy. In common usage, it is also called a twin-prime Hadamard array, because of the two dimensional pixel $p \times q$, where p and q are prime and $|p - q| = 2$.

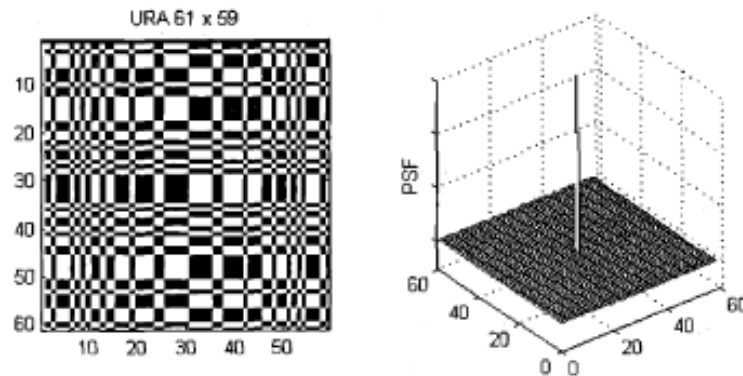


Figure 16: 61 × 59 Uniformly Redundant Array.

Modified Uniformly Redundant Array (MURA)

The Modified Uniformly Redundant Arrays (MURAs) are generated with the same algorithm used for the URAs. They have $p \times q$ pixels (p and q are prime), with different properties depending on whether $q-p = 0, 2, 4$, or 6 . However, G cannot be equal to A because the autocorrelation is not a δ function. Gottesman and Fenimore [71] pointed out that a slight modification in the definition of A gives the G resulting in ideal imaging properties.

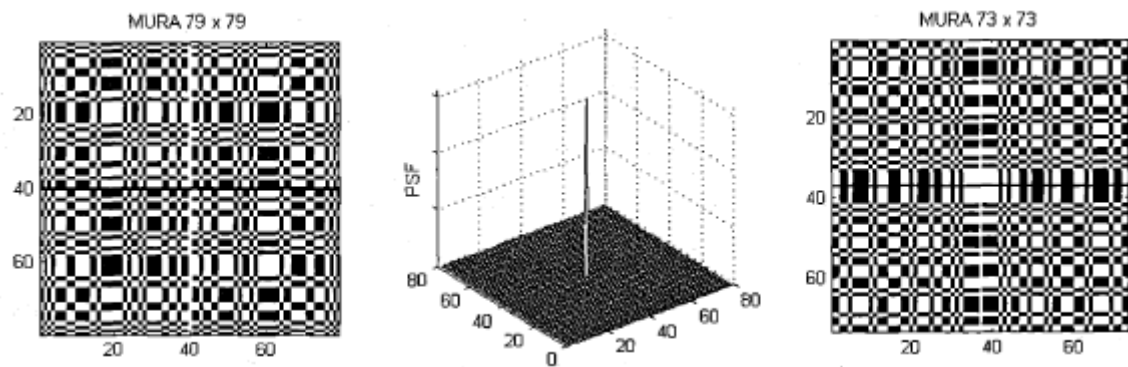


Figure 17: MURA pattern, while the 79×79 is anti-symmetric about its center, the 73×73 is symmetric.

2.1.3 Coded Aperture Geometry

The most important parameters in coded aperture imaging are shown in Figure 18. The analysis can be carried out from 1D to 2D, because it is assumed that the mask and detector plans are parallel.

The image or detector pixel size is:

$$P_{xDet} = \frac{d_1 + d_2}{d_1} \times P_{xMask} \quad (2-10)$$

where P_{xDet} and P_{xMask} are pixel size of detector and mask respectively, d_1 is the distance between object (source) to mask and d_2 is the distance between detector (image) to mask.

The resolution of system is defined by object pixel size (P_{xObj}):

$$P_{xObj} = \frac{d_1 + d_2}{d_2} \times P_{xMask} \quad (2-11)$$

The Image size can be recognized by assumption of an ideal (dimensionless) pinhole:

$$\text{Image size} = \frac{d_2}{d_1} \times \text{Object size} \quad (2-12)$$

The linear dimensions of mask and detector determine the CAI Field of View (FOV) which is the maximum object size that can be fully coded on the detector:

$$FOV = \frac{d_1 L_D - (d_1 + d_2) L_M}{d_2} \quad (2-13)$$

where L_D and L_M are the lengths of the detector and mask sides respectively. With $L_M = 0$ the pinhole's field of view will be:

$$FOV = \frac{d_1}{d_2} L_D \quad (2-14)$$

This means that the FOV for pinhole imaging system is bigger than that of a CAI system with the same size of detector. In applications such as the plasma focus in which the source is relatively compact, the reduction in the FOV (by comparison with a pinhole) is not a problem.

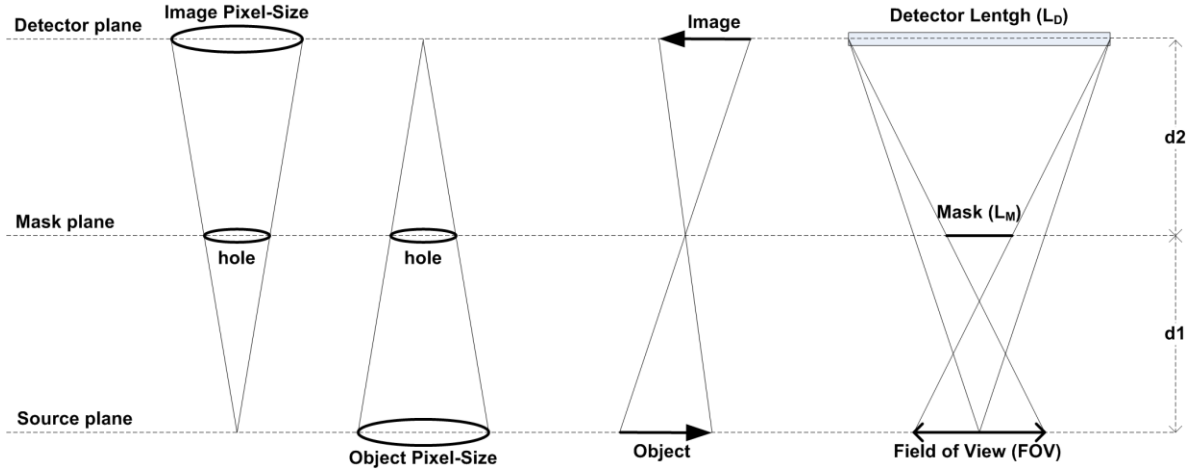


Figure 18: CAI geometry to identify the magnification, object pixel size (resolution), image pixel size and field of view.

2.1.4 Singer set array, a new approach for coded aperture imaging:

As it was mentioned before, Coded Aperture Imaging is a linear technique. If $O(x,y)$ is the two-dimensional object distribution, $A(x,y)$ is the aperture transmission function and $N(x,y)$ is the signal independent noise, then the spatial distribution of the detected fluence (or coded image) is given by:

$$R(x,y) = A(x,y) * O(x,y) + N(x,y) \quad (2-15)$$

where $*$ is the correlation operator. Then using the correlation method of de-coding, $\hat{O}(x,y)$, the estimate of the object or reconstructed image is defined by:

$$\begin{aligned} \hat{O}(x,y) &= G(x,y) * R(x,y) \\ &= G(x,y) * A(x,y) * O(x,y) + G(x,y) * N(x,y) \end{aligned} \quad (2-16)$$

where $G(x,y)$ is the decoding function. From this equation it can be seen that the Point Spread Function, $PSF = G * A$, should ideally be a delta function for the best image reconstruction.

A Cyclic Difference Set is a set of h positive numbers $[a_1=0, a_2, a_3, \dots, a_s] < p$, with the property that all differences $(a_i - a_j)$ modulo p for $i \neq j$, occur the same number of times (λ), corresponding to a flat auto-correlation side-lobe. The $i = j$ condition clearly occurs h times, corresponding to the auto-correlation peak. The numbers p and h are referred to as the modulus and size of the CDS, respectively. So the important parameters for any CDS can be abbreviated to (p, h, λ) . Among the Cyclic

Difference Sets there are two subclasses which are particularly suitable for generating masks for multiplex imaging:

1) Singer sets [72, 73] for which

$$p = \frac{t^{m+1} - 1}{t - 1} \quad h = \frac{t^m - 1}{t - 1} \quad \lambda = \frac{t^{m-1} - 1}{t - 1} \quad (2-17)$$

with t being a prime number, or a prime power,

2) Hadamard sets [74, 75] for which

$$p = 4t - 1 \quad h = 2t - 1 \quad \lambda = t - 1 \quad (2-18)$$

with t being an integer.

To form a two dimensional mask the CDS is wrapped onto a rectangular array, where the sides of the rectangle are factors of p . Hence, p represents the total number of pixels in the mask, and h is the number of open pixels (holes). Considering these two sets, it can be seen that Hadamard masks have an open fraction, very close to 50%.

$$\rho_H = h/p = (2t - 1)/(4t - 1) \quad (2-19)$$

Whereas those based on Singer sets have a variable open fraction

$$\rho_S = (t^m - 1)/(t^{m+1} - 1) \approx 1/t \quad (2-20)$$

Since t is a prime number, or prime power, the obtainable open fractions with Singer sets are:

$$\rho_S \cong \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{7}, \frac{1}{8}, \frac{1}{9}, \frac{1}{11} \dots \quad (2-21)$$

(open fractions of $\frac{1}{6}$ and $\frac{1}{10}$ etc. are not obtainable).

The simplest modulo is “Binary Field” or modulo 2 which produce a ~50% open area Singer set pattern. The irreducible polynomial for modulo 2 is $x^3 + x + 1$ and the parameters of this set are (7, 3, 1) which can be found by putting $t=2$ and $m=2$. Figure 19 shows a 1 Dimensional analogue of the CAI technique using the $(v, k, \lambda) = (15, 7, 3)$ set. It includes 15 pixels, 7 holes and 3 side lobes which represents about 50% open area.

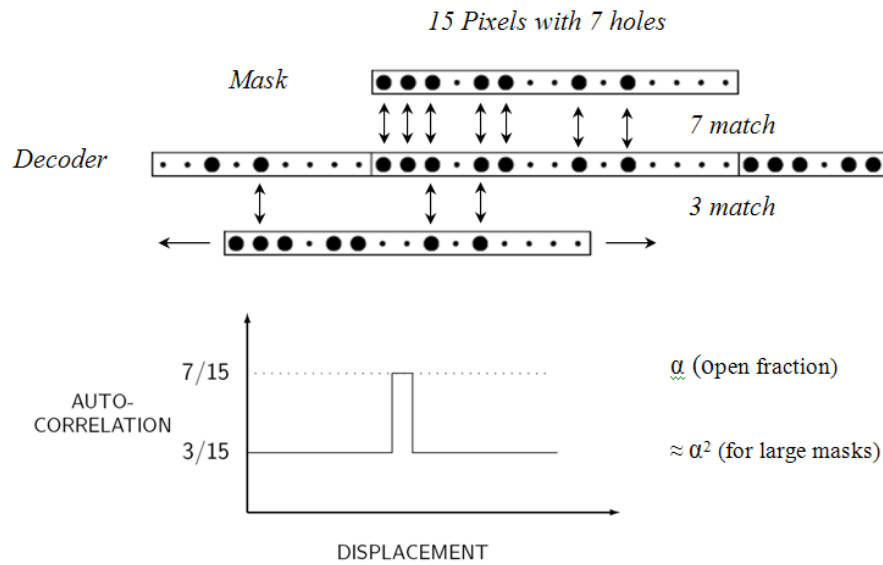


Figure 19: 1D Cyclic Difference Singer Set in modulo 2

Figure 20 also shows the final image reconstruction procedure which is called “Deconvolution” algorithm.

Image Reconstruction

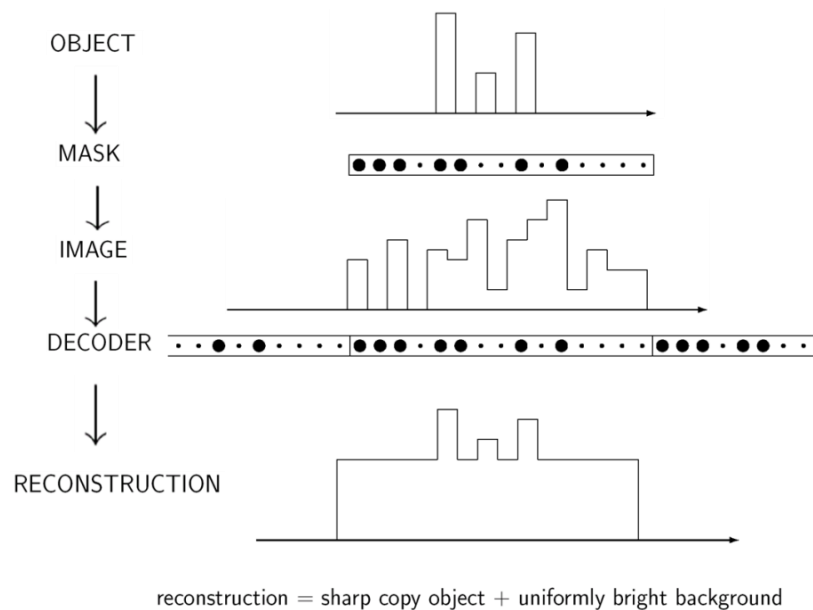


Figure 20: schematic of reconstruction algorithm (Deconvolution)

If with the same modulo (modulo 2 with 50% open area) one wants to increase the number of pixels and holes, one has to apply an irreducible polynomial of a higher order. For making the Singer set with parameter (15, 7, 3) ($t=2$ and $m=3$) the polynomial $x^4 + x + 1$ must be used as the irreducible factor. For our first mask

design, a modulo 7 with parameter (400, 57, 8) has been used ($t=7$ and $m=4$) where the irreducible polynomial is $x^4 + 2x^3 + 6x^2 + 3x + 6$. From these parameters, it is clear that the mask comprises 400 pixels in a 2D pattern (20×20), 57 holes and a side lobe value of 8. The open fraction is $57/400 \cong 14.3\%$.

It is known in the literature that for maximum SNR, the optimum mask open fraction decreases as the spatial extent of the source increases [76, 77]. However, attempts to reduce the effective open fraction of Hadamard set masks have not been successful [78]. Hence for imaging extended sources, the wide range of open fractions attainable with Singer sets is an important advantage, enabling the mask to be designed to achieve near optimum SNR.

2.1.5 Signal to Noise Ratio (SNR) in Coded Aperture Imaging

Fenimore [76] and Accorsi et al. [78] have compared the SNR for different Hadamard families and pinholes for a series of simulated sources using the standard “balanced” method of decoding. However, their definition of SNR corresponds to the variation in brightness of a given fixed pixel between hypothetically many exposures for a source with assumed constant intensity, which simplifies the statistical analysis at the cost of a loss of intuition. Also, balanced decoding, while relatively simple in the Hadamard case (the matrix G consists of +1 where A has a hole and -1 where it does not) becomes more complicated in the Singer case. In this work, it was considered more meaningful to define SNR in terms of the variation in brightness across different pixels within a single exposure, more precisely as follows:

$$SNR = \frac{\text{signal from given pixel}}{\sqrt{\text{variance of signal across image}}} \quad (2-22)$$

As well as being closer to our intuition, this within-exposure definition of SNR has the added advantage of being more readily compared with the results of experiment. And in cases of non-repeatability, where a single-exposure is the only thing available, it may be the only possibility. We shall also find it more convenient to use direct decoding, where the decoding matrix $G = A$ the aperture itself, which, although differing from balanced decoding by overall constants and scaling, separates the terms involved in a way which makes the statistical analysis more straightforward.

Direct decoding, where we take $G = A$, the aperture itself, uses the defining property of a CDS based coded aperture, namely that if two copies of the aperture are

cyclically displaced relative to each other then the resulting open fraction, that is, where the holes of the two copies happen to coincide, is ρ^2 independent of the displacement, except of course when the two copies themselves coincide, in which case the open fraction is just ρ again. In other words, as the aperture A passes across the encoded image R we will pick up the signal from the object pixel matching that position but attenuated by a factor ρ , together with the signal from all other object pixels but attenuated by a factor ρ^2 . More specifically, let N be the total number of particles incident on the mask, and $N' = \rho N$ the number which pass through the mask, and let n be the number of particles incident on the mask which arrive from a direction associated with a particular object pixel, and $n' = \rho n$ those passing through the mask, and let the object comprise m bright (illuminated) pixels. Then, when the decoding matrix is aligned with that particular pixel, the decoded signal will be $n' + \rho(N' - n') = \rho n + \rho^2(N - n)$, while if the decoding matrix is aligned with any other pixel in the field of view not corresponding to a pixel on the object the decoded signal will be $\rho N' = \rho^2 N$. So direct decoding produces a sharp image of the object but superimposed on an approximately uniformly bright background, where the average background brightness is $\rho N'$. Subtracting this average background brightness we recover a signal $\rho n + \rho^2(N - n) - \rho^2 N = \rho n - \rho^2 n = \rho(1 - \rho)n$ against an approximately uniformly zero background. It is then not difficult to show that, up to an overall scaling factor of $1/(1-\rho)$ the resulting decoded image is identical with that obtained from balanced decoding.

Including this overall $1/(1-\rho)$ scaling, in which case the reconstructed signal is just ρn has the advantage that the total particle count in the reconstructed image is identical with the number of particles actually detected. Since the SNR is invariant under overall scaling of the data, however, direct decoding allows us to neglect it for the purposes of the statistical analysis. An additional advantage of direct decoding, at least in the case where SNR is defined within-exposure, is that the overall constant $\rho N'$ subtracted also does not affect the computation of the variance and can be neglected, whereas in the case where SNR is defined exposure-to-exposure it does vary. To compute the SNR within-exposure it only remains therefore to compute the variance of the signal $n' + \rho(N' - n') = \rho n + \rho^2(N - n)$. It is a property of CDS based coded apertures that the second term represents a pseudo-random sampling of the detected particles with probability ρ and so follows binomial statistics. If we make the simplifying assumption that the m object pixels are uniformly bright, then the first term

also follows binomial statistics with probability ρ . Neglecting the covariance between the two terms, and using the standard result that a $B(N, \rho)$ random variable of N trials with probability success p and failure $q=1-p$ has expectation Np and variance Npq , the variance of the directly decoded signal as the decoder moves across the coded image is

$$\rho(1 - \rho)n + \rho(1 - \rho)(N' - n') = \rho(1 - \rho)n + \rho(1 - \rho)\rho(N - n) \quad (2-23)$$

Substituting this into our earlier formula defining the within-exposure SNR we obtain:

$$SNR = \frac{\rho(1 - \rho)n}{\sqrt{\rho(1 - \rho)n + \rho^2(1 - \rho)(N - n)}} = \frac{\sqrt{\rho(1 - \rho)} n}{\sqrt{n + \rho(N - n)}} \quad (2-24)$$

$$\text{Then putting } m = \frac{N}{n}: \quad SNR = \sqrt{\frac{N\rho(1-\rho)}{m(1+\rho(m-1))}} \quad (2-25)$$

This expression happens to coincide with the formula derived by Fenimore [76] although under quite different assumptions i.e. exposure-to-exposure variance, and later shown by Accorsi [78] to be in error in that case. These formulae are, however, approximations. A truly complete SNR analysis for the within-exposure case has, to our knowledge, not yet been reported.

Again using the simplification that the m bright pixels are considered to be equally bright, and differentiating the SNR function with respect to ρ , gives the optimum open fraction in terms of the number of bright pixels:

$$\rho_{\text{opt}} = \frac{1}{1 + \sqrt{m}} \quad \text{and} \quad SNR_{\text{max}} = \rho_{\text{opt}} \sqrt{\frac{N'}{m}} = \rho_{\text{opt}} \sqrt{n'} = \sqrt{\rho_{\text{opt}} n} \quad (2-26)$$

Hence, for a mask of constant size (and therefore constant N'), both ρ_{opt} and SNR_{max} decrease as the source becomes more extended and illuminates more pixels; or alternatively, as the pixel-size shrinks and the number of pixels increases. Figure 21 shows a plot of SNR versus ρ for different numbers of bright pixels $m=8, 16, 32$ and 64 . This calculation is for a mask of the dimensions used for our first experimental study ($8 \times 8 \text{ mm}^2$) with a total of 400 pixels and an isotropic source of 10^8 particles located at a distance of 45 mm from the centre of the mask.

The solid angle subtended by the mask is 0.0314 sr, and the number of particles hitting the mask is $N' = 2.5 \times 10^5$. Open circles represent values of ρ_{opt} for each curve. The vertical lines on the plot represent the values of $\left(\rho_s \cong \frac{1}{t} = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{7}, \frac{1}{8}, \frac{1}{9}, \frac{1}{11}\right)$ attainable with Singer set masks, while the circle symbols represent values of ρ_{opt} for

each curve. Hadamard masks, being restricted to $\rho_H \cong 0.5$, are best suited to situations with very few bright pixels.

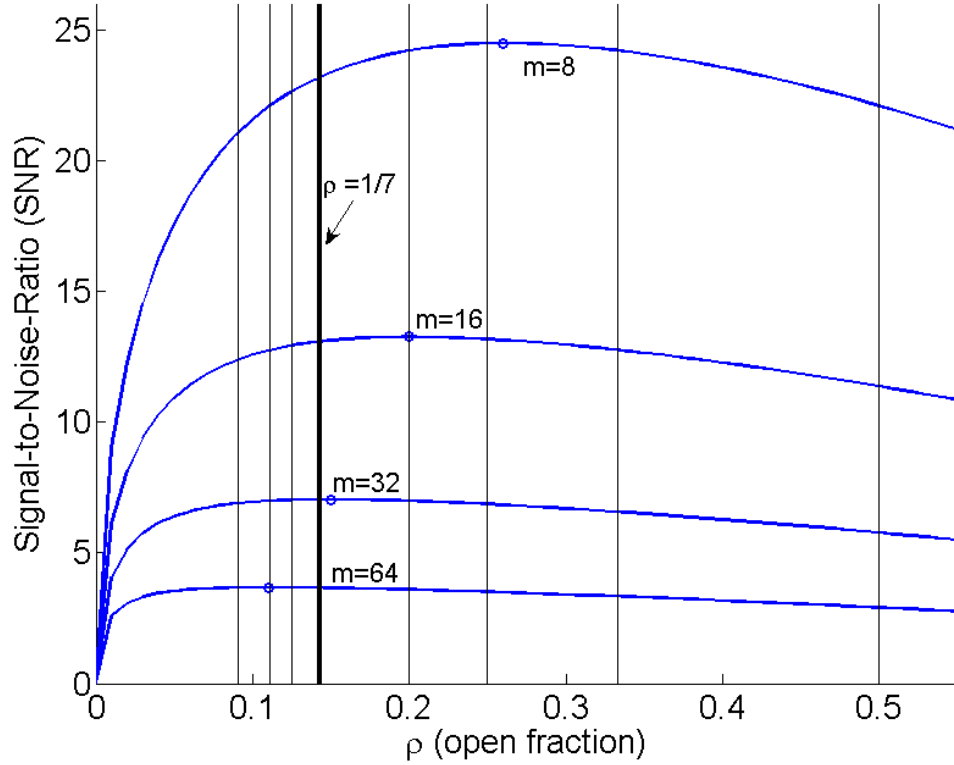


Figure 21: SNR vs. open fraction (ρ) for objects (of the same total brightness) which illuminate different numbers of pixels (m) the vertical black lines represent the possible open fraction for Singer Set.

Also for pinhole imaging, the SNR is simply:

$$SNR_{ph} = \frac{n}{\sqrt{n}} = \sqrt{n} \quad (2-27)$$

It should be remembered that the value of n for a CAI is h times greater than that for a pinhole of the same angular resolution. Hence the optimum enhancement of a CAI mask over a pinhole (i.e. always with a mask of optimum open fraction ρ_{opt}) is given by

$$(SNR_{max}/SNR_{ph})_{opt} = \sqrt{h\rho_{opt}} = \rho_{opt}\sqrt{p} = \frac{\sqrt{p}}{1 + \sqrt{m}} \quad (2-28)$$

It is clear from this formula that the SNR advantage of CAI over pinhole imaging decreases as the source becomes more extended, until the limit at which the source fills the entire field of view ($m \approx p$) when CAI offers essentially no SNR advantage over pinhole imaging. As a rule of thumb, CAI can be considered advantageous, and worth the additional complexity, in cases where the source occupies no more than 10% of the field of view (FOV).

Experimentally it is usually the case (and certainly so for the plasma focus) that the precise number of bright pixels is not known prior to the exposure. It is probably best therefore to use a mask with ρ somewhat larger than ρ_{opt} for the anticipated value of m . As can be seen from Figure 18, the SNR penalty for using a moderately larger value of ρ is slight. Also, for experiments using CR-39 detectors, the signal independent noise $N(x,y)$ in equations (1) and (2) is primarily due to flaws, scratches, alpha tracks from background radon, which are always present to some extent on the etched CR-39 surface. With proper handling of the material this background is relatively low and uniform across the detector surface, resulting in a slightly degraded SNR in the decoded image.

2.1.6 Mask Design

Based on the geometry of the experimental setup, the first prototype mask was designed with $t = 7$ (open area fraction $\rho = 1/7$) and $m = 3$ giving CDS characteristic values of $(p = 400, h = 57, \lambda = 8)$. The $8 \times 8 \text{ mm}^2$ mask has 57 square holes shown in Figure 22.

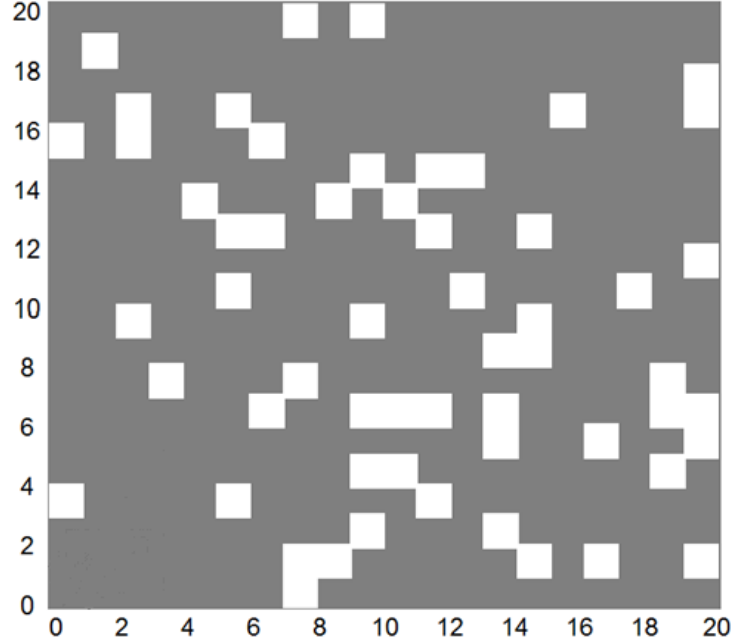


Figure 22: mask-400 (20×20) pattern with $\rho = 1/7$ open fraction

For the next generation mask, it was decided to design larger masks with more holes and more particle throughput. One candidate option is mask-1365 (91×15) with $\rho = 1/4$, $t = 4$ and $m = 5$, so $(p = 1365, h = 341, \lambda = 85)$, shown in Figure 23. Another option is mask-4680 (104×45) with $\rho = 1/8$, $t = 8$ and $m = 4$, so: $(p = 4681, h = 585, \lambda = 73)$. As it can be seen p is 4681 which cannot be folded it in 2D mask. So by ignoring 1 pixel, the mask-4680 can be folded to 104×45, shown in Figure 24.

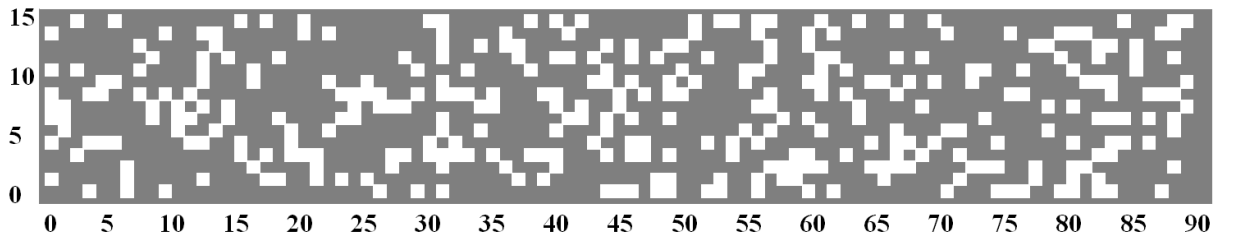


Figure 23: mask-1365 (91×15) pattern with $\rho = 1/4$ open fraction

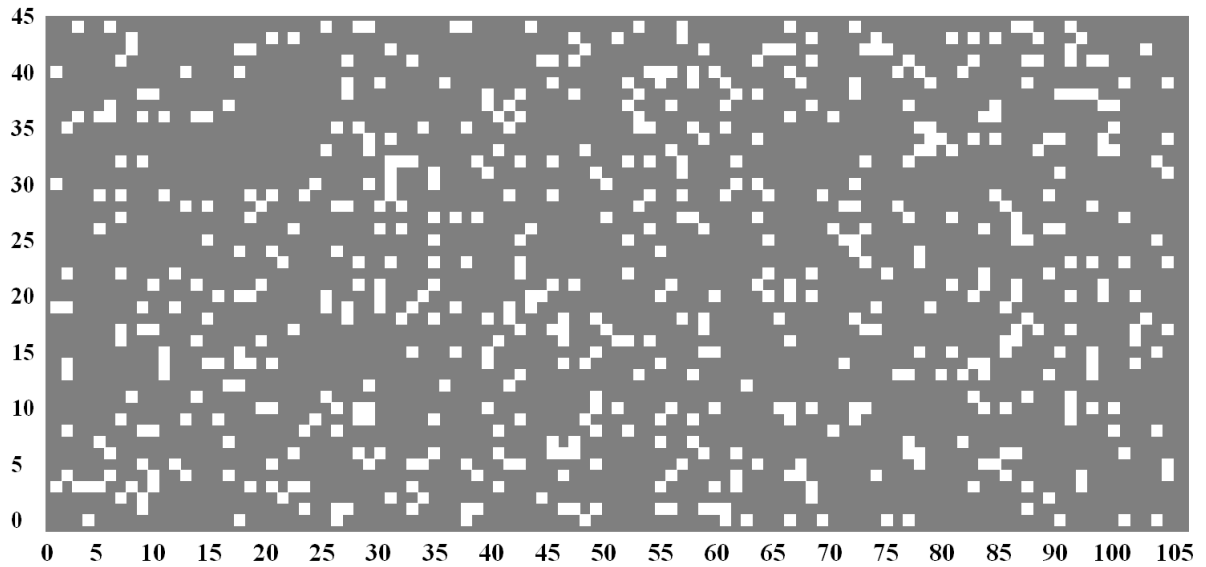


Figure 24: mask-4680 (104×45) pattern with $\rho = 1/8$ open fraction, with one pixel truncated from the complete CDS length of 4681.

Spaced Mask

If one looks carefully at mask-1365 and mask-4680, it can be seen that some of the opaque pieces have no support (except at the corners), or are poorly supported (at one edge only). So after machining the physical mask, the unsupported pixels will simply fall out. And the poorly supported pixels are likely to be damaged by the plasma jet emitted during plasma focus shots. We therefore investigated row-spaced masks as a means of ensuring all opaque pixels in the mask pattern are well supported. The idea is simply to separate the pattern rows by inserting an integer number ($k-1$) of solid rows between them. In imaging terms this effectively distributes all the bright source-pixels between k interlaced sub-images. And as will be seen from the signal-to-noise (SNR) ratio analysis later in this chapter, for an elongated source, such as the plasma focus pinch, there are advantages to this row-spacing in terms of avoiding excessive overlap of proton tracks on the CR-39 detectors whilst maintaining a good SNR. Figure 25 shows the spaced mask for mask-1365 with space number $k=6$. Figure 26 also shows the spaced mask for mask-4680 with $k = 2$. The reason of selection these space number is the final mask structure be approximately square.

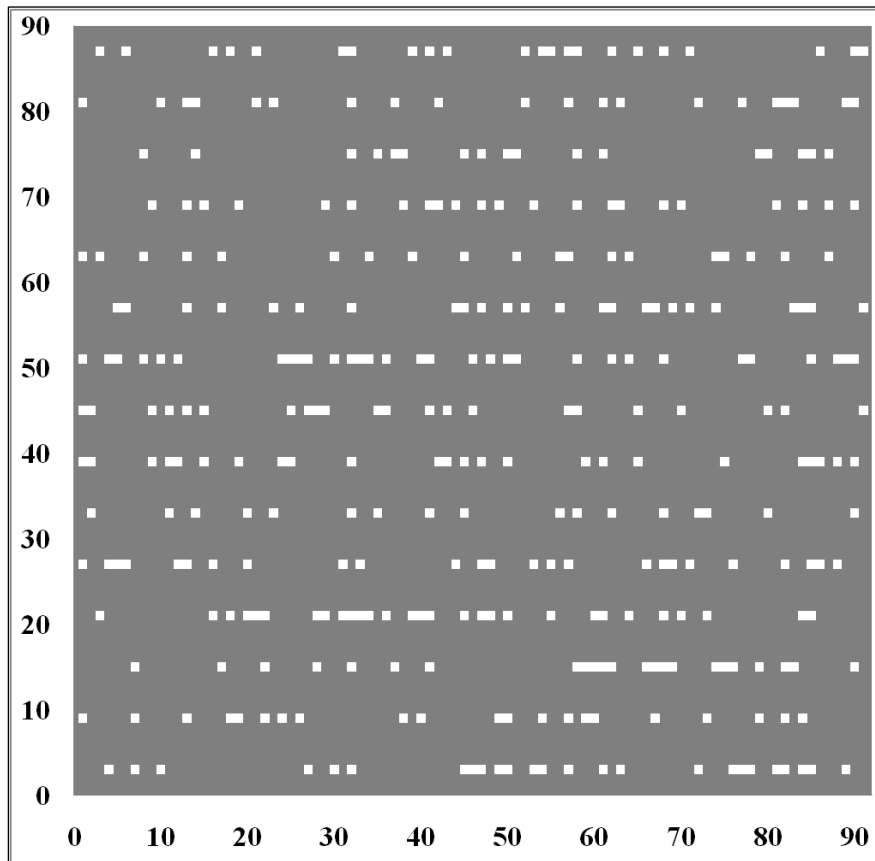


Figure 25: mask 1365 (91×15) spaced by factor of $k=6$

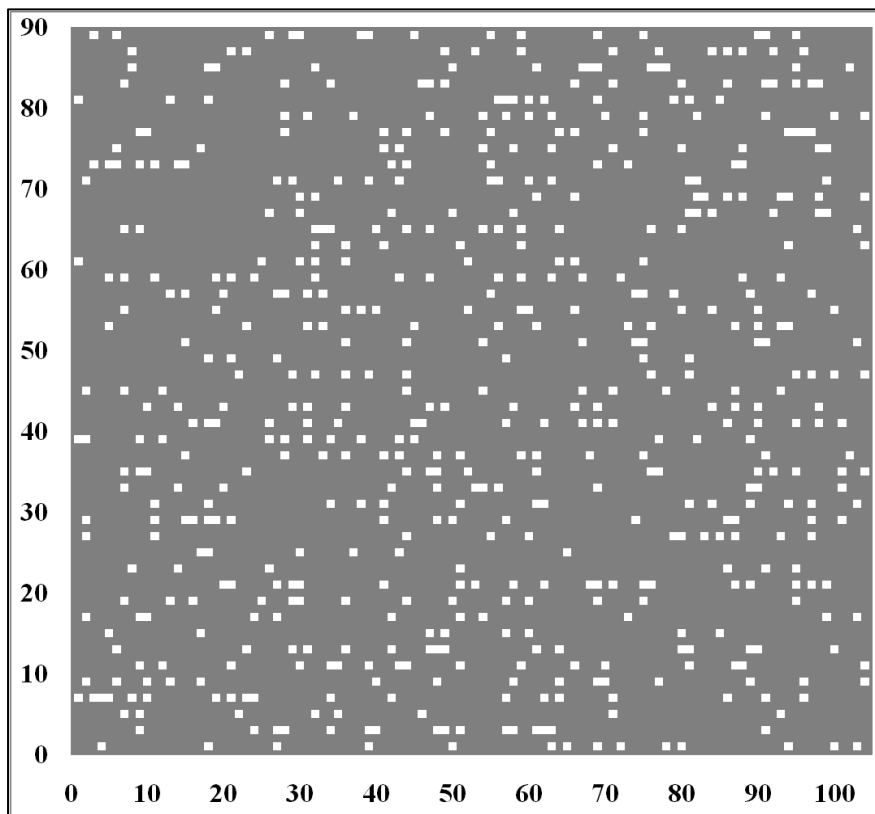


Figure 26: mask-4680 (104×45) spaced by factor of $k=2$

To calculate the SNR for spaced mask, we should consider that the mask is elongated k times bigger, makes the total hitting particle (N and N') k times more. So for background noise calculation, N or N' must be divided by k . Regarding to formula (2-23), for spaced mask, the total background will be:

$$\rho(1 - \rho)n + \rho(1 - \rho)\left(\frac{N'}{k} - n'\right) = \rho(1 - \rho)n + \rho(1 - \rho)\rho\left(\frac{N}{k} - n\right) \quad (2-29)$$

Repeating the same procedure leads us to SNR formula for spaced mask by factor of k .

$$SNR = \frac{\sqrt{\rho(1 - \rho)} n}{\sqrt{n + \rho\left(\frac{N}{k} - n\right)}} = \sqrt{\frac{N\rho(1 - \rho)}{m(1 + \rho\left(\frac{m}{k} - 1\right))}} \quad (2-30)$$

Figure 27 and Figure 28 show the plot of SNR versus ρ for mask-1365 and mask-4680 respectively. The blue curves are for basic mask pattern and red curves are for spaced mask with different numbers of bright pixels $m=32, 64$ and 128 . For both masks, the pixel size is assumed $300 \mu\text{m}$. This calculation is for two masks which are used for our second experimental study with an isotropic source of 10^8 particles located at a distance of 44 mm from the centre of the mask.

It shows very clearly the advantage of spaced mask in term of SNR. Although for this mask, a bigger detector is needed following by longer time for scanning and analysis.

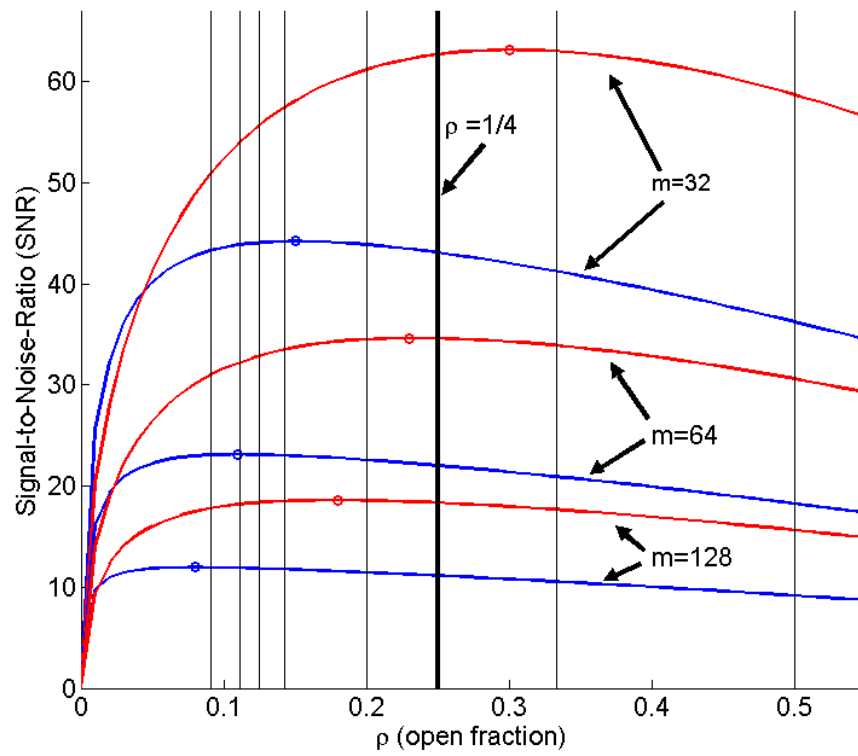


Figure 27: SNR of mask-1365; basic pattern (blue lines) and spaced mask (red lines) for different object size ($m=32$, 64 and 128). Bold black line represents $\rho=1/4$ (the optimum open fraction for $m=64$).

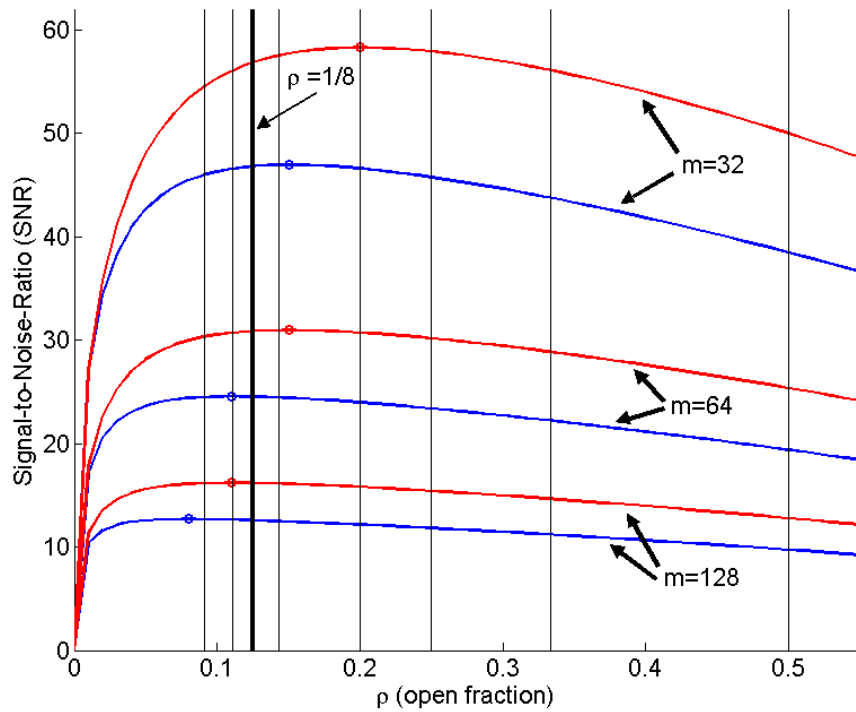


Figure 28: SNR of mask-4680; basic pattern (blue lines) and spaced mask (red lines) for different object size ($m=32$, 64 and 128). Bold black line represents $\rho=1/8$.

2.1.7 Image Reconstruction

After exposing the detector by charged particles, it is etched in an aqueous solution. Then the surface images are acquired by an optical microscope and analyzed automatically to record the all tracks (x,y) position. The coded image was then processed by our convolution-based decoding algorithm. The first step in this algorithm is to pixelize the detector plane by two-dimensional binning of the track positions in a $(2m-1) \times (2n-1)$ histogram array. Standard CAI shifting and folding operations reduced this to $m \times n$ fully-coded image (pink colour), regarding to Figure 29-right, the left part (green colour) must be shifted up by 1 pixel and then shifted left by m pixels and finally the upper section (blue colour) shifted down by n pixels. The experimental setup and detailed procedure will be discussed in later sections.

Then cyclic convolution (rotation) of this array with the identified mask pattern as it is shown in Figure 29-left, followed by side-lobe subtraction produces the final de-coded image.

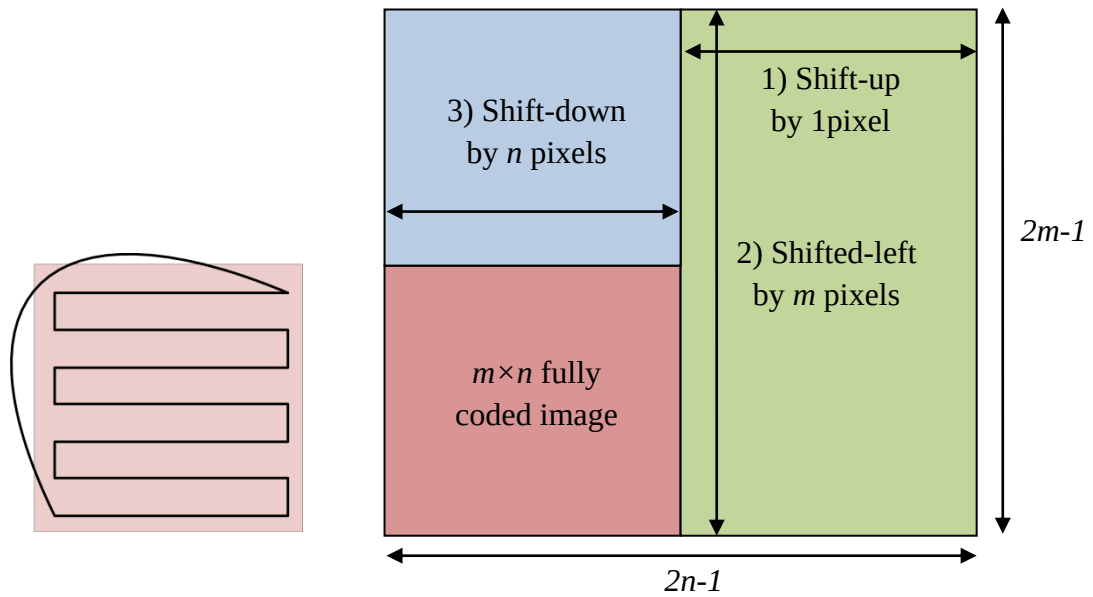


Figure 29: schematic diagram for reconstruction algorithm (cyclic convolution)

2.2 CAI Experimental Procedure

2.2.1 Mask Fabrication

The different masks with the different patterns and open area fraction have been made and applied for experiments.

The first mask was made based on Singer CDS with $t = 7$ and $m = 3$ which has overall dimension (8mm×8mm) comprising 20×20 pixels (400 μm side) and 57 circular holes (300 μm diameter), so the open area fraction for this mask was just 6.3% (4.03 mm²). The chosen material was 50 μm thick HAVAR alloy because of its strength and high melting point (1480 °C). A Micro-EDM (Electrical Discharge Machining) system, model MG-ED72W (Precision Engineering & Nanotechnology Center, NTU) was used for drilling holes which utilizes the spark erosion phenomenon for removing the material from the work piece immersed in a dielectric medium. Figure 30 shows the physical mask and its pattern.

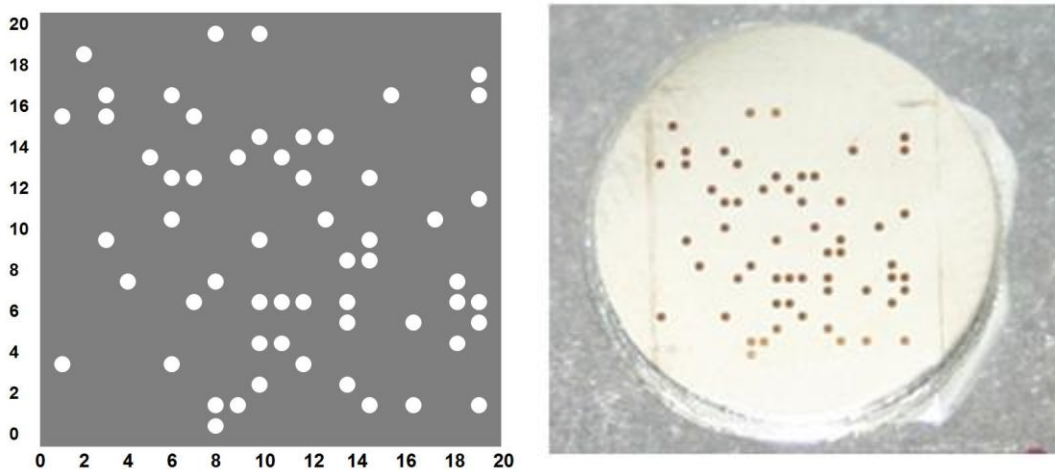


Figure 30: mask-400, Physical mask (right) made by Micro-EDM drilling and its pattern (left)

It soon became clear, however, that the micro-EDM method was too slow for the production of large masks, and also unsuited to producing square holes (as the EDM wire has too large a diameter). Laser machining was therefore employed for the fabrication of later masks, which made it possible to machine much larger open area masks (due to a larger number of holes) thereby increasing the number of protons in the recorded image. A Q-switched third harmonic Nd:YVO₄ laser (Avia-X) with a wavelength of 355 nm with the average power of 10 W and repetition of 40 kHz (SIMTech-NTU) was used. The laser spot diameter was $\sim 20 \mu\text{m}$ and the pattern is fed to control system and the beam move around the pixel pattern and drills it completely,

so with this technique we have 14.3% open area fraction (9.12 mm^2) and the throughput of system is more than double of the old mask. This mask and its pattern are shown in Figure 31.

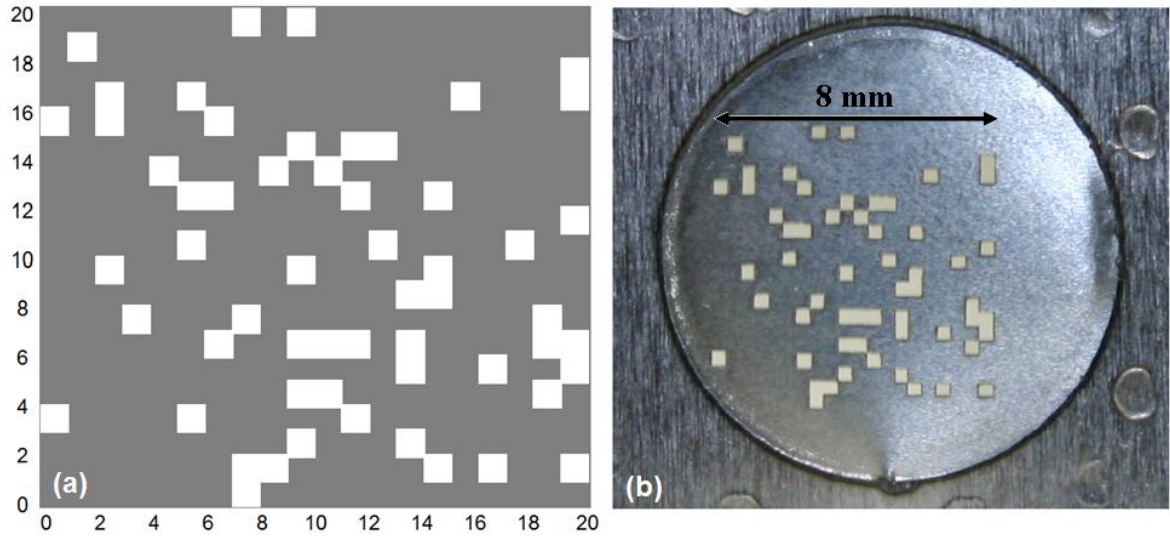


Figure 31: The mask-400 pattern (a), and a photograph of the fabricated mask (b) made by laser drilling.

The alpha source imaging, in order to test the coded aperture technique and 3-dimensional fusion source imaging in plasma focus device has been done with this mask, which will be explained in next chapter. However we tried to design new masks to achieve a better resolution and precision. To have a better resolution, the pinhole size must be smaller and for better precision (more particle throughput), the mask must have more open area, leads us to have a bigger mask. Regarding to our discussion in the introduction section, two new spaced masks were designed and fabricated. The same laser system was used to drill $300 \mu\text{m}$ side square holes on $50 \mu\text{m}$ stainless steel.

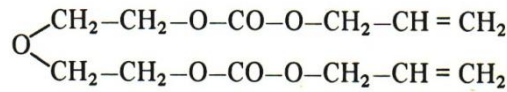
Figure 25 shows mask-1365 (91×15) which is designed based on Singer set ($p=1365$, $h=341$, $\lambda=85$) with $t=4$ and $m=5$, giving the open fraction $\rho=1/4=25\%$, so the total open area via 341 holes is 30.69 mm^2 . Also Figure 26 shows mask-4680 (104×45) ($p=4681$, $h=585$, $\lambda=73$) with $t=8$ and $m=4$, giving the open fraction $\rho=1/8=12.5\%$ and total open area with 585 holes is 52.65 mm^2 .

2.2.2 Charge Particle Detection

CR-39 Nuclear Track Detector

Poly-Allyl-Diglycol-Carbonate (PADC), marketed under the commercial name CR-39 is an excellent material in the detection and measurement of low and medium energy alphas due to its dielectric properties, which makes it resistive to annealing [79]. With the advent of research in plasma technologies, it has been used as a Solid State Nuclear Track Detector (SSNTD) in detecting charged particles emitted from plasma devices. It is worth emphasizing that linearity of detector response and high spatial resolution (ideally sub-pixel) are key factors for the successful application of the CAI technique. In both these respects CR-39 is an excellent detector: track-counting ensures very good linearity and the spatial resolution is of the order of microns.

When hit by heavily ionizing particles, invisible latent tracks of cleaved polymer chains are created. These latent tracks are developed by chemically etching the PADC, hence fixing the tracks for subsequent studies. The formation of tracks through etching is dependent on the track etch rate V_T , being greater than the bulk etch rate V_B which produces tracks or pits at the position of the latent tracks. CR-39 is a transparent thermo-set polymer with chemical formula $C_{12}H_{18}O_7$. The structure of the monomer is



given below:

To compute the Stopping Power for a compound using the Bethe-Bloch formula, we must sum the contributions for each element in the compound as below:

$$S(E) = \left(-\frac{dE}{dx}\right)_{tot} = \frac{z^2 e^4}{4\pi \epsilon_0^2 m_0 v^2} \sum_i n_i Z_i \ln\left(\frac{2m_0 v^2}{I_i}\right) \quad (2-31)$$

where Z_i is the atomic number for the i^{th} element, I_i is the mean ionization and excitation potential for that element (given by N. Tsoulfanidis -1995) and n_i is the associated atomic number density (1.3 g/cm^3) (see Table 2).

Table 2: characteristic of CR-39 elements for stopping power calculation

CR-39 (PM-355) with $\rho=1.3 \text{ g/cm}^3$				
Element Symbol	Atomic Number.(Z)	Formula Number	Ionization pot. (eV)	No. Density $n \text{ (cm}^{-3}\text{)}$
H	1	18	20.4	5.14E+22
C	6	12	73.8	3.43E+22
O	8	7	115.7	2.00E+22

Stopping Powers were calculated from eq. 2-31 for proton energies in the range 1 to 18 MeV, and the value plotted in Figure 32.

The Stopping Power calculation shows the expected approximately inverse relationship between proton energy and stopping power. Actually we just consider a kinetic energy of 3 MeV as characteristic of $\text{D(d,p)}^3\text{H}$ fusion protons with 14.83 keV/ μm stopping power. In our experiments, CR-39 (PM-355, Page Mouldings Ltd., UK) polymer nuclear track detectors registered the arrival positions of the particles passing through the mask. The detection efficiency of C-39 for charge particles incident at near normal incidence is effectively 100%.

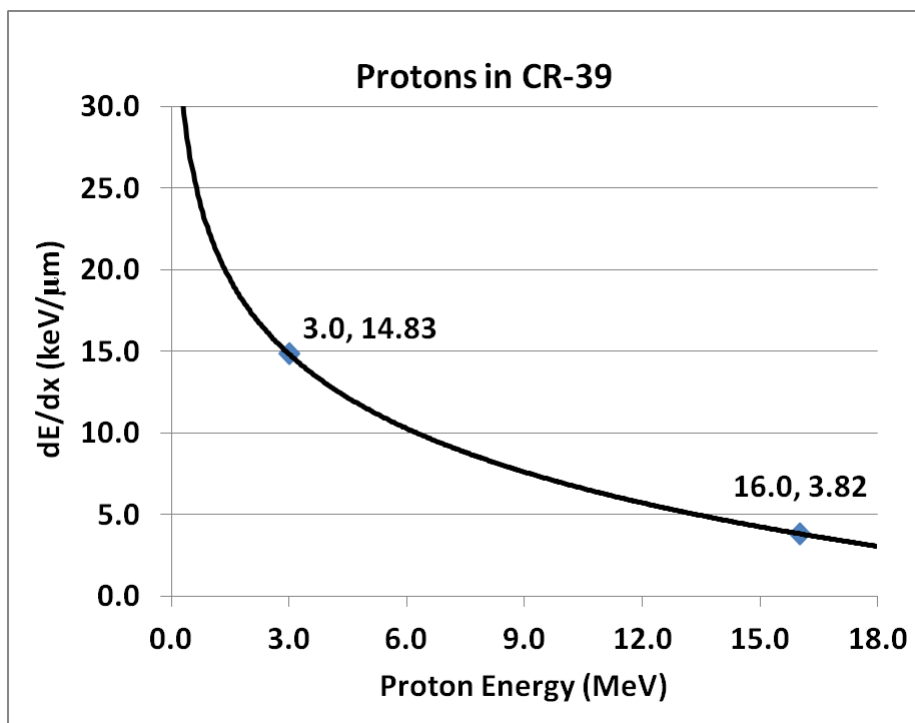


Figure 32: Stopping power graph for protons in CR-39

SRIM Simulation

I have also used SRIM (the Stopping and Range of Ions in Matter) Monte Carlo Simulation for calculating the stopping power which demonstrates good agreement with graph. By using this simulation, we can determine the depth penetration range of the various charged particles emitted from the plasma focus. Figure 33 shows the output screen of SRIM program which is run by 3 MeV protons pass through a CR-39 layer. It illustrates that these protons can go 118 μm deep in CR-39 detectors. As CR-39 characteristics is not in SRIM library, we have to include it in a new layer with element formula number as Table 2 with density= 1.31 g/cm^3 .

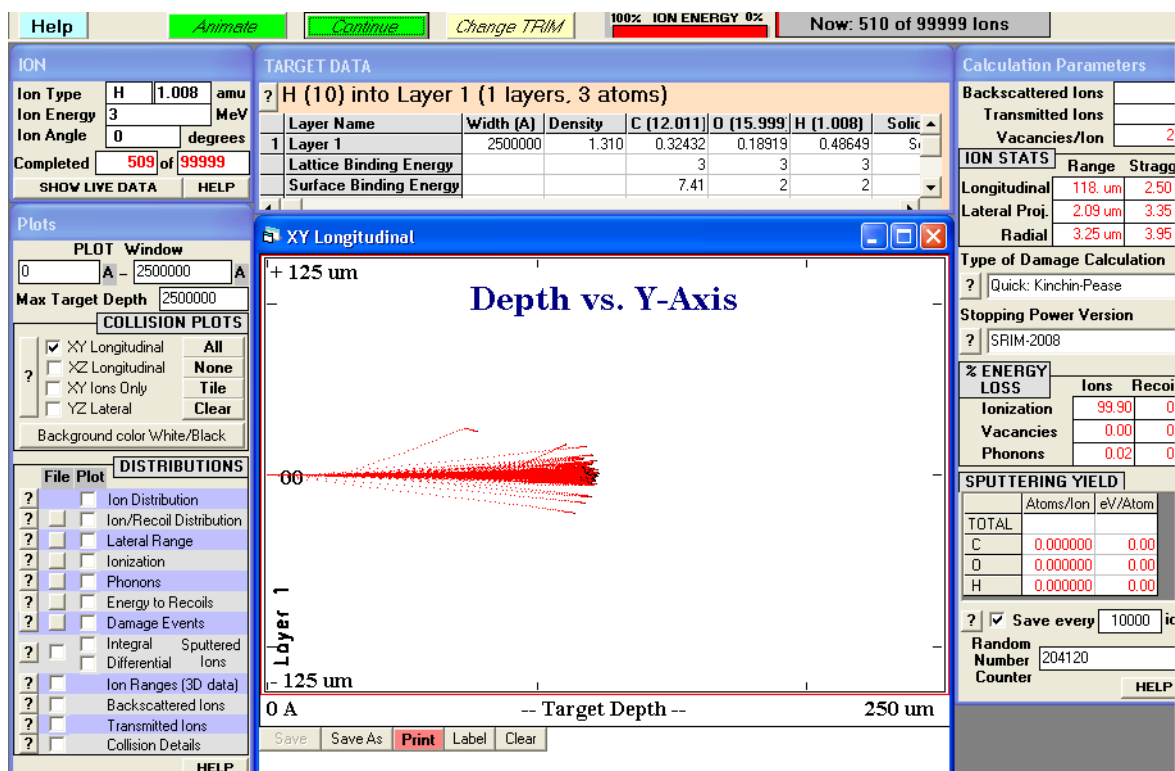


Figure 33: The SRIM simulation result by 3 MeV proton pass through the CR-39 with 250 μm thickness

The main problem is that all charged particles would enter the detector and make a track in the detector, so after etching, the desirable protons position and number cannot be recognized. It leads us to use a proper filter to prevent passing the other charge particles mostly deuterons.

For this matter, Kapton was chosen to filter the energetic deuterons. Table 3 shows the minimum energy of charge particles (proton and deuteron) to pass the Kapton filter layer with different thickness (50 μm , 75 μm , 100 μm) using by SRIM simulation.

Although the results show the best thickness is 100 μm for 3 MeV protons, however in plasma focus device, the D-D reaction mostly relies on beam target model and so because of collision and momentum conservation, the fusion product protons have a wider energy range (roughly $\pm 10\%$). To avoid missing some lower energy range protons, 75 μm Kapton is preferred.

Table 3: minimum energy of proton and deuteron to pass the Kapton filter

	proton min. energy (MeV)	deuteron min. energy (MeV)
Kapton (50 μm)	1.8	2.3
Kapton(75 μm)	2.3	3.0
Kapton(100 μm)	2.7	3.6

Using the same method it can be realized that the residue energy of 3 MeV protons after passing the Kapton is 1.6 MeV and they can penetrate 41 μm in CR-39, instead all deuteron particles with energy less than 3 MeV will be stopped by Kapton. It is worth emphasizing that the 4 MeV deuterons (if any) can go the same depth as 3 MeV protons and make the same track shape on CR-39. Fortunately, the magnetic spectroscopy results predict the quantity of such high energy deuterons drops dramatically, much less than the fusion protons in NX2 plasma focus device [43].

It is worth to emphasizing that sometimes it has been seen a bunch of high energy deuterons pass through the Kapton filter and create a very dense track quantity on the detector (see Figure 34). They were taken with $\times 50$ (left side) and $\times 5$ (right side) object lens magnifications. This detector was placed at 45 degree related to plasma focus axis using the mask-400 (20 $\times$ 20). Moreover the image on the detector shows a pattern quite similar to the mask which evidences the existence of a point (or almost a point) source in pinch plasma. Figure 35 shows the mask-400 pattern and the scanned detector surface by the conventional laser scanner. Also, as it can be seen in this figure, there are some clear shadows in the left-upper side and down side which indicates the existence of the different point sources which are coming like a narrow beam.

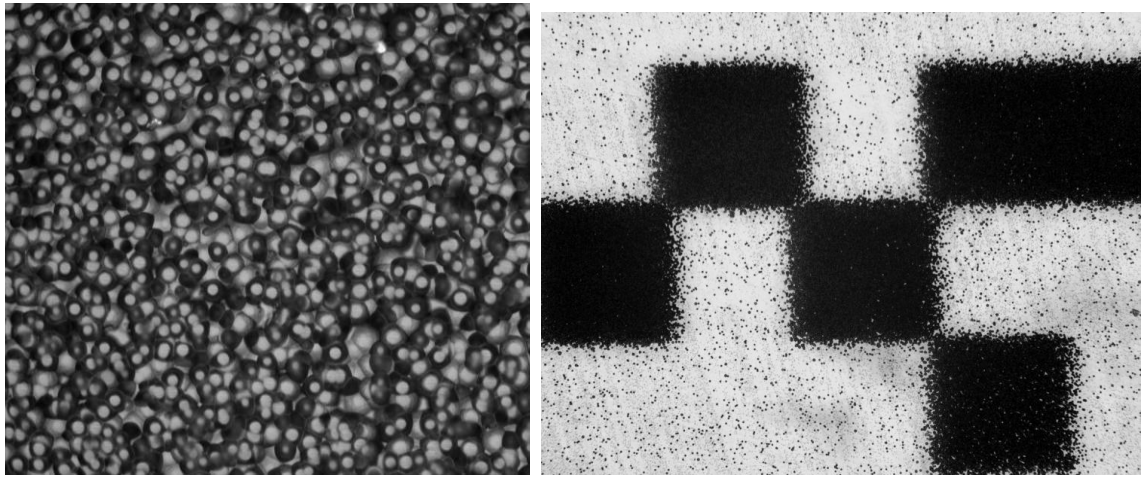


Figure 34: very high dense deuteron tracks on the detector (occasionally) evidence the existence of extremely high energy deuterons in plasma focus device. The deuteron tracks can be seen in images, taken by different microscope object lens: $\times 50$ at the left side and $\times 5$ at the right side.

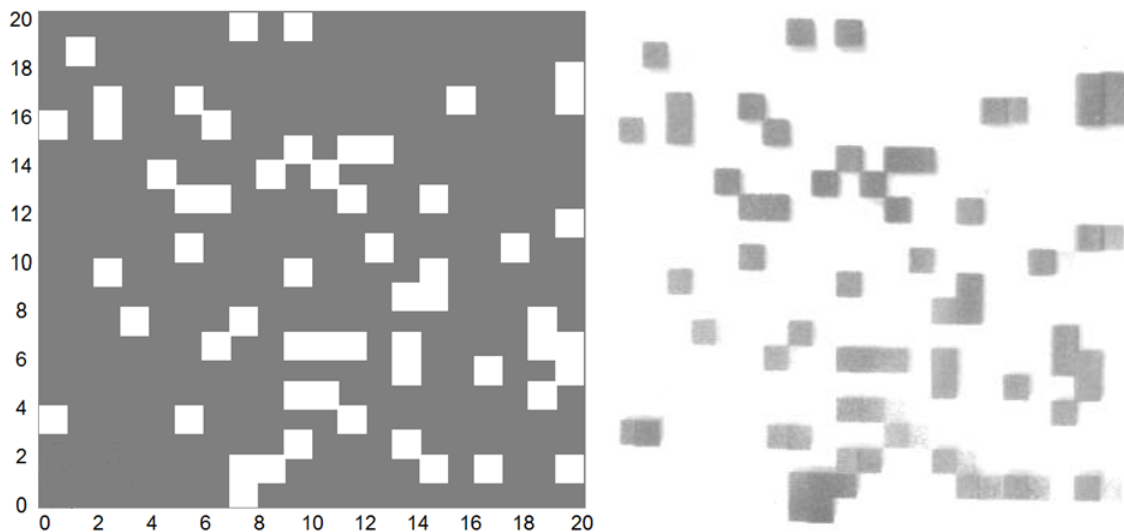


Figure 35: the scanned image of detector (right) shows the same pattern as the mask (left) which shows the present of a point source. The shadows of mask pattern in up-left and down parts of image also evidence that there are more than one point source which are coming like a narrow beam.

Etching Criteria

After exposure, the CR-39 detector is chemically etched to reveal an etch-pit (also referred to as a track) where the energetic charged particle entered the detector. This enables us to extract an ‘image’ of the fusion within the plasma focus in a reliable and efficient fashion. For explaining the nature of charged particle tracks we can use the Restricted Energy Loss (REL) model. When charge particles pass through nuclear tracks detectors, they leave trails of damage along their tracks. This damage (in the form of a trail of broken polymer chains and free radicals) represents a latent track for each

incident particle. The amount of local damage is related to the local rate at which energy is lost by the particle. Non-local forms of energy loss (e.g. bremsstrahlung) are insignificant for protons at these energies; hence REL is essentially equivalent to the stopping power as given by the Bethe-Bloch formula. From Figure 32 and also the Bragg peak in Figure 36 it is clear that the damage along the trail is lowest as the particle enters the detectors, and highest at the end of its track. Hence, in principle, it is evident that the rate of chemical attack on the damaged trail will not be strictly constant. However, to simplify the details of the etch pit geometry we can assume the rate of chemical attack to be constant. It is common that this rate of chemical attack on the damage trail to be called the *track etch rate*, V_T . For areas that have not been damaged by incident charged particles the chemical attack rate on the surface is constant and is often called the “bulk etch rate”, V_B . The main etching parameters are:

- Etchant chemistry [NaOH, KOH, Ba(OH)₂ aqueous solution]
- Molarity
- Temperature
- Etching time

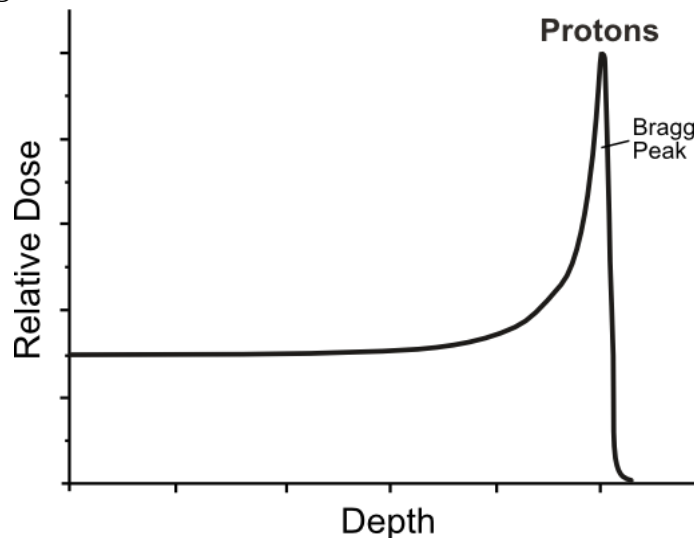


Figure 36: Bragg curve for proton particle. The relative dose (y-axis) and depth (x-axis) are arbitrary unit.

Using a simplified model, when the track etch rate is greater than the bulk etch rate, we can predict that a conical pit will be formed if the particle is incident normally to the surface. This is shown in Figure 37. The more general and complicated case is when the particle is not normally incident on the surface which has been discussed in [80].

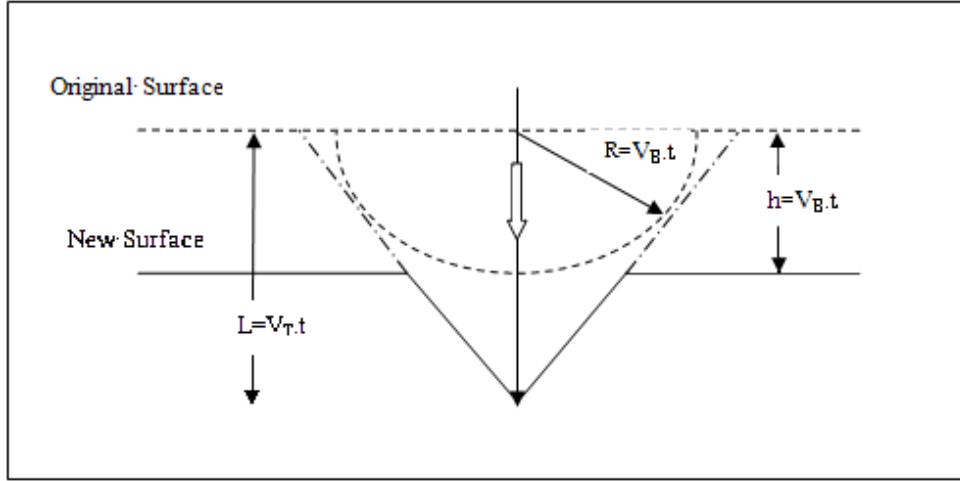


Figure 37: Etching speed and normal track shape

As we have mentioned earlier about stopping power, the protons with the higher energy (like 14.7 MeV for the case of the ${}^3\text{He}(d,p){}^4\text{He}$ reaction.) lose at the beginning of the track much less than that of lower energy (like 3MeV for the case of the ${}^2\text{H}(d,p){}^3\text{H}$ reaction.). This is based on the predictions of the Bethe-Bloch formula. The formula predicts that when non-relativistic particles have the same charge, the specific energy loss varies approximately inversely with the particle energy. As the amount of damage in the material is directly related to the track etch rate, we can infer that the V_T in the case of the higher energy protons is lower.

But the sensitivity, $V = V_T/V_B$ has much more important role to identify the etching solution; higher sensitivity is depended to higher V_T and lower V_B . The response function of V has been defined by three groups:

$$\text{Green et al: } V = 1 + (11.45e^{-0.339R} + 4.0e^{-0.044R}) \times (1 - e^{-0.58R}) \quad (2-32)$$

$$\text{Brun et al: } V = 1 + e^{-0.1R+1} - e^{-R+1.27} + e^{1.27} - e^1 \quad (2-33)$$

$$\text{Nikezic et al: } V = 1 + e^{-0.06082R+1.119} - e^{-0.8055R+0.8055} \quad (2-34)$$

where R is residual range.

Based on these formulas, a program (NRU) has been written by D. Nikezic and K.N. Yu [81] (City University of Hong Kong) which calculates the track parameters, major and minor axes and depth. Then it plots the track profile and track opening contour. By specification of detector (CR-39 or LR-115), α -particle energy (MeV), etching time (hour), bulk etch rate ($\mu\text{m/h}$), incident angle and choosing a formula, it shows the tracks shape and its specification Figure 38 demonstrates the result of a track

for 5 MeV α particle which is etched 5 h and bulk etch of solution is 1.6 $\mu\text{m/h}$ (for 6.25 n NaOH which we use it as usual) [47, 82], incident angle 90° by choosing the formula number1 (from 3 suggested empirical formulas in NRU program).

It indicates a nice conic track shape which it can be seen a dark cone under microscope, however if we increase the etching time, the shape goes to be shallow and shiny end point under microscope. Figure 39 shows this problem with 8 hours etching. Moreover, suppose we have different energy ranges of incident particle, in our case 2.7-3.3 MeV protons and <3.5 MeV deuterons. If the etching time is defined properly for proton tracks, they will appear like a dark end cone circle whilst deuterons tracks will be a bright end cone circles, because they have been over etched. This is a good approach to distinguish the genuine proton tracks and high energy deuteron tracks. Figure 40 (a,b) has been taken by Confocal microscope which shows the normal and over etched tracks and (c) shows the difference between proton and deuteron tracks.

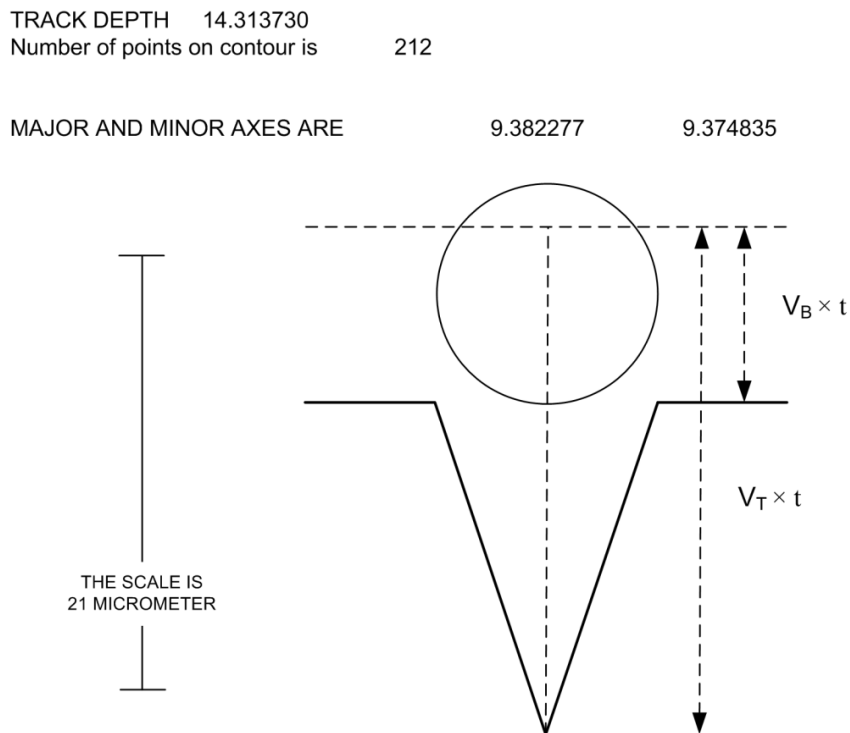


Figure 38: NRU program output shows the 5 MeV α track size after 5 h etching.

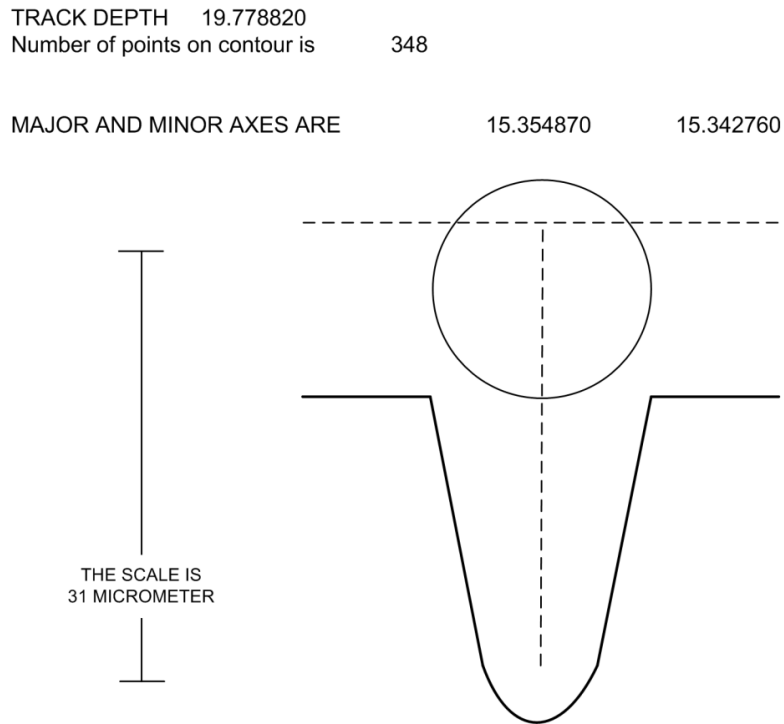


Figure 39: Over etched track shape is shallow and appears like bright circle

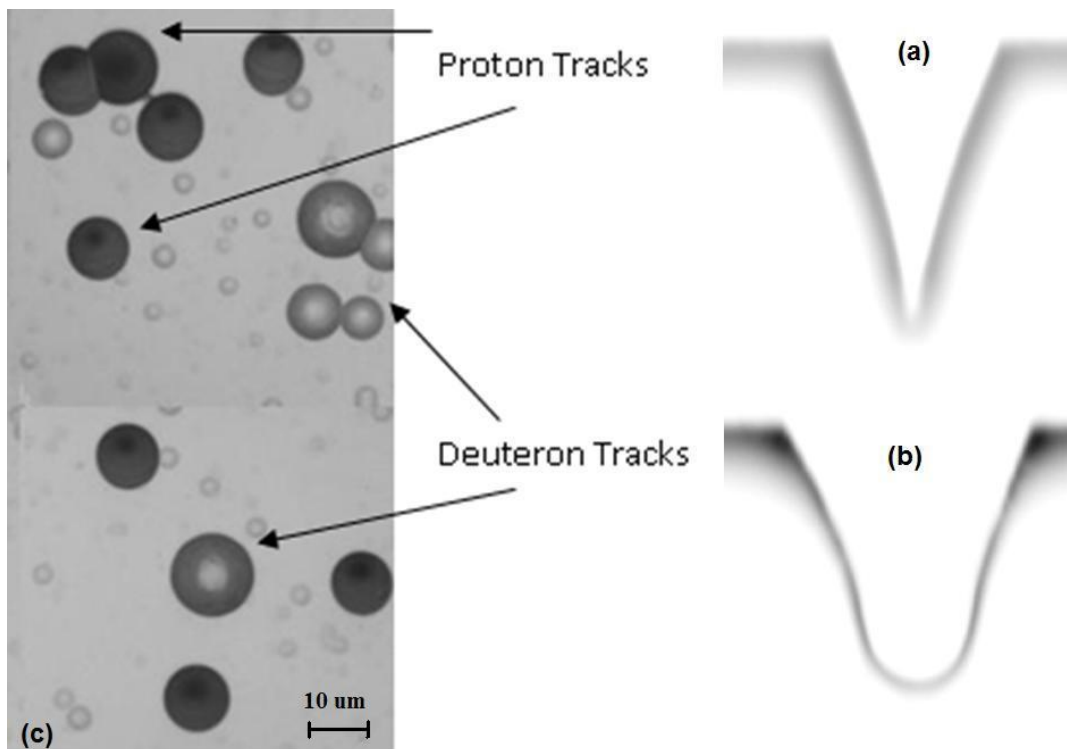


Figure 40: normal track (a) and over etched track (b) taken by confocal microscope.
(c) Distinguishing between usual proton and deuteron tracks under microscope

2.2.3 Scanning

To acquire the images of CR-39 surface, an Optical Microscope (Olympus BX51) has been used together with a motorized stage (PRIOR Scientific H138) controlled by Stage-Pro software (Media Cybernetics). Figure 41 shows the detail of different sections of microscope. It includes four objective lenses ($\times 5$, $\times 10$, $\times 20$ and $\times 50$) which usually MPlan $\times 20$ objective lens has been used. The detector scanning system also comprises a (PRIOR Scientific) motorized microscope stage controlled by Image-Pro software (Media Cybernetics). For experiments with small mask (mask-400) the 1.4 Mpx (1360 $\times$ 1024) CCD camera model CFW-1310M in full 10-bit depth was used with resolution of 0.38 $\mu\text{m}/\text{pixel}$ and 528 $\times$ 388 μm^2 field of view for each frame. However, for the last part of experiment, using the bigger mask and naturally bigger detector scan area, a new 4.9 Mpx (2560 $\times$ 1920 pixels) CCD camera was employed with the bigger field of view (726 $\times$ 544 μm^2) and better resolution (0.28 $\mu\text{m}/\text{pixel}$). The resolution of image is 300 dpi with Bit-depth of 8. The camera interface used was IEEE 1394 (FireWire).

To obtain the best focusing on detector, we have two choices; automatic laser-focusing and software auto-focusing. In auto-focus option, the program allows the system to scan the frame in different focusing state (z-position) and choose the best one (maximum local contrast) and goes to new position (if it is set). The number of stage and step size for finding the best focus must be identified by user. In laser focusing option (using Prior Scientific LF100 laser auto focus) the system sends a laser wave to detector surface and detects the reflected wave and finds the best focus stage. Although it is much faster than software auto-focusing but for using this option the detector surface must be highly reflective, so it needs to be coat by Silver or Indium before scanning. Actually Silver has the best reflectivity, but it will be oxidize very soon and the tracks will be disappeared. The problem of Indium is that it cannot be bonded on CR-39, so, it must be coated first by Chromium because of its good adhesiveness. In our case, we coat the detectors with 200 μm Chromium and 250 μm Indium by using the thermal evaporator.

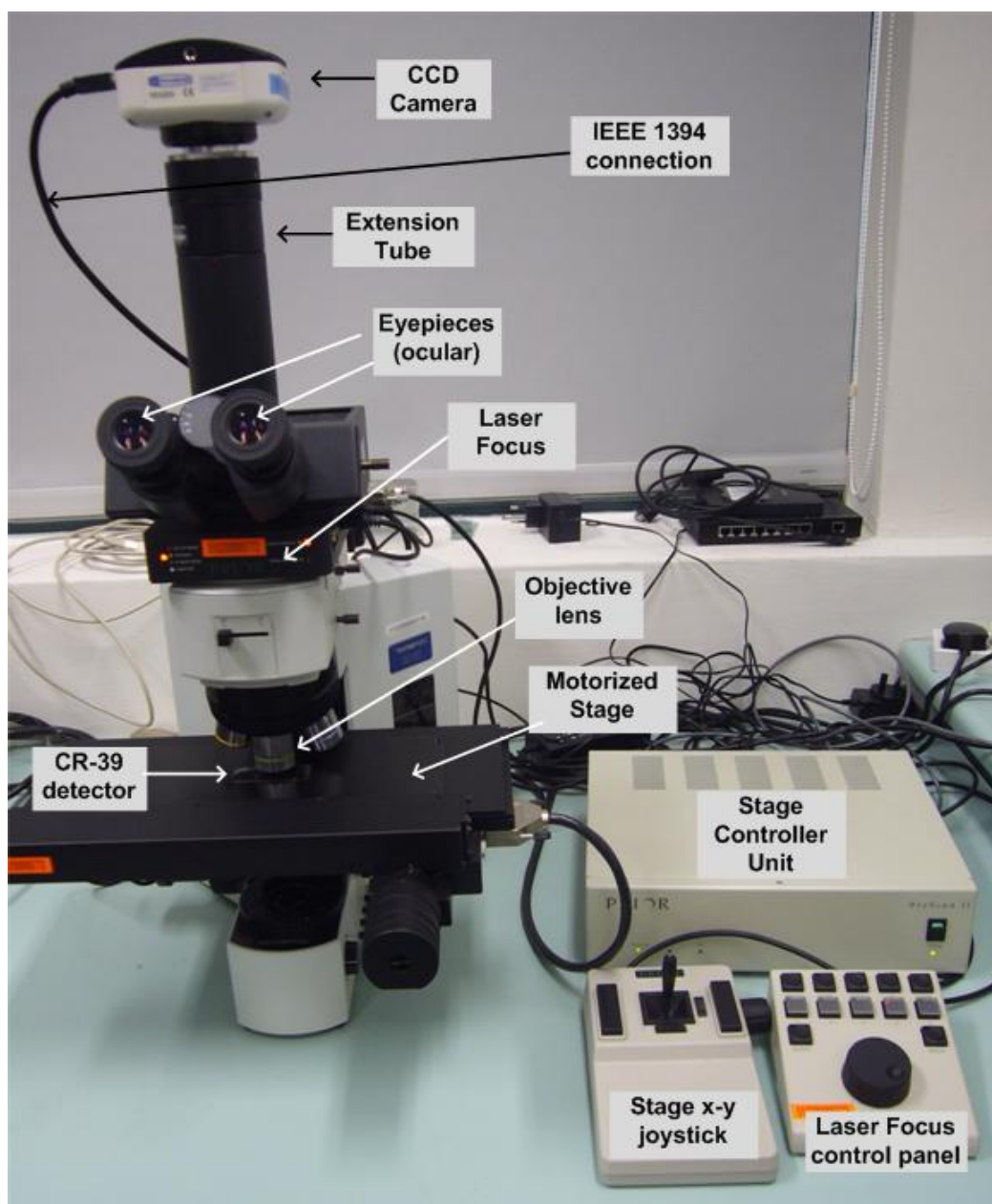


Figure 41: Microscope Olympus BX51 and its attachments.

2.2.4 Image-Pro for Track Analysis

Filtering operations reduce or increase the rate of change that occurs in the intensity transitions within an image. Areas in which there are sudden or rapid changes in intensity appear as hard edges in an image. Areas where there are gradual changes produce soft edges. Filtering acts to detect and modify the rate of change at these edges. It can increase the intensity differences in a soft edge to make it appear sharper, or reduce the intensity differences in a hard edge to smooth and soften it. Filtering operations produce their effect by modifying a pixel's value based upon the values of the pixels that surround it. For example, blurring is accomplished by averaging all of the pixel values in a specified region, and replacing the center pixel with the averaged value. This produces reduced variation among neighboring pixels, which visually softens the image. A sharp black/white edge would be softened with intervening levels of gray.

Filtering techniques are divided into two categories: convolution filters (linear filters) and nonconvolution (nonlinear) filters. Both techniques accomplish their results by examining and processing an image in small regions, called pixel "neighborhoods." A neighborhood is a square region of image pixels, typically 3×3 , 5×5 , or 7×7 in size. Two of most important filters which have been used in our image analysis are explained here:

Flatten: Using this filter causes to even out background variations. This is often one to prepare an image for a count/size operation if its objects are difficult to isolate because the background contains pixels of the same intensity as the objects of interest. Flatten reduces the intensity variations in the background pixels.

Watershed: Using this filter causes to separate objects that are touching. The Watershed filter erodes objects until they disappear, then dilates them again, but will not allow them to touch. Figure 42 shows the effect of Watershed filter to split the overlap tracks.

After filtering the frame, the tracks are counted by proper pre-criteria to avoid having a huge output file, e.g. unusual too small or big tracks or too bright tracks are considered as fake tracks and are omitted in initial analysis. Finally the desired track parameters are measured and record in output data file. These parameters are: area, average brightness (gray-value), roundness and (x,y) position.

An Image-Pro macro (see appendix A) was written to automatically analyze all the stored images, and write a data file containing five measured parameters for each recognized dark candidate track.

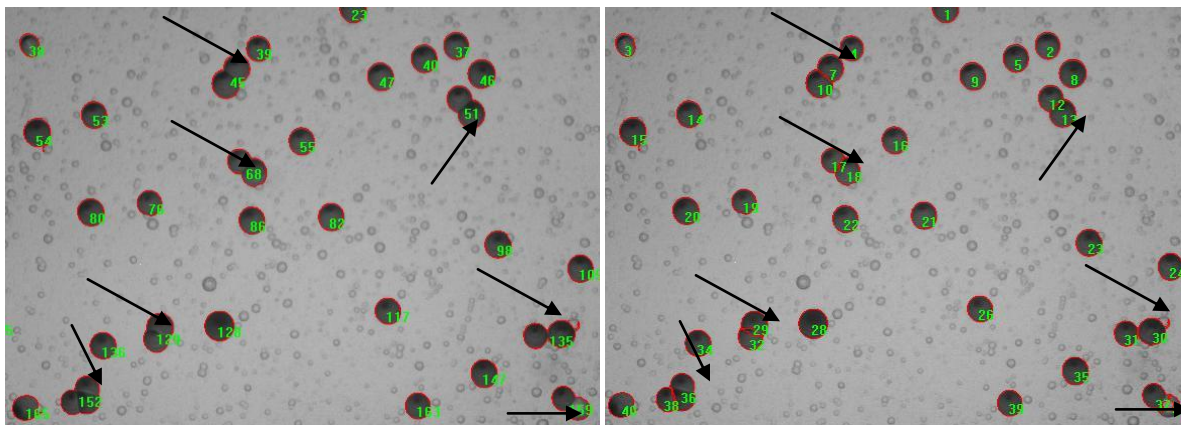


Figure 42: overlapped tracks (left) have been separated by watershed filter (right)

2.2.5 Data Analysis

To discriminate between genuine particle tracks and spurious dark features, within-polygon criteria are applied to scatter plots of (i) brightness (gray-value) vs. area, (ii) roundness vs. area, and (iii) roundness vs. brightness, as shown in Figure 43 (a-c). Genuine particle tracks are selected by applying the logical AND of these three criteria, and the (x,y) positions of all selected tracks (representing the coded image) are written to a new file. A macro was written in MATLAB, allows the user to select the desired region. (see appendix B)

However usually the Image-Pro output data file is too big to be uploaded and processed in MATLAB, so before that, the tracks are selected by simple rectangular criteria (more generous) on three parameters (area, average brightness and roundness). It was done by writing a simple program in Visual Basic (see appendix C) with the simple criteria like $A_{\min} \leq \text{Area} \leq A_{\max}$, $Br_{\min} \leq \text{Brightness} \leq Br_{\max}$ and $R_{\min} \leq \text{Roundness} \leq R_{\max}$. To select the boundaries of each parameter, first, the file size must be reduced by factor of n which can be defined by user, (see appendix D) and then scatter plots of measured parameter let us to find the proper boundaries to apply on data.

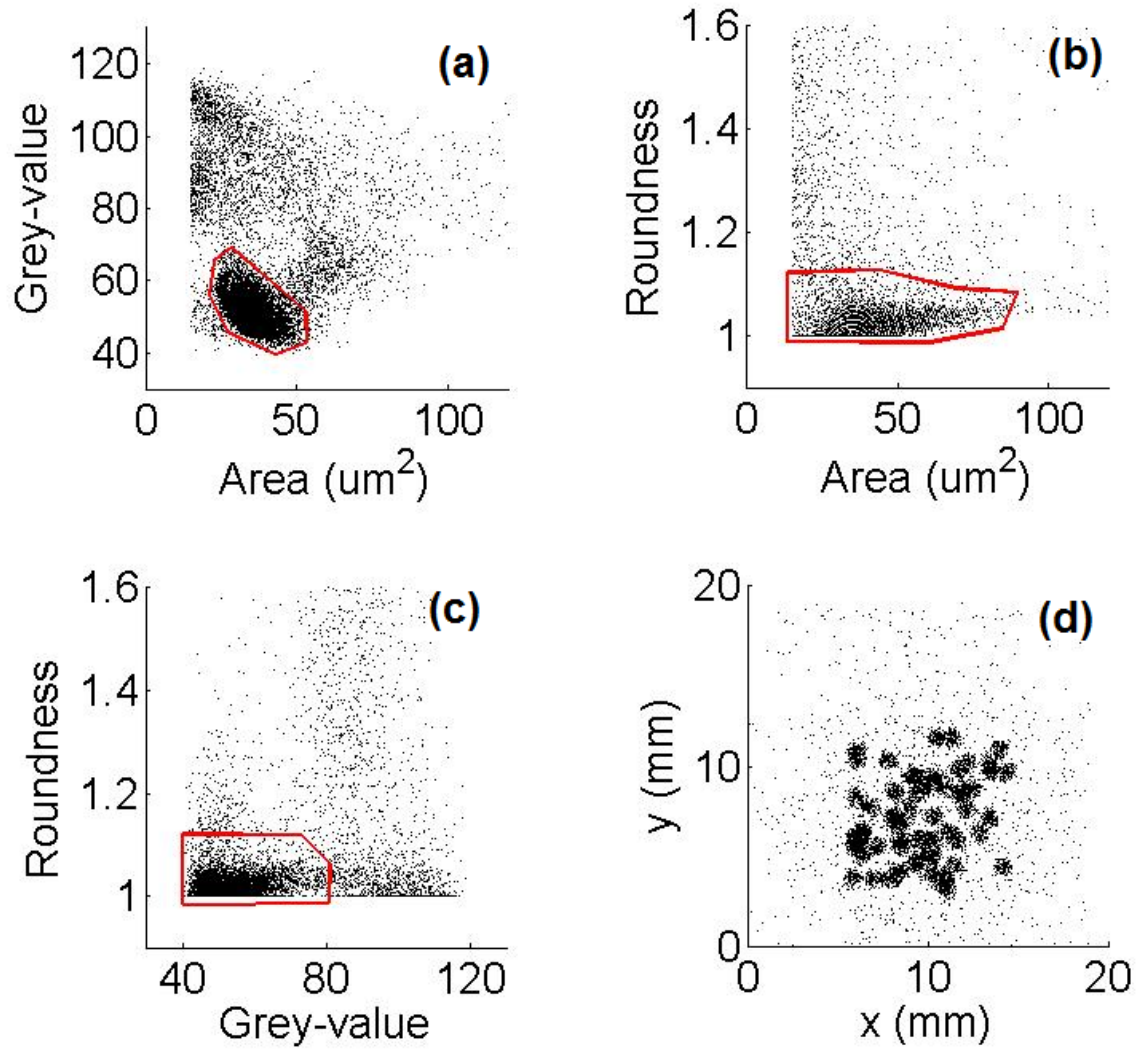


Figure 43: Scatter plots of measured parameters of dark features on CR-39 detector surface: (a) grey-value vs. area, (b) roundness vs. area, and (c) roundness vs. grey-value, with user-defined polygon for each. (d) Positions (x,y) of genuine particle tracks (for 1-spot source) selected by applying the logical AND of within-polygon criteria shown in a-c. This (x,y) data represents the coded image

Finally the coded data was then processed by our convolution-based decoding algorithm (see appendix E) as it was explained in section 2.1.7. The coded data file (filename.caid) comprises the (x,y) positions of all selected tracks. The mask data file (maskname.cam) includes the hole's (x,y) pixel position which must be introduced in experiment geometry file (geometry.caix). It also includes the mask transformation (the direction of scanning) and the distances between Source-to-Mask and Detector-to Mask. Other important problems in imaging are detector/mask misalignment (rotation), inaccurate distance measurement (magnification) and inaccurate tracks' center of mass. In decoding algorithm, it has been tried to correct these problem to achieve the best Signal to Noise Ratio (SNR) by simplex method. With this method, the up to ~10% of

inaccuracy in these parameters will be solved. All program output information like SNR, rotation correction, magnification correction, center of mass correction and other useful results are reported in a log file (filename.calg). The output convoluted data (filename.conv) also comprises the folded coded image before side-lobe subtraction. And finally the output reconstructed file (filename.dat) is the final result of program which can be plot in proper software. Figure 44 shows the final reconstructed contour plot in MATLAB.

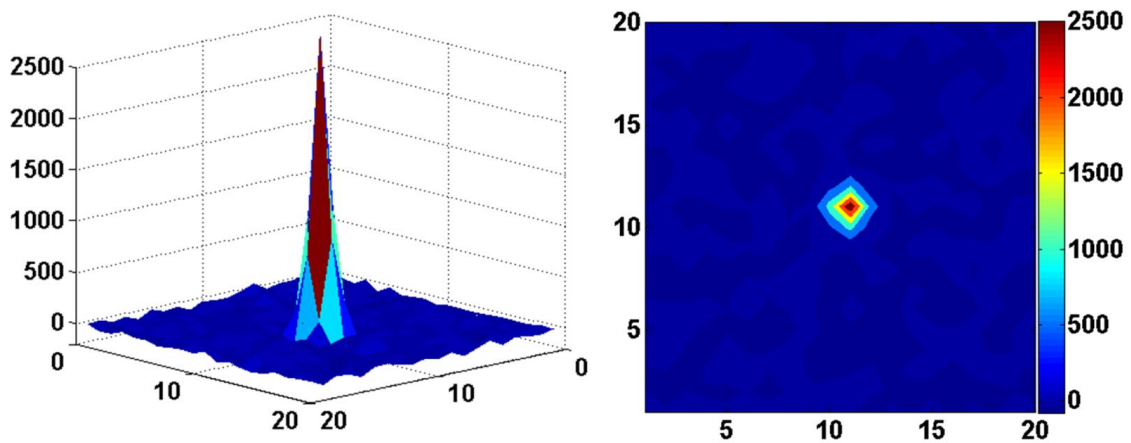


Figure 44: reconstructed image of 1-spot α source with mask-400

2.3 Monte Carlo Numerical Simulation of CAI System

In order to check the correctness of the theoretical analysis and the decoding algorithm, a Monte Carlo program was written in Visual Basic to simulate the results to be expected for various experimental CAI arrangements. In effect, the program generates numerical data which is equivalent to the coded image data extracted by scanning CR-39 detectors in the experimental situation. Inputs to the program include the definition of the source distribution and intensity, the coded mask pattern, and the geometry of the CAI camera. For each point in the source distribution emitted rays were then generated isotropically within a cone which fully illuminates the CAI mask. Rays impinging on the mask plane at the position of an open hole were transmitted to detector plane and the (x,y) arrival position in the detector plane was written to a file; while rays encountering opaque regions of the mask were terminated. In this way a coded image file is produced which is the equivalent of the experimental coded image file. The image

decoding was then performed with the same program as for the experimental data. The number of rays emitted from the simulated source is the equivalent of anticipated experimental exposures (as determined by the detector track count). SNR values in these images were calculated in the same way as for the experimental images. In these simulations, the single pinhole and Singer set mask were compared for the three source geometries. The SNR also are calculated for them and compared.

After definition of the source coordinates, the particles are distributed in whole over a circular surface which its area must be bigger than mask surface to ensure to cover it. For homogenous random distribution:

$$\theta = \cos^{-1}(1 - (1 - \cos \theta_{\max}) \times \text{rnd}()) \quad (2-35)$$

where the $\text{rnd}()$ function returns a pseudo-random real number between 0.0 and 1.0 and θ is the angle of particle emission within a cone of half-angle $\theta_{\max}$.

To approach a better random number in our simulation, the “Twister algorithm” was used. Mersenne Twister is a long period, fast, high quality pseudorandom number generator. Its period is $2^{19937}-1$ (approximately 10^{6002}). It has 623-dimensional equidistribution. The Mersenne Twister algorithm, by Makoto Matsumoto and Takuji Nishimura[83], has a sound theoretical basis and has been tested thoroughly. This Visual Basic .NET code has functions to produce 32-, 64- and (in version 3.1 which we have used in our programming) 128-bit integers, as well as 32- and 53-bit precision real. In all, `mt19937ar.vb` provides sixteen equidistribution generator functions and five initializing methods.

As this represents an isotropic distribution within a cone of half-angle $\theta_{\max}$, the density of rays across the surface of the mask will be slightly non-uniform (see Figure 45).

Visual Basic does not have inverse cosine function, so the inverse tangent function is used

$$\theta = \tan^{-1} \frac{\sqrt{1 - k^2}}{k} \quad (2-36)$$

where:

$$k = 1 - (1 - \cos \theta_{\max}) \times \text{rnd}() \quad (2-37)$$

And also:

$$\varphi = 2\pi \text{rnd}() \quad (2-38)$$

and then: $r_{\text{Mask}} = (z_{\text{Mask}} - z_{\text{Src}}) \times \tan \theta \quad (2-39)$

$$x_{\text{Mask}} = r_{\text{Mask}} \cos \varphi + x_{\text{Src}} \quad \text{and} \quad y_{\text{Mask}} = r_{\text{Mask}} \sin \varphi + y_{\text{Src}} \quad (2-40)$$

where $(z_{\text{Mask}} - z_{\text{Src}})$ is distance between source and mask and r_{Mask} ranges from 0 to r_{Max} according to the distribution $r_{\text{Mask}} = r_{\text{Max}} \sqrt{\text{rnd}()}$ and φ ranges from 0 to 2π . So x_{Mask} and y_{Mask} are the coordinates at which the rays intercept the mask.

The input data file for De-convolution is produced by a program written in Visual Basic (see Appendix F). This data file should be the same as real track distribution on CR-39 detector.

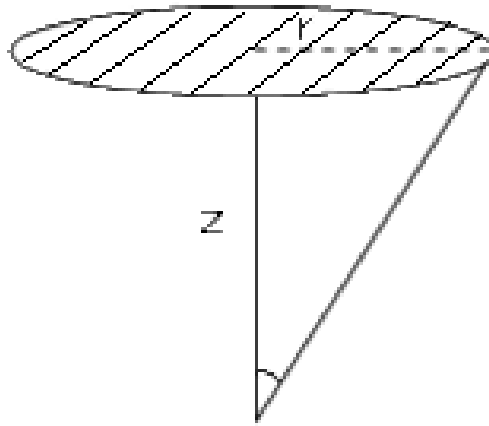


Figure 45: isotropic particles distribution within a cone from a point source

2.3.1 Simulated Source Distributions

In these Monte Carlo simulations, the single pinhole and Singer set mask (mask-400 with 57 square holes) were compared for the three source geometries. In the absence of background noise, the result of pinhole system shows zero background but weak precision (Figure 46-a1), whereas the coded aperture demonstrates better precision with the much higher particle number which is proportional to the number of holes (Figure 46-b1). From the 1.38×10^6 radiated particles emitted within the 1° cone, about 143 particles will pass through the pinhole, whereas for the CAI mask about 8,150 (~57 times more than the pinhole) particles reach the detector.

For the 1-spot source, the SNR shows an improvement by factor of 6.8, from 7.0 for pinhole to 47.7 for CAI (see Table 4). The 2-spot source is simulated as two 1-spot sources separated by 24 mm distance (center to center). Figure 46-a2 and b2 show the simulation results for the pinhole and mask, respectively. The SNR for pinhole and CAI are 7.1 and 44.8 respectively; a factor of 6.3 improvement. Finally for the ridge source, the centres of random uniform disc sources were distributed randomly along a line of 2.4 cm length. Figure 46-c1 shows the simulated pinhole image and it can be seen that the height and width of the imaged ridge varies irregularly along its length. By contrast the simulated CAI image, shown in Figure 46-c2 reproduces the shape of the ridge far better. As the ridge source is the most extended of the three sources, the SNR for both pinhole and CAI mask are lower than the other two sources; being 4.7 for the pinhole and 22.9 for CAI mask. In this case the SNR for CAI mask is a factor of 4.9 better than that of the pinhole.

Table 4: Track number and SNR for CAI and pinhole system from experimental results and theoretical calculation

Simulation	No. of tracks (pinhole)	No. of tracks (CAI)	Sim. SNR (pinhole)	Sim. SNR (CAI)	Theory SNR (CAI)
1-point circular source	149	8,127	7.0	47.7	44.9
2-point circular source	292	16,299	7.1	44.8	40.9
Ridge source	144	8,051	4.7	22.9	21.1

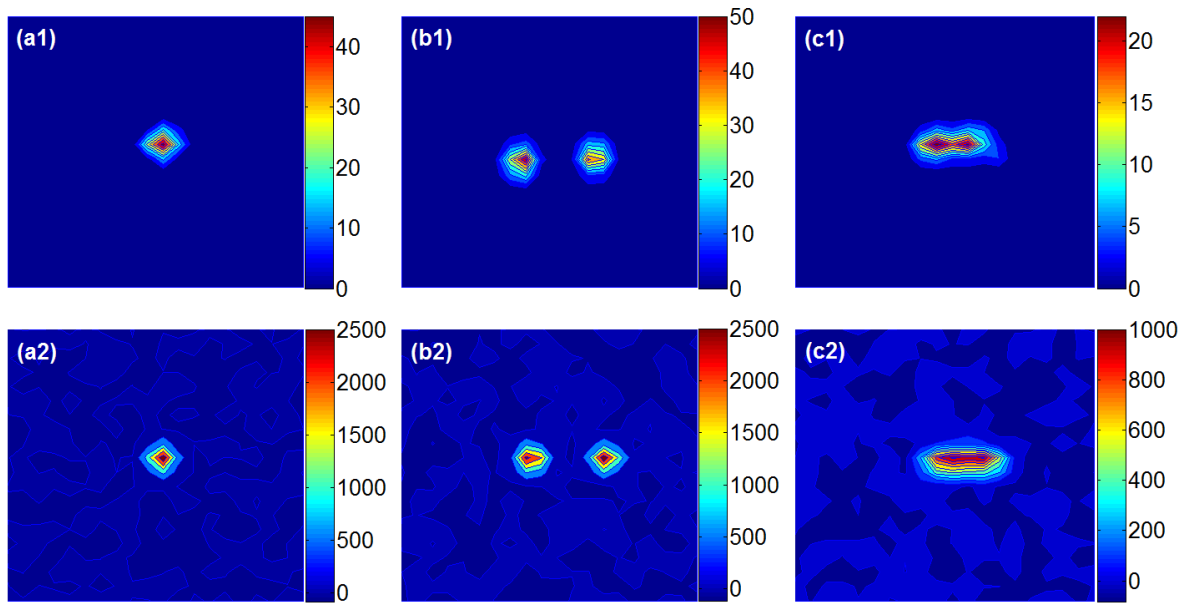


Figure 46: Reconstructed images from Monte Carlo generated data for different source shapes with pinhole (top row) and 20x20 CAI mask (bottom row). All images are of 10 cm side (in the source plane). 1-point source for pinhole (a1) and mask (a2); 2-point source for pinhole (b1) and mask (b2); ridge source for pinhole (c1) and mask (c2)

2.3.2 Fusion Source Simulation with mask-1365 and mask-4680

For the simulation of fusion source in NX2 plasma focus device, two masks were applied: mask-1365 ($p = 1365$, $h = 341$, $\lambda = 85$) and mask-4680 ($p = 4681$, $h = 585$, $\lambda = 73$), the same as that we used for the experiments. The source was generated pseudo-randomly and uniformly across a (10 mm×2 mm) rectangle as the proton source, placed at 42 mm from the mask. The directions of emitted rays were then generated isotropically within a 40° cone, fully illuminating the CAI mask. Then the same procedure as alpha source simulation was applied for detector plane and reconstructing the image. The number of rays emitted from the simulated source is the equivalent of the average experimental exposures; 1.5×10^7 emitted particles within the 40° cone is similar to one of the real PF shot with the neutron yield equals 2×10^8 and the neutron anisotropy equals 1.7 to get the same amount of the exposed particles on detector. SNR values in these images were calculated in the same way as for the experimental images.

The final exposed proton track number and SNR for mask-1365 were 160,000 and 33.7 respectively, whereas for mask-4680, they were 272,000 and 32.4. It can be seen, despite of increasing the total exposed particle number by 70% with mask-4680, the SNR decreased by 4%. This reality comes from our previous calculation of SNR in section 2.1.6 and shows in Figure 27 and Figure 28.

Figure 47 and Figure 48 show the reconstructed images from Monte Carlo generated data for a 10 mm×2 mm rectangular source coded with mask-1365 and mask-4680 respectively where the right axis is the particle count density.

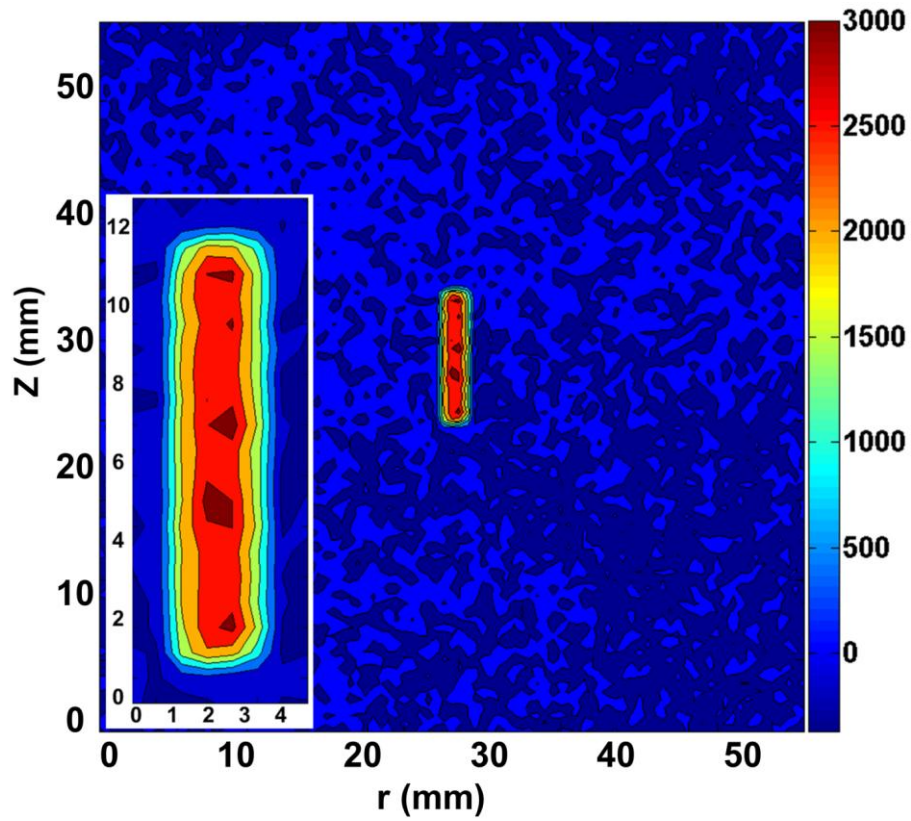


Figure 47: mask-1365: reconstructed image from Monte Carlo generated data for a 10 mm×2 mm rectangular source.

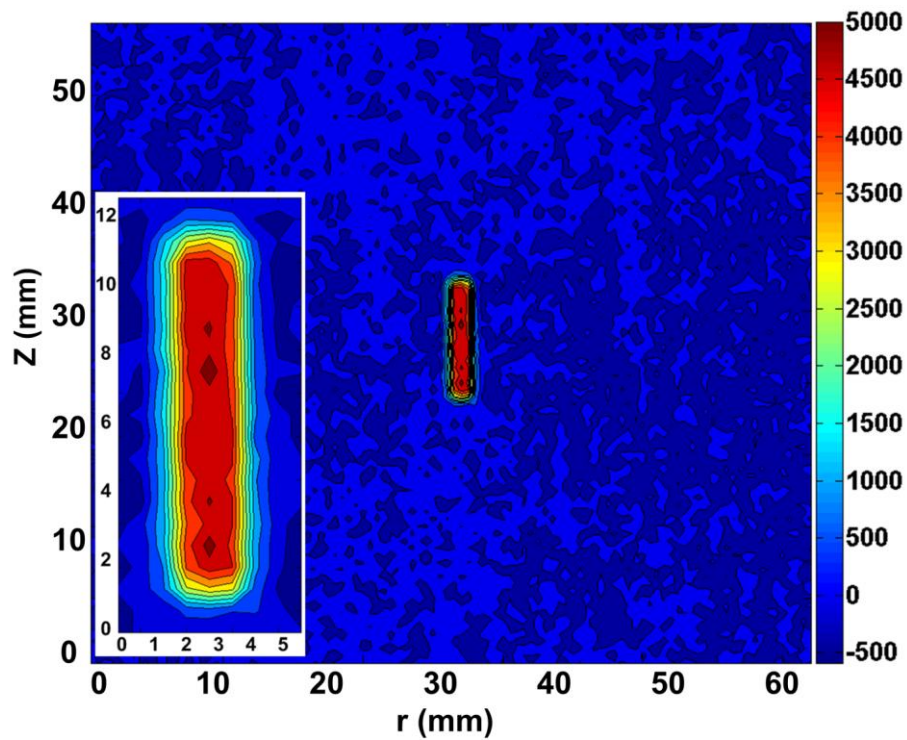


Figure 48: mask-4680: reconstructed image from Monte Carlo generated data for a 10 mm×2 mm rectangular source.

Chapter 3

Beryllium Activation Neutron Detector

3 Beryllium Activation Neutron Detector

3.1 Overview

In 1932 James Chadwick bombarded beryllium with α particles leading to the emission of a previously unknown particle. In order to explain his experimental results he postulated the existence of a "new" neutral particle, which he named the neutron. Hence the nuclear reaction he observed was ${}^9\text{Be} + \alpha \rightarrow {}^{12}\text{C} + \text{n}$. In the present day, neutron measurements have become an essential part of any experiment or technique involving nuclear fission or fusion.

Neutron detection techniques are based on indirect methods. Typically, a nuclear reaction with, or a neutron scattering from, a target nucleus produces one or more energetic charged particles, and the ionization or excitation resulting from the transit of the charged particles is processed by the detection system. Detectors which rely on neutron absorption are generally more sensitive to thermal neutrons, and are orders of magnitude less sensitive to high-energy neutrons. On the other hand, organic scintillation detectors, which rely on elastic scattering of fast neutrons to transfer kinetic energy to hydrogen nuclei, have difficulty registering low-energy neutrons. This illustrates the principle that different approaches are required for detecting neutrons of different energy groups. Thermal neutrons are usually detected by absorption/capture methods which can be divided into prompt detectors employing ${}^3\text{He}$, ${}^6\text{Li}$, ${}^{10}\text{B}$ (examples of these are ${}^3\text{He}$ filled proportional counters, lithium-iodide (LiI) scintillator crystals, and ${}^{10}\text{B}$ -enriched-boron-trifluoride (${}^{10}\text{BF}_3$) filled proportional counters); and activation detectors employing silver (Ag) [84], indium (In), or rhodium (Rh) [85]. Fast neutrons can be detected by organic scintillation [86]; activation by inelastic scattering from certain chemical elements, such as indium (In) [87, 88], arsenic (As) [89] or yttrium (Y) [90]; or by moderation to thermal energies followed by capture by a suitable chemical element that emits charged particles.

Neutron activation methods are well suited to make accurate measurements of neutron fluence in environments with high backgrounds of gamma-rays or other forms of radiation. For many chemical elements, neutron-induced nuclear reactions lead uniquely to a specific radioactive product which no other form of radiation can produce. The half-life of the radioactive product allows measuring (in some way) of both the

neutron fluence and the background radiation; and then the essence of the activation technique is that the neutron fluence during exposure can be inferred from a measurement of the activity of the product nuclide after exposure. In this way the influence of even an intense background of non-neutron radiation is very effectively eliminated.

Several different chemical elements and activation schemes have been employed in a wide variety of experimental situations. In practical terms, the half-life of the induced activity must be long enough so that the target can be recovered (if necessary) and counted before the activity decays away, but not so long as to make the counting interval impractically long. Many other factors come into the selection of a particular target element and reaction, including: the natural abundance of the relevant isotope, the reaction cross-section, the chemical and physical forms of the element available, the type and energy of the decay radiation, and the branching ratio for the relevant decay mode. For some targets (e.g. rhodium) cost can also be an important consideration. The type and energy of the decay radiation strongly influences its detection efficiency, discrimination against interfering radiations, and self-absorption in the activation target. To reduce self-absorption of the decay radiation, the target is often in the form of a foil or sheet, and its lateral dimensions are chosen to match the active area of the detection system. In situations where the high background radiation is more or less continuous, the activated target is typically transported to a shielded low-background area where it is “counted”. Automated foil activation systems with pneumatic target transfer (or sometimes activation of a flowing liquid or gas) are important neutron diagnostics for large fusion devices [91-94]. If relatively long half-life activations are chosen, then several different foils may be activated over the same exposure period and measured individually with a single remote counting system.

For pulsed neutron sources, it is often advantageous to measure the activity *in situ* after the burst of background radiation associated with the neutron pulse has died away. The detector and activation target are therefore kept in close proximity, and together form a single instrument referred to as an “activation counter” [86]. Unlike foil activation schemes, no transfer-time is involved, and short half-lives (typically milliseconds to minutes) are preferred. Both thermal neutron and fast neutron activation detectors have been developed. Rhodium and silver, for which (n, γ) neutron capture reactions lead to β radioactive products, are the most suitable elements for thermal

neutron activation counters. Naturally, with the addition of a moderator fast neutrons can also be detected. However, the bulky nature of the moderator, mutual interference between nearby moderating detectors, and the lack of neutron energy discrimination are potential disadvantages of such detectors.

For fast neutron activation counters a considerably wider group of elements have been employed, including: arsenic [89], beryllium [95], fluorine [96], indium [87, 88], lead [97], neon [94], oxygen [93], praseodymium [98], sodium [99], yttrium [90] and zirconium [97]. Fast neutron activation of these elements occurs via one of the reactions: (n,n') , $(n,2n)$, (n,p) or (n,α) . In each case, the reaction has a definite threshold energy which is useful for differentiating between neutrons of different energy groups, such as D-D and D-T fusion neutrons, or discriminating against room scattered neutrons (which will have lower energies than the source). For fast D-D neutron activation detectors the most suitable elements (with half-lives) are: arsenic (17.6 ms), beryllium (807 ms), indium (4.49 h) and yttrium (15.7 s). The half-life for arsenic activation is too short to properly exclude the influence of capture gamma-rays from the room, making beryllium a good choice of activation target for moderate repetition-rate sources. Fast neutron activation detectors have proven well suited for neutron fluence measurements on pulsed devices such as ICF, Z-pinch, and plasma focus sources.

${}^9\text{Be}(n,\alpha){}^6\text{He}$ is the relevant nuclear reaction for the activation of beryllium by fast neutrons, and Figure 49 shows a plot of the cross-section from the Evaluated Nuclear Data File (ENDF), National Nuclear Data Center (NNDC) [100], where the main source for experimental cross-section data for energies up to 4.4 MeV is [101]. The reaction has threshold energy of 670 keV, but the cross-section remains negligible for neutron energies below 1 MeV, then climbs to a maximum of 105 mb between 2.8 and 3.0 MeV. Consequently the detector has good sensitivity to ~ 2.5 MeV neutrons from D-D fusion, while largely rejecting room-scattered neutrons, which are predominantly < 1 MeV. The ${}^9\text{Be}(n,\alpha){}^6\text{He}$ cross-section at 14 MeV is approximately 10 mb, so the detector has much lower sensitivity for the detection of DT fusion neutrons. The ${}^6\text{He}$ beta decay ($100\% \beta^-$ to ${}^6\text{Li}$) has a half-life of 0.807 second and end-point energy of 3.5 MeV. This high end-point energy means that a relatively thick beryllium sheet (5 mm in our case) can be used as the activation target with a substantial fraction of the emitted beta particles still being able to escape from the metal sheet. So in terms of detector sensitivity, the high beta end-point energy and low atomic number

of the target compensate for the relatively modest reaction cross-section. Moreover, the half-life of ${}^6\text{He}$ enables this detector to be used with moderate repetition rate sources (<0.2 Hz) such as plasma focus devices.

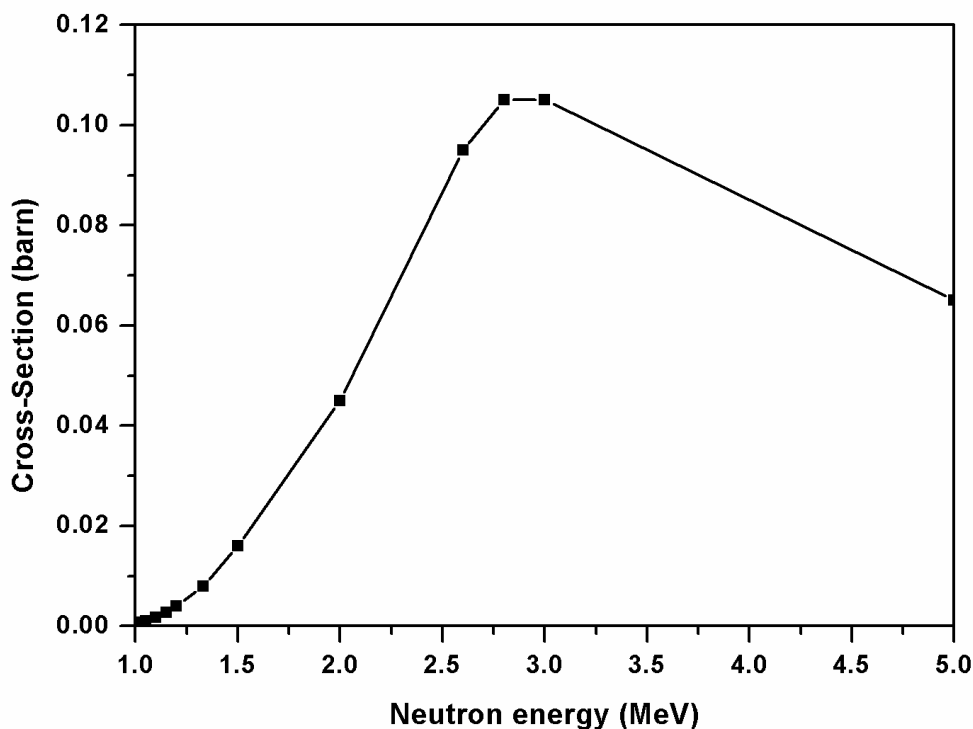


Figure 49: The ${}^9\text{Be}(n,\alpha){}^6\text{He}$ reaction cross-section as function of neutron energy.

A particular difficulty with neutron detectors is that, unless a well-characterized neutron source with the same energy characteristics as the intended application is available, it is usually difficult to obtain a reliable absolute calibration of the detector response. A method of achieving this for beryllium activation detectors is presented in this chapter.

Moreover, one of the most important parameters for D-D fusion in the deuterium-filled plasma focus device is neutron anisotropy which yields important information on the different neutron production mechanisms responsible for the fusion reactions [102, 103]. Some models such as the moving boiler model suggests that the anisotropy of the neutron yield results from plasma Center of Mass (CM) motion where ions and electrons are thermalized at energies of a few keV [104]. Conversely, the beam-target model considers the fusion to be due to beams of charged deuterons accelerated by electric fields [105-108] within the pinch column. The neutron anisotropy has been investigated by different groups on plasma focus devices [44, 109-

114] of widely different beam energies. It is worth emphasizing that the neutron emission from the plasma focus exhibits both fluence and energy anisotropy. Moreover any displacement (or varying distribution) of the fusion source along the PF axis will influence anisotropy measurements. It is therefore necessary to study the fusion source position.

A prototype of the beryllium detector was previously constructed in our lab [46], with a 2 mm thick Be sheet and proportional counters which required a flowing gas mixture (90% Ar + 10% CH₄) at a rate of about 1.2 L/min. For the new detector used in these experiments, the beryllium sheet is sandwiched between two sealed xenon-filled proportional counters. There are two main advantages of xenon- over argon-filling for the proportional counters. Firstly, xenon has a higher electron density (a factor of 3 for the same gas pressure) and therefore higher energy loss for the beta particles traversing the gas layer. Secondly, xenon has much higher x-ray absorption than argon enabling the proportional counters to be energy-calibrated using standard x-ray sources. Of course, the xenon-filled proportional counters are also sensitive to the hard x-rays emitted during and after the maximum compression of the plasma focus pinch. However, since this is an activation detector, a delay of a few milliseconds following the shot eliminates the influence of the hard x-rays and is sufficient time for the proportional counters to recover (i.e. to clear the gas-ionization and re-charge to the applied bias voltage) before counting of beta particles begins.

The goals for developing the new Be activation detector were: (i) to eliminate the inconvenience of flowing gas in the proportional counters, (ii) to obtain a high β particle count (typically several thousand counts for NX2 PF shots of $\sim 10^8$ neutrons), and thereby achieve good statistical accuracy in the measured neutron fluence, (iii) attain an absolute calibration using the method described in the next sections. In order to perform anisotropy measurements on the PF device, two identical beryllium activation detectors were fabricated.

3.2 Construction

The beryllium sheets used are of dimensions: $210 \times 106 \times 5 \text{ mm}^3$, with a mass of 205.9 g. Then for the maximum cross-section of 105 mb, the macroscopic cross-section for the (n, α) reaction is $\Sigma = N_{\text{Be}} \times \sigma = 1.43 \text{ cm}^2$. Where N_{Be} is the number of beryllium atoms ($N_{\text{Be}} = m_{\text{Be}} N_A / w_{\text{Be}}$). Then for a β escape probability of 40 % (will be explained in next section) the effective cross-section of the detector would be $\Sigma_{\text{eff}} = 0.57 \text{ cm}^2$. And hence the response factor for a beam of 1 n/cm^2 fluence would be $R = 0.57$ counts. This estimation clearly neglects the influence of elastic and inelastic scattering, and attenuation of the neutron beam due to $(n, 2n)$ and (n, α) reactions. These effects are accounted for in the Monte Carlo calculations described in next section.

The proportional counters (PCs) used are LND 49727 beta-gamma counters filled with xenon gas at 800 torr, with an active area of 200 cm^2 , active depth of 1.27 cm, and with a titanium window of 5 mg/cm^2 thickness. The manufacturer of these PCs (LND Inc., Oceanside, NY, USA) specifies a minimum lifetime of 10^{12} counts, and a recommended operating voltage range of 1350 to 1600 V. The beryllium metal sheet and two proportional counters are held in the desired relative positions by a machined aluminum frame. Wrapping this “sandwich” in adhesive tape then forms a robust and compact assembled detector.

The β particles are detected by the PCs enclosing the Be sheet (either of them). The minimum energy deposited by a β particle in the proportional counter can be estimated from the electron stopping power in xenon (at 800 torr) for the minimum path length of 1.27 cm, using electron stopping power values from the ESTAR online database from the National Institute of Standard and Technology (NIST) [115].

$$\Delta E = \left(-\frac{dT}{dx} \right) (E_e = 1.0 \text{ MeV}) = \left(1.122 \text{ MeV} \cdot \text{cm}^2/\text{g} \right) \left(5.75 \times 10^{-3} \text{ g/cm}^3 \right) (1.27 \text{ cm}) = 8.2 \text{ keV} \quad (3-1)$$

The above value of 8.2 keV corresponds to the minimum energy deposited for the Continuous Slowing-Down Approximation (CSDA). In reality however, the β particles deposit energy by making many discrete collisions with Xe atoms, and since this is a stochastic process there will be a spread in energies deposited by individual beta particles (even for the same path length). Moreover, the beta particles traverse the Xe layer over a wide range of angles, and sometime scatter through large angles in the Xe gas. Hence the path length of the β particles in the gas layer varies considerably. As

will be seen in the next section, numerical simulations indicate that 2.5 keV is a suitable threshold level for counting of the β particles in the PCs.

The xenon-filled proportional counters are, of course, sensitive to x-ray and gamma-ray backgrounds present during neutron pulses (e.g. during plasma focus shots). However, a delay of 20 ms following the pulse eliminates the influence of this electromagnetic radiation and is sufficient time for the proportional counters to recover (i.e. to clear the gas-ionization and re-charge to the applied bias voltage) before counting of beta particles begins. This delay also eliminates the potential for detecting background gamma-rays due to neutrons thermalized and captured in room materials, as these neutrons have a die-away time of a few milliseconds.

Other than elastic scattering, the only neutron-induced reaction in Be for D-D neutrons is the $(n, 2n)$ reaction, which does not lead to activation. Above 2.75 MeV, the cross-section for this reaction becomes larger than that for (n, α) , but it has a negligible effect on the response of the activation detector. Other reactions have threshold energies which are much higher than those associated with D-D fusion. For example, the ${}^9\text{Be}(n, p){}^9\text{Li}$ reaction has threshold energy of 14.3 MeV. As regards the possibility of activation of the detector materials by the room-thermalized neutrons, only aluminium and beryllium are present in sufficient quantity to be of concern. Activation of beryllium by thermal neutrons is completely negligible. Aluminium, however, has a thermal capture cross-section of 0.23 b, leading to β radioactive ${}^{28}\text{Al}$ with a half-life of 134.5 s. So if moderating materials are near the beryllium activation detector, some interfering ${}^{28}\text{Al}$ activation can be expected. An experimental investigation of interfering activations was therefore performed, as described in section 3.6.2.

3.3 Energy Calibration and Procedure for Setting the Energy Threshold

To maintain the compactness of the detector, the signal from each PC is processed by a PDT10A unit (Precision Data Technology, Inc., WA, USA) (see Figure 50(b)). This unit (subsequently referred to as a PDT) contains a precision wide-band charge sensitive pre-amplifier, pulse shaping amplifier and discriminator circuit. The PDT has an adjustable electronic gain and threshold level, and outputs a TTL (Transistor-Transistor Logic) pulse (of adjustable duration) for input charge pulses above the set threshold level. The minimum detectable charge pulse is specified to be

0.08 pC. The PDT also has a TTL input, enabling them to be daisy-chained (i.e. the last TTL output represents the logical AND of all the PDT units in the chain). In this way, a single output signal is obtained for both PCs (positioned either side of the Be sheet). The width of the TTL pulse was adjusted to be 200 ns. The PDT unit does not, however, have an energy output: the lack of such an output necessitated the use of an alternative pre-amplifier (Ortec 142PC) and some NIM-based electronics for part of the calibration procedure. Two Ortec MCS-pci Multi-Channel-Scalers (one for each beryllium activation detector) and MCS-32 software were used to acquire the data.

An image of the constructed beryllium “sandwich” activation detector with amplifier/discriminator units attached is shown in Figure 50(a).

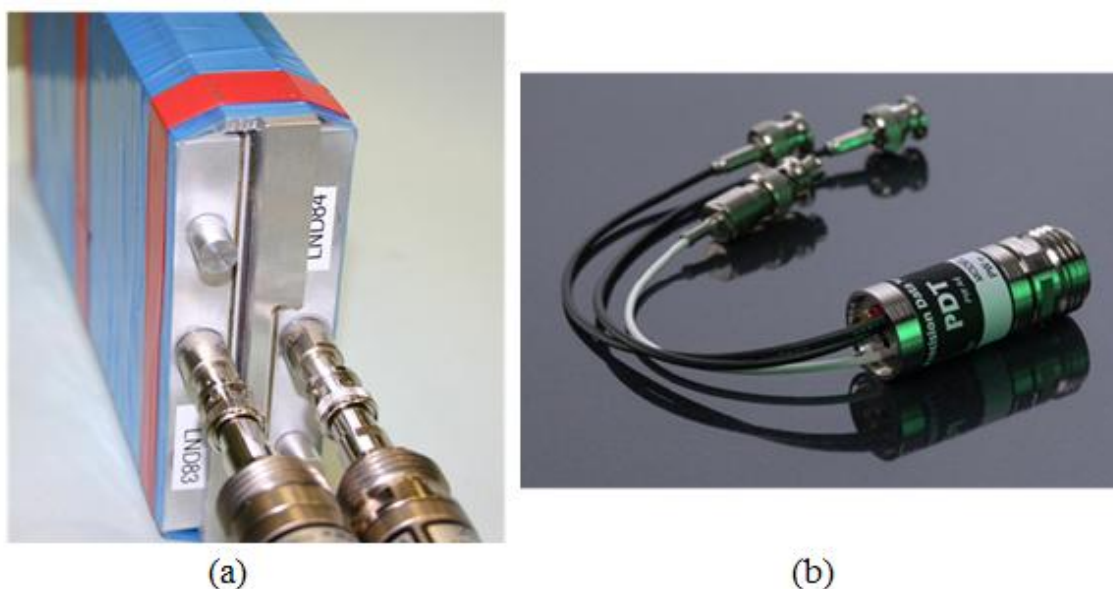


Figure 50: (a) Assembled activation counter comprising beryllium sandwiched between two xenon-filled proportional counters, aluminium holding plate, and two PDT10A units attached. (b) One PDT10A unit providing preamplifier, shaping amplifier and discriminator functions.

An essential step in the calibration process is the measurement of the relative gas-gain of the LND 49727 proportional counters using two x-ray sources: ^{55}Fe and ^{109}Cd . This provides a means of setting the PDT thresholds at the desired level of 2.5 keV. The relevant x-ray energies for these sources are: 5.90 and 6.49 keV from ^{55}Fe , and 22.16 keV from ^{109}Cd [116]. Since the PDT units do not have an energy output, the pulse processing electronics used for these gas-gain measurements used conventional NIM-based units: Ortec 109PC pre-amplifier, Ortec 855 shaping amplifier and Ortec pci-MCA multi-channel analyzer system. The PC bias voltage was supplied by an Ortec

556 unit. Each of the four LND-PCs was connected in turn to this set of pulse processing electronics, and the channel centroid of the peaks for the 5.90, 6.49 and 22.16 keV x-rays were measured as a function of the applied PC bias voltage over the range: 1400 to 1600 V.

For one of the LND proportional counters, a plot of the MCA channel-centroid versus applied bias voltage for each of the three x-ray peaks is shown in Figure 51. As expected the gas-gain of the proportional counter increases with increasing bias voltage. An energy-calibrated MCA spectrum for the region of the ^{55}Fe x-ray peaks is also shown in this figure.

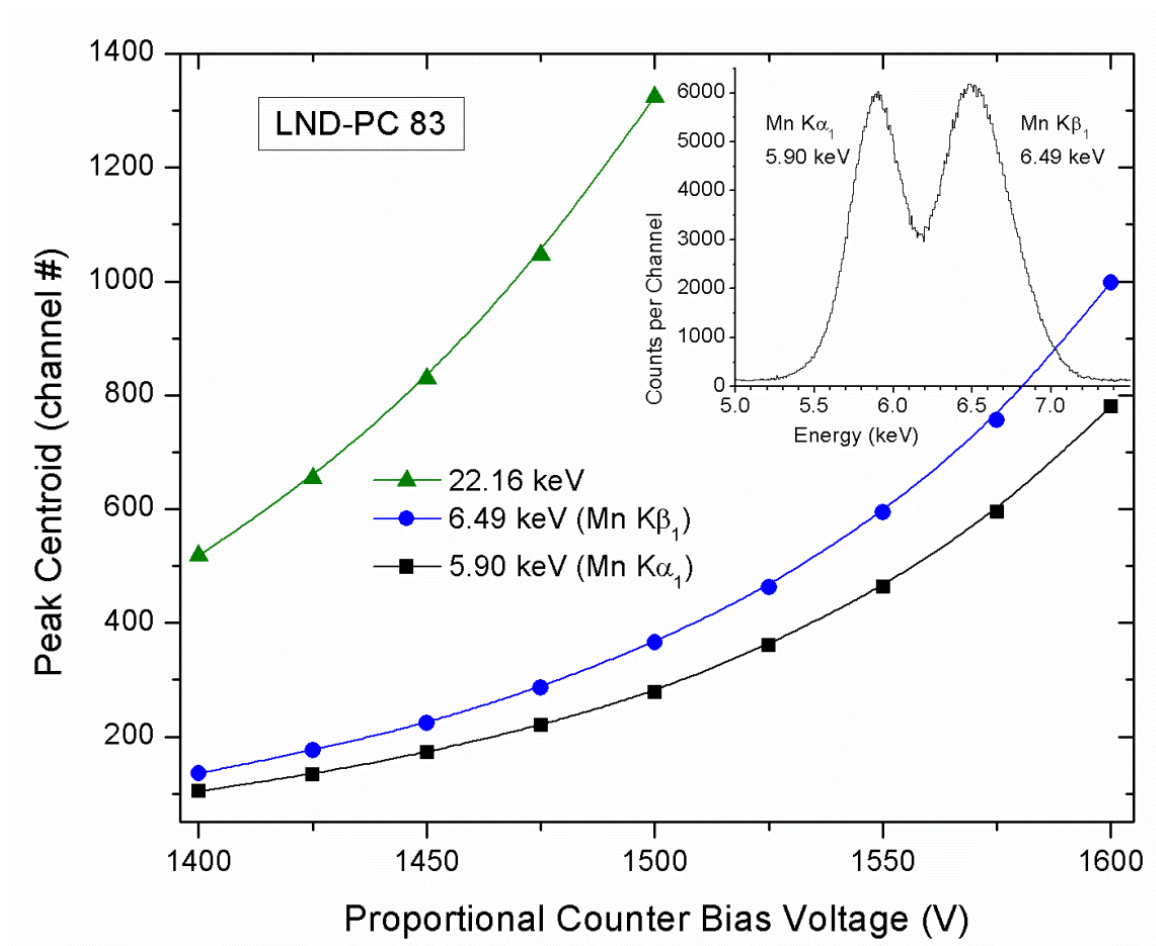


Figure 51: Plot of channel centroid for 5.90, 6.49 and 22.16 keV x-rays for LND-PC 83 as a function of applied bias voltage. Overlaid on this plot is the energy spectrum in the vicinity of the 5.90 and 6.49 keV x-rays from ^{55}Fe acquired with Ortec pulse processing electronics.

The electronic gain and threshold was set individually for each LND-PC/PDT pair using ^{55}Fe x-rays as the energy standard. For this step, the relative gas-gain for the LND-PCs is the important parameter to be measured. With the choice of 1500V bias voltage to represent the normalization point, the relative gas-gain measured for each of the four LND-PCs are plotted in the Figure 52 (where each LND-PC is labeled according to the last two digits of its serial number). The Ortec signal train electronics was used throughout these gas-gain measurements.

As can be seen from this figure the gas-gain trend for the four LND-PCs are practically identical. Over the bias voltage range 1400 to 1600 V, the relative gas-gain is well-fitted by the cubic equation:

$$gg(V) = 2.1678 \times 10^{-7} V^3 - 9.189 \times 10^{-4} V^2 + 1.30308 V - 617.736 \quad (3-2)$$

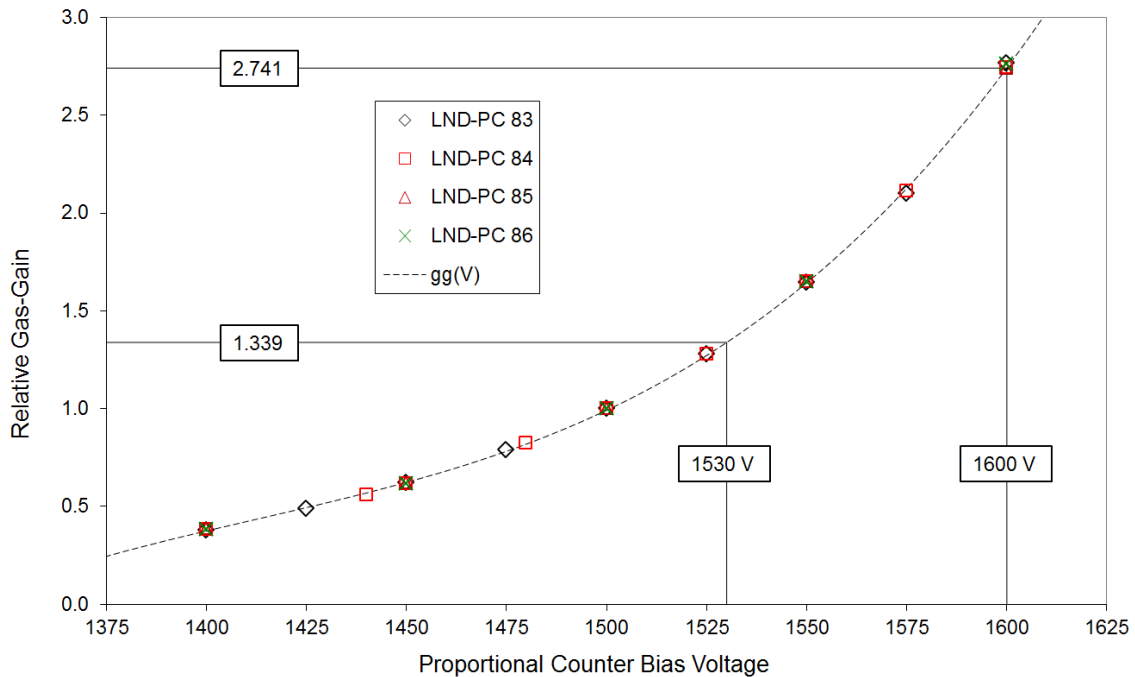


Figure 52: Plot of relative gas-gain for each of the four LND proportional counters, normalized to unity at 1500 V for each LND proportional counter. The dashed curve represents the empirical function $gg(V)$.

Using this empirical formula, it is found that the gas-gain increases by a factor of $2.74 / 1.34 = 2.05$ as the applied bias voltage is increased from 1530 and 1600 V. This is an important point as the method for adjusting the PDT electronic thresholds to correspond to a deposited energy of 2.5 keV in the xenon gas of the LND-PCs relies on this factor. As the 5.90 keV x-rays from ^{55}Fe represents the lowest energy mono-energetic x-rays which can be practically used, it is not possible to adjust the electronic

threshold to 2.5 keV in a single step. Instead the electronic threshold was adjusted to correspond to 5.0 keV at the bias voltage of 1530 V. Then during normal detector operation at 1600 V, the gas-gain of the PC is a factor of 2.05 higher, and the effective threshold level is reduced to 2.5 keV.

The procedure followed for each LND-PC/PDT pair was to adjust the electronic gain of the PDT unit to its maximum value, then for PC bias voltage of 1530 V, the electronic threshold of the PDT unit was gradually lowered until the full-width of the 5.90 keV x-ray peak was above threshold. This was determined by obtaining a near identical count rate from the PDT TTL output, to that obtained with the Ortec signal train for the ^{55}Fe x-ray source and PC kept in the same relative positions, and summing all counts above 5.0 keV in the MCA spectrum. As can be seen in Figure 51, the lower side of the 5.90 keV peak flattens out in the 5.0 to 5.5 keV region, and the estimated uncertainty in the final threshold level (at 1600 V) is ± 0.5 keV. So unlike the multilayer beryllium and plastic scintillator detector used by Rowland et al. [95, 117], the present Be activation detector has a well-characterized threshold level for detection of the β particles.

3.4 Rate count and capability

Let N_{exp} be the experimental count accumulated over the time interval from t_1 to t_2 , and let D be the total number of *detectable* beta-particles entering either PC over all-time following the shot (i.e. D would be the measured count if $t_1 = 0$, $t_2 = \infty$ and detector dead-time were zero). If detector dead-time can be neglected, then:

$$D \cong N_{\text{exp}} / (e^{\lambda t_2} - e^{\lambda t_1}) \quad (3-3)$$

where λ is the radioactive decay constant ($= 0.859 \text{ S}^{-1}$ for ^6He), and the measured neutron fluence is $\Phi = D / R(En)$. This approximation is satisfactory (i.e. $\sim 3\%$ too low) up to $D = 5.0 \times 10^4$ and $\Phi \approx 10^5 \text{ n/cm}^2$. A more accurate and complete analysis is given below.

The dead-time for the LND 49727 PC and PDT10A combination was measured using the two source method and formulae, as described in ref. [86]. For this measurement, 10 individual ^{204}Tl beta sources of 5 μCi (nominal) activity, and 5 dummy sources were used. The 10 active sources were split into two groups; the dummy sources represented a third group. These groups of “sources” were placed, and replaced, on the metal grid protecting the thin titanium window of the proportional

counter, following the sequence set out in the reference. Hence, measurements were made with ^{204}Tl source activities of: 25, 50, 25 and 0 μCi (i.e. background only). For each measurement the Ortec MCS-pci accumulated the gross count over a 100 s interval. Applying the standard (non-paralyzable model) formulae gave a dead-time of $\tau_d = 2.5 \mu\text{s}$ for one PC-PDT. (The measured count rate for the 50 μCi case was $m = 9.23 \times 10^4 \text{ s}^{-1}$ giving $m\tau_d = 0.23$, which is >0.2 as recommended in the reference for accuracy).

Since the detector output is the logical AND of the pulses from two PDTs, there is a further loss of pulses due to random overlaps of the $\tau_w = 0.2 \mu\text{s}$ wide TTL pulses. The effective width of this overlap window is $2\tau_w$. Then making the simplifying assumption that the true count rate n is split equally between the two PCs, analysis gives the relationships between the *measured* and *true* rates, m and n , respectively, as:

$$m = \frac{2(\tau_d - \tau_w + 2/n)}{(\tau_d + 2/n)^2} \quad (3-4)$$

$$n = \frac{2m}{1 - m\tau_d + \sqrt{1 - 2m\tau_w}} \quad (3-5)$$

Substituting $n = \lambda D e^{-\lambda t}$ into Eqn. (3-4), and integrating $m(t)$ from t_1 to t_2 , gives the expected measured count:

$$M_{\text{exp}} = \int_{t_1}^{t_2} m(t) dt = 2 \int_{t_1}^{t_2} \frac{(\tau_d - \tau_w + 2e^{\lambda t}/\lambda D)}{(\tau_d + 2e^{\lambda t}/\lambda D)^2} dt. \quad (3-6)$$

Then:

$$\begin{aligned} \frac{M_{\text{exp}}}{2} &= \int_{t_1}^{t_2} \frac{1}{(\tau_d + 2e^{\lambda t}/\lambda D)} dt - \int_{t_1}^{t_2} \frac{\tau_w}{(\tau_d + 2e^{\lambda t}/\lambda D)^2} dt \\ \frac{M_{\text{exp}}}{2} &= \lambda D \int_{t_1}^{t_2} \frac{dt}{(\lambda D \tau_d + 2e^{\lambda t})} - \lambda^2 D^2 \tau_w \int_{t_1}^{t_2} \frac{dt}{(\lambda D \tau_d + 2e^{\lambda t})^2} \end{aligned}$$

Make use of two results from tables of Integrals:

$$\begin{aligned} \int \frac{dx}{(p + qe^{bx})} &= \frac{x}{p} - \frac{1}{bp} \ln(p + qe^{bx}) \\ \int \frac{dx}{(p + qe^{bx})^2} &= \frac{x}{p^2} - \frac{1}{bp^2} \ln(p + qe^{bx}) + \frac{1}{bp(p + qe^{bx})} \end{aligned}$$

Give us the Integral as:

$$\frac{M_{\text{exp}}}{2} = \left[\lambda D \frac{t}{p} - \frac{\lambda D}{bp} \ln(p + qe^{bt}) - \frac{\lambda^2 D^2 \tau_w}{p} \left\{ \frac{t}{p} - \frac{1}{bp} \ln(p + qe^{bt}) + \frac{1}{b(p + qe^{bt})} \right\} \right]_{t_1}^{t_2}$$

Where: $p = \lambda D \tau_d$, $q = 2$ and $b = \lambda$ and so:

$$\begin{aligned} \frac{M_{\text{exp}}}{2} &= \left[\frac{t}{\tau_d} - \frac{1}{\lambda \tau_d} \ln(\lambda D \tau_d + 2e^{\lambda t}) \right. \\ &\quad \left. - \frac{\lambda D \tau_w}{\tau_d} \left\{ \frac{t}{\lambda D \tau_d} - \frac{1}{\lambda^2 D \tau_d} \ln(\lambda D \tau_d + 2e^{\lambda t}) + \frac{1}{\lambda(\lambda D \tau_d + 2e^{\lambda t})} \right\} \right]_{t_1}^{t_2} \\ \frac{1}{2} M_{\text{exp}} \lambda \tau_d &= \left[\left(1 - \frac{\tau_w}{\tau_d} \right) (\lambda t - \ln(\lambda D \tau_d + 2e^{\lambda t})) - \frac{D \lambda \tau_w}{(\lambda D \tau_d + 2e^{\lambda t})} \right]_{t_1}^{t_2} \end{aligned}$$

Now putting: $s_1 = \lambda D \tau_d + 2e^{\lambda t_1}$ and $s_2 = \lambda D \tau_d + 2e^{\lambda t_2}$

And so: $a = s_2 - s_1 = 2(e^{\lambda t_2} - e^{\lambda t_1})$ and $\Delta t = t_2 - t_1$

$$\begin{aligned} \frac{1}{2} M_{\text{exp}} \lambda \tau_d &= \left(1 - \frac{\tau_w}{\tau_d} \right) \left(\lambda \Delta t - \ln \left(\frac{s_2}{s_1} \right) \right) + \lambda D \tau_w \left(\frac{1}{s_1} - \frac{1}{s_2} \right) \\ \frac{1}{2} M_{\text{exp}} \lambda \tau_d &= \left(1 - \frac{\tau_w}{\tau_d} \right) \left(\lambda \Delta t - \ln \left(\frac{a + s_1}{s_1} \right) \right) + \lambda D \tau_w \left(\frac{a}{s_1(a + s_1)} \right) \end{aligned}$$

Re-arranging this, it becomes:

$$\begin{aligned} \frac{1}{2} M_{\text{exp}} \lambda \tau_d - \left(1 - \frac{\tau_w}{\tau_d} \right) (\lambda \Delta t - \ln(1 + a/s_1)) - \lambda D \tau_w \left(\frac{a}{s_1(a + s_1)} \right) &= 0 \\ s_1(a + s_1) \left(\frac{1}{2} M_{\text{exp}} \lambda \tau_d - \left(1 - \frac{\tau_w}{\tau_d} \right) (\lambda \Delta t - \ln(1 + a/s_1)) \right) - a \lambda D \tau_w &= 0 \\ s_1(1 + s_1/a) \left(\frac{1}{2} M_{\text{exp}} \lambda \tau_d^2 - (\tau_d - \tau_w)(\lambda \Delta t - \ln(1 + a/s_1)) \right) - \lambda D \tau_w \tau_d &= 0 \end{aligned}$$

But since: $D = (s_1 - 2e^{\lambda t_1})/\lambda \tau_d$ then:

$$s_1(1 + s_1/a) \left(\frac{1}{2} M_{\text{exp}} \lambda \tau_d^2 - (\tau_d - \tau_w)(\lambda \Delta t - \ln(1 + a/s_1)) \right) + (2e^{\lambda t_1} - s_1) \tau_w = 0$$

Then simply replacing the s_1 with x , gives the final equation as:

$$\begin{aligned} x(1 + x/a) \left[\frac{1}{2} M_{\text{exp}} \lambda \tau_d^2 - (\lambda \Delta t - \ln(1 + a/x)) (\tau_d - \tau_w) \right] + (2e^{\lambda t_1} - x) \tau_w \\ = 0 \end{aligned} \quad (3-7)$$

Hence, in the experimental situation M_{exp} is the measured net count (i.e. after background subtraction). Then Eqn. (3-7) can be solved numerically for x as all other factors are known. The experimental neutron fluence Φ is then determined from:

$$D = (x - 2e^{\lambda t_1})/\lambda\tau_d \quad (3-8)$$

and

$$\Phi = \frac{D}{R(E_n)} = \frac{x - 2e^{\lambda t_1}}{\lambda\tau_d R(E_n)}. \quad (3-9)$$

For example, assuming D-D neutrons ($R = 0.461$) and summing the detector counts over a 3 s interval ($t_1 = 0.02$ s and $t_2 = 3.02$ s during which interval: 90.8% of ${}^6\text{He}$ atoms decay), an accumulated count of $M_{\text{exp}} = 3.304 \times 10^5$ is found (by numerically solving eqn. (3-7) and applying eqns. (3-8) & (3-9)) to give $D = 4.612 \times 10^5$ and $\Phi = 10^6$ n/cm². (The approximation represented by eqn. (3-3) is 21% too low in this case). And the statistical uncertainty Φ is only 0.2%. At the beginning of the counting interval (i.e. t_1) the value of $m\tau_d$ is ~ 0.33 for one PC, and this situation can be taken as a suitable upper limit for M_{exp} .

However, a very useful attribute of activation detectors is that the upper measurable limit can be increased almost arbitrarily by interposing a known delay before the counting interval begins. This allows the target activity to decay (by a well determined factor) to the point where the fraction of dead-time is manageable. And the short half-life of ${}^6\text{He}$ is helpful in this regard. For example, by interposing a 6 half-life delay: setting $t_1 = 4.86$ s and $t_2 = 7.86$ s, the upper measurable fluence is increased by a factor of 2^6 and an experimental count of $M_{\text{exp}} = 3.304 \times 10^5$ now corresponds to $\Phi = 6.4 \times 10^7$ n/cm². A potential disadvantage of this delay-method is that there will be relatively greater interference from (longer-lived) activated nuclides, but as will be seen in section 3.6.2, there is very little interfering activation of the detector materials. Lastly, the lowest fluence for which a 1% statistical uncertainty can be obtained with this beryllium activation detector is 3×10^4 n/cm². Hence the detector has a very wide dynamic range for measurement of pulsed D-D neutron fluences with low statistical uncertainties.

3.5 Numerical Calibration of Detector Response

3.5.1 General Purpose Calibration

Simulations necessary for the absolute efficiency calibration of the activation detector were performed with the Monte Carlo radiation transport code MCNP5. The overall simulation of the detector response was conducted in two parts: firstly, the neutron interaction in the Be and the production of ${}^6\text{He}$, and secondly, the path of the emitted β particle and its energy-deposition (if any) in one of the Xe gas layers. The simulated geometry was simplified to three cuboids representing beryllium and xenon cells with the same dimensions and densities as for the physical detector. A parallel neutron beam of 1 n/cm^2 fluence was taken to be normally incident on the (largest area) face of the detector. The probability of a β particle escaping from the Be sheet and entering one of the xenon cells is clearly greater if it originates close to the surface of the Be sheet. For this reason, in the MCNP simulation, the beryllium sheet is divided into 5 layers (sub-cells), each of 1 mm thickness, and the neutron-induced production of ${}^6\text{He}$ within each layer is determined separately. Simulations were performed over the range of neutron energies 2.0 to 4.0 MeV in steps of 0.2 MeV. The neutron energy 2.45 MeV which is characteristic of D-D thermonuclear fusion was also simulated. This energy range is sufficiently wide to cover neutrons from almost any D-D fusion device. From the first part of the simulation η -values ($\eta_1(E_n), \dots, \eta_5(E_n)$), representing the number of ${}^6\text{He}$ atoms produced in each layer for a neutron beam of energy E_n and 1 n/cm^2 fluence, were calculated. As examples, η -values found for a selection of four of the neutron energies simulated are shown in Table 5. In these MCNP5 simulations the neutron beam is incident on the “layer 1 side” of the Be cell.

In the second part of the MCNP simulation, the probability of a β particle, originating in a given layer, reaching a Xe gas cell and depositing energy E_{dep} is tallied. The ${}^6\text{He}$ β energy spectrum used for this simulation, and plotted in Figure 53, was taken from the NNDC database [100]. The probability distributions obtained from this part of the simulation are plotted in Figure 54. The summed probabilities for $E_{\text{dep}} > 2.5 \text{ keV}$ (denoted $\kappa_1, \dots, \kappa_5$) is shown in the first part of Table 6. Since however, as the xenon cells are placed symmetrically either side of the Be cell, this probability distribution can be folded as shown in the second part of Table 6. It can be seen that approximately 40 % of the ${}^6\text{He}$ β particles are detected in one of the proportional counters.

Table 5: Beryllium layers (each 1 mm) contribution to ${}^6\text{He}$ production

Layer	$E_n = 2.0 \text{ MeV}$	2.2 MeV	2.4 MeV	2.45 MeV	2.6 MeV	2.8 MeV
η_1	0.1279	0.1763	0.2246	0.2366	0.2776	0.3037
η_2	0.1274	0.1751	0.2219	0.2334	0.2732	0.3022
η_3	0.1264	0.1733	0.2185	0.2294	0.2676	0.2993
η_4	0.1251	0.1709	0.2146	0.2249	0.2612	0.2953
η_5	0.1234	0.1679	0.2099	0.2198	0.2536	0.2898
η_{tot}	0.6302	0.8636	1.089	1.144	1.333	1.490

Layer	$E_n = 3.0 \text{ MeV}$	3.2 MeV	3.4 MeV	3.6 MeV	3.8 MeV	4.0 MeV
η_1	0.3033	0.2921	0.2806	0.2697	0.2589	0.2478
η_2	0.3019	0.2905	0.2788	0.2679	0.2569	0.2457
η_3	0.2991	0.2877	0.2759	0.2649	0.2538	0.2425
η_4	0.2952	0.2838	0.2719	0.2609	0.2497	0.2384
η_5	0.2900	0.2787	0.2669	0.2557	0.2445	0.2334
η_{tot}	1.489	1.433	1.374	1.319	1.263	1.208

Table 6: Sum of frequencies of β particle originating in each 1 mm layer of the Be cell entering a Xe cell and depositing energy $> 2.5\text{keV}$.

Beryllium layer	κ_1	κ_2	κ_3	κ_4	κ_5
Sum of frequency	0.416	0.264	0.167	0.102	0.058

Folded Be layer	$\kappa_1 + \kappa_5$	$\kappa_2 + \kappa_4$	$2\kappa_3$
Folded frequency	0.473	0.366	0.335

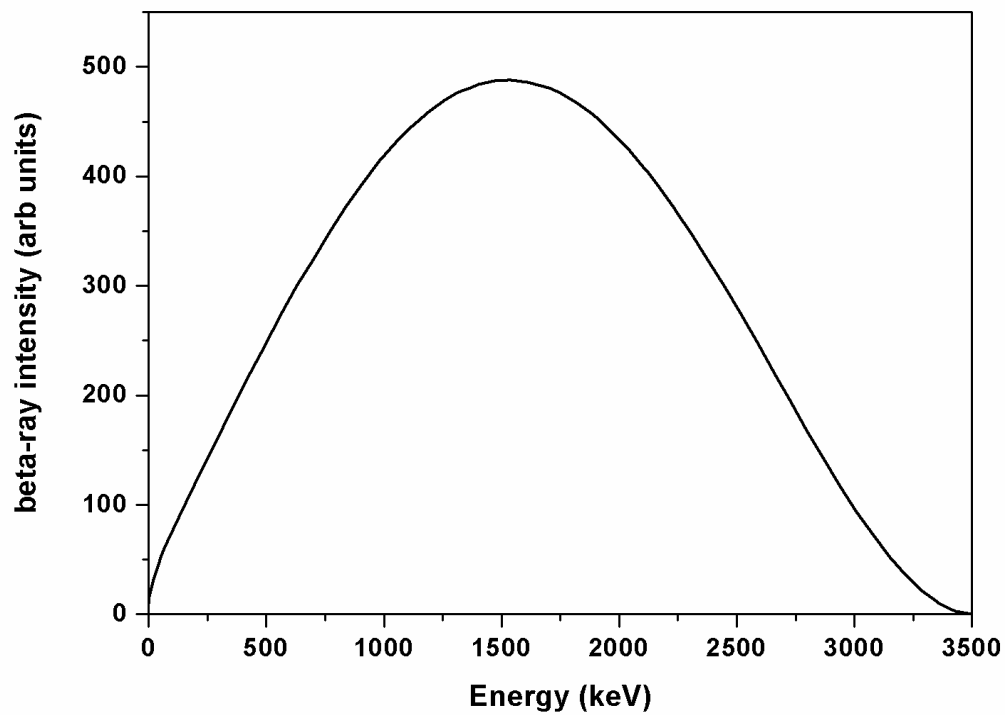


Figure 53: β -particle energy spectrum for the decay of ${}^6\text{He}$.

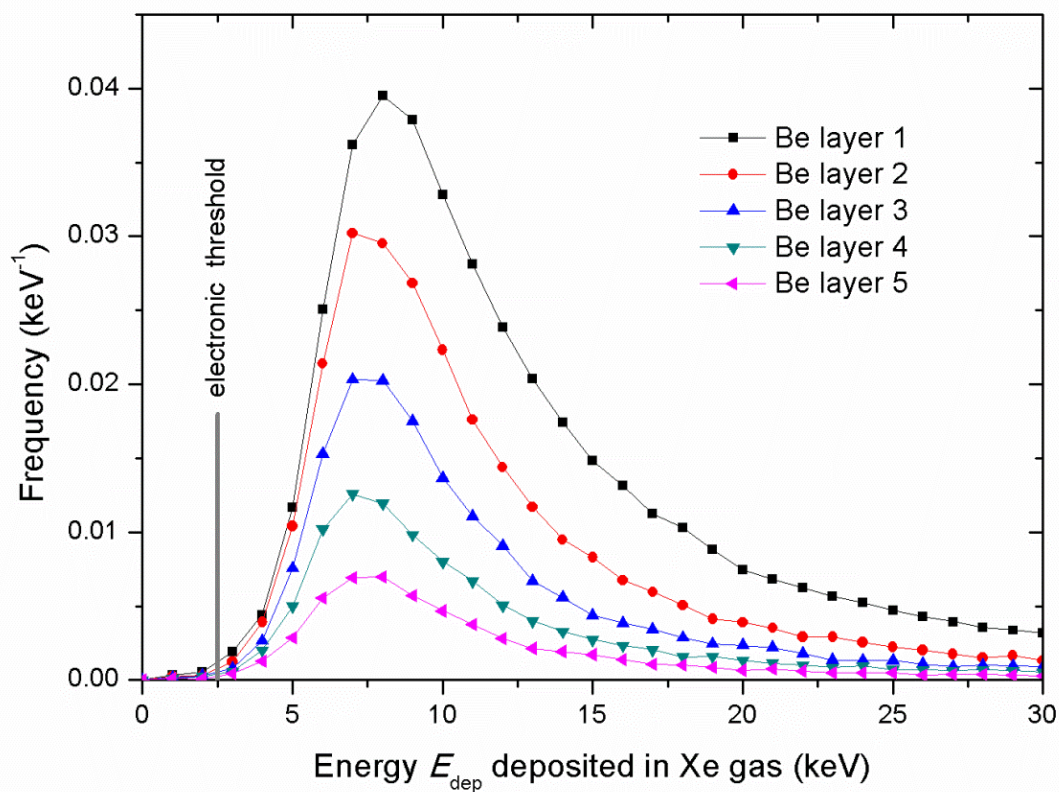


Figure 54: MCNP simulation results of frequency distribution of energy deposition in PC xenon gas layer (12.7 mm depth) by ${}^6\text{He}$ β particles which escape from Be sheet and enter PC. Separate curves are for ${}^6\text{He}$ β particles originating in each 1 mm thick layer of the 5 mm thick Be sheet. In this simulation a PC is located on one side only of the Be sheet, and adjacent to layer 1.

Then the overall calibration, or response function, of the detector is given by:

$$R(E_n) = \sum_{l=1}^5 R_l(E_n) = \sum_{l=1}^5 \eta_l(E_n) \times (\kappa_l + \kappa_{6-l}) \quad (3-10)$$

Where η_l is the neutron energy dependent production of ${}^6\text{He}$ in layer l of the Be sheet, and κ_l is probability of a β particle originating in layer l depositing more than 2.5 keV of energy in one xenon gas cell.

Making use of Eqn. (3-10) with values from Table 5 (expanded for all E_n) and Table 6: the calculated response function $R(E_n)$ of the detector for a parallel beam of 1 neutron/cm² is presented in Table 7. These values of $R(E_n)$ are also plotted in Figure 55, along with the contributions $R_L(E_n)$ from each of the 5 layers modeled in MCNP5. The largest contribution to the uncertainties in these values comes from the experimental cross-section measurements. From Fig. 2 of ref. [101] it is apparent that the shape (or relative cross-section vs. energy) of the ${}^9\text{Be}(n,\alpha){}^6\text{He}$ cross-section is very well determined. Table II of the same reference lists the six absolute cross-section points around the 2 to 4 MeV region relevant to this work, and the mean relative uncertainty for these cross-sections is 9.6%. Consequently the shape of the response function in Figure 55 (this work) should be correspondingly accurate, but the absolute values across the curve as a whole, have an estimated uncertainty of $\pm 10\%$.

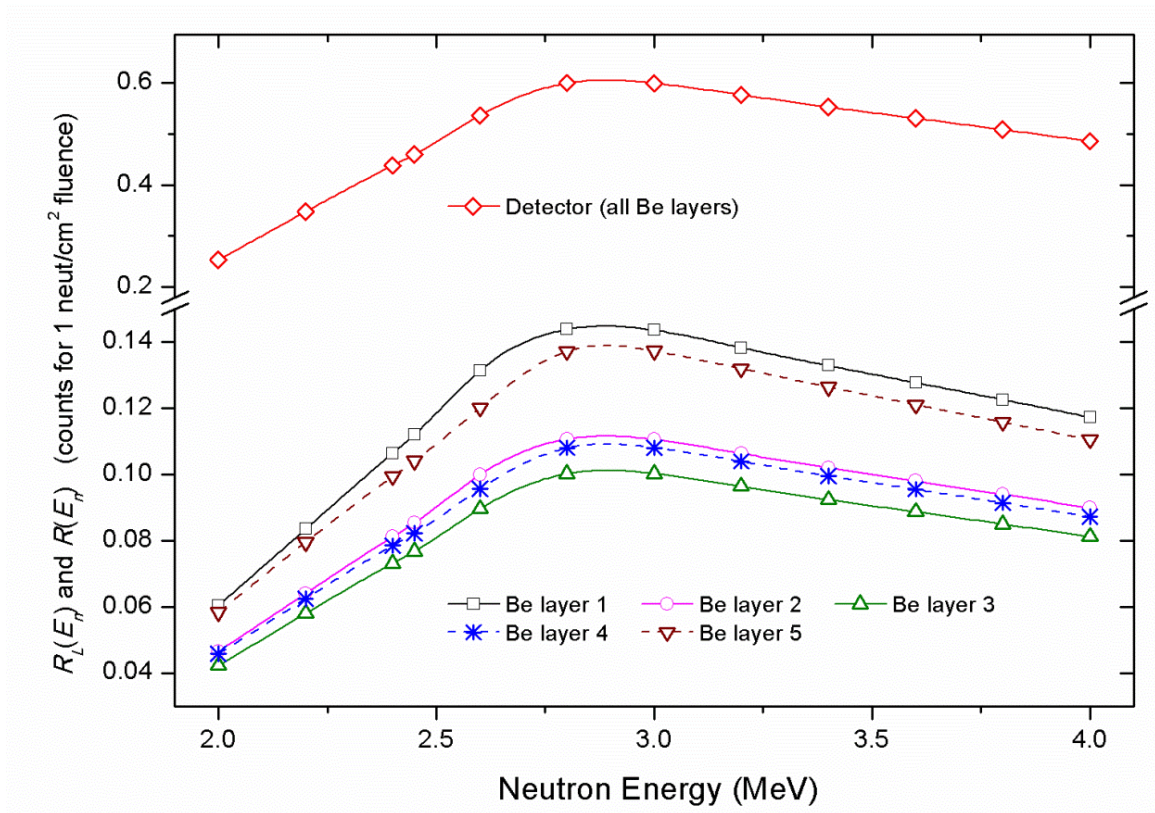


Figure 55: Response function $R(E_n)$ of beryllium neutron activation detector for a parallel neutron beam of fluence 1 neutron/cm², and contributions $R_L(E_n)$ from each 1 mm thick layer of the Be sheet. Note that the vertical axis is broken and the two parts are scaled differently. The absolute $R(E_n)$ values across the curve as a whole, have an estimated uncertainty of $\pm 10\%$.

Table 7: Counts from layers for a parallel neutron beam fluence of 1 neutron/cm²

Layer	$E_n = 2.0$ MeV	2.2 MeV	2.4 MeV	2.45 MeV	2.6 MeV	2.8 MeV
R_1	0.0605	0.0835	0.1064	0.1121	0.1314	0.1438
R_2	0.0466	0.0641	0.0812	0.0854	0.0999	0.1106
R_3	0.0423	0.0580	0.0732	0.0768	0.0896	0.1002
R_4	0.0458	0.0625	0.0785	0.0823	0.0956	0.1080
R_5	0.0584	0.0795	0.0994	0.1041	0.1201	0.1372
R_{total}	0.2537	0.3476	0.4386	0.4607	0.5367	0.5999

Layer	$E_n = 3.0$ MeV	3.2 MeV	3.4 MeV	3.6 MeV	3.8 MeV	4.0 MeV
R_1	0.1436	0.1383	0.1329	0.1277	0.1226	0.1173
R_2	0.1105	0.1063	0.1020	0.0980	0.0940	0.0899
R_3	0.1002	0.0964	0.0924	0.0887	0.0850	0.0812
R_4	0.1080	0.1039	0.0995	0.0955	0.0914	0.0872
R_5	0.1373	0.1320	0.1264	0.1211	0.1158	0.1105
R_{total}	0.5996	0.5770	0.5532	0.5311	0.5088	0.4862

To consider the response function for point sources, it is convenient to keep the number of neutrons incident on the Be plate the same as for the parallel beam of 1 neut/cm², and therefore numerically equal to the plate area (in cm²) of 21×10.6 = 222.6. Then the response factor is affected only by the spread in angles of incidence. Also useful is the formula for the solid angle subtended by a right rectangular pyramid:

$$\Omega_{rrp} = 4 \arctan \left(\frac{ab}{2d\sqrt{4d^2 + a^2 + b^2}} \right) \quad (3-4)$$

where a and b are the dimensions of the Be plate, and d is the distance from the point source to the midplane of the plate. Hence, for the above scaling, the emittance of the source is $(222.6/\Omega_{Be})$ n/sr. MCNP simulations were performed for a point source of 2.45 MeV neutrons located at $d = 15, 20, 25, 30, 40$, and 50 cm. The simulation results were processed in an analogous way to that described above, and scaled to 222.6 neutrons incident on the plate midplane. Figure 56 shows a plot of point source response function $R_{ps}(E_n = 2.45 \text{ MeV})$ versus distance d . It can be seen that for larger d , the response tends asymptotically towards the value (0.4607) obtained for a parallel beam. To obtain the response function in terms of neutrons per steradian, an R_{ps} value can be interpolated from the graph and multiplied by the factor $(\Omega_{Be}/222.6)$.

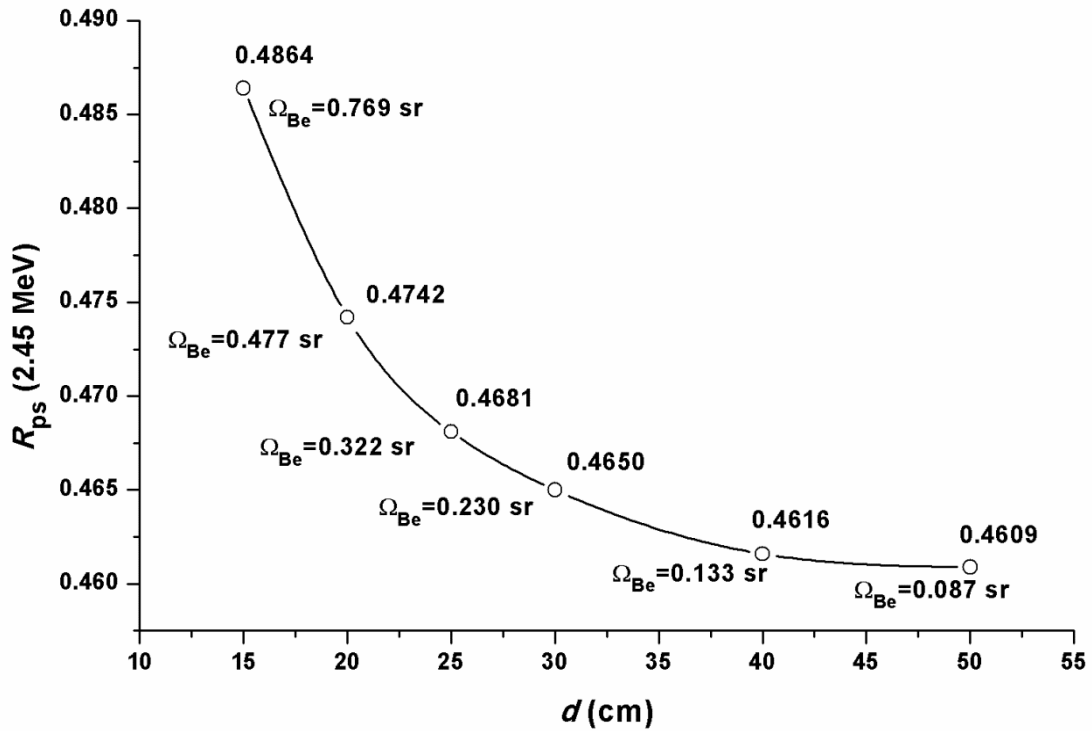


Figure 56: Beryllium neutron detector response function $R_{ps}(E_n = 2.45 \text{ MeV})$ for a point source at distance d .

3.5.2 Special Calibration for NX2 Plasma Focus Device

To approach the numerical absolute efficiency calibration of the activation detector, the accurate reaction cross section is needed which is related to the incoming neutron energy and this in turn is determined by the kinematics of the D-D fusion reaction: e.g. for beam-target reactions, it depends on the energy of the ingoing deuteron and the angle of neutron emission.

For the first step, we need the angular differential cross-section of $D(d,n)^3\text{He}$ reaction as a function of deuteron energy. The most suitable references are IAEA Technical Report [118] and Brown-Jarmie [119]. Both data have been fitted by using a polynomial function of $\cos(\theta)$, because the D-D differential cross-sections are symmetric about 90° . We have chosen the IAEA result in the form of Legendre polynomial:

$$\sigma(\theta) = S_0 \left[A_0 + \frac{A_2}{2} (3\cos^2\theta - 1) + \frac{A_4}{8} (35\cos^4\theta - 30\cos^2\theta + 3) \right] \quad (3-5)$$

where S_0 is Center of Mass (CM) differential cross-section at 0° , A_0 , A_2 , and A_4 are reduced Legendre coefficients. Figure 57 shows the angular distribution of CM cross section of D-D reaction.

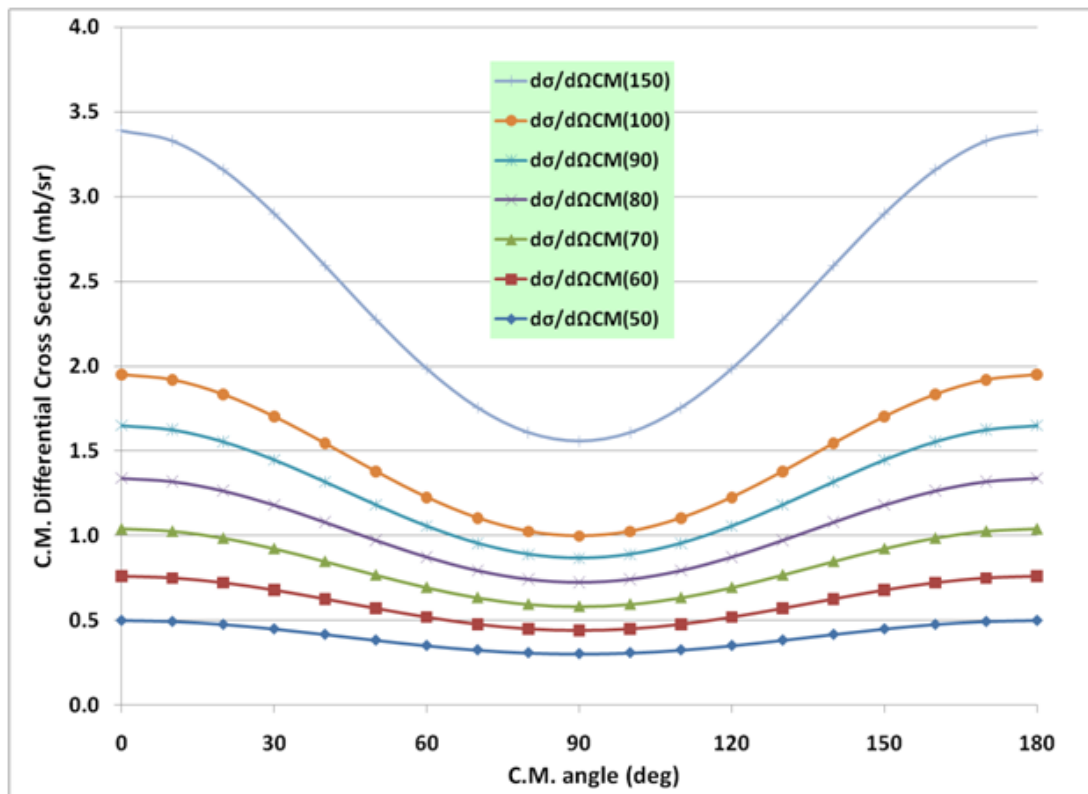


Figure 57: Center of Mass (C.M.) cross-section for D-D reaction with the deuteron energy (in keV) as a parameter

But as the measurement is performed in the Lab Frame (LF), so the data must be changed to the lab frame regarding to these followed formulas:

$$\theta_{\text{Lab}} = \tan^{-1} \left(\frac{\sin \theta_{\text{CM}}}{\gamma + \cos \theta_{\text{CM}}} \right) \quad \text{where: } \gamma = \left(\frac{E_d}{3E_d + 6Q} \right)^{1/2} \quad \text{and } Q = 3.27 \text{ MeV} \quad (3-6)$$

$$\text{so,} \quad \left(\frac{d\sigma}{d\Omega} \right)_{\text{Lab}} = \frac{(1 + 2\gamma \cos \theta_{\text{CM}} + \gamma^2)^{3/2}}{|1 + \gamma \cos \theta_{\text{CM}}|} \left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} \quad (3-7)$$

The result is shown in Figure 58.

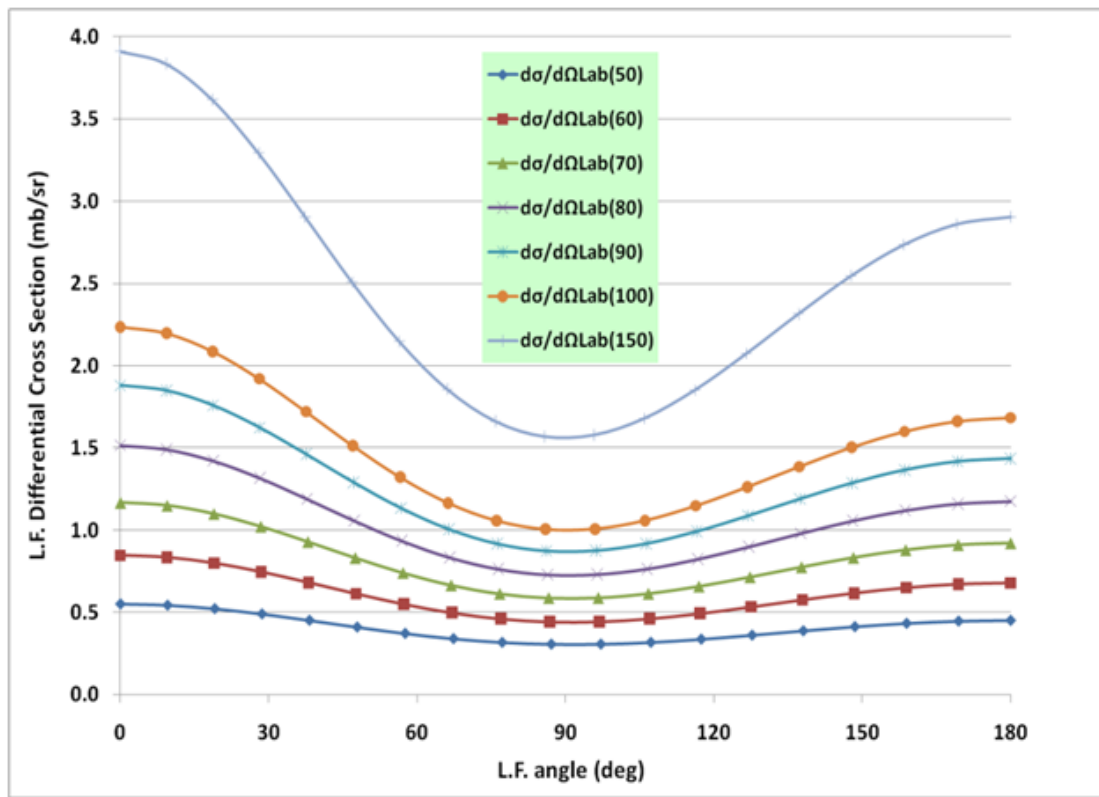


Figure 58: Lab frame cross-section for D-D reaction with the deuteron energy (in keV) as a parameter

In the second step, we need the angular distribution of neutron energy from $D(d,n)^3\text{He}$ reaction with different deuteron energy. Using accepted atomic masses of $m_D = 2.014102$ (u), $m_n = 1.008665$ (u), and $m_{\text{He}3} = 3.016029$ (u), the energy released in the $D(d,n)^3\text{He}$ reaction is $Q = \Delta m \times 931.5 = 3.27$ MeV and the neutron energy is given by the kinematic formulae below, and plotted in Figure 59 for the deuteron energy range $E_D = 50$ to 150 keV.

$$E_n = \left(u + \sqrt{u^2 + w} \right)^2 \quad (3-8)$$

Where

$$w = \frac{Q \times M_{\text{He3}} + E_D(M_{\text{He3}} - M_D)}{M_{\text{He3}} + M_n} = E_D(90^\circ)$$

$$u = \frac{(M_D M_n E_D)^{1/2}}{M_{\text{He3}} + M_n} \cos \theta_{\text{Lab}}$$

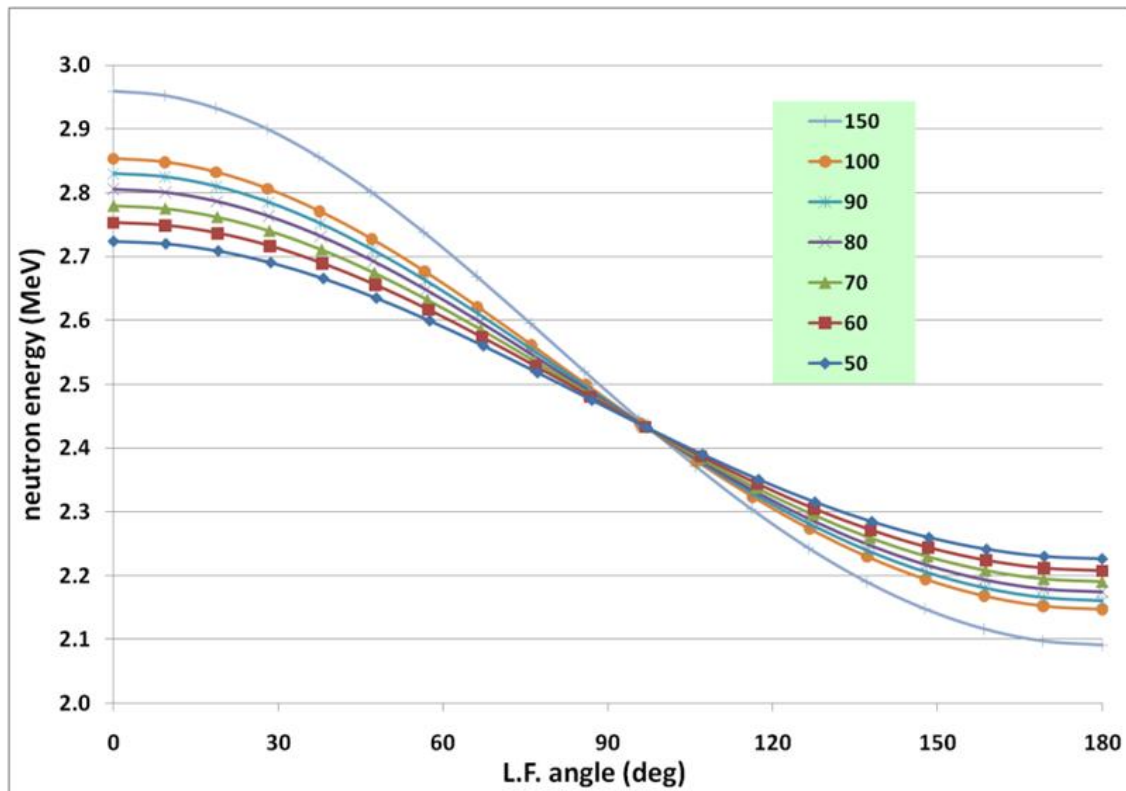


Figure 59: Lab frame (L.F.) neutron energy distribution for D-D reaction with the deuteron energy (in keV) as a parameter

To find the accurate beryllium count factor or efficiency for a special device like NX2 plasma focus device, the system geometry has been simulated in Monte-Carlo N-Particle code (MCNP5-visual version). The chamber's size and material, all flanges and plates, xenon proportional counters and beryllium plate of neutron detector have been arranged carefully. The distances between detectors and anode tip are 25.0 cm as it is in real experiment. Figure 60 shows the 3D image of simulation design.

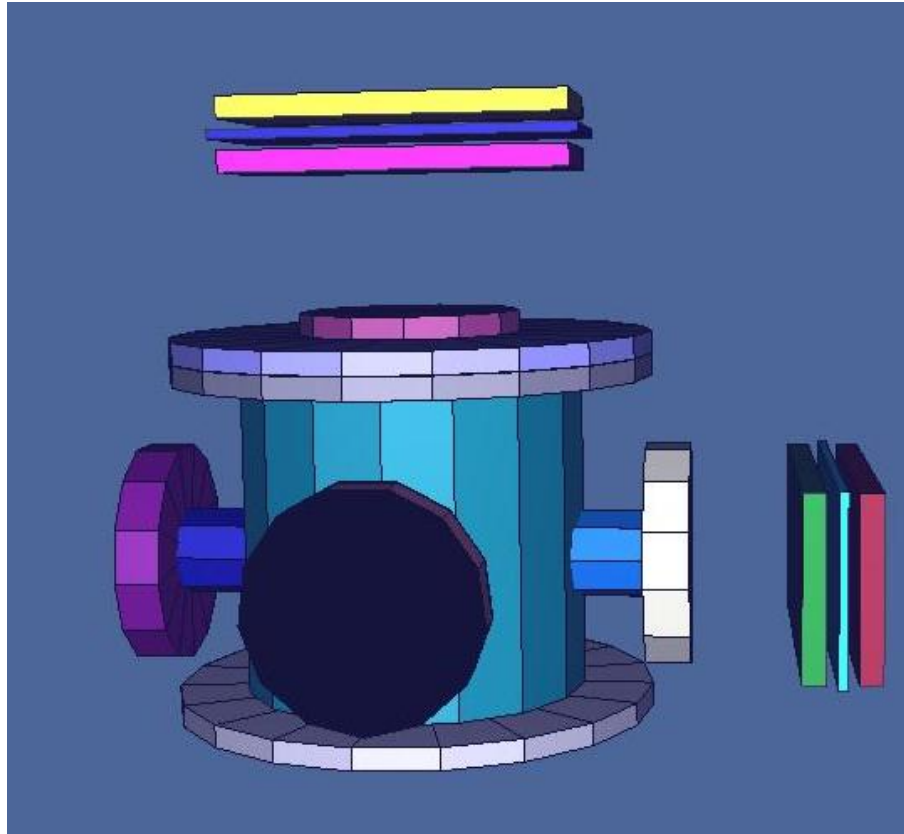


Figure 60: Simulation of NX2 chamber and Beryllium detectors in MCNP-visual

Moreover, the source has been considered as a cylinder with 10 mm height and 2 mm diameter (the same as deuteron source shape or pinch region in NX2 plasma focus device). The angular distribution of lab-frame cross-section and neutron energy have been calculated (as it is explained in introduction) and put as SDEF input data for MCNP. The SDEF inputs include: Source Information (SI) ($\cos(\theta_{\text{Lab}})$), Source Probabilities (SP) (Cross Section) and Distribution Source (DS) (Energy).

The real deuteron angular and energy distributions within the pinch column are not known, but as a model we assumed the physically reasonable case [48] of deuterons with directions distributed isotropically within a forwardly directed 30° cone, and with positions distributed uniformly throughout a cylinder of 2 mm diameter and 10 mm length (at the pinch location). The deuterons were assumed to be mono-energetic, and this value of energy was varied as a parameter. The D-D kinematics were entered into SDEF cards in the MCNP simulation as outlined above, so that the outgoing neutron energy and relative angular fluence were in accordance with the differential cross-section (for the selected energy) and the reaction kinematics. Since the response of the

beryllium detectors varies with neutron energy, the purpose of this simulation was to find the beryllium detector count/anisotropy as a function of deuteron energy within the D-D fusion model described above. In order to make the correspondence with the experimental situation as close as possible, the simulation includes the effects of neutron scattering and absorption, in the chamber wall and top-plate, etc.

The simulation was performed for a range of deuteron energies as shown in Table 8. The count factors given in this table represent the number of counts per source neutron for the beryllium detectors located at 0° and 90° , based on the source model described above. It can be seen that the modeled anisotropy increases as the deuteron energy increases. It should be emphasized that values in the last column represent a mixture of fluence anisotropy and energy anisotropy as it affects the count from the beryllium detectors. The first row of the table, labeled '0 keV' represents thermonuclear fusion, for which the neutron yield is perfectly isotropic and the neutron energy is 2.45 MeV. The simulated anisotropy value of 1.140 is therefore the result of neutron scattering and absorption in the chamber materials. For comparison, the simulation was also performed in absence of the chamber and all other materials (done by setting their densities to zero). In this case, the thermonuclear '0 keV' modeled anisotropy is 1.00, as expected.

Table 8: Efficiency and anisotropy for different deuteron energy

E_D (keV)	Be count factor (0°) [/n]	Be count factor (90°) [/n]	Anisotropy
0	4.34×10^{-5}	3.81×10^{-5}	1.140
50	7.08×10^{-5}	3.49×10^{-5}	2.030
60	7.36×10^{-5}	3.44×10^{-5}	2.140
70	7.64×10^{-5}	3.42×10^{-5}	2.235
80	7.96×10^{-5}	3.43×10^{-5}	2.324
90	8.24×10^{-5}	3.40×10^{-5}	2.423
100	8.24×10^{-5}	3.38×10^{-5}	2.435
150	8.72×10^{-5}	3.32×10^{-5}	2.624

Figure 61 shows the efficiency or response factor for both side-on and end-on situations. Also Table 9 compares the effect of scattering in different deuteron energy. It can be seen that counts in both beryllium detectors are reduced as a result of neutron scattering, but more so for the side-on detector (at 90°) than for the end-on (0°) detector. Figure 62 shows the anisotropy versus deuteron energy in presence and absence of the

chamber. It is worth mentioning that for calculating the efficiency factor for different device, the simulation must be repeated with regarding the actual geometry and material of device. Even for NX2 plasma focus device, to measure the neutron yield at different distances, the simulation must be repeated for them.

Table 9: Changing percentage of efficiency and anisotropy by scattering effect

E_D (keV)	eff.(0°) %	eff.(90°) %	anisotropy %
0	-22.3	-32.4	14.9
50	-25.4	-29.5	5.8
60	-25.6	-29.9	6.1
70	-25.5	-29.2	5.2
80	-24.7	-28.4	5.2
90	-24.1	-28.9	6.8
100	-25.5	-28.6	4.5
150	-26.4	-28.1	2.3

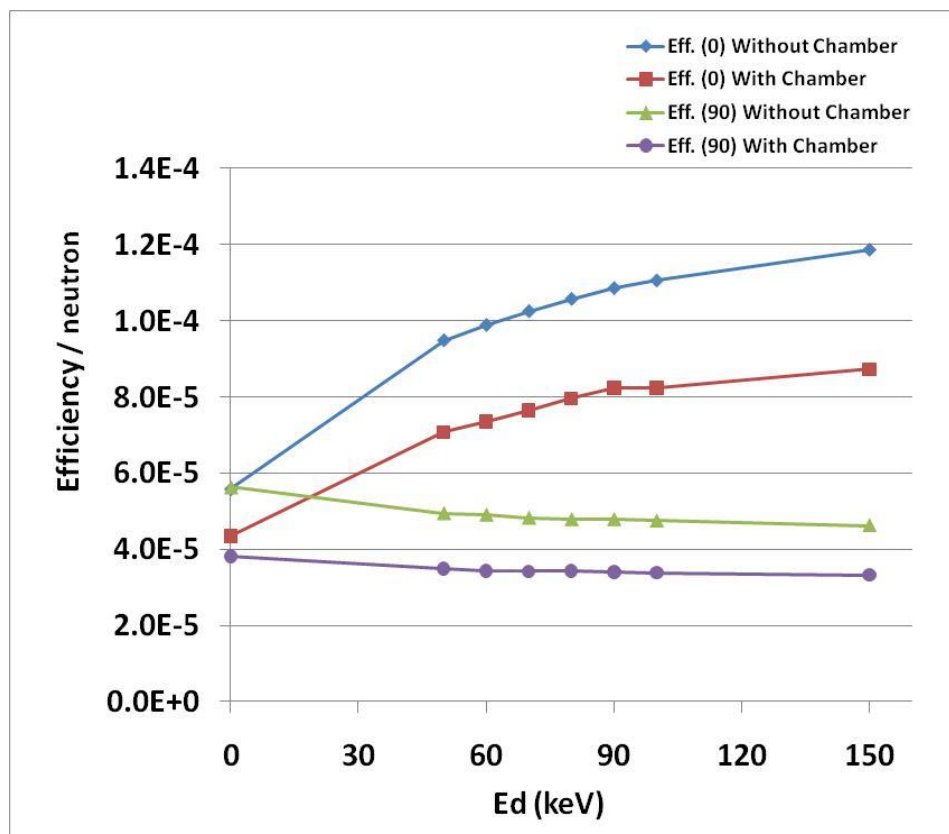


Figure 61: total efficiency of neutron detector in axial and radial direction at 25.0 cm from anode tip

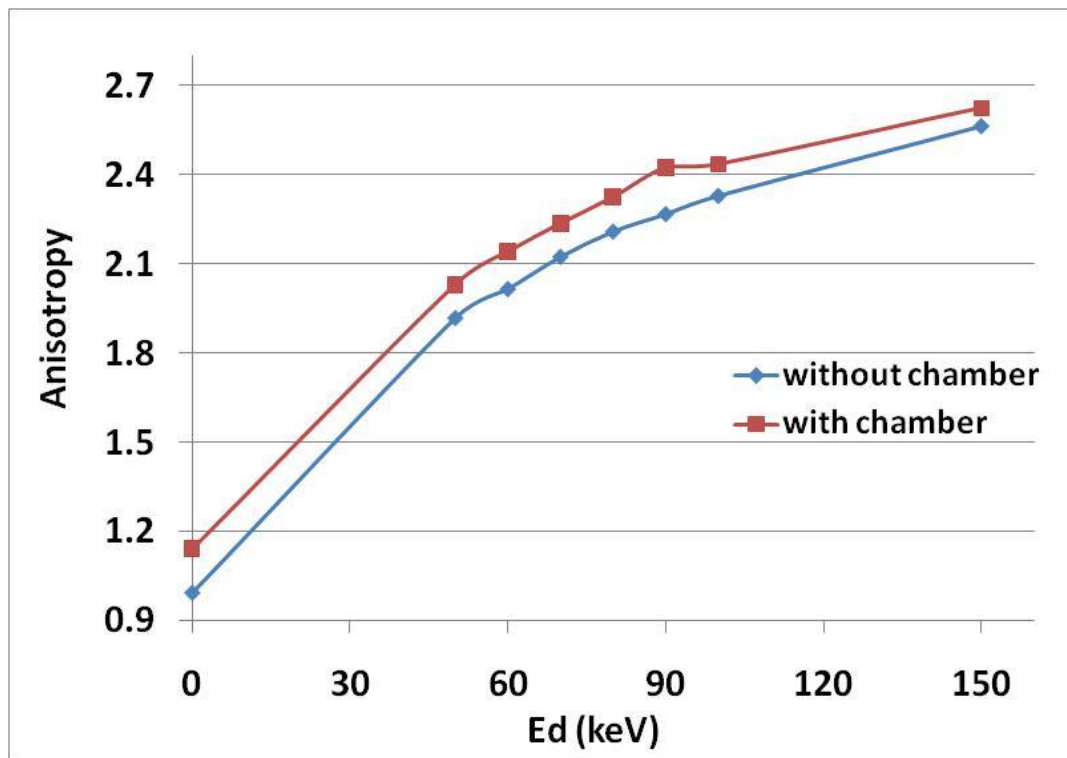


Figure 62: the result of neutron anisotropy versus deuteron energy by MCNP

MCNP simulations also confirmed that for the best estimation of neutron yield, both detectors should be positioned at the same distances from the PF pinch, and for all experimental measurements this distance was 25 cm. Since anisotropy refers to particular deuteron energy, with calculating the ratio and regarding to Figure 62, the average deuteron energy in that experiment can be found. Finally to refer the efficiency plot in Figure 61, and find the proper count factor (or just refer to Table 8 and find it by interpolation), the final neutron yield can be calculated by dividing the output count by efficiency. With this assumption, the results from both angles give almost the same numbers.

Moreover, MCNP simulation was applied to find the neutron fluence versus energy at different angles in the range of 0-180 degree (with 30° interval) to find the neutron energy spectrum in plasma focus device. It was assumed that 30° forward cone deuteron beam is coming from pinch region with constant energy (100 keV) based on beam target mechanism for D-D reaction. The point detector was kept at 1 m from center of pinch.

The simulation was run for two different situations, without chamber (void command) and with complete NX2 chamber. The result clearly shows the shifting of

neutron energy by changing the angle for both situations. Regarding to the thermonuclear D-D reaction, neutron energy which is 2.45 MeV, at side-on (90°) the neutron fluence will be remained almost constant in the range of 2.3-2.65 MeV, however, for 180° it has a peak around 2.18 MeV (~ 0.3 MeV less) and for end-on (0°) it has a peak around 2.85 MeV (~ 0.4 MeV more). As the beryllium neutron reaction cross section is depended to the neutron energy, this energy spectrum must be taken in account for response factor calculation.

It is worth referring to the simulation which was done by Bernstein (1972) [35]. In Figure 4 of mentioned paper, there are 2 sharp spikes at both sides of neutron energy spectra for θ_n (neutron detector is placed at 90°). It is in contrast to our simulation which shows zero at the end sides. The reason is in his assumption, the deuterons, all are emitting with 90° according to the PF axis. However, we have assumed that deuterons are going via a 30° cone to the PF axis.

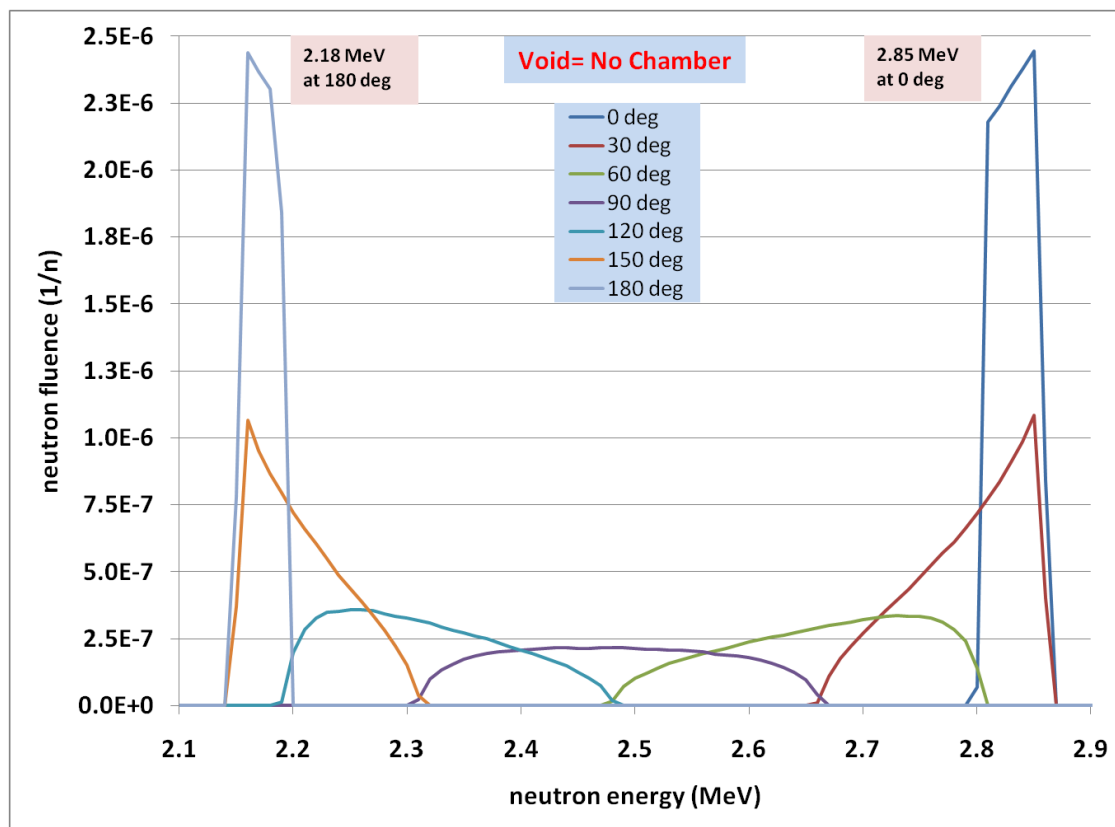


Figure 63: neuron fluence vs. energy at different angles with 100keV deuteron energy 30° forward cone at $r = 100\text{cm}$ from centre of pinch, without chamber.

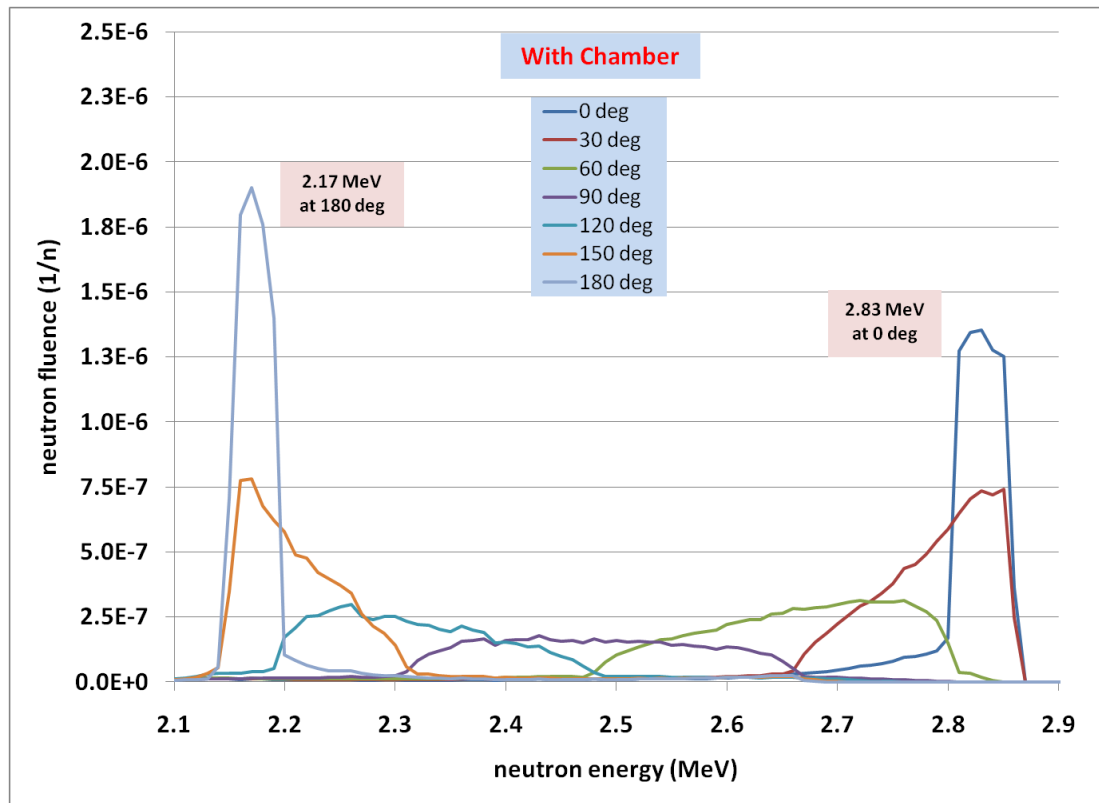


Figure 64: neutron fluence vs. energy at different angles with 100keV deuteron energy 30° forward cone at $r=100\text{cm}$ from centre of pinch, with NX2 chamber.

3.6 Experimental Setup and Results

3.6.1 Data Acquisition System

Two sets of Ortec MCS-pci Multi Channel Scalar (Figure 65) were used to acquire the data from the two beryllium detectors. Each PCI board has a BNC connector for input signal from the PDT10A units, and 25 pins D-shell connector for various control signals e.g. MCS start/stop, channel advance, etc.

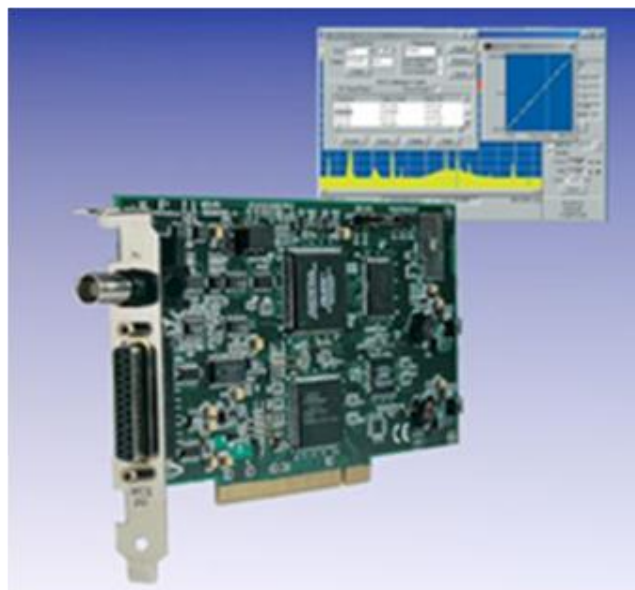


Figure 65: Ortec MCS-pci board

The MCS-32 software (Figure 66) is similar application suitable for time-resolved single-photon counting, Phosphorescence lifetime spectrometry, Scanning mass spectrometers, time-of-flight spectrometry, etc. Some of major features of this system are:

- dwell time selectable from 100 ns to 1,300 seconds per channel
- accepts counting rates up to 150 MHz at the fast input signal
- zero dead time between channels: no lost counts and no double counting at channel boundaries
- up to 1,073,741,823 counts per channel in single or multiple passes
- start of the scan can trigger the experiment, or the experiment can trigger the start of the scan.

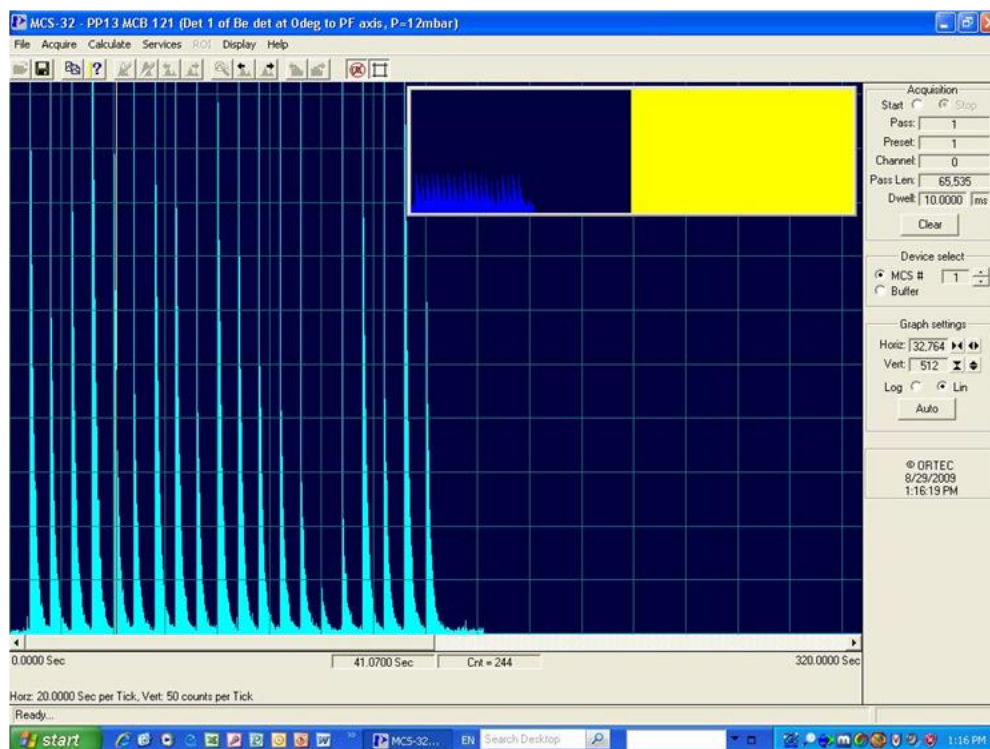


Figure 66: MCS-32 software for Data Acquisition System – showing ^6He decay plot for a series of 20 NX2 shots fired at 0.125 Hz repetition rate.

In the external trigger mode, a positive TTL logic pulse delivered to the Start Input will initiate the scan in the MCS-pci but in the internal mode, the scan starts on the first edge of the internal 50-MHz clock following the rising edge of the Start Input pulse.

Some measurements require changes in the data acquisition conditions as different spectra are acquired. The Start Job command under the Services menu allowed us to define a stream of Job Commands that varies the instrument settings and controls acquisition of multiple spectra. The Job Stream can be simple or sophisticated. Once the Job Stream is defined and implemented, data acquisition proceeds automatically under the Job Control facility.

Two Ortec pci-MCS multi-channel-scalers record the TTL pulse counts from the two Be activation detectors over time. A dwell-time 10 ms per MCS bin was selected. In analyzing the acquired MCS data, the detector counts were summed over a 3 s interval (therefore 300 MCS bins, and approximately 4 half-lives) following the neutron pulse. Then the background count ($\sim 5\%$ of the net count) over this interval was subtracted from the net count to give the foreground count for each plasma focus neutron pulse.

An ORTEC 556 power supply unit (with two SHV outputs, Figure 67) was used to provide the high voltage for all four proportional counters (two per Be activation detector). This unit is designed for use with photomultiplier tubes, micro-channel plates, proportional counters, and Geiger-Mueller tubes. The ORTEC 556 is housed in a double-width NIM module (per DOE/ER-0457T). The low-noise output is continuously adjustable from ± 10 to ± 3000 V dc with 0 to 10 mA load current. The noise on the output HV is <10 mV peak-to-peak, which ensures a stable gas-gain for the proportional counters.

An ORTEC 771 Counter-Timer was used for setting the interval time between plasma focus shots in repetitive mode. By running the Start-Job macro in MCS-32, a signal will be sent to Timer. Then e.g. every 8 seconds, the Timer sends a pulse to trigger the plasma focus device and data acquisition system as well. With Stop-Job routine we can stop the program.



Figure 67: Ortec 556 H.V. power supply

3.6.2 Interfering Activation

In order to experimentally check for the presence of any interfering activations (i.e. other than ^6He), the beryllium plate of one, of the two, detectors was replaced by an aluminium plate (mass: $m_{\text{Al}} = 191.0$ g) having the same lateral dimensions as the Be plate. With this mass, the Al plate has very nearly the same number of electrons (and electrons/cm²) as the Be plate it replaces, and therefore its Compton-scattering and electron-stopping properties are closely equivalent to that of the Be plate. The time-averaged background rates for these two cases were: 73.7 s^{-1} (Be) and 78.8 s^{-1} (Al).

MCS data were recorded for both detectors (with Al & Be plates) over a 650 s time period: a series of 30 PF shots were fired at 2 s intervals during the third minute (122 to 180 seconds) of this time period. The cumulative neutron fluence, at the detectors, over this series of 30 shots was $5.6 \times 10^5 \text{ cm}^{-2}$. To check for the presence of ^{28}Al activation, the MCS data after the shots were summed for 3 consecutive ^{28}Al half-life (134.5 s) intervals. For the Al-plate detector the first of the 3 consecutive intervals started immediately after the series of PF shots, but for the Be-plate detector the 3 intervals

were each shifted 12 s later (allowing effectively all ${}^6\text{He}$ to decay away). Background subtractions were performed. The net count rates averaged over each of the intervals were then: 45.2 ± 0.4 , 22.2 ± 0.3 , $12.6 \pm 0.2 \text{ s}^{-1}$ for the Al-plate detector; and 42.9 ± 0.5 , 20.6 ± 0.3 , $11.3 \pm 0.2 \text{ s}^{-1}$ for the Be-plate detector. These results are consistent with the expected decay of ${}^{28}\text{Al}$, indicating that its production via thermal neutron capture by ${}^{27}\text{Al}$ is the predominant (though small) interfering activation. However, aluminium is also a main structural material for the NX2 plasma focus device, and the decay of ${}^{28}\text{Al}$ is associated ($\sim 100\%$) with the emission of a 1.78 MeV gamma-ray. Hence, given the similarity of the net counts for the 3 consecutive half-life intervals for the Be-plate and Al-plate cases, it seems likely that most of the interference is due to Compton scattering of the 1.78 MeV gamma-rays emitted from the structural aluminium in the NX2, rather than β decay within the detector itself. Since the ${}^{28}\text{Al}$ interference for a single plasma focus shot is negligibly small ($\sim 1 \text{ s}^{-1}$), more exact location of its origin was not pursued.

3.6.3 Comparison of Response for Two Be Activation Detectors

To test the relative response of the two Be activation detectors, they were tested experimentally using the NX2 plasma focus as a pulsed neutron source. As shown in Figure 68, both of the Be activation detectors were positioned in the radial direction (90° to the PF axis) at a distance of 25.0 cm from, but on opposite sides of, the PF source. In the radial direction the D-D neutrons from the PF have energies centred around 2.5 MeV [35]. For this purpose the NX2 was operated with 11 kV charging voltage (1.6 kJ capacitor bank energy) with deuterium filling gas at 12 mbar. Two Ortec pci-MCS multi-channel-scalers accumulated the TTL pulses from the two Be activation detectors with bin dwell-time of 10 ms. To implement the 20 ms delay for background elimination (as described in section 3.2), the contents of the first two MCS bins associated with each PF shot were ignored. Then the pulse-counts in the following 300 bins (i.e. a 3.0 second interval) were summed, and the measured time-averaged detector background for a 3.0 s interval ($= 3 \times 73.7$ counts) was subtracted, to give the net count for both MCSs (i.e. both detectors) for each PF shot.

A series of 60 plasma focus shots were fired, and Figure 69 shows a scatter plot of the shot-to-shot detector foreground counts and associated neutron yields (calculated on the assumption of isotropic emission). The slope of the trend-line is 1.003, indicating that the response of the two detectors is, for practical purposes, identical. The

correlation between the counts from the two detectors is good, but the scatter is larger than that associated with statistical uncertainty alone. This is taken to be an indication that there are small but genuine variations between the neutron fluence in the two directions, most likely associated with shifting of the position of the plasma pinch column from shot to shot. Finally the average count M_{exp} for the 60 shots shown in Figure 69 is about 4.7×10^3 (over 3 s) and the background rate of $\sim 74 \text{ s}^{-1}$ is about 5% of the foreground count, representing a reasonably high signal-to-background ratio.

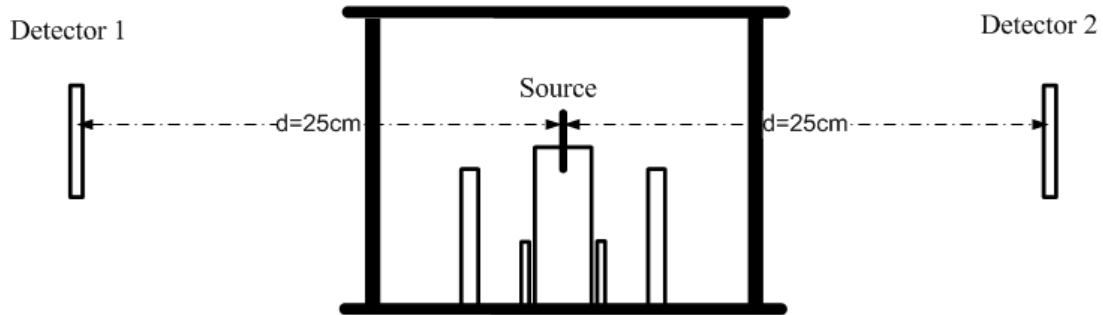


Figure 68: Experimental setup for the NX2 plasma focus neutron fluence measurements and detectors' cross comparing.

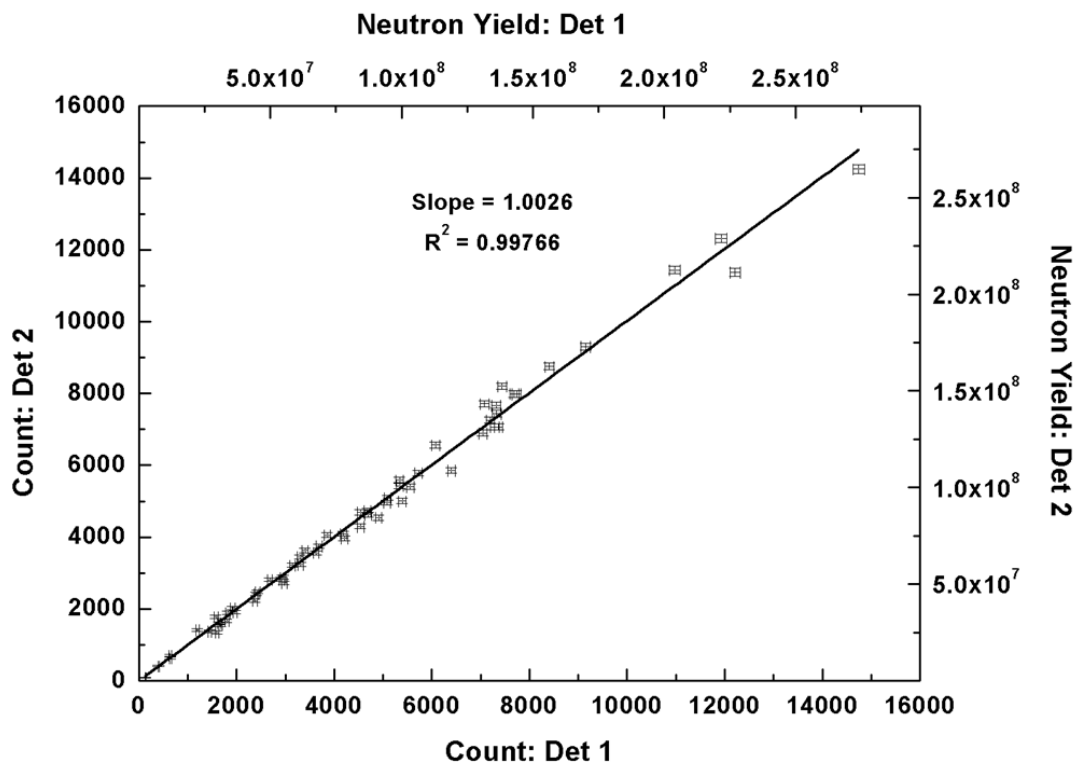


Figure 69: Experimental data from the two beryllium detectors placed in radial directions, for 60 plasma focus shots.

3.6.4 NX2 neutron yield and anisotropy measurements

To measure the neutron yield and anisotropy for the main experiments, the two beryllium neutron detectors were positioned in the end-on and side-on positions, i.e. at 0° and 90° to the PF axis respectively. The distance between the anode tip and the front surface of the detectors was 25.0 cm for both detectors. Figure 70 shows the experimental setup with the beryllium activation detectors in position.

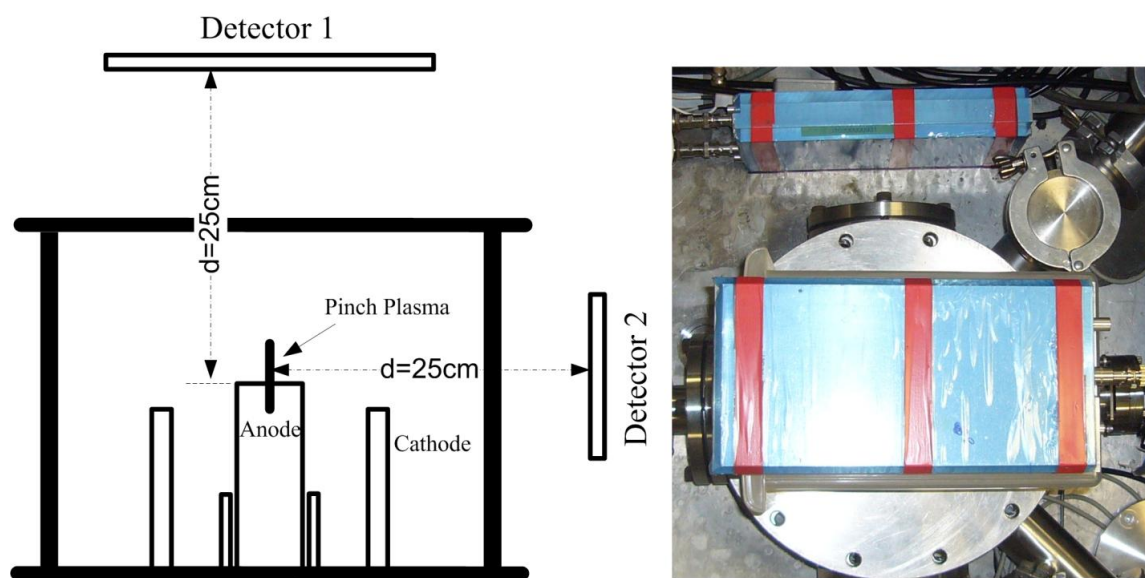


Figure 70: Experimental setup for neutron anisotropy measurements using both beryllium activation detectors.

In the first experiment, the deuterium gas pressure was scanned from 6 to 16 mbar to find the correlation between neutron yield and anisotropy with pressure. 20 shots were performed at each pressure.

Figure 71 presents an overview of the results from an investigation of the relative neutron yield and anisotropy over a range of deuterium gas pressures. The top-right figure shows that the optimum pressure for highest neutron yield is ~ 12 mbar. The top-left graph shows the neutron anisotropy drops linearly with increasing gas pressure. This trend may be explicable in terms of the relationship between anisotropy and deuteron energy.

For higher pressure, the deuterons have lower average kinetic energy within the pinch, which results in lower anisotropy. Besides, other graphs of Figure 71 (anisotropy versus neutron count for different energy) show a relative correlation between neutron yield and anisotropy which can be interpreted by the mean deuteron beam energy.

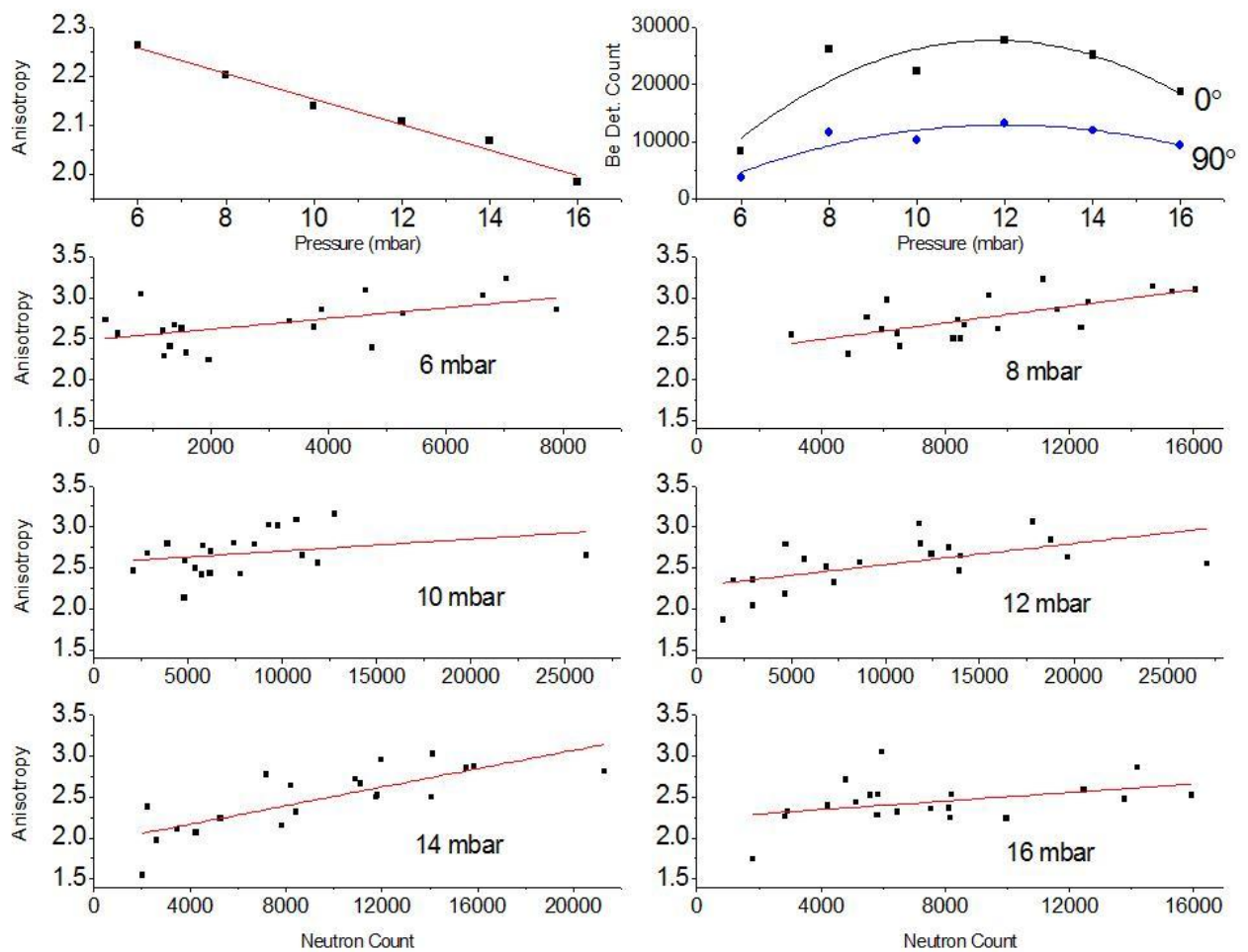


Figure 71: Average neutron anisotropy and beryllium count (0° and 90°) vs. pressure (top plots) neutron anisotropy vs. neutron yield in different filling gas pressure

In order to compare the proton anisotropy with the neutron anisotropy, a simple experiment was performed where two pieces of CR-39 (PM-355) Nuclear Track Detector were placed behind a $\sim 1 \text{ mm}^2$ open area pinhole camera in the 0° and 90° directions. The distances of pinholes to anode tip were both 95 mm, and the distances of the CR-39 detectors to pinholes were 65 mm and 35 mm respectively. The detectors were covered by 75 μm Kapton to prevent the high energy deuterons, up to 3 MeV, from reaching to detector. The detectors were exposed to 2 PF shots, followed by 8 hours etching in NaOH (6.25n at 70°), scanning, processing by Image Pro and counting of the proton tracks. The two beryllium neutron detectors were in their standard positions at 0° and 90°.

The results of the proton track distribution on the detector at 0° (end-on) and 90° (side-on) are shown in Figure 72-a and Figure 72-b, respectively. The total number proton tracks on each detector are 1890 for x-y plane and 667 for x-z plane. The accurate pinhole open area is 0.96 mm^2 for 0° and 1.00 mm^2 for 90° . Assumption of isotropic emission into 4π steradian (sr) results 2.2×10^8 for 0° and 7.6×10^7 for 90° . However as the hollow anode has been used and approximately half of the fusion source will be inside the anode and proton emission will be obstructed by anode wall, so for 90° , the count must be doubled i.e. 1.52×10^8 proton in radial direction. Although the proton gyration in plasma self-magnetic field also must be taken in account. So, very roughly, the results show the proton average yield 1.86×10^8 and the anisotropy 1.42 comparing to the neutron yield 1.55×10^8 and the anisotropy 1.63 from the same shot.

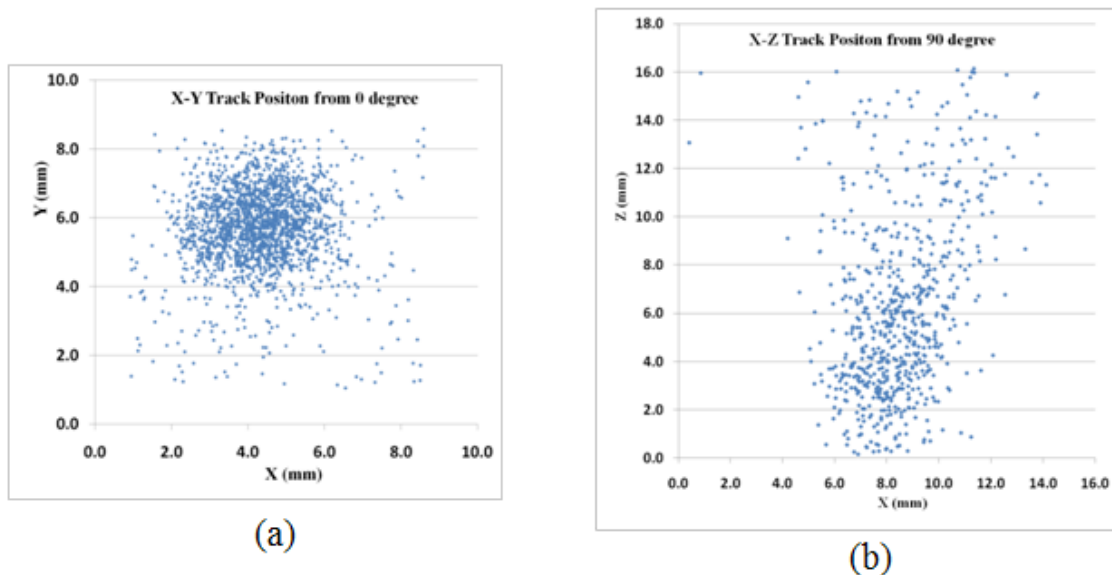


Figure 72: proton tracks distribution on PM-355 detector with pinhole imaging at
a) AP1 (0°) and b) AP2 (90°)

In the second step, the fusion source position was considered. As it is mentioned previously, the neutron emission exhibits both fluence and energy anisotropy. However, if the source is not a point source, the neutron yield measurement is related to the distance between source and detector as well. In plasma focus device, deuterons move out the pinch region and fuse together in some place which maybe is nearer to one detector. So the anisotropy will be changed because of the different distance. Usually in plasma focus device, some fraction of fusion source takes place in front of the pinch like a cone. For considering that how much fraction of neutron source is out of pinch, a Pyrex plate was put in different distances from anode top to prevent the passing of high energy deuteron to restrict the fusion source position (see Figure 73). The Pyrex glass

was put in 3 cm and 6 cm from anode top and 40 shots were performed for 2 different settings inclusive without Pyrex for comparison.

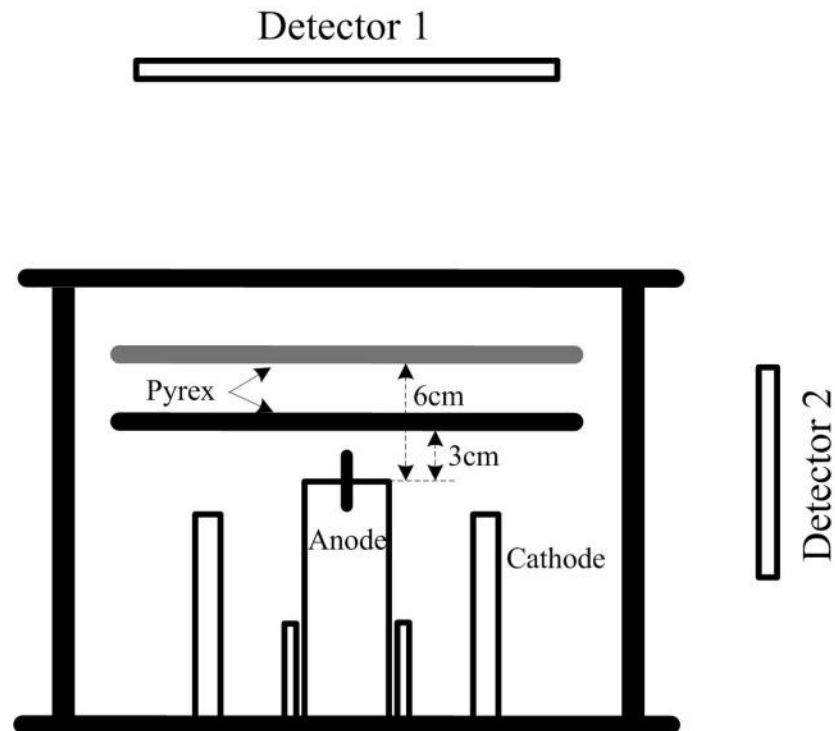


Figure 73: Experiment setup for anisotropy with Pyrex plate (6 mm thick and diameter slightly less than chamber)

Figure 74 shows the shot-to-shot variation in neutron anisotropy and the average anisotropy value for 40 shots for without-Pyrex and with-Pyrex at $d = 3$ cm and 6 cm. Table 10 also shows the average of neutron count and anisotropy for the same experiment. Experiment with Pyrex indicates that around 90% of fusion source is inside or around the pinch region. It is worth mentioning that the given anisotropy needs to be corrected by Solid Angle Ratio Correction, because the surface of detector covers an area instead of a point.

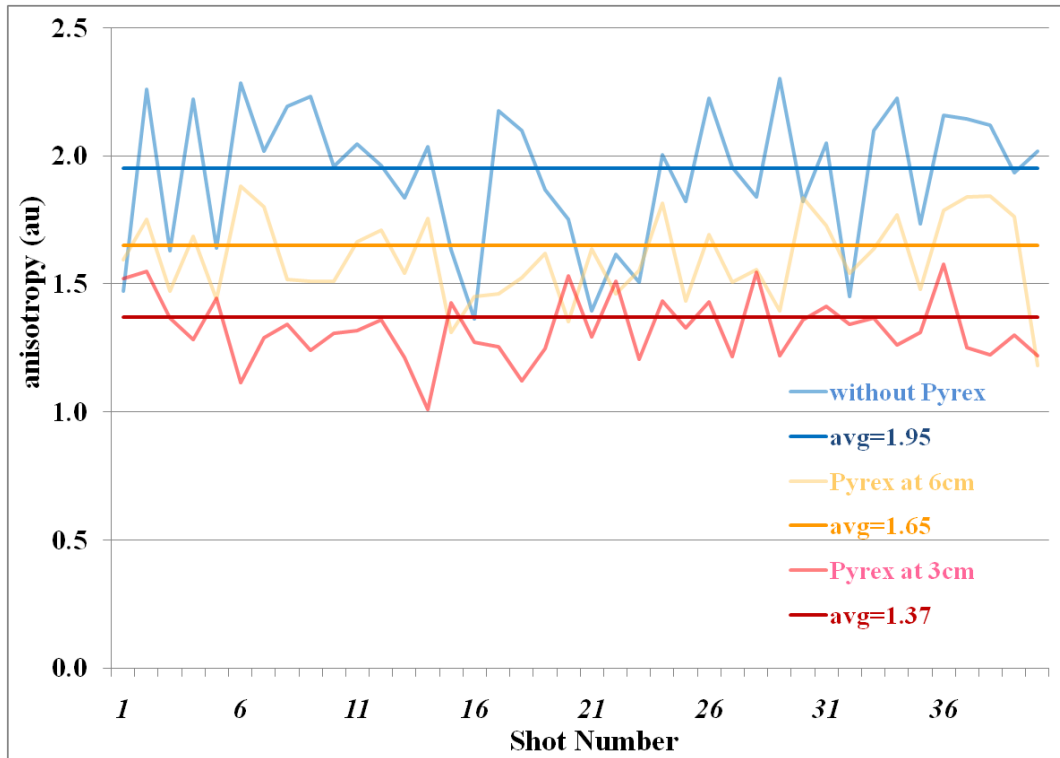


Figure 74: anisotropy variation with Pyrex which prevents the deuterons reaching the upper part of the chamber.

Table 10: average neutron count and anisotropy for different distances of Pyrex

#40 shots	without Pyrex	with Pyrex	
		d = 6 cm	d = 3 cm
avg. neutron count (0°)	24,621	14,642	17,557
avg. neutron count (90°)	12,596	8,867	12,813
avg. anisotropy	1.95	1.65	1.37

3.7 Conclusion

Two sensitive beryllium activation detectors for fast neutrons (especially for D-D fusion neutrons) have been constructed and tested. An absolute calibration, or response function, for the detectors has been derived without the availability of a calibrated neutron source. The reliability of this calibration method relies primarily on: (i) the evaluated ${}^9\text{Be}(n,\alpha){}^6\text{He}$ reaction cross-section data from the ENDF database, (ii) energy calibration of the xenon-filled proportional counters and the measurement of their relative gas-gain as a function applied bias voltage using ${}^{55}\text{Fe}$ and ${}^{109}\text{Cd}$ x-ray sources, (iii) the accuracy of the neutron and electron transport physics incorporated in the MCNP5 code. We consider the dependability of these three antecedents to be high.

The response function of the beryllium activation detector has been determined over the energy range 2.0 to 4.0 MeV, and has the form of a broad peak which reaches a maximum in the 2.8 to 3.0 MeV region. For D-D thermonuclear neutrons, of 2.45 MeV energy, the response function is about 23 % below the peak value; while for energies below 1 MeV the response is negligible. This is a useful attribute of the detector as it is insensitive to the majority of room scattered neutrons.

The two constructed detectors are fairly compact, enabling them to be positioned close to the pulsed neutron source with negligible mutual interference. And the relevant half-life of ~ 0.8 s is shorter than that for most types of neutron activation detector, enabling it to be used for moderate repetition rate sources (up to ~ 0.2 Hz). And the experimental test with both detectors positioned at 90° to the NX2 plasma focus axis shows that the response of the two detectors is practically identical.

For investigation of fusion source place in plasma focus device, a Pyrex plate was positioned in different distances from anode tip to prevent of passing the high energy deuteron to restrict the fusion source position between anode and Pyrex plate. The results show that in NX2 plasma focus device, $\sim 90\%$ of fusion source is around or near the pinch region.

Chapter 4

Other Diagnostics

4 Other Diagnostics

To investigate the correlation between fusion source image with other parameters such as x-ray source image which illustrates the bright x-ray region (pinch), the x-ray pinhole imaging system was applied with a CCD image sensor. For time resolved comparisons, three different scintillation detectors were employed simultaneously. Plastic scintillator was used for hard x-ray and neutron detection, while two different fast inorganic scintillator crystals were employed for hard x-ray measurements, these are: undoped cesium-iodide (CsI(pure)) and barium-fluoride (BaF₂). Concurrently with these, a Rogowski coil was used to measure the current derivative signal. The signals from these instruments were captured by a high-bandwidth digital storage oscilloscope. This chapter describes the applied diagnostic systems, their place in the experimental setup and their operation.

4.1 X-Ray Pinhole Imaging

An x-ray pinhole camera was employed to record the time-integrated image of the hot (~keV) plasma x-ray source for each shot. It comprised a 1/3" CCD (Charged Coupled Device) sensor with 768×576 pixels, and a laser-drilled pinhole through 50 µm thick tungsten foil. The pinch-to-pinhole and pinhole-to-CCD distances are 192 mm and 43 mm respectively, so the demagnification for this pinhole camera was ~4.5.

The video output is directly connected to a digital video camcorder (25 frame/sec) which records the plasma image continuously. To synchronize the frame capturing with plasma focus discharge (exposure) time, a synchronization circuit was designed which is shown in Figure 75. By pushing the external trigger, it opens a time gate for 20 ms (the CCD frame rate is 50 frame/sec) and waits for video synchronization signal from CCD. When both of them are ready, it sends a trigger signal with 0.2 ms delay to be sure that CCD started the video capturing.

For calibration of CCD camera, 2 mercury (Hg) lamps were applied standing parallel at 63.8 cm from CCD. The captured image shows 508 blank pixels (no bright) between them according to 4.7 cm actual distance which leads us to calculate the object pixel size at 64 cm, 92 µm. With due attention to the pinhole-CCD distance, 4.2 cm, the physical pixel size of CCD camera can be calculated ~6 µm. So the full active area is

5.5 mm (diagonal), 4.4 mm (horizontal) $\times$ 3.3 mm (vertical), and the object pixel size for our experiment on plasma focus device with pinch-pinhole distance $\sim$ 24 cm is 30 μ m.

For pure deuterium operation, the pinhole was covered with a 20 μ m Be filter to cut all visible light, UV, and x-rays with energies below $\sim$ 700 eV. For deuterium-krypton operation, a 12.5 μ m Ti plus 50 μ m Be filter combination was used. Both sets of filters transmit the Cu K_{α} and K_{β} x-rays of energies $\sim$ 8 and 8.9 keV, which permits the position of the copper anode tip to be observed in many of the x-ray images.

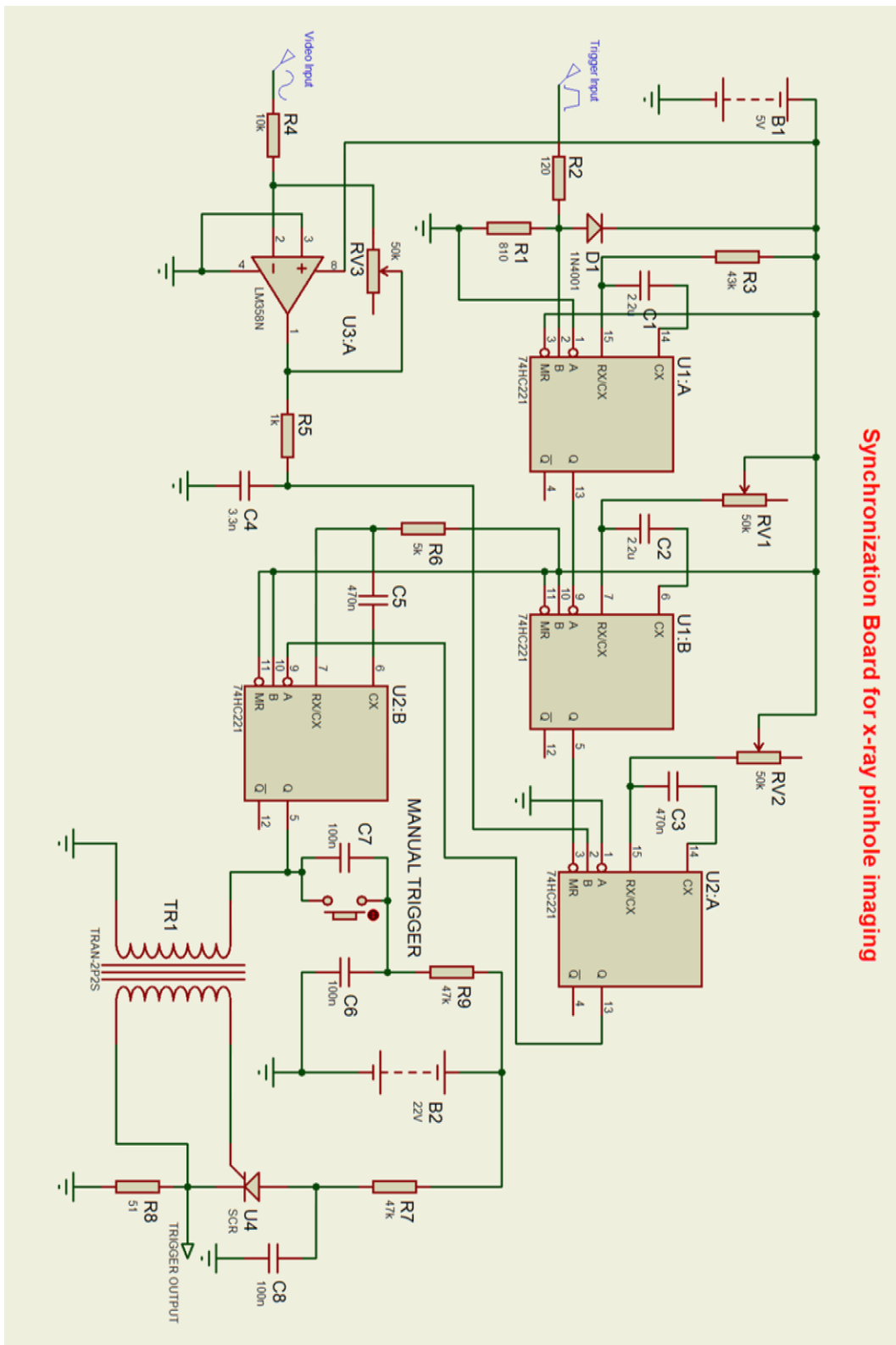


Figure 75: Circuit diagram of Synchronization Board using for x-ray imaging.

4.2 Plastic Scintillator

The time-resolved neutron detection was done using a detector consisting of a plastic scintillator and a photomultiplier tube. The detector was placed radially at the distances of 3.0 m from the pinch plasma column. To cut off most of the medium energy hard x-rays, lead sheets (5 cm thick) were placed in front of detector. It is possible to obtain a few nanoseconds resolution with the combination of a fast multiplier and a selected scintillation material. The detection efficiency for fast neutrons in a plastic scintillator depends upon the neutron energy, the geometry and the scintillation material. In our experiments, a 14-stage high gain photo-multiplier tube, Thorn EMI 9813BK coupled with NE-102A plastic scintillator (of 40 mm length and 52 mm diameter) having decay time constant of ~ 2.4 ns has been used. A high voltage power supply has been applied for providing -1500 V bias to the photo-multiplier tube. Figure 76 shows the various components of assembled Scintillator Photo-Multiplier-Tube combination.

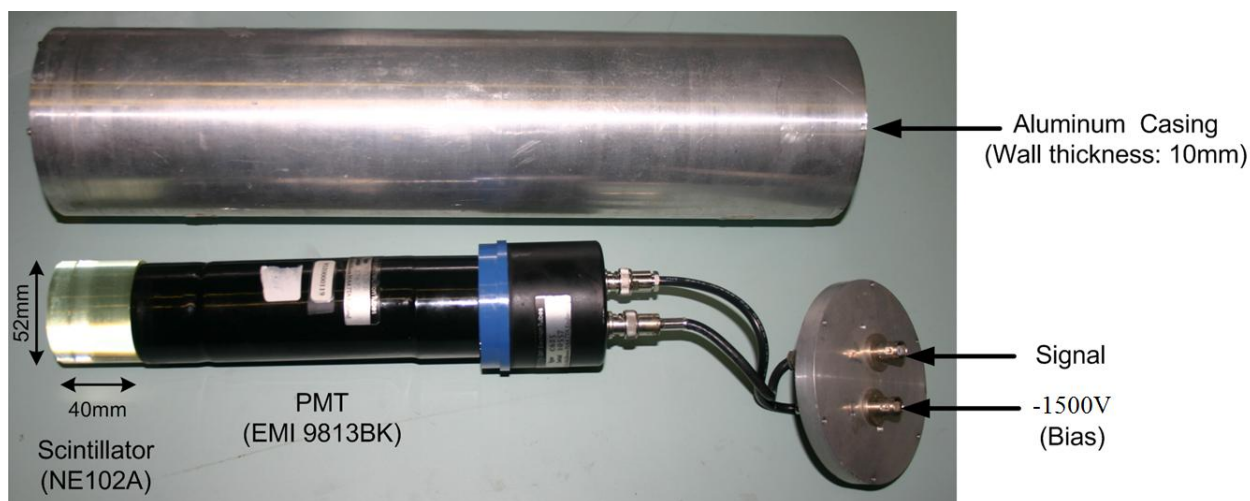


Figure 76: components of Scintillator-PMT combination [PhD thesis: Rishi Verma]

4.3 Cesium Iodide (CsI)

Pure cesium iodide (CsI) crystal is a fast scintillation crystal for γ ray and hard x-ray detection. With the density of 4.51 g/cm^3 , and atomic numbers $Z=55$ and 53 for its constituent elements, it is an efficient absorber of x-rays and γ -rays. The crystal has refractive index 1.95 at the emission maximum. The maximum wavelength of emission and the lower wavelength cutoff are 315 nm and 260 nm respectively, having the primary decay time 16 ns . With doping an impurity such as thallium (Tl) and sodium (Na), the maximum wavelength of emission can be longer (420 nm and 550 nm for Tl and Na respectively), however the primary decay time will increase (630 ns and $1 \mu\text{s}$ for Tl and Na) which is a big disadvantage for our application, plasma focus device.

In our experiments, a 14-stage high gain photo-multiplier tube, EMI 9813BK coupled with pure CsI Scintillator (of 50 mm length and 50 mm diameter) has been used. A high voltage power supply has been applied for providing -1300 V bias to the photo-multiplier tube.

4.4 Barium Fluoride

Barium Fluoride is an inorganic scintillator with a high atomic number $Z=56$ for its barium constituent, and a density of 4.88 g/cm^3 . BaF_2 therefore also has high detection efficiency for x-rays and γ -rays. The scintillation light from BaF_2 has two components, one fast the other slow. The decay time for the ‘fast component’ emitted in the short wavelength region ($170 - 220 \text{ nm}$) of the spectrum is only $0.6\text{-}0.8 \text{ ns}$. However the ‘slow component’ in the longer wavelengths has a much longer decay time of 630 ns . The fast component is within the ultraviolet region of the spectrum and is therefore difficult to detect with most PMTs. In order to detect the short wavelength the PMT requires the use of a quartz end window tube or some other non-standard light detector. The main advantage of using BaF_2 is its very short decay time a full order of magnitude shorter than any other scintillation crystal in common use. This makes BaF_2 an extremely useful detector for our application with x-ray emission time period around $50\text{-}100 \text{ ns}$. The transmission range of BaF_2 is from 150 nm to $11.5 \mu\text{m}$ with refractive index 1.46 at $3.2 \mu\text{m}$ and the reflection loss of 6.8% at $3.0 \mu\text{m}$ (2 surfaces). Figure 77 shows the scintillation emission spectrum of BaF_2 for the fast and slow components.

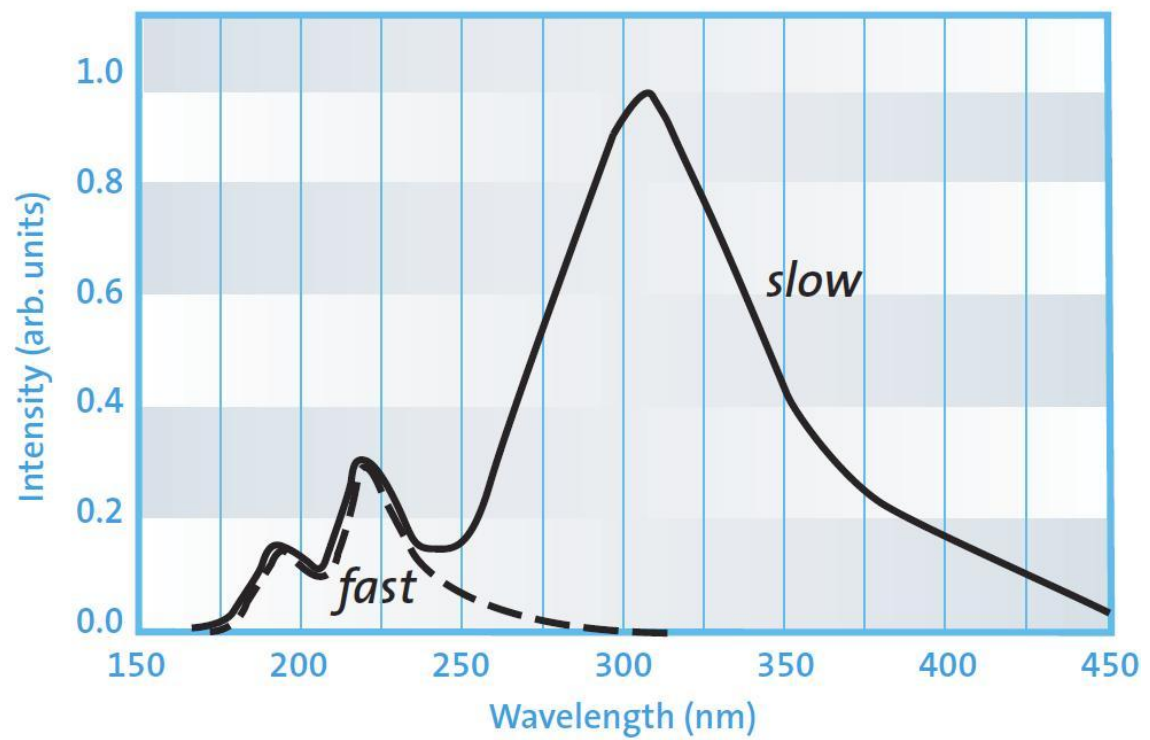


Figure 77: BaF₂ scintillation emission spectrum [SAINT-GORBAIN CRYSTALS]

Chapter 5

Experiments and Results

5 Experiments and Results

In experiment section, two different setups are explained. First, to test the idea of coded aperture imaging, reliability of De-convolution algorithm and comparing with pinhole imaging, a more controllable alpha source was applied to produce different object shape. The results show the advantage of CAI system and its validity.

Second, the fusion product proton imaging was done on NX2 plasma focus device with different geometries and different mask patterns to achieve the desired resolution and precision. Also the other diagnostics were used to find the correlation between fusion source image and other plasma parameters.

5.1 Alpha Source Imaging

To test the CAI idea, we used a coded aperture mask based on the Singer set $(p, h, \lambda) = (400, 57, 8)$ giving a $\rho = 1/7 = 14.3\%$ open fraction mask with 20×20 pixels. It is explained in section 2.1.6 and shown in Figure 22. A ^{226}Ra (half-life 1602 y) source of 5 μCi activity and 9 mm diameter (Figure 78-left) was used as the α -particle source to be imaged. The decay chain of ^{226}Ra involves the emission of 5 α particles with energies lying in the range 4.6 MeV to 7.7 MeV (Figure 78-right).

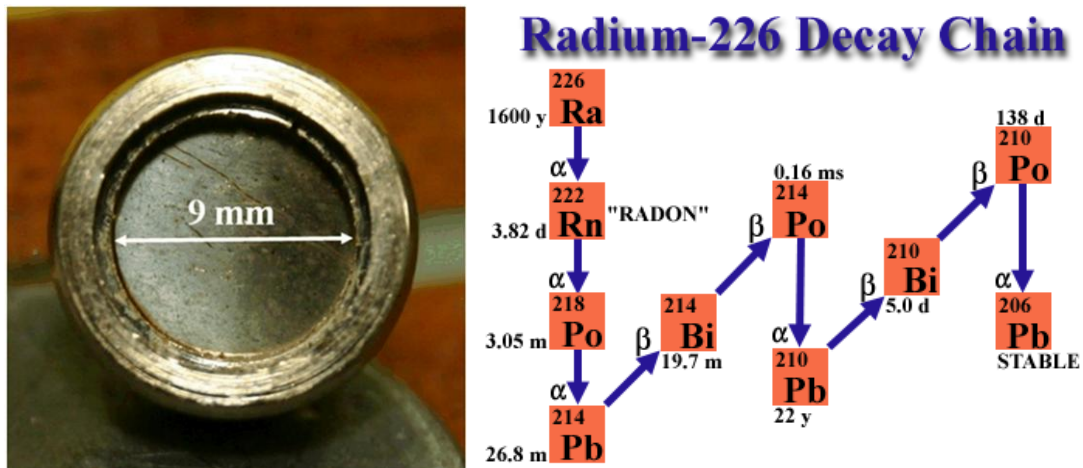


Figure 78: Radium-226 alpha-particle source (left) and its Decay Chain (right)

5.1.1 Experimental Setup

A straightforward arrangement of vacuum pipes, rotary pump and a linear translator, was used to perform imaging experiments for several simple source geometries. The ^{226}Ra disc-source and CR-39 detectors were located at opposite ends of

a vacuum pipe, with the coded aperture positioned between them. The source-to-mask and mask-to-detector distances were $d_1 = 127$ cm and $d_2 = 11$ cm respectively (see Figure 79). The de-magnified image of the ^{226}Ra disc-source has a diameter of approximately 2 pixels in the image plane, and this small feature will be referred to as a “spot”. The rotary pump maintained the pressure below 1 mbar during exposures, ensuring that the alpha-particles have ranges $\gg (d_1 + d_2)$ within the enclosure.

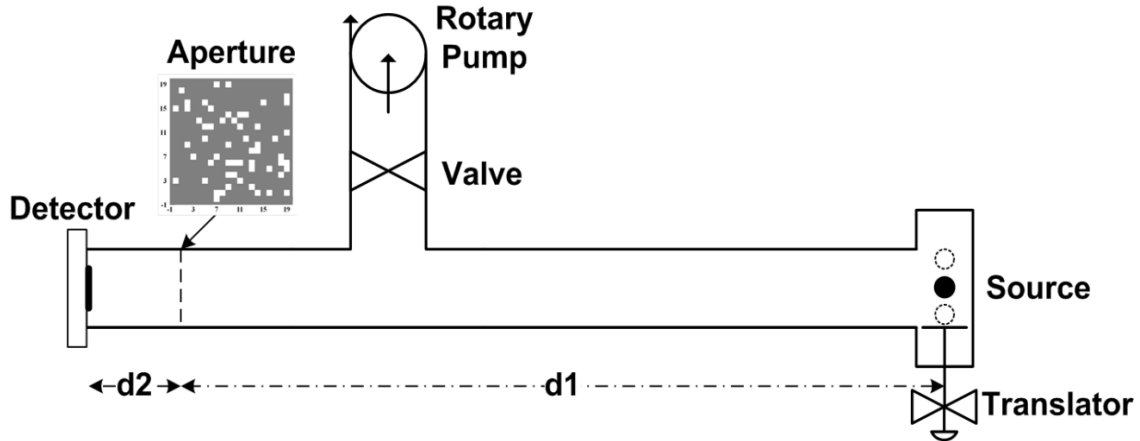


Figure 79: Experimental arrangement of CR-39 detector, CAI mask and Ra-226 α source within vacuum enclosure.

A 1-spot image was obtained for an exposure of 6.5 h duration. Using the linear translator we also synthesized a 2-spot image and an extended “ridge” image. For the 2-spot image, two 6.5 h exposures were performed, with ^{226}Ra source translated through 2.4 cm between exposures. The ridge-image comprised 13 individual exposures of 0.5 h duration, with 12 translations of 0.2 cm between exposures. Hence the brightest part of the ridge source is of length 2.4 cm, while its full length is 3.3 cm and its width is 0.9 cm. For the sake of comparison, 1-spot, 2-spot and ridge exposures were also performed with the coded aperture mask replaced by a conventional pinhole of the same size as one pixel of the mask (400 μm side). The duration of each pinhole exposures was the same as the corresponding mask exposure.

The linear dimensions of mask and detector determine the CAI Field of View (FOV) which is the maximum object size that can be fully coded on the detector:

$$FOV = \frac{d_1 L_D - (d_1 + d_2) L_M}{d_2} \quad (5-1)$$

where L_D and L_M are the lengths of the detector and mask sides respectively. The CR-39 detectors used were 2.5×3.5 cm², while L_M for the mask was 0.8 cm, giving a FOV in

the source plane of $18.8 \times 30.4 \text{ cm}^2$. So our simulated sources (full length $\sim 3.3 \text{ cm}$) were easily within the fully coded FOV. Also the FOV for the pinhole was still wider than that of the CAI camera. The resolution of a CAI or pinhole camera is effectively the magnified pixel size in the source plane. For the present geometry Resolution is $P_M \times \frac{d_1+d_2}{d_2} = 5.0 \text{ mm}$, where P_M is the mask pixel size of $400 \text{ }\mu\text{m}$.

CR-39 (PM-355, Page Mouldings Ltd., UK) polymer nuclear track detectors registered the arrival positions of the ~ 5 to 8 MeV α particles passing through the mask. The detection efficiency of C-39 for α particles incident at near normal incidence is effectively 100%. Suitable exposure times were calculated in the basis of obtaining a sufficiently high track count with minimal track overlap. Alpha particles are emitted isotropically from the ^{226}Ra source at a rate of $5 \times 5 \times 3.7 \times 10^4 \text{ s}^{-1} = 9.25 \times 10^5 \text{ s}^{-1}$.

With the mask having an open area of $57 \times (0.4)^2 = 9.12 \text{ mm}^2$ and positioned 1270 mm from the source, the rate of α particle transmittance by the mask was 0.42 s^{-1} . So for an exposure of 6.5 h duration, the expected number of α particle tracks on the CR-39 is $9,740$. Then for typical track densities of $\sim 900 \text{ mm}^{-2}$, the fraction of overlapped tracks was found to be below 3%. A small correction factor was applied in the analysis to take account of overlapped and unrecognized α particle tracks. For the case of the pinhole the expected number of tracks (with the same exposure time) is simply $1/57$ of that for the mask; therefore about 140 tracks. Track overlap for the pinhole case was negligible.

After exposure, the CR-39 detectors were etched in an aqueous solution of 6.25 n NaOH , at 70°C for 5 hours . The bulk etch rate under these conditions is $1.6 \text{ }\mu\text{m/h}$. A magnified image of etched tracks on the detector surface is shown in Figure 80. Making use of a program for calculating the parameters of etched tracks, the track diameter and depth for 5 MeV α particles were predicted to be $\sim 14 \text{ }\mu\text{m}$ and $\sim 9 \text{ }\mu\text{m}$ respectively (see Figure 38). The diameter value was found to be in good agreement with the experimentally measured average track diameter. The software auto-focus option was used to scan the entire exposed area on the CR-39, storing contiguous images of the CR-39 surface to the computer hard disk. For CAI exposures an array of $36 \times 48 = 1,728$ microscope images were captured, while for pinhole case it was $4 \times 6 = 24$ images.

Automatic analyzing of all the stored images with Image Pro macro, discriminating between genuine α particle tracks and spurious dark features, within polygon criteria and finally processing the data by the convolution-based decoding algorithm, gives us the reconstructed image.

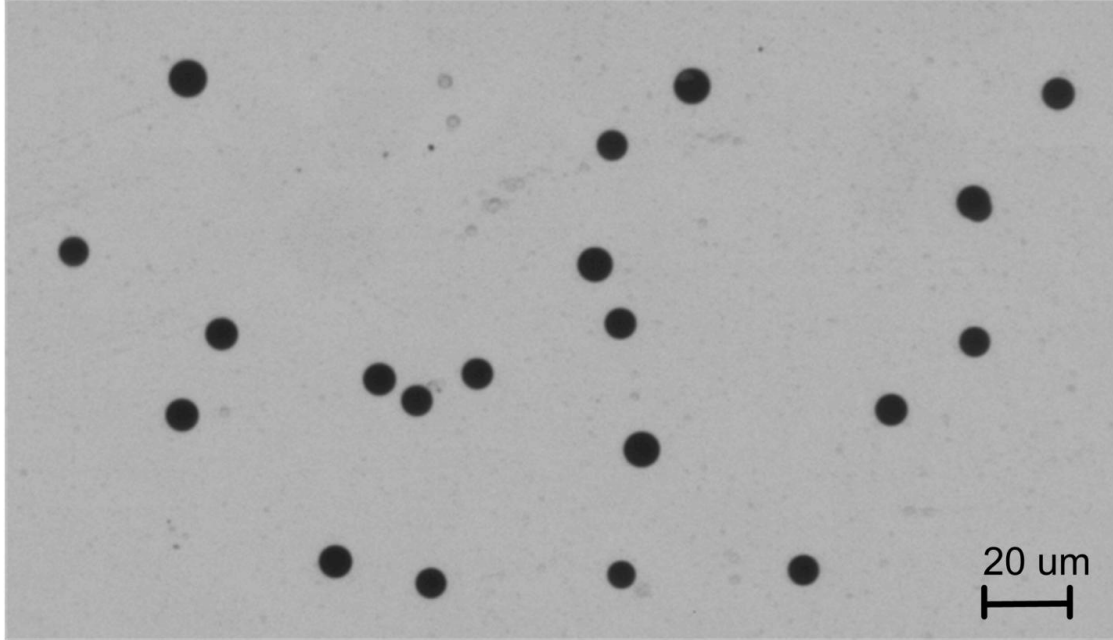


Figure 80: Optical micrograph of etched alpha-particle tracks on CR-39 surface. The overall dimensions of the image are $262 \times 152 \mu\text{m}^2$.

5.1.2 Results and Discussion

Experiments with 1-spot, 2-spot and ridge sources were performed with both a CAI mask and a pinhole having the same angular resolution. Table 11 presents the number of recognized alpha-particle tracks and SNR values for each exposure.

For CAI decoded images, the SNR was calculated by application of Eqn. (2-22). Since every pixel in the image effectively has a different signal-to-noise ratio, the values given in column 5 of Table 11 are those for the brightest pixel (n_p) in the image. Then with σ being the calculated standard deviation of pixel values outside of the illuminated region(s): $\text{SNR} = n_p / \sqrt{n_p + \sigma^2}$. This formula represents contributions from self-noise and sidelobe noise. The theoretical SNR values given in column 6 of Table 11 are calculated from Eqn. (2-24) using the total track number N in the reconstructed image and n_p the brightest pixel value. For pinhole images, only self-noise is present, and in accordance with Eqn. (2-27): $\text{SNR} = \sqrt{n_p}$.

Table 11: Track number and SNR for CAI and pinhole system from Experimental results

Experiment	No. of tracks (pinhole)	No. of tracks (CAI)	Exp. SNR (pinhole)	Exp. SNR (CAI)	Theory SNR (CAI)
1-point circular source	142	8,086	6.2	26.6	38.0
2-point circular source	317	16,648	7.4	18.8	33.5
Ridge source	129	8,040	3.9	13.3	19.3

For the 1-spot CAI mask exposure, the total number of α particle tracks was about 8,000, and the coded image (as a scatter plot of tracks on the detector surface) is shown in Figure 43 (d). The de-coded image derived from this data is shown in Figure 81 (a2). The corresponding pinhole image for 1-spot, as shown in Figure 81 (a1), has 142 tracks. Both images show the disc shape of the source but the CAI image reproduces it more faithfully, and as shown in Table 11, the CAI image has a superior SNR of 26.6 by comparison with 6.2 for the pinhole.

For the 2-spot images, the detectors were exposed for 13 hours (twice the duration of the 1-spot exposures). Figure 81 (b1), for pinhole, shows two separated regions with no background; however the number of tracks in each spot differs significantly. With the same source, exposure times, and distances, the track counts per spot are 130 and 155. So they are respectively one standard deviation above and below their mean. This is a reflection of the large self-noise associated with the low-brightness pinhole image. By comparison, Figure 81 (b2) shows the CAI reconstructed 2-spot image is much brighter and has a SNR value of 18.8 (a factor of 2.5 higher than that for the pinhole image). Here the number of counts in each spot is 5,656 and 5,759 respectively.

Lastly, Figure 81 (c1) and (c2), show the pinhole/CAI comparison for the ridge source. Again the CAI system produces a much brighter and more faithful image of the source, as reflected in the considerably larger SNR for the CAI vs. pinhole image: 13.3 and 3.9, respectively. However, despite the better SNR exhibited by the CAI, the values are lower compared to the theoretical SNR values predicted by equation (2-25).

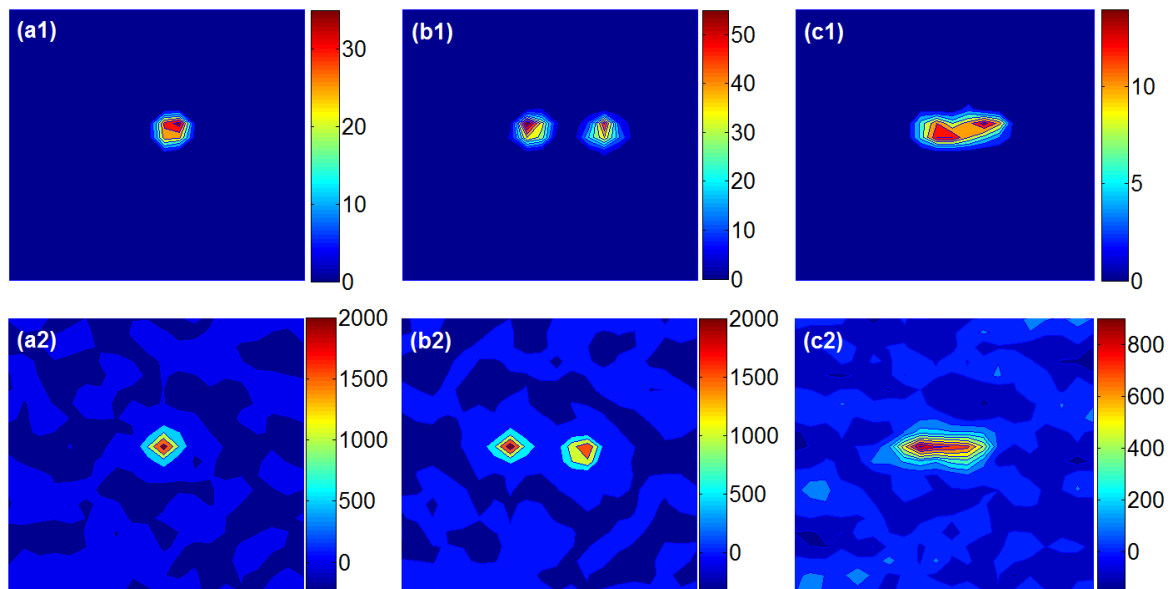


Figure 81: Reconstructed image from experimental track data for different source shapes with pinhole (top row) and 20×20 CAI mask (bottom row). All images are of 10 cm side (in the source plane). 1-point source for pinhole (a1) and mask (a2); 2-point source for pinhole (b1) and mask (b2); ridge source for pinhole (c1) and mask (c2)

One factor which degrades the SNR in the experimental situation is the background of spurious tracks due to microscopic flaws and scratches on the CR-39. This roughly uniform background is evident in the 1-spot coded image in Figure 43 (d). A second factor is the spread in emitted alpha-particle energies, from 4.6 to 7.7 MeV, from members of the ^{226}Ra decay chain. This spread in alpha-particle energies necessarily produces a spread in etched track diameters and average grey-values, which complicates track-recognition and discrimination between genuine alpha-particle tracks and spurious tracks. As an indication of such problems: a total of 32,774 tracks were recognized and counted for the combined 26 hours of CAI exposure time (see Table 11), whereas the expected number (based on the activity calculation) is $4 \times 9,740 = 38,960$. Therefore the recognized number of tracks is about 16% lower than the expected value. Uncertainty in the ^{226}Ra source activity calibration may account for part of this difference. However, a 20% reduction would result from none of the highest energy alpha-particles (7.7 MeV from ^{214}Po), which produce the smallest tracks, being counted. More work aimed at optimizing detector etching conditions and image processing for track recognition would alleviate such problems. However, the present measurements are sufficient to validate the CAI technique with Singer set based masks before proceeding to image fusion protons from the plasma focus device.

5.2 Fusion Source 3-D Imaging

In this section, the first application of the CAI technique to obtaining high precision fusion proton images for a small plasma focus device is reported which presents the spatial structure of fusion reaction source by fusion product protons. By using the conventional pinhole imaging, we cannot get such high precision image, because of its small open area and naturally fewer transmitted protons. In one sense the imaging properties of a pinhole are ideal, as its autocorrelation is a delta-like function; however it suffers greatly from low collection efficiency. In practice the diameter chosen for the pinhole ought to represent the best tradeoff between image resolution and image noise; as the pinhole diameter decreases the image resolution improves while the relative image noise enlarged. Instead, Coded Aperture Imaging achieves high precision images by using many small pinholes arranged in very specific patterns.

5.2.1 Experimental Setup

The NX2 device used for these experiments is a Mather-type plasmas focus. A hollow copper anode (Figure 82) was used in order to minimize impurities in the working gas caused by ablation of anode material by strong electron beams emitted from the pinch. Five CAI cameras were employed to image the NX2 fusion source from different directions.

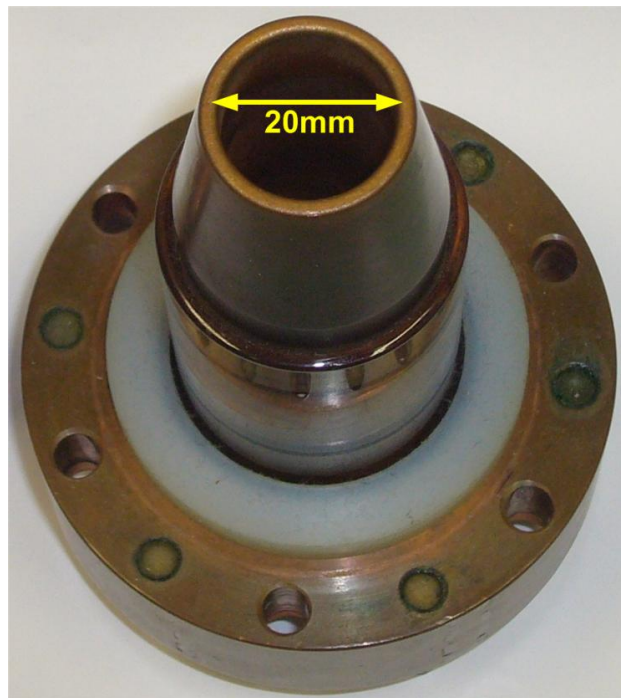


Figure 82: The hollow anode using in NX2

As can be seen in Figure 83, one CAI camera is located at 0° (axial direction, designated camera_0) and four others are at 45° “compass points” (designated camera_1 to camera_4). For the 0° camera, the source-to-mask and mask-to-detector distances were $d_1 = 10.3$ cm and $d_2 = 6.5$ cm, respectively. While for the 45° cameras, the corresponding distances were $d_1 = 4.9$ cm and $d_2 = 3.4$ cm. The open solid angles for these cameras are 8.58×10^{-4} and 3.77×10^{-3} sr, respectively.

The linear dimensions of mask and detector determine the CAI Field of View (FOV) which is the maximum object size that can be fully coded on the detector:

$$FOV = \frac{d_1 L_D - (d_1 + d_2) L_M}{d_2} \quad (5-2)$$

where L_D and L_M are the lengths of the detector and mask sides respectively. The dimension of the CR-39 detectors used are 3.0×3.0 cm² for 0° and 4.5×3.0 cm² for 45° , while L_M for the mask was 0.8 cm (square), so the FOVs in the source plane were 6.8×6.8 cm² for 0° and 4.5×2.4 cm² for 45° . Based on numerical simulations of PF dynamics, the expected dimensions of the PF pinch column are about 10 mm length and 2 mm diameter [15], which are easily within the fully coded FOV. The resolution of a CAI camera is effectively the magnified pixel size in the source plane. For the present geometry, the object pixel size or resolution is $P_M \times \frac{d_1 + d_2}{d_2} \cong 1.0$ mm for both directions, where P_M is the mask pixel size of 0.4 mm.

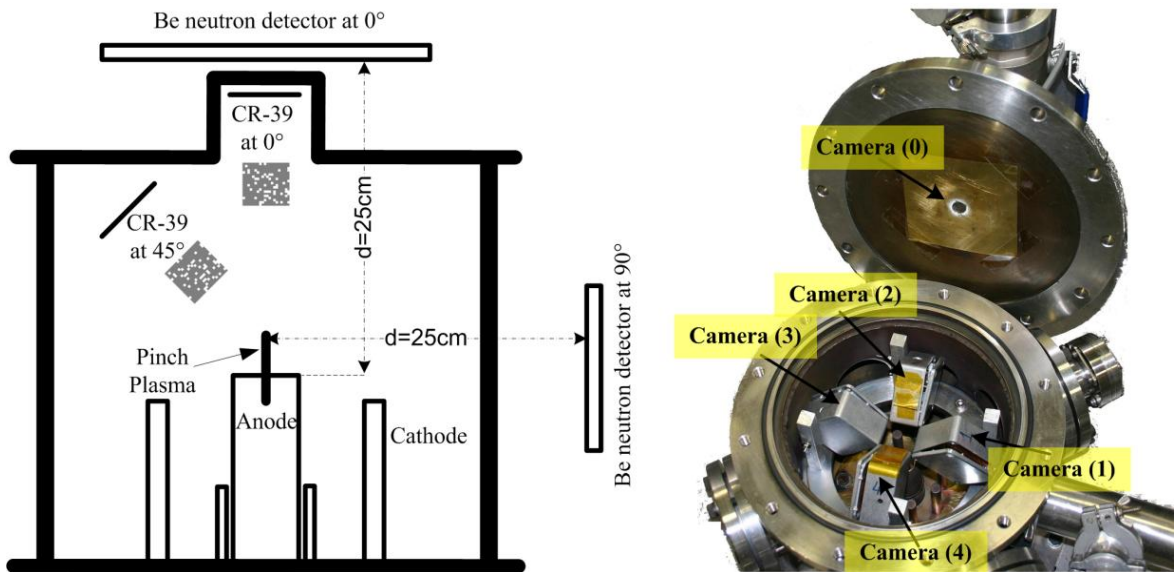


Figure 83: experimental arrangement for 3-D fusion imaging in NX2

It was anticipated that the fusion source dimension would be somewhat larger than the size of the pinch: the reason being that medium energy deuterons (~ 20 to 50 keV) gyrate within, and around, the pinch column due to the high magnetic field. On this basis, we roughly predicted the size of the fusion source to be about 3 mm in diameter and 11 mm in length. As the object pixel size for all five CAI cameras is ~ 1.0 mm, so the image of the pinch source will fill about $m = 3 \times 11 = 33$ pixels which determines the near optimum SNR would be obtained for a mask open fraction of $\rho = 1/7$. We therefore based our coded aperture pattern on a Singer set with 20×20 pixels and 57 open holes (Figure 21).

Coded aperture mask based on the Singer set $(p, h, \lambda) = (400, 57, 8)$ with $\rho = 1/7$ was folded to 20×20 pixels. The $50 \mu\text{m}$ thick Havar alloy with overall dimension $8 \text{ mm} \times 8 \text{ mm}$ was laser-machined to produce 57 square holes ($400 \mu\text{m}$ side). The mask pattern and fabricated sample are shown in Figure 31. CR-39 (PM-355, Page Mouldings Ltd., UK) polymer nuclear track detectors registered the arrival positions of the ~ 3 MeV proton passing through the mask. The detector is covered by $75 \mu\text{m}$ Kapton film to prevent the high energy deuterons up to 3 MeV from reaching the detector.

For proton imaging it is essential to consider the curvature of proton trajectories due to the magnetic field in and around the pinch column, and its influence on the images recorded by the CAI camera. For this purpose, a program was written in MATLAB to calculate trajectories of D-D fusion protons that impinge on the CAI mask, and to calculate the apparent source position. It was assumed that current through the PF plasma column is parallel to the PF axis and uniformly distributed throughout the column (i.e. disregarding breaking and kink instabilities), and also that the current flows uniformly out of the pinch in a fountain shaped current sheath. Then the magnetic field is purely toroidal and given by:

$$\begin{aligned} B_\phi &= \frac{\mu_0 I_p r}{2\pi R_p^2} & (r \leq R_p) \\ B_\phi &= \frac{\mu_0 I_p}{2\pi r} & (R_p < r \leq R_{cs}(z)) \\ B_\phi &= 0 & (r > R_{cs}(z)) \end{aligned} \tag{5-3}$$

where R_p is the radius of the pinch column and I_p is the current flowing through it. The radius of the current sheath (CS) is given by an empirical function $R_{cs}(z)$ which follows the observed shape of the CS. The acceleration of protons in the magnetic field is

$$\frac{d\vec{v}}{dt} = \frac{\vec{F}}{m} = \frac{e}{m} \vec{v} \times \vec{B} = (v_r \hat{z} - v_z \hat{r}) e B_\phi / m \quad (5-4)$$

and this ordinary differential equation is solved in MATLAB by numerical integration (Runge-Kutta method) to determine the proton trajectory for a given point of origin and initial velocity. When the proton exits from magnetic field region ($r \leq R_{cs}(z)$) it follows a straight line path. For simplicity, a 3 MeV proton source was assumed to be positioned on-axis ($x = y = 0$) and the z position within the plasma pinch column was selected randomly ($0 \leq z \leq l$) where l is the length of the column. The program then follows the trajectories of protons emitted over a range of angles, to find those that are incident on the coded mask. Figure 84 shows calculated proton trajectories for the geometry of the CAI exposures, and a pinch current $I_p = 300$ kA.

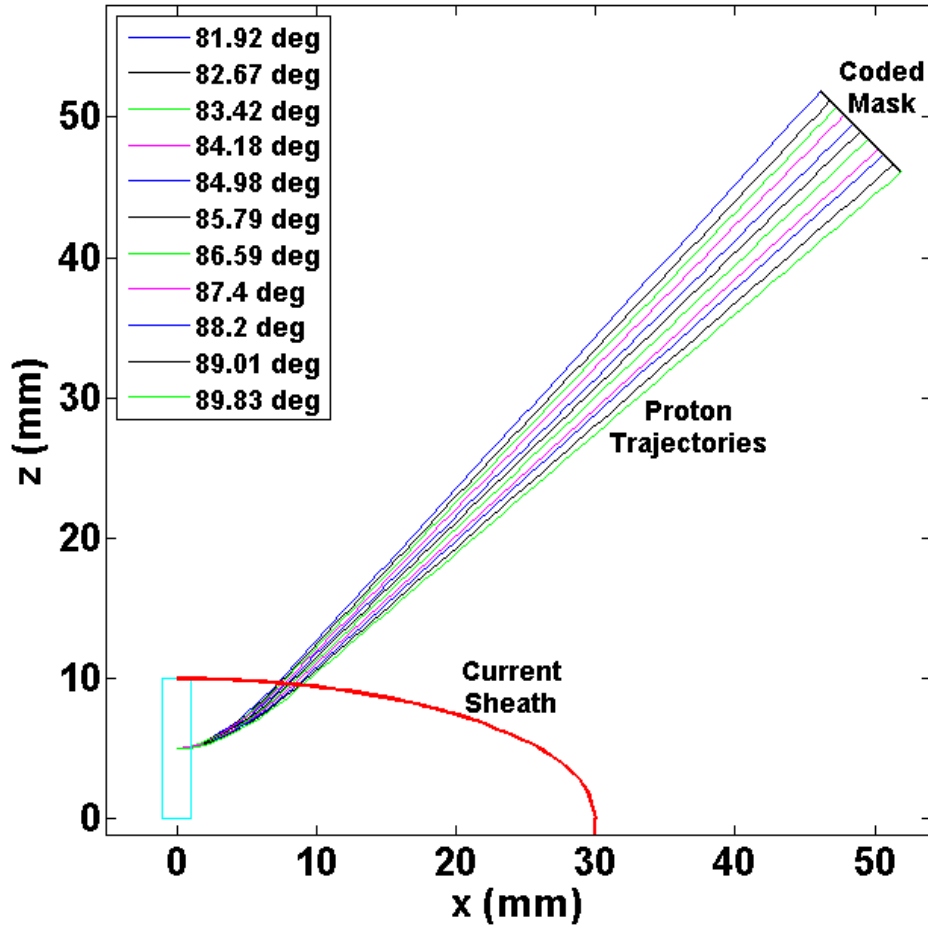


Figure 84 Trajectories of 3 MeV protons (which impinge on coded mask at 45 degree) in plasma pinch magnetic field.

For these mask-incident trajectories, the program determines the apparent position of the proton source by extrapolation of the straight section of the trajectory back to the pinch. Table 12 shows the z displacements of protons emitted over a range of angles which impinge on the coded mask (with the same 45° positioning as used experimentally). The mean z displacement δz_m is -3.0 mm. More importantly, the range of z displacements is 0.08 mm which is only 8% of the object pixel size (~ 1 mm) and the standard deviation is 0.02 mm. Hence for a mono-energetic proton point source, the effect of the magnetic field amounts simply to a displacement (in the negative z direction) of the apparent source position. Figure 85 shows $|\delta z_m|$ as a function of the point source height z_s within the pinch column, and it can be seen that the relationship is almost linear. Since δz_m is negative, the overall affect of magnetic curvature of proton trajectories is that the fusion source image is linearly stretched (by a factor of $S_z = 1.6$) in the axial z direction, but no other significant distortion of the image occurs.

Table 12: Displacements of apparent proton source position versus emission angle, for trajectories which impinge on coded mask at 45 degree. Actual source position is mid-pinch: $z = 5$ mm.

Angle (deg)	Mask Position (mm)	Displacement (mm)
81.92	4.1	-3.11
82.67	3.1	-3.06
83.42	2.5	-3.09
84.18	1.7	-3.09
84.98	0.8	-3.06
85.79	0	-3.08
86.59	-0.8	-3.09
87.4	-1.8	-3.05
88.2	-2.3	-3.13
89.01	-3.3	-3.09
89.83	-4.1	-3.11

Average Displacement	-3.09
Maximum Displacement Range	0.08
Standard Deviation	0.02

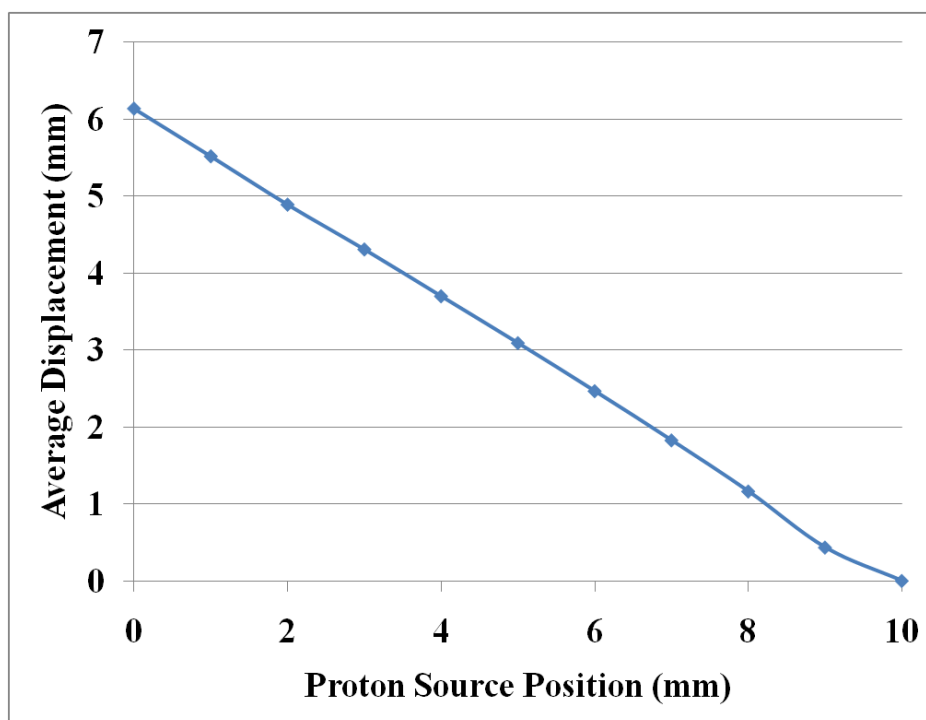


Figure 85: Magnitude of mean displacement of apparent source position $|\delta z_m|$ versus actual z position of source within the pinch for mask-400 at 45 degree.

In addition, two beryllium activation fast-neutron detectors (explained in 3.6.4) were positioned in the axial and radial direction (0° and 90° to the PF axis), both at a distance of 25.0 cm from anode tip to measure the time-integrated neutron yield and anisotropy.

5.2.2 Experimental Results

Single shot exposures were performed on the NX2 plasma focus device at 11 kV charging voltage (1.6 kJ capacitor bank energy) with deuterium filling gas at 13 mbar (which is the neutron optimized regime). For the shot discussed below, the neutron counts for the end-on and side-on beryllium detectors were 22,950 and 8,100 respectively, giving anisotropy of 2.83 and a neutron yield of 2.8×10^8 . The five detectors were etched for 8 hours followed by scanning and image processing. The proton track numbers and calculated proton yield (with isotropic distribution assumption) of each detector are shown in Table 13. Figure 87 shows the de-convoluted images obtained for each CAI camera. For the 45° camera images, the two effects of foreshortening due to the viewing angle (Figure 86 and Figure 85) and stretching due to magnetic curvature of proton trajectories are corrected by scaling the z image dimension

by a factor of $1/(s_z \sin 45^\circ) = 0.884$. Then the 45° camera images are dimensionally equivalent to an undistorted 90° view.

The results show a proton 45° -anisotropy of 2.6, which is about 10% less than neutron anisotropy. Anisotropy is normally defined in terms of angles 0° and 90° , but a CAI camera placed at 90° would be obscured by the anode wall, so such a measurement is not feasible with a hollow anode. Based on the shapes of the $D(d,n)\text{He}$ and $D(d,p)\text{T}$ differential cross-sections, the 0° vs. 90° proton anisotropy should be greater than the 0° vs. 45° value, but not so large as the neutron anisotropy. Also considering the proton counts from detectors 1 to 4, it appears that the location of the pinch was not precisely on the anode axis, but shifted away from camera 1 and towards camera 3.

Table 13: proton count and yield for five coded aperture imaging cameras

	Camera-1	Camera -2	Camera -3	Camera -4	Camera -0
Proton Count	27,900	30,400	32,400	30,400	17,900
Proton Yield	0.92×10^8	1.0×10^8	1.1×10^8	1.0×10^8	2.6×10^8

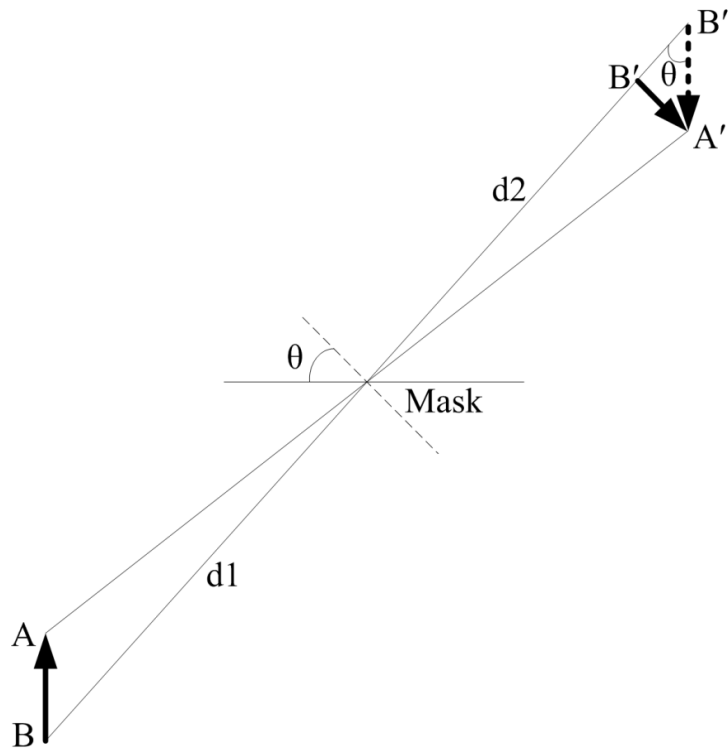


Figure 86: elongation of image for 45° view

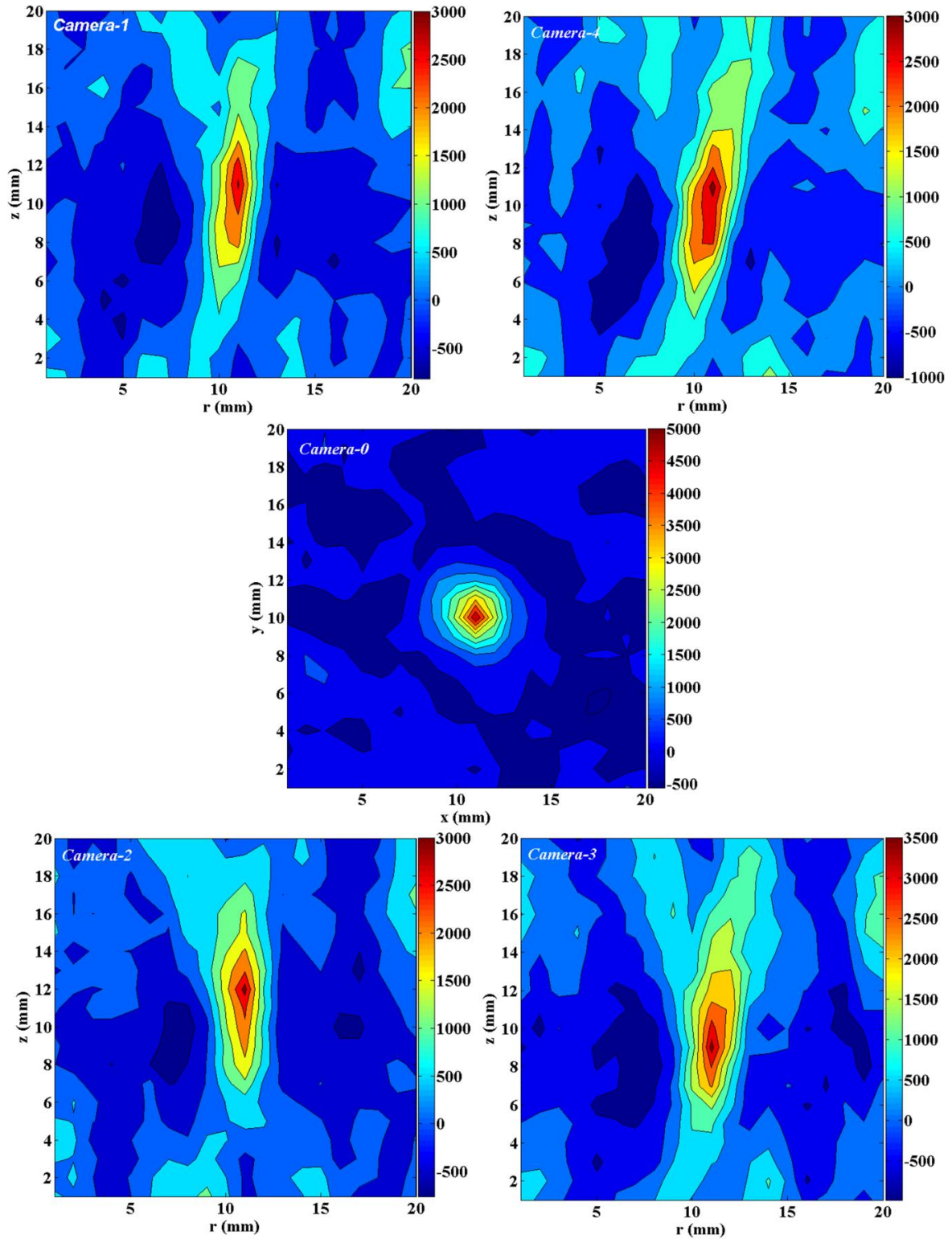


Figure 87: reconstructed images: Camera-1 to Camera-4 at 45° and Camera-0 at 0°

5.3 Fusion Imaging with Spaced Mask

For investigation of fusion source image to finding out its correlation with other diagnostics, two spaced masks (mask-1365 and mask-4680) with better resolution than mask-400 (masks' pixel size are 0.3 mm and 0.27 mm respectively) and higher intensity (341 and 585 holes) were used. NX2 plasma focus device was fired in optimum neutron emission yield (13 mbar and 11 kV/1.6kJ). In order to consider different fusion source behavior in operation with pure and admixture gas, two series of shot were performed, first with pure deuterium and then deuterium plus 2% volumetric krypton admixture. Each time the detectors were exposed only by one shot. All other diagnostics such as beryllium activation neutron detector, soft x-ray image, current derivative, plastic Scintillator, CsI and BaF₂ detectors were applied and recorded to compare their signals and shape with fusion source image.

5.3.1 Experimental Setup

For final stage of experiment, the NX2 plasma focus device with new designed anode was used in order to see the whole pinch region and fusion source. In fact, a cooper insert anode with 2 mm diameter hole (Figure 88-middle) was put on top of the previous hollow anode.

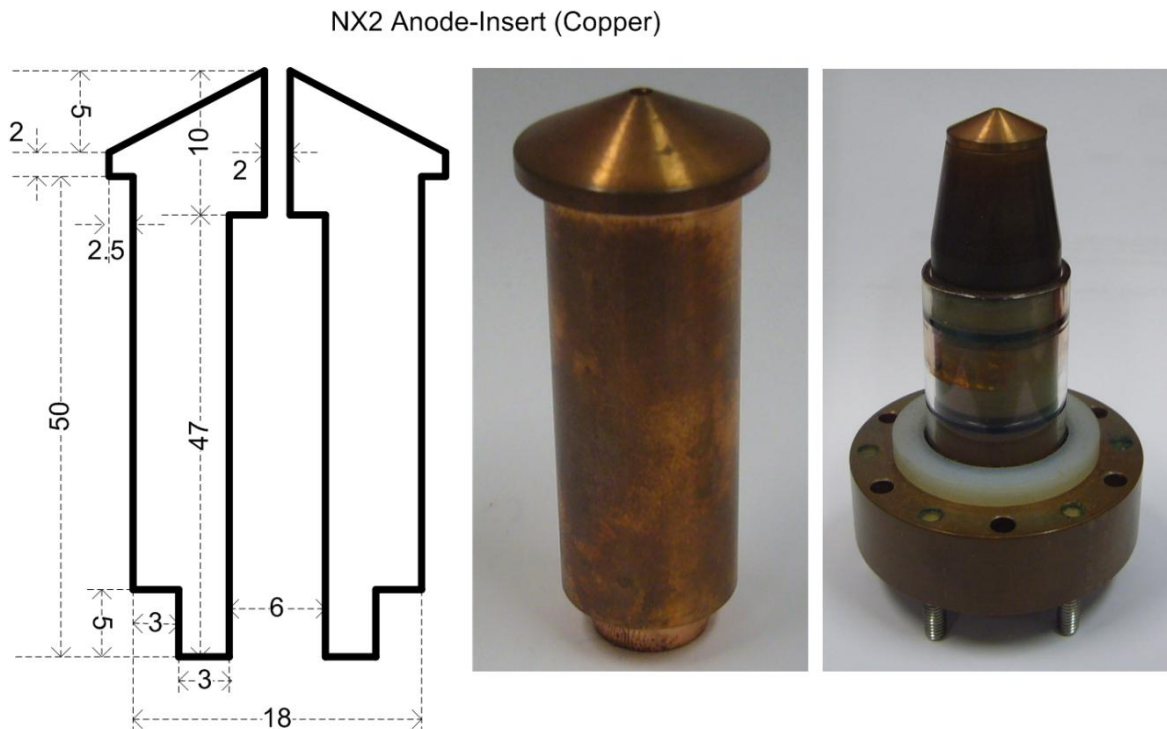


Figure 88: insert tapered anode with 2 mm diameter hole (middle) and complete anode shape (right).

Two CAI cameras with the different properties were positioned at 90° to image the NX2 fusion source from side-on view. Mask-1365 (91×15) with $h = 341$ holes ($300 \mu\text{m}^2$), $\rho = 1/4$ open area fraction, $k = 6$ space factor has 30.69 mm^2 absolute open area (Figure 89) and Mask-4680 (104×45) with $h = 585$ holes ($270 \mu\text{m}^2$), $\rho = 1/8$ open area fraction, $k = 2$ space factor has 42.65 mm^2 absolute open area (Figure 90). Both masks and detectors were housed within the base as shown in Figure 91.

The source-to-mask and mask-to-detector distances were $d_1 = 44 \text{ mm}$ and $d_2 = 40 \text{ mm}$, respectively, so the open solid angle for these cameras is $1.315 \times 10^{-2} \text{ sr}$ for mask-1365 and $2.154 \times 10^{-2} \text{ sr}$ for mask-4680. The dimension of the CR-39 detectors used is $67 \times 77 \text{ mm}^2$, while L_M for the masks was $\sim 2.7 \text{ cm}$ (square), so according to formula (5-2), the FOV in the source plane is $2.8 \times 1.7 \text{ cm}^2$ and an expected $1.0 \times 0.3 \text{ cm}^2$ fusion source of NX2 is easily within the fully coded FOV. The resolutions of these CAI cameras (the magnified pixel size in the source plane) for the present geometry are 0.63 mm for mask-1365 and 0.57 mm for mask-4680. The total detector scan area comprises $100 \times 120 = 12000$ frames which covers $72.6 \times 65.3 \text{ mm}^2$.

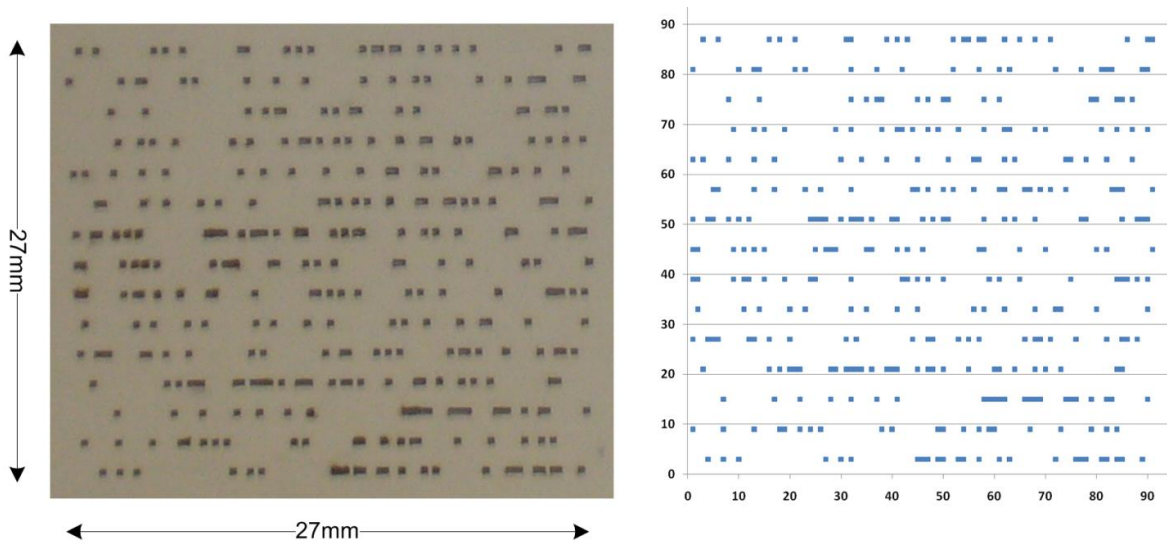


Figure 89: mask-1365 (left) and the pattern (right)

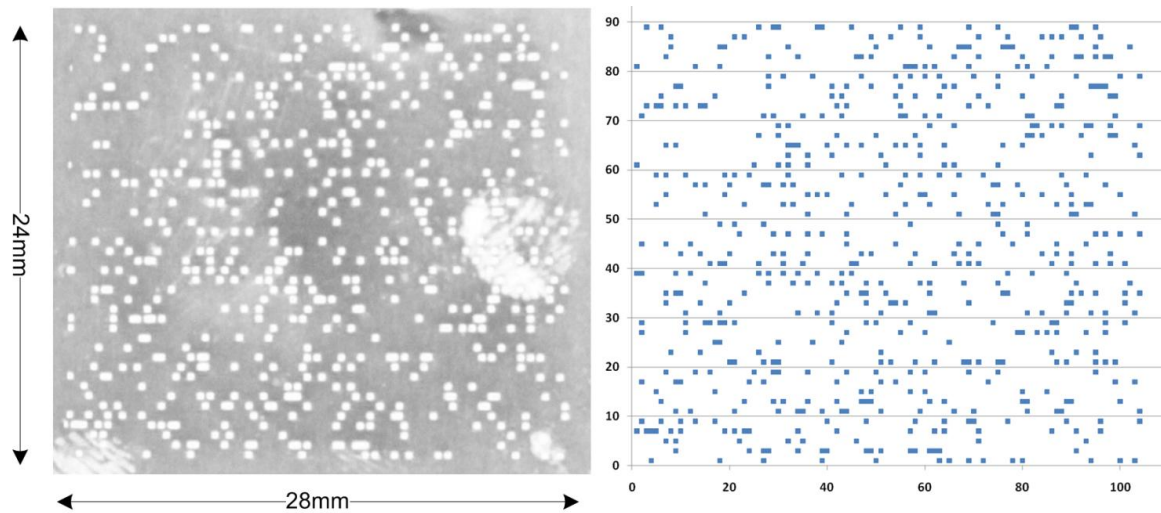


Figure 90: mask-4680 (left) and the pattern (right)

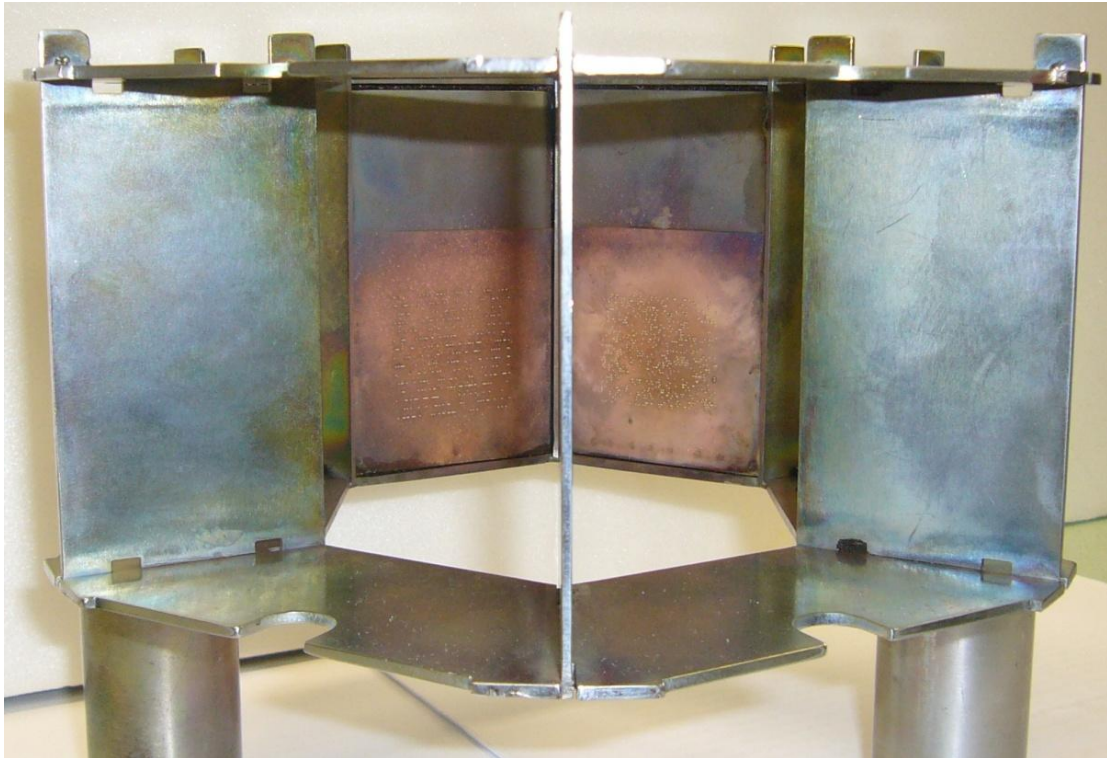


Figure 91: both masks were housed on the base

Based on our first experiment with mask-400, we have roughly measured the size of the fusion source about 2.5 mm in diameter and 10 mm in length. According to the object pixel size for both CAI cameras, the image of the pinch source will fill about $m = 4 \times 16 = 64$ pixels for mask-1365 and $m = 4.4 \times 17.5 = 77$ pixels for mask-4680 which determine the near optimum SNR for each of them would be obtained for a mask open fraction of $\rho = 1/4$ and $\rho = 1/6$ respectively (refer to Figure 27 and Figure 28). For mask-1365 we have exactly $\rho = 1/4$, however for mask-4680, the open area fraction is $\rho = 1/8$ which causes the lesser SNR.

Moreover, the same proton trajectory calculation was done for this new geometry. Figure 92 shows calculated proton trajectories for the geometry of the CAI exposures, and a pinch current $I_p = 300$ kA. Comparing to previous geometry for mask-400 at 45° , some improvement has been done in term of current sheath which is more realistic than before.

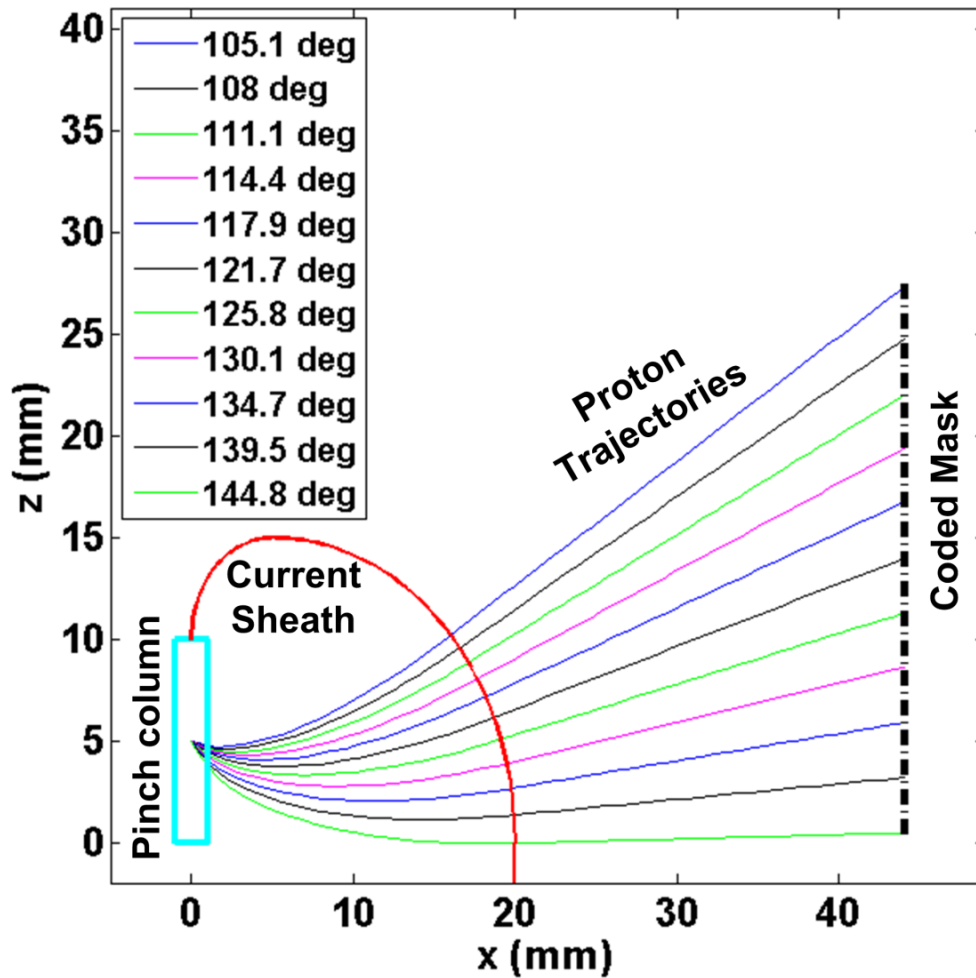


Figure 92: Trajectories of 3 MeV protons (which impinge on coded mask) in plasma pinch magnetic field for mask-1365 and mask-4680 at side-on.

Also Table 14 shows the z displacements of protons emitted over a range of angles which impinge on the coded mask-1365 and mask-4680 which are positioned at 90° (side-on). The mean z displacement δz_m is -4.76 mm. The range of z displacements is 0.93 mm which is bigger than the object pixel size (~ 0.6 mm) and the standard deviation is 0.30 mm. Although this variation makes the reconstructed image a bit blurred, however, still it can be applied for fusion source imaging. Figure 93 shows $|\delta z_m|$ as a function of the point source height z_s within the pinch column. Since δz_m is negative, the overall affect of magnetic curvature of proton trajectories is that the fusion source image is linearly stretched slightly (by a factor of $S_z = 1.08$) in the axial z direction.

Table 14: Displacements of apparent proton source position versus emission angle, for trajectories which impinge on coded mask at side-on. Actual source position is mid-pinch: $z = 5$ mm.

Angle (deg)	Mask Position (mm)	Displacement (mm)
105.11	13.3	-4.57
107.98	10.8	-4.58
111.08	8	-4.5
114.4	5.4	-4.55
117.86	2.7	-4.57
121.74	0	-4.6
125.8	-2.7	-4.68
130.12	-5.4	-4.81
134.68	-8.1	-4.95
139.5	-10.8	-5.16
144.82	-13.5	-5.43

Average Displacement	-4.76
Maximum Displacement Range	0.93
Standard Deviation	0.30

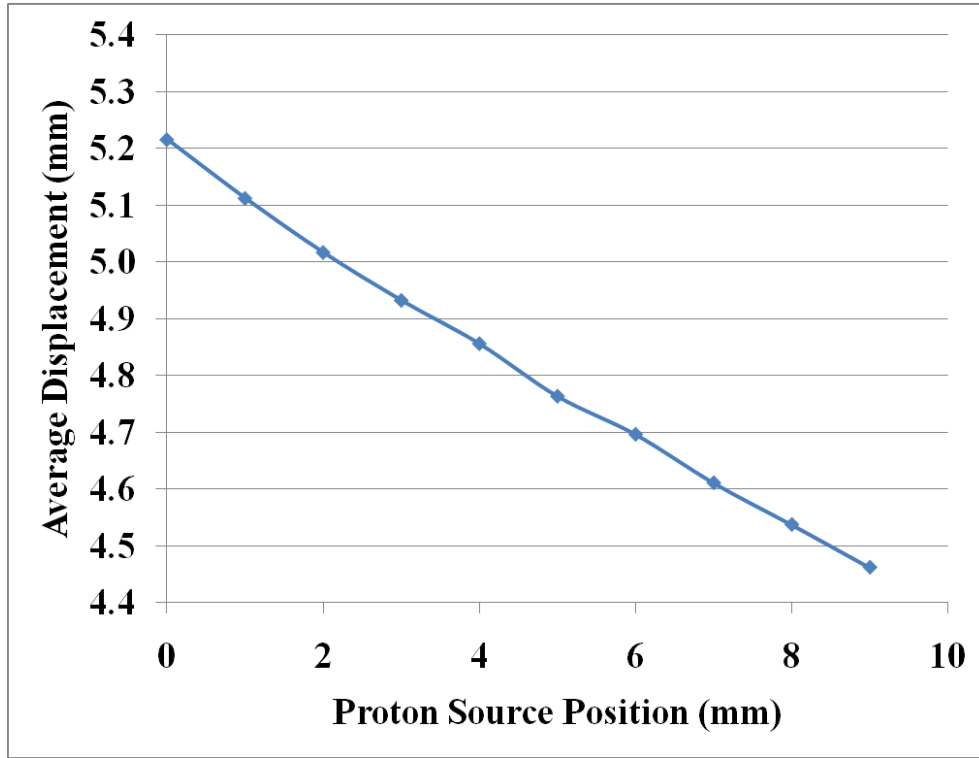


Figure 93: Magnitude of mean displacement of apparent source position $|\delta z_m|$ versus actual z position of source within the pinch for masks at side-on.

It has to be emphasized that the above calculation still may not be realistic as we do not know the exact current sheath shape (it needs to be performed the shadowgraphy for individual shot to recognize it). Also the exact current passing through the pinch is not identified. So the above corrections have not applied on the reconstructed fusion source images. Moreover, only a fraction of fusion source takes place inside the pinch which needs more intelligent programming to correct the source position.

The CAI camera system with both mask were located in NX2 chamber somehow that masks are almost at 90° related to axial direction. The lowest level of mask is just above the anode tip level which it is shown in Figure 94. As it is mentioned before, the source-to-mask and mask-to-detector distances were $d_1 = 44$ mm and $d_2 = 40$ mm, respectively.

The CR-39 nuclear track detectors (Intercast Europe Srl-Parma-Italy) were covered by $75 \mu\text{m}$ Kapton and kept always the same position for all experiments. The exposed detectors were etched with NaOH (6.25 n at 70°C) for 15 h. Then they were coated by thermal coater (HIND HIGH VACUUM Company Pvt. Ltd.) with 200 nm

chromium and 250 nm indium to make them shiny enough for laser focus scanning. Each detector then was scanned by a motorized stage controlled by Stage-Pro software to obtain $100 \times 120 = 12,000$ frames, a total area of $72.6 \times 65.3 \text{ mm}^2$, following by analyzing with Image-Pro software to obtain the tracks' characteristics. Selecting the genuine proton tracks within-polygon criteria (explained in section 2.2.5) and finally applying de-convolution algorithm give us the fusion source images for each shot via two different masks.



Figure 94: the position of masks in NX2 chamber.

As usual, two beryllium activation fast-neutron detectors were positioned in the axial and radial direction, both at a distance of 25 cm from anode tip to measure the time-integrated neutron yield and anisotropy. Also a plastic Scintillator was employed to detect the time-resolved neutron yield at a distance of 3 m from anode in radial direction. To cut off most of the medium energy hard x-rays, lead sheet (5 cm thick) was placed in front of the detector. At the same position two more detectors were used for x-ray emission, Cesium-Iodide and Barium-Fluoride which all can be seen in Figure 95. These signals plus current derivative were recorded by a YOKOGAWA oscilloscope (model: DL9140 – 5 GSample/sec – 1 GHz) (Figure 96).

Moreover the pinhole x-ray imaging system was applied to record the x-ray source image within a shot. It comprises a CCD camera (covered with 20 μm beryllium for pure deuterium and 50 μm beryllium plus 12.5 μm titanium for krypton admixture

gas), Sony video recorder and synchronization circuit board. All experimental setup has been shown in Figure 97.

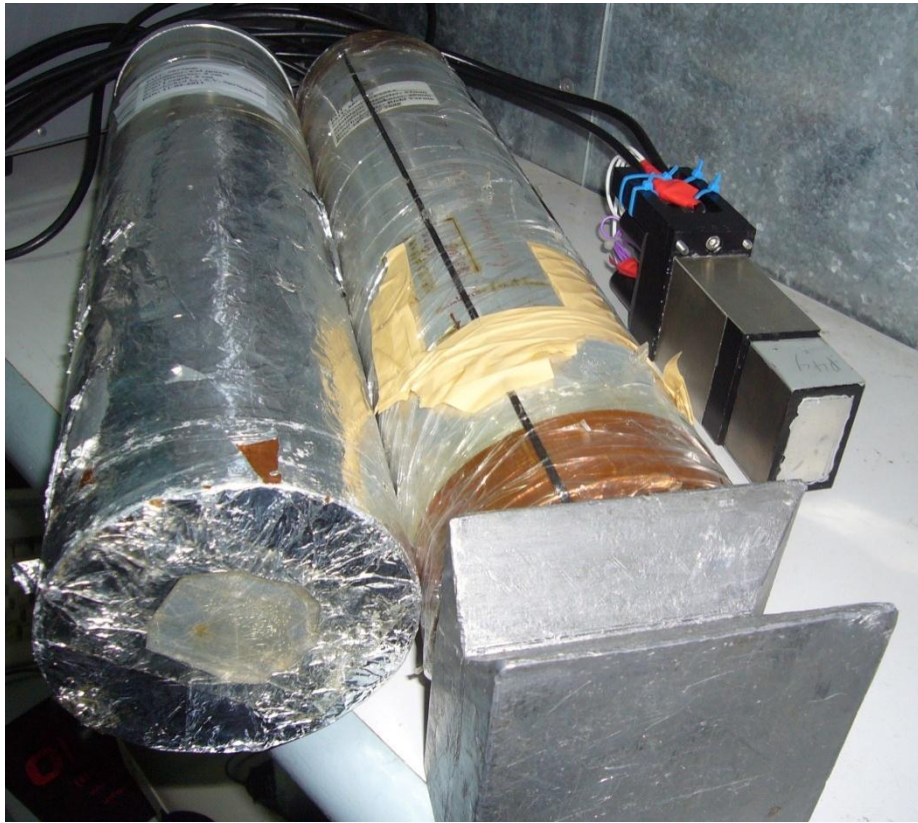


Figure 95: plastic Scintillator with lead sheet in front, Cesium-Iodide and Barium-Fluoride, with included PMTs.



Figure 96: YOKOGAWA 4-channels digital oscilloscope

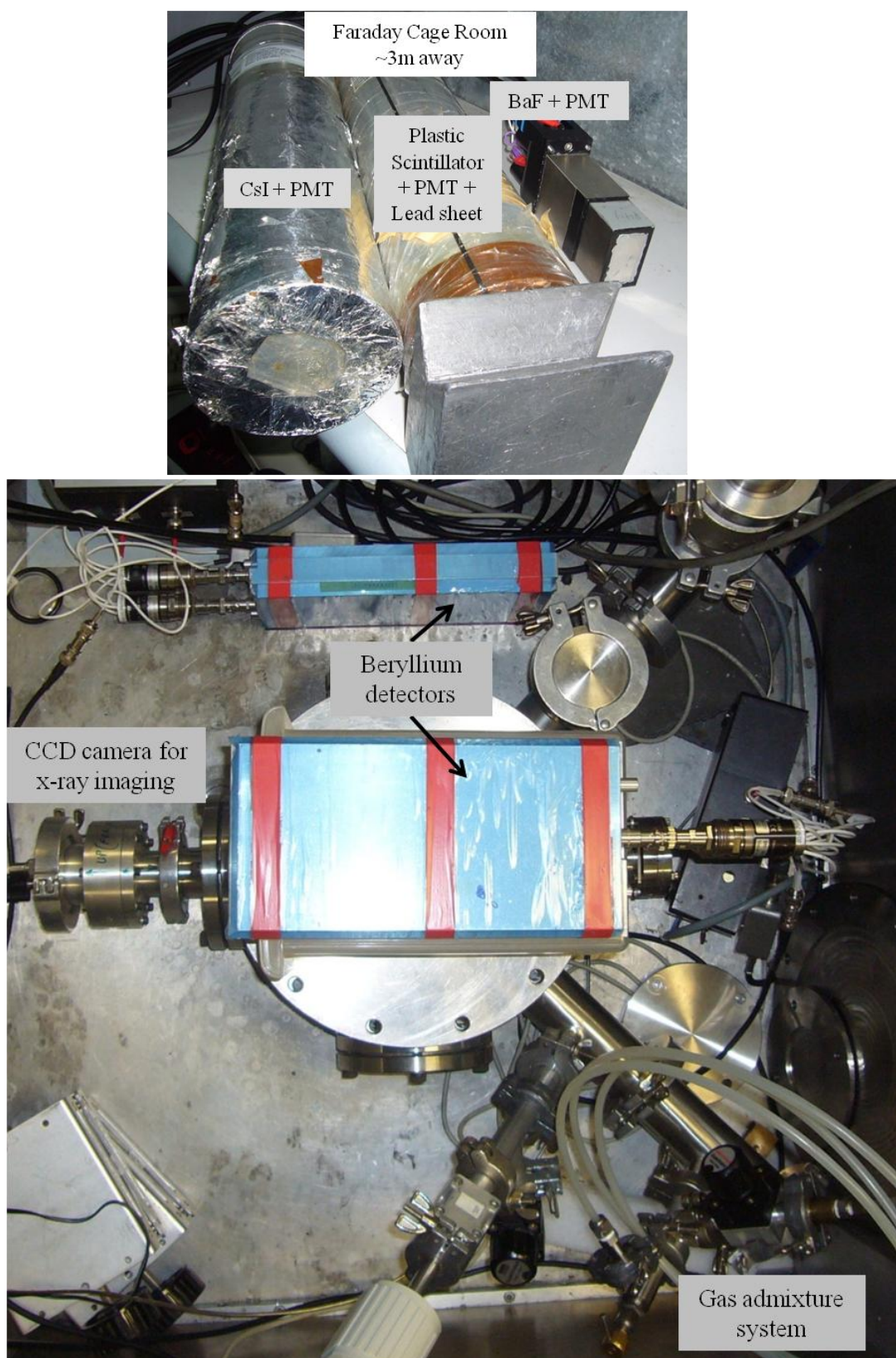


Figure 97: The experimental setup on NX2: Be detector, pinhole x-ray imaging and admixture gas valves.

5.3.2 Experimental Results of operation in pure deuterium

Before doing the main experiment with imaging system, two series of shots were done to investigate the performance of other diagnostics such as x-ray imaging, beryllium activation neutron detector, plastic Scintillator, cesium-iodide and barium-fluoride with different operating gas, pure deuterium and deuterium-krypton (D_2 -Kr) admixture gas.

For operation with pure deuterium as working gas, 50 shots were fired at 13 mbar and 11 kV (1.6 kJ). The proper filter for x-ray imaging is 20 μm beryllium foil which cuts off all visible light and x-ray with energy less than 700 eV. Figure 98 shows the transmission of filter versus x-ray energy from CXRO website, x-ray interaction with matter [120].

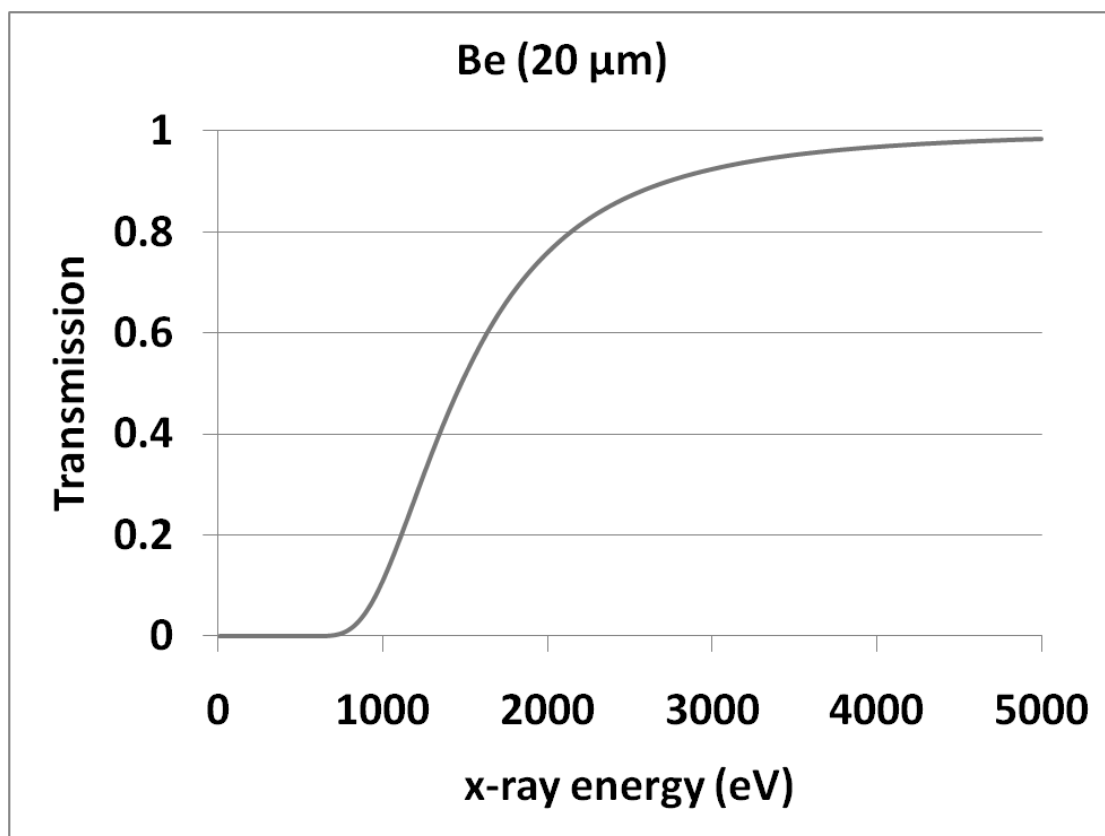


Figure 98: Beryllium filter transmission for x-ray starts from ~ 700 eV

The recorded x-ray images show the time-integrated distribution of bright x-ray region in deuterium working gas. While the left image (Figure 99) shows a strong pinch at middle of anode, the right image does not have any particular shape. Also they show clearly the x-ray of copper anode (the Cu K-edge is ~ 9 keV) which is because of anode bombardment by high energy electron.

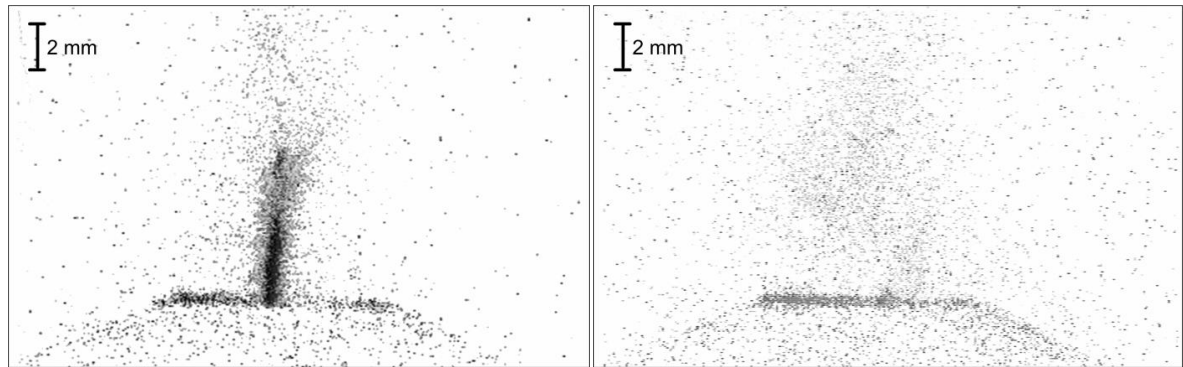


Figure 99: x-ray images in pure deuterium. While the left image shows a strong pinch at middle of anode, the right image does not have any particular shape.

The current derivative signal and also time resolved neutron yield and hard x-ray signals were recorded by oscilloscope which is shown in Figure 100.

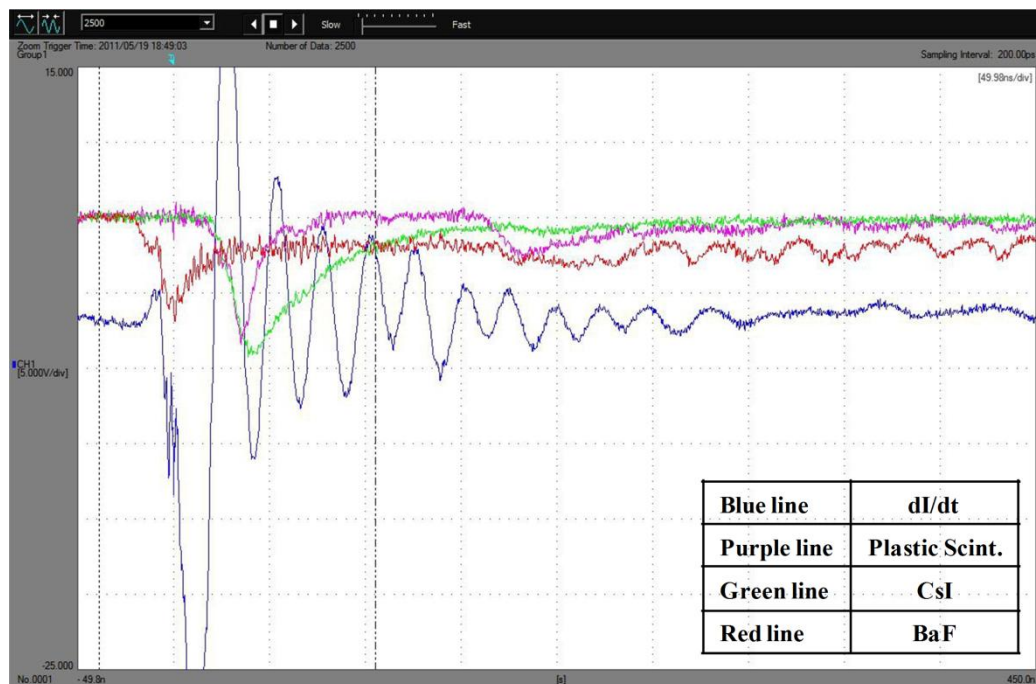


Figure 100: recorded signals: current derivative, plastic Scintillator (neutron and hard x-ray), CsI and BaF₂ (hard x-ray) signals.

The triggering time is the current derivative signal falling down and the time per division is 50 ns. The barium fluoride (BaF_2) has a very fast response time (~ 0.6 ns) which is a good indication of x-ray emission start time, but because of its second slow reaction response time (630 ns) does not have a good tail, rather than, cesium iodide (CsI) shows a better signal for time resolved hard x-ray signal. So integration of CsI signal in a time period of 200 ns, gives us a relative number to compare the strength of hard x-ray in different shots. Moreover, the time difference between the first bump of plastic Scintillator signal (belong to hard x-ray) and second bump (belong to neutron yield) is related to average neutron energy. An approximately time difference of 130 ns is consequence of ~ 2.5 MeV neutron energy.

The scatter plot of anisotropy versus neutron yield (from beryllium detector) shows a relative trend (Figure 101). According to our simulation and experimental result in section 3.6.4, there is a positive correlation between anisotropy and deuterons average energy. The cross section of D-D reaction will increase by increasing of deuteron energy which increases the neutron yield naturally.

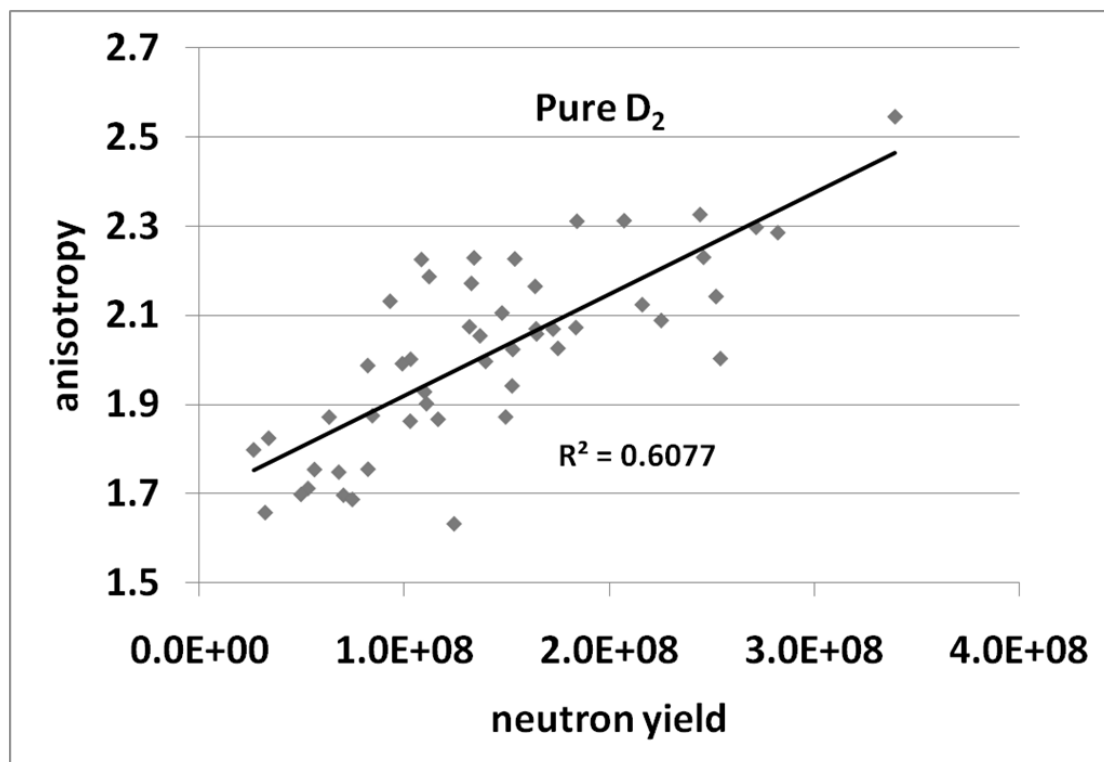


Figure 101: anisotropy versus neutron yield with pure deuterium gas at 13 mbar

The plasma focus routinely produces hard x-rays indicating the presence of bremsstrahlung radiation from high-energy electrons colliding with nuclei [121]. There is the same mechanism for acceleration of electrons and deuterons in pinch region. So as long as, there is a correlation between deuteron average energy and neutron yield (and anisotropy), it has to be the same correlation between hard x-ray and neutron yield (and anisotropy). Figure 102 and Figure 103 show these trends in scatter plots of hard x-ray (from CsI) versus neutron yield and anisotropy.

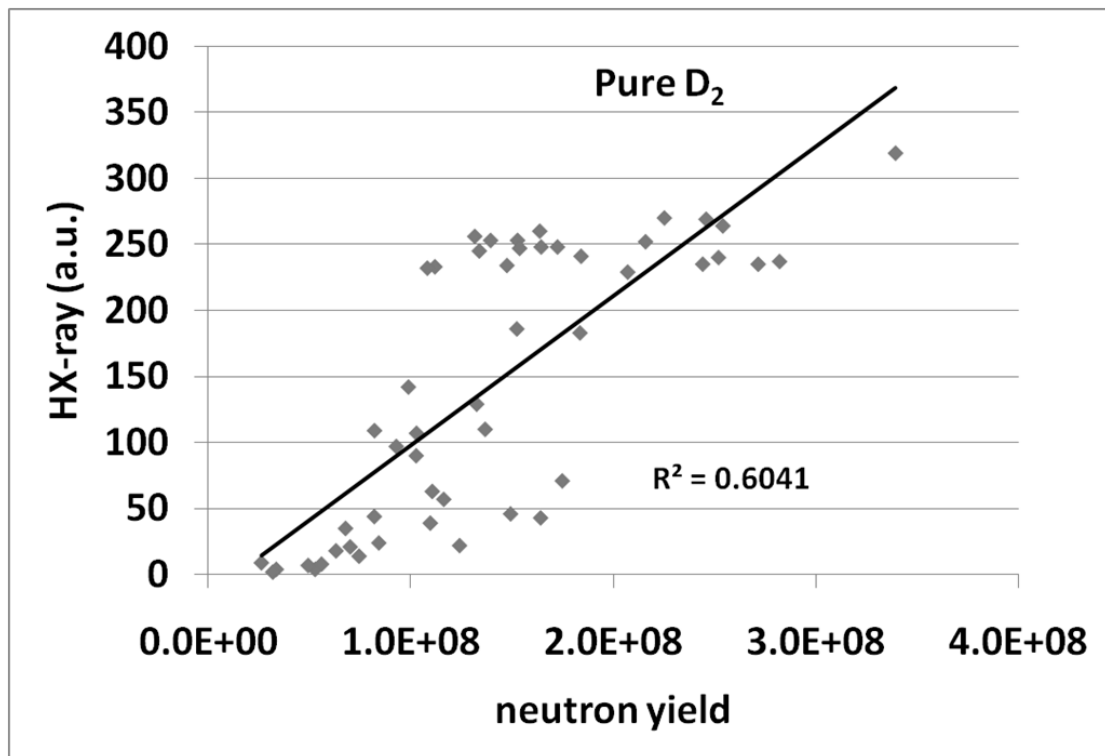


Figure 102: hard x-ray (integral of CsI signal) versus neutron yield with pure deuterium gas at 13 mbar

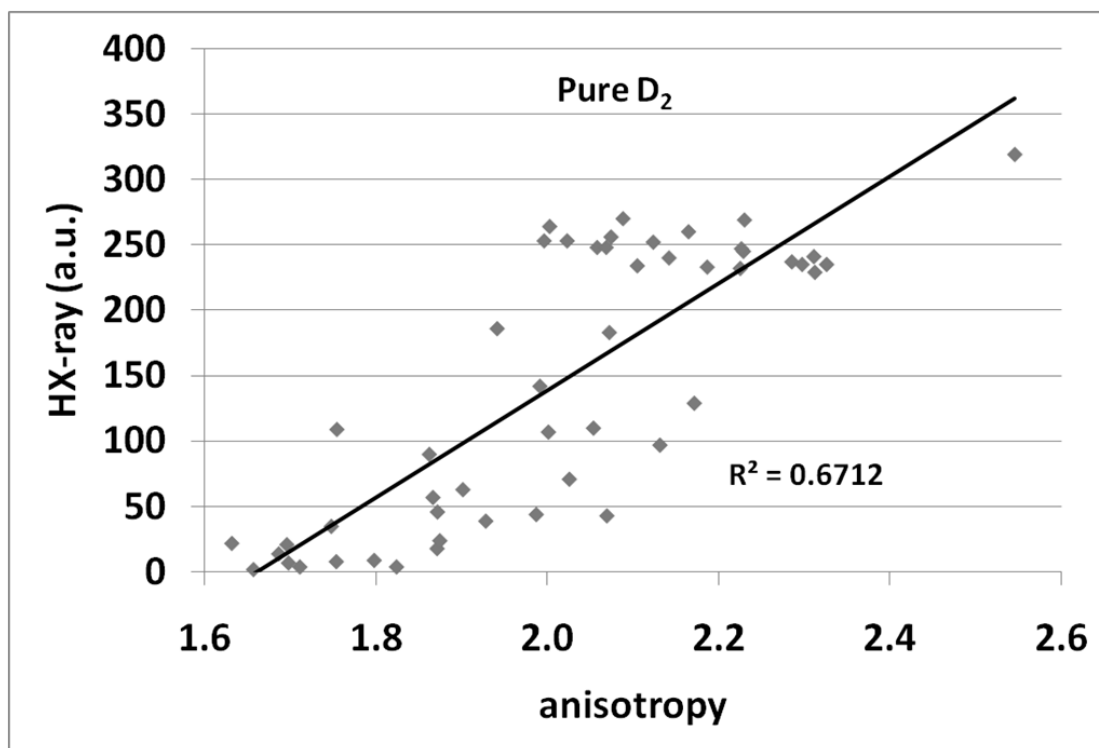


Figure 103: hard x-ray (integral of CsI signal) versus neutron anisotropy with pure deuterium as working gas at 13 mbar

For these 50 shots, the average neutron yield, anisotropy and hard x-ray were 1.4×10^8 , 2.0 and 144 (a.u.) respectively.

Although the correlation between the neutron yield, anisotropy and hard x-ray is not very strong, but it is interesting that the highest neutron yield (3.4×10^8) in shot number “Run3-Shot9” has the highest anisotropy (2.55) and highest hard x-ray radiation (319) as well. The recorded signal has been shown in Figure 104 which blue line represents a very strong dip in current derivative signal. The soft x-ray image (Figure 105) also shows a quite dense pinch on the anode top center and distributed everywhere as well.

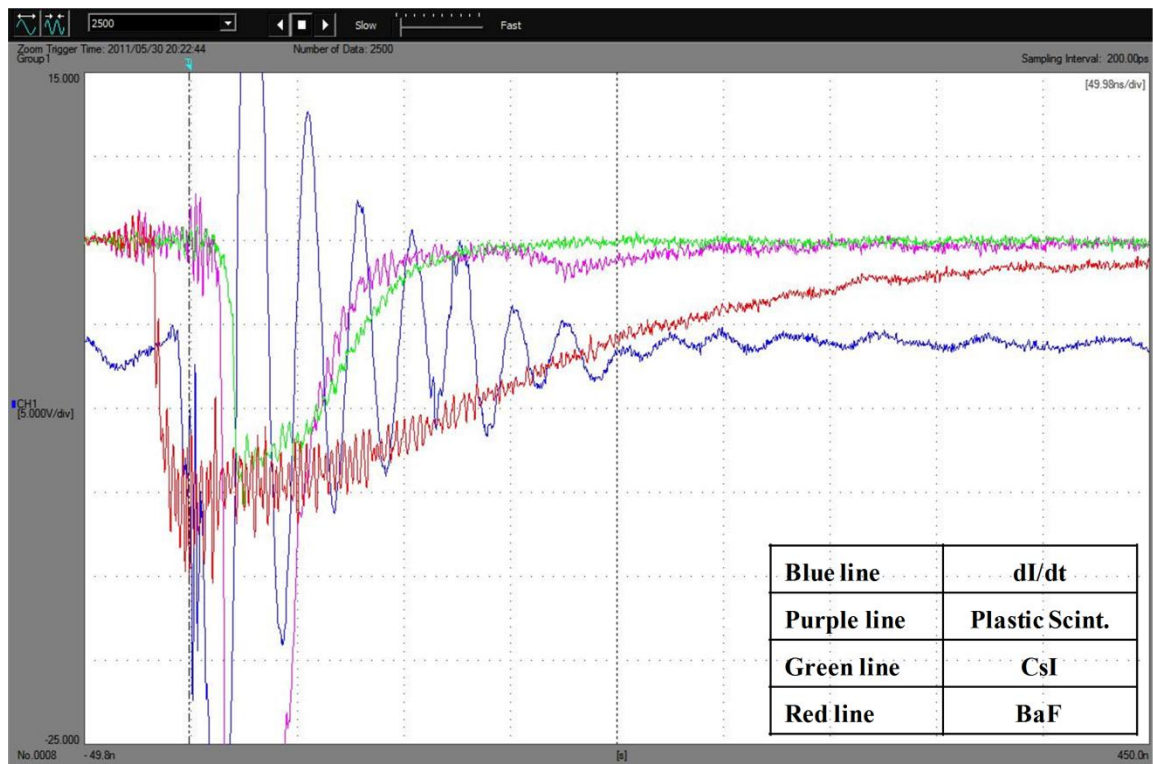


Figure 104: recorded signals: current derivative, plastic Scintillator, CsI and BaF₂ signals for the best shot.

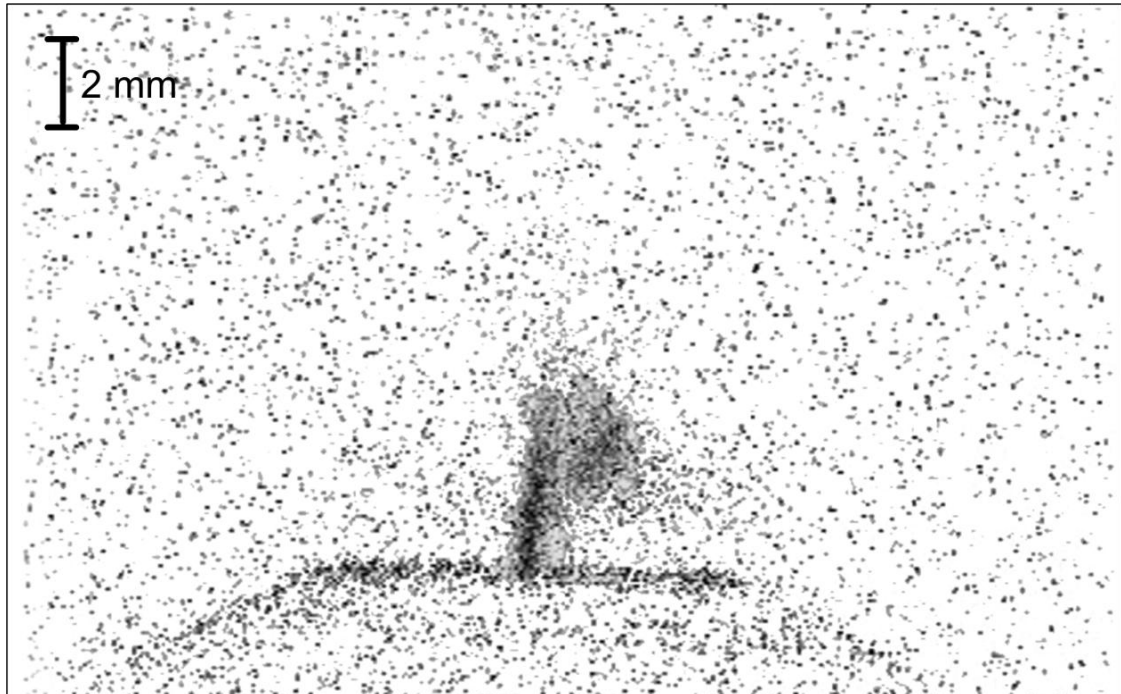


Figure 105: x-ray image in pure deuterium for the best shot shows a dense and strong pinch.

5.3.3 Experimental Results of operation in admixture gas

For operation with deuterium-krypton (D_2 -Kr) admixture gas (with Kr volumetric ratio of 2%), 50 shots were fired at 13 mbar and 11 kV (1.6 kJ). With the same filter (20 μm beryllium foil), the x-ray image was saturated because of high x-ray emission rate of krypton seeding pinch (Figure 106-left). Then we increased the beryllium sheath thickness to 50 μm and although the saturation became less, but still the pinch area was not clear (Figure 106-right). So finally, the suitable filter for krypton seeding was chosen as a combination of 50 μm beryllium and 12.5 μm titanium. Titanium plays the main role in this filter because of its high-Z atomic number and represents a transmitted x-ray energy range of bigger than ~ 1 keV with a huge reduction at 2.5 keV (Figure 107). A typical x-ray image of admixture gas discharge has been shown in Figure 108 which clearly proved the existence of hot spots in pinch region.

Just to comparing with previous experiment, the same filter was applied for x-ray image while the pure deuterium was used as working gas. As it can be seen in Figure 109 there is no hot spot formation or even not strong x-ray in range of energy more than 1 keV. It means that almost all x-ray emission in pure deuterium, shown in images is in the range of 0.7 keV-1 keV.

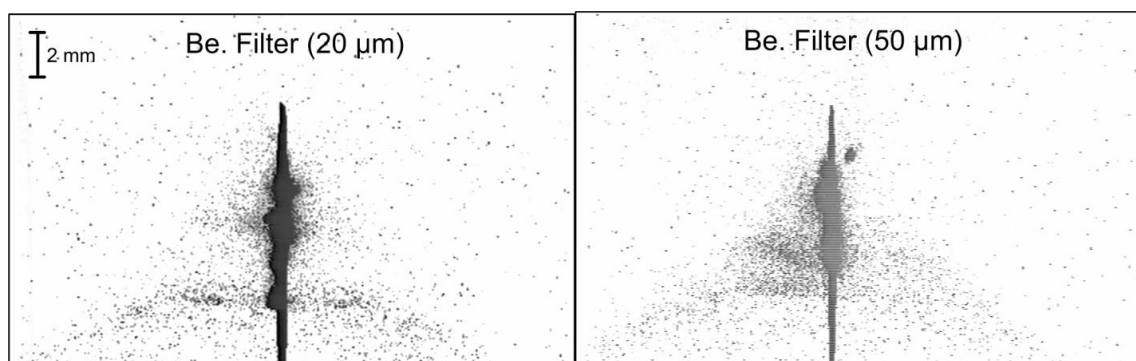


Figure 106: saturation of x-ray image in krypton seeding gas admixture with beryllium filter thickness: 20 μm (left) and 50 μm (right)

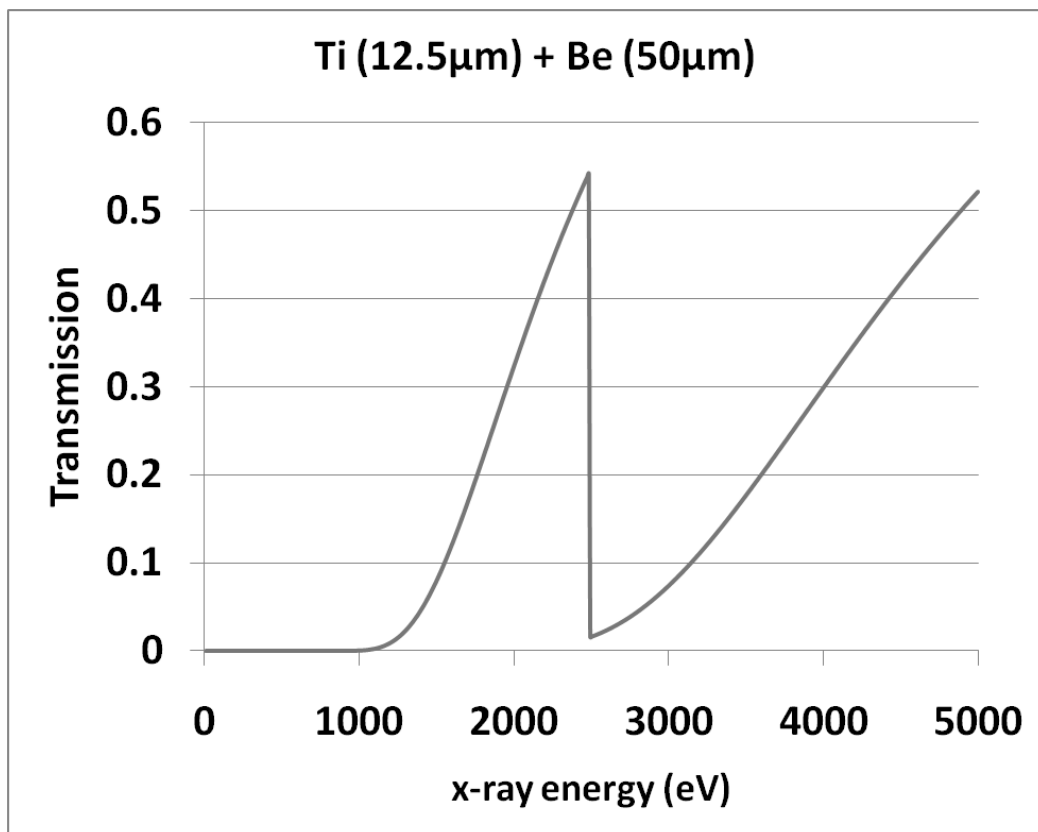


Figure 107: x-ray transmission of combination filter: titanium (12.5 μm) and beryllium (50 μm) for admixture gas

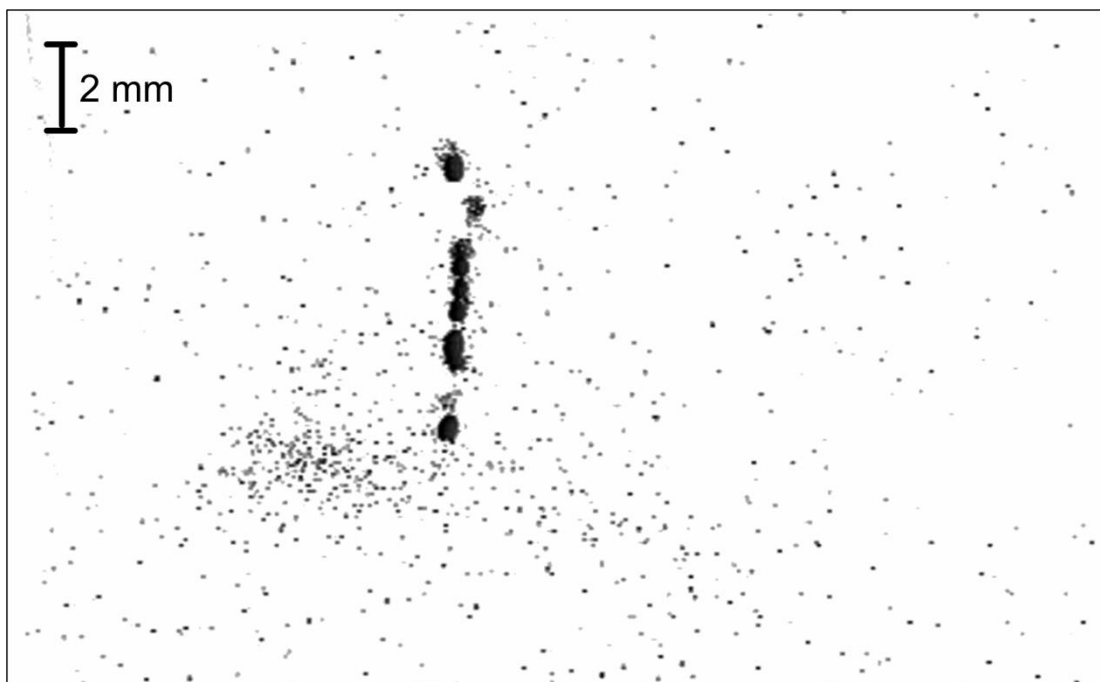


Figure 108: x-ray images in $D_2+2\%$ Kr admixture gas with combination filter: titanium (12.5 μm) and beryllium (50 μm).

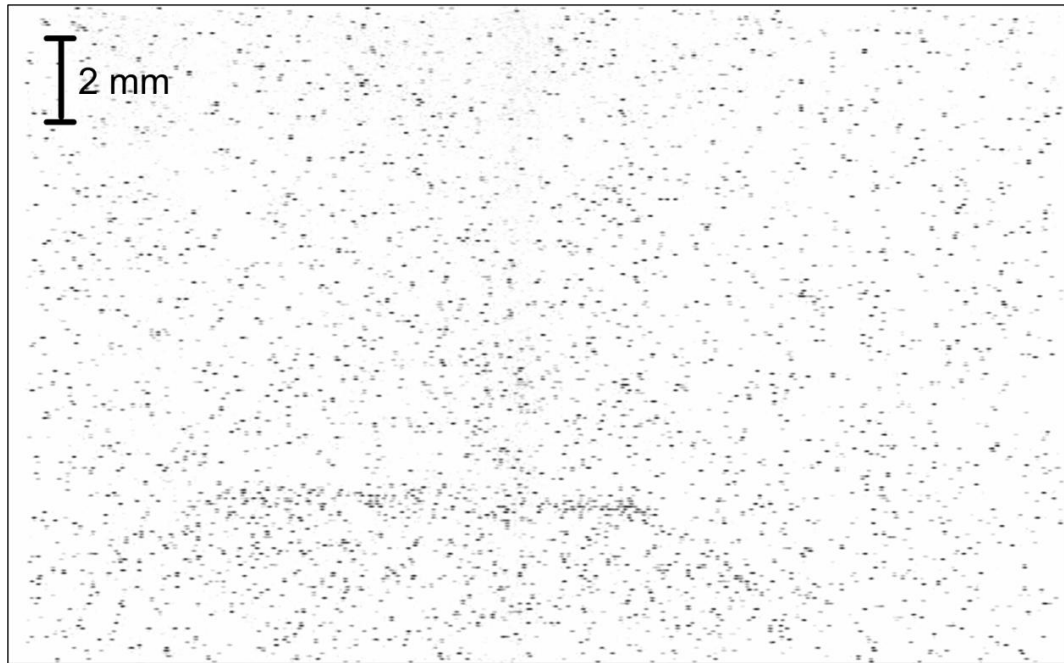


Figure 109: x-ray images in pure deuterium gas with combination filter: titanium (12.5 μm) and beryllium (50 μm).

The scatter plot of anisotropy versus neutron yield (Figure 110) shows that for krypton seeding admixture gas, there is less strong correlation between neutron yield and anisotropy in comparison to pure deuterium. Much the same, the same poor correlation can be seen in the plots of hard x-ray versus neutron yield (Figure 112) and hard x-ray versus anisotropy (Figure 111).

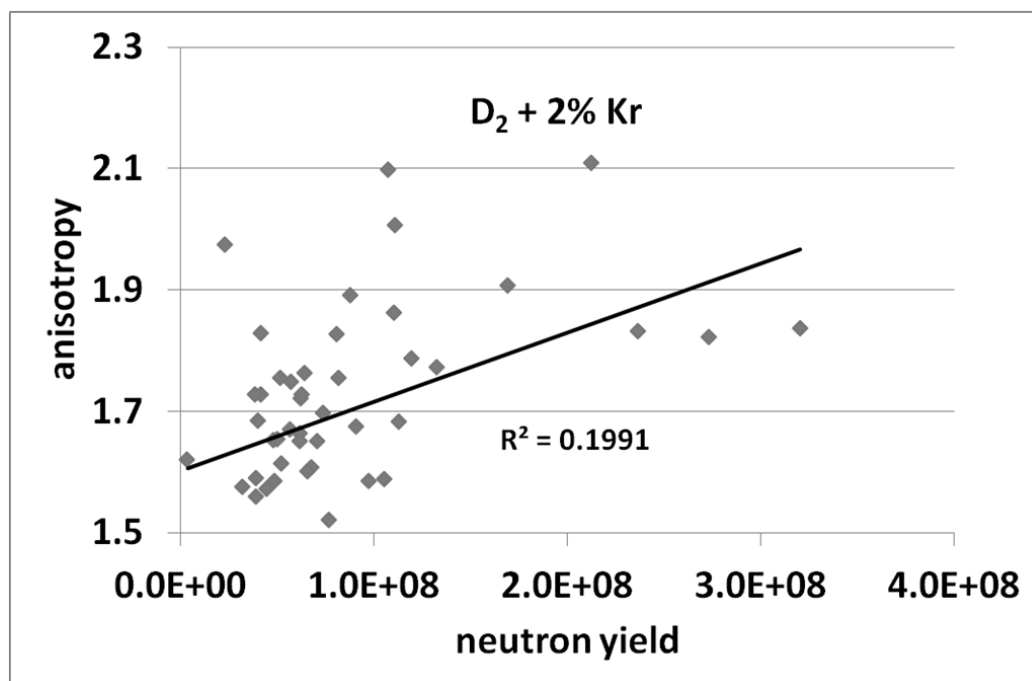


Figure 110: anisotropy versus neutron yield with admixture gas

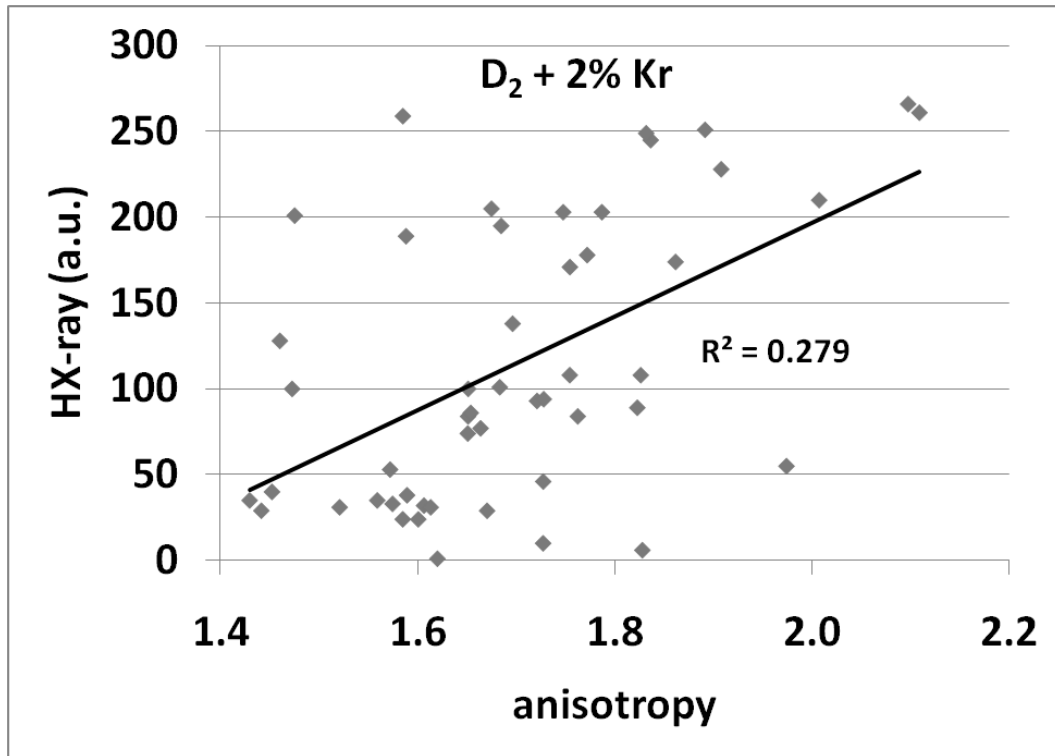


Figure 111: hard x-ray (integral of CsI signal) versus neutron anisotropy with admixture gas

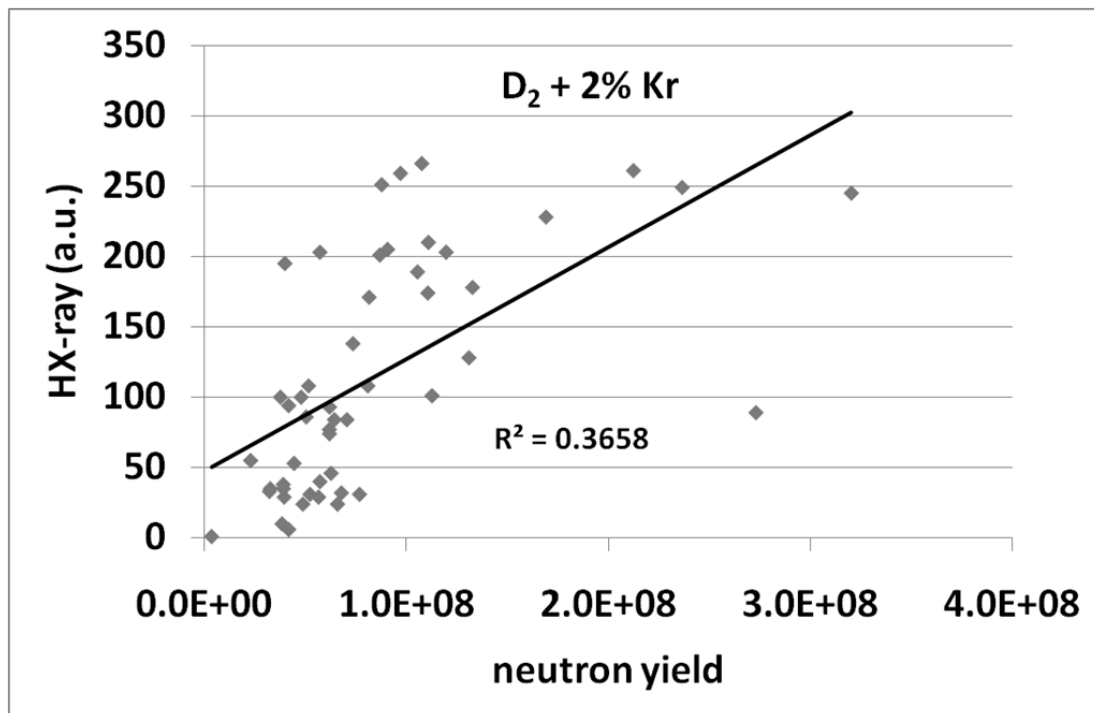


Figure 112: hard x-ray (integral of CsI signal) vs. neutron yield with admixture gas

For these 50 shots firing with deuterium-krypton admixture gas, the average neutron yield, anisotropy and hard x-ray were 0.84×10^8 , 1.7 and 114 (a.u.) respectively.

Reducing the anisotropy is predictable and matched with the experimental results from other devices. The high-Z and high mass doped gas (in our case, krypton) move slightly behind the deuterons and so in final compression region, the inner side of pinch is occupied by deuterons which are surrounded by krypton. This causes more confinement of plasma pinch and less chance for deuterons to accelerate and increasing their energy. The lesser deuteron energy is correlated to the lesser neutron anisotropy as it has been discussed in section 3.5.2.

The same explanation can be carried out for decreasing the hard x-ray emission in admixture gas discharge. The same mechanism for acceleration can be applied for electron and deuteron. So with the same reason, electrons cannot reach such high energy as they have in pure deuterium regime and on its consequence, less bremsstrahlung radiation and so less hard x-ray emission.

It is very interesting that in contrary of the almost no correlation between the neutron yield, anisotropy and hard x-ray in admixture gas discharge, however the same as pure deuterium, one the highest neutron yield (2.9×10^8) in shot number “*Run1-Shot2*” has the highest anisotropy (2.22) and highest hard x-ray radiation (400) as well. The recorded signal has been shown in Figure 114 which blue line represents a very strong dip in current derivative signal. The soft x-ray image (Figure 113) also shows a combination of multi hot spot on anode top center and distributed everywhere as well.

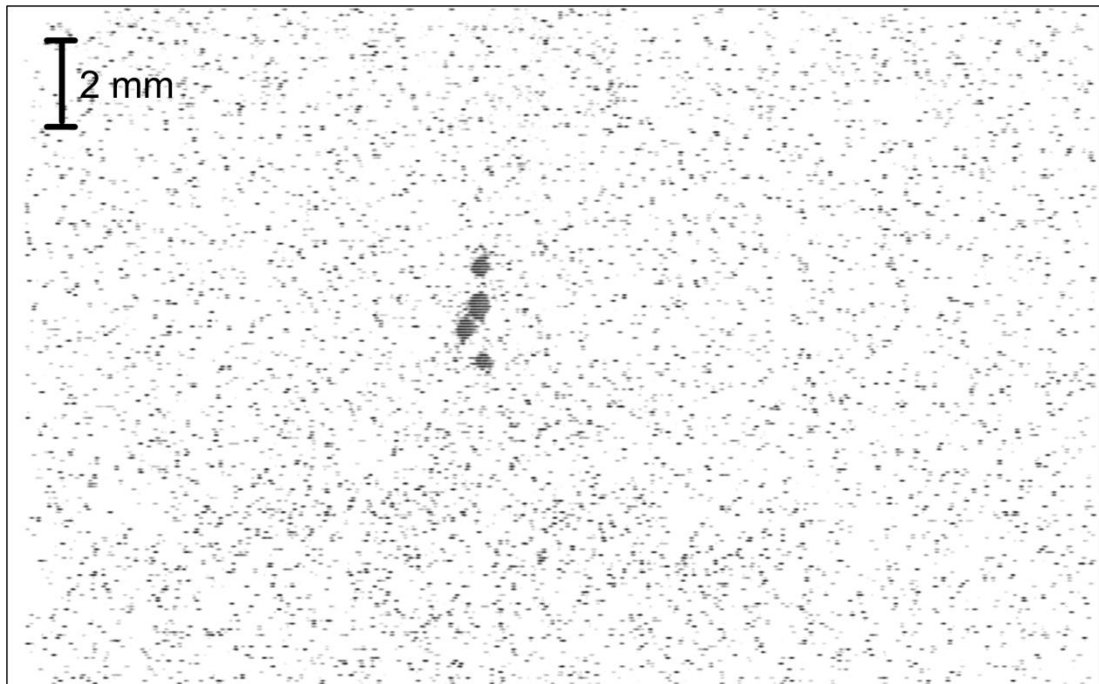


Figure 113: x-ray image in pure deuterium for one of the best shots shows a multi-hot spot pinch.

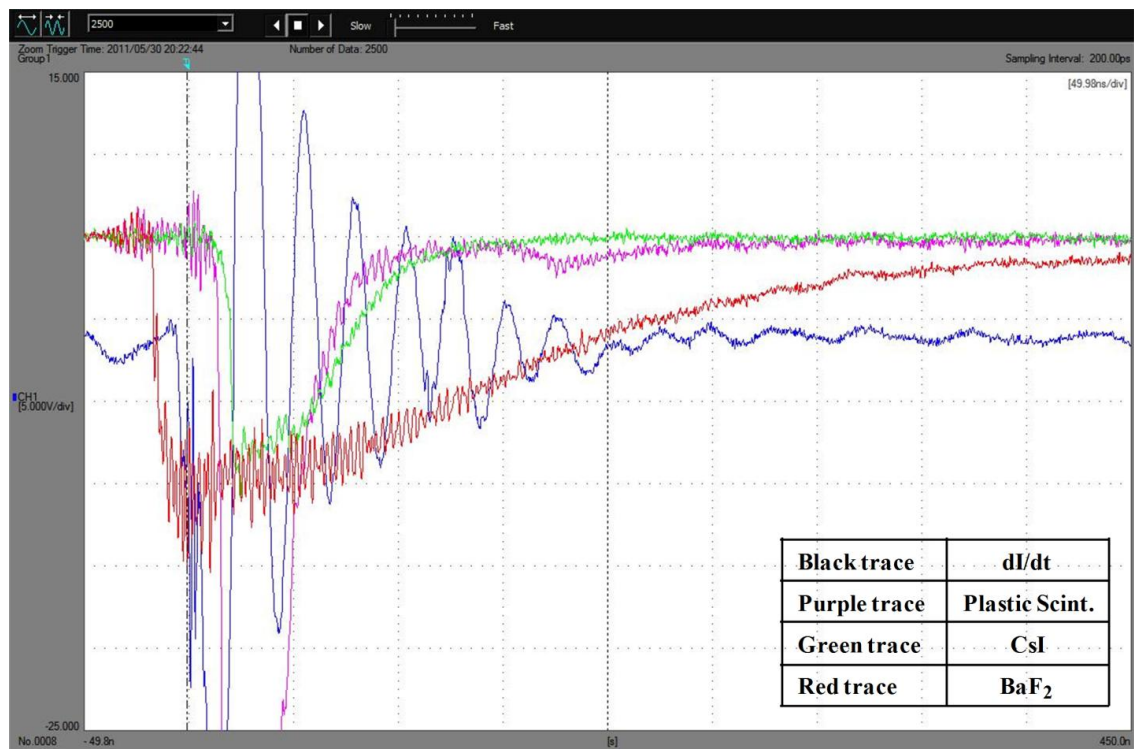


Figure 114: recorded signals: current derivative, plastic scintillator, CsI and BaF₂ signals for one of the best shot.

Imaging the fusion source:

For fusion source imaging, the NX2 plasma focus device was operated in its optimized regime (13 mbar, 11 kV/1.6 kJ) for highest neutron emission. For pure deuterium gas operation, 17 separate shots were performed. Both masks (mask-1365 and mask-4680) were used simultaneously, so as to compare the reconstructed images obtained for each. There was, however, a 60° difference between their viewing angles (around the z-axis). For deuterium-krypton admixture gas operation ($D_2+2\%$ volumetric krypton), 7 shots were fired using only mask-1365. The signal to noise ratio (SNR) for each image is reported from reconstruction program. To avoid squandering the detector, both sides were used which have been identified as side “A” and “B”. It should be mentioned that some of the exposed detectors were not considered for some reasons such as loss of corresponding x-ray image (because of video recorder malfunction), missing oscilloscope signals (no triggering because of weak current signal dip) and simply because it was a very weak shot (low neutron and proton yield).

In the following discussion, the final reconstructed image(s) along with associated x-ray image and other diagnostic signals will be elaborated. The oscilloscope was triggered by pinch current (actually dI/dt) and covered an overall time of 500 ns (starting at almost compression time) which shows the details of signal more clearly. Four signals can be seen in different colours: blue colour (channel 1) represents the discharge derivative current signal, pink colour (channel 2) is the output from plastic scintillator/photomultiplier for hard x-ray and neutron detection, green colour (channel 3) is CsI hard x-ray detector signal and red colour (channel 4) is BaF_2 fast hard x-ray detector signal. For hard x-ray yield, the integral of CsI trace over 200 ns is reported. As usual, two beryllium activation neutron detectors were used for neutron yield and anisotropy measurements. As a simple classification, a neutron yield less than 0.5×10^8 will be called a “weak shot”, and in this order, $Y_n = (0.5-1) \times 10^8$ a moderate shot, $Y_n = (1-2) \times 10^8$ a good shot and Y_n more than 2×10^8 will be considered as a very good shot.

It should be remembered that for x-ray imaging, two different filters were used: for pure deuterium operation, a 20 μm beryllium (for x-ray energy range above 700 eV) and for admixture gas operation, the combination of 50 μm beryllium and 12.5 μm titanium (for x-ray energy range above 1 keV).

5.3.4 Image registration

As a general definition, Image registration is the process of transforming different sets of data into one coordinate system. Data may be multiple photographs, data from different sensors, from different times, or from different viewpoints [122]. Registration is necessary in order to be able to compare or integrate the data obtained from different measurements. Image registration or image alignment involves spatially transforming the target image to align with the reference image.

In our system, the fusion source (or a part of that) may be shifted down by proton trajectory in magnetic field of current sheath (see Figure 92). However, this magnetic field is not well known, as firstly, we do not know the amount of current passing through the pinch or sheath and secondly, the shape of current sheath must be defined by shadowgraphy for individual shot. Moreover protons carry a range of kinetic energy which causes a variety of gyration curve.

So the image registration helps us to find a relative fusion source position in plasma focus device. It should be mentioned that the anode (insert anode) tip was changed after firing hundreds shots (see Figure 115 and compare with Figure 88).

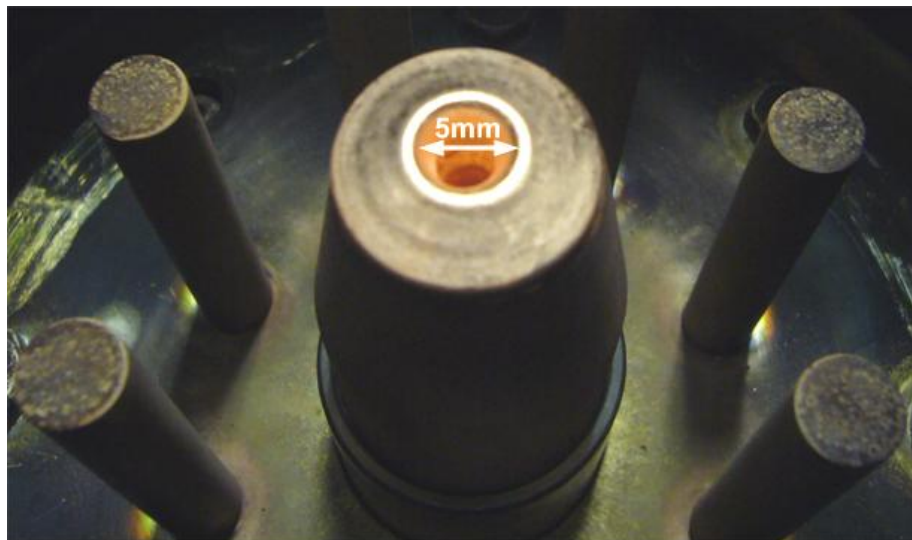


Figure 115: anode tip has been changed after 100s shots

Figure 116 shows one of the extended fusion source images from shot number 1365-I23-A (mask-1365) which is result of deconvolution algorithm. The final image includes 91×90 pixels and needs to be zoomed-in, registered and calibrated by the real size of source. The final image is illustrated in the left down corner.

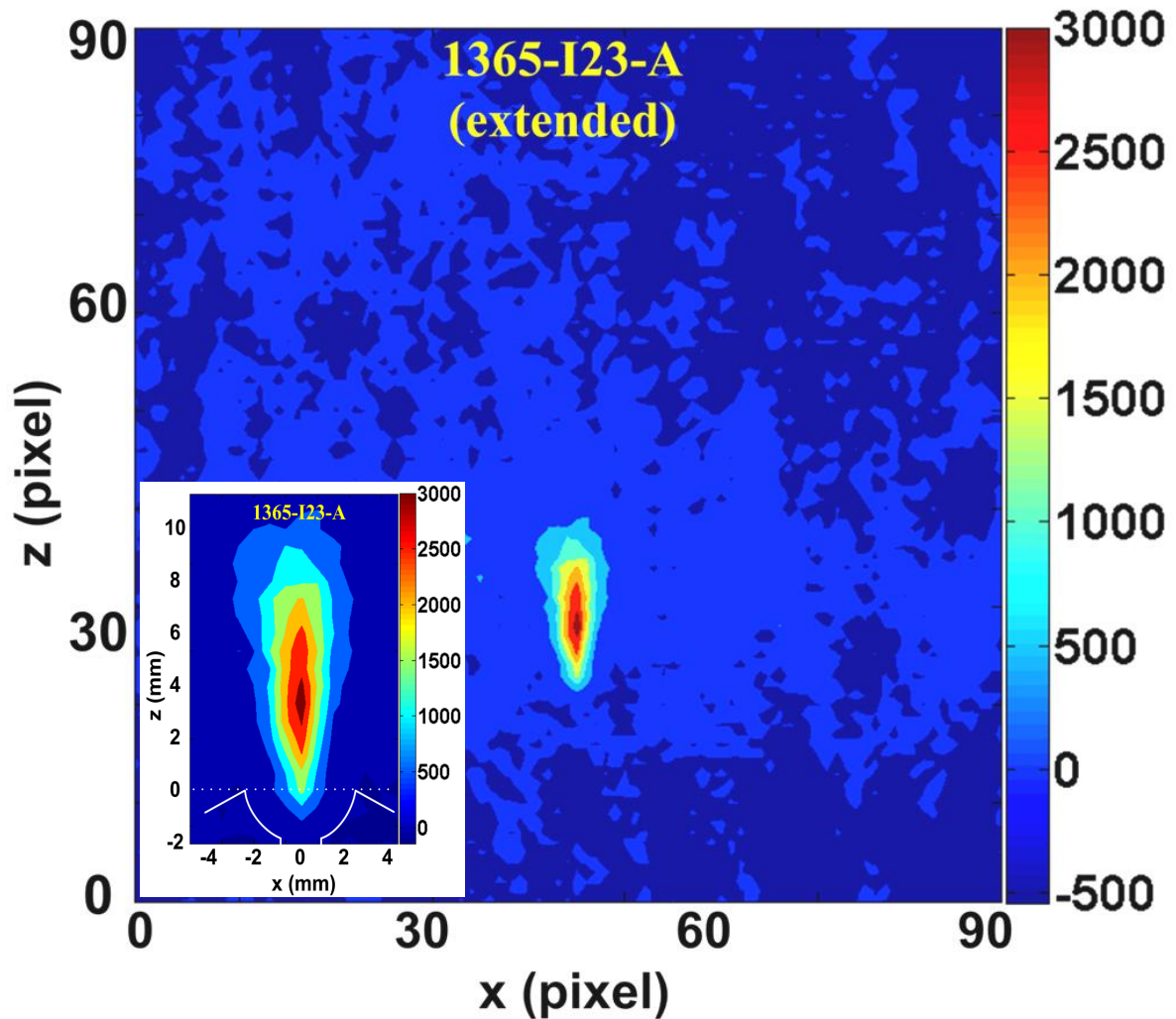


Figure 116: The reconstructed image of plasma focus fusion source from mask-1365 (91×90 pixels). The actual fusion source position must be zoomed-in, calibrated and registered by referring to anode tip place (left down image).

To achieve this, an Americium-241 α source (Figure 118 left) was used which is directly obtained from plutonium upon absorption of one neutron. It decays by emission of a 5.486 MeV α -particle to Np-237 with the half-life of 432.2 years. With activity of 5 μCi , the neutron emission rate is 1.85×10^5 per second, so for 30 min exposure, the total yield will be $\sim 3.3 \times 10^8$ neutron which is the same neutron yield of a good plasma focus shot.

The camera was set up with mask-1365 and CR-39 detector (of course without Kapton filter) in as same geometry as the real shot and α source was put on top of anode. After exposure, the detector was etched for 6 hours. Figure 117 shows these mono-energetic α tracks.

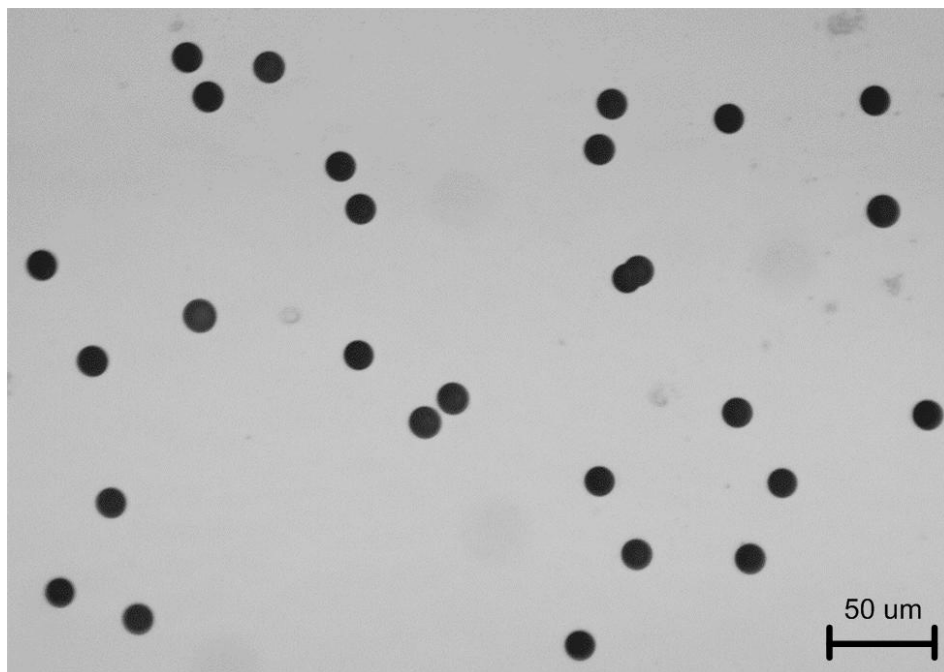


Figure 117: 5.5 MeV α tracks on the detector after 6 hours etching

The same procedure (metal coating, scanning and analyzing) was done to obtain the reconstructed image. Figure 118 right shows the detail of source and those sections which are covered by metal mesh. Regarding to the shape of anode tip, the center of source must be at position of $z \approx 6.8$ mm.

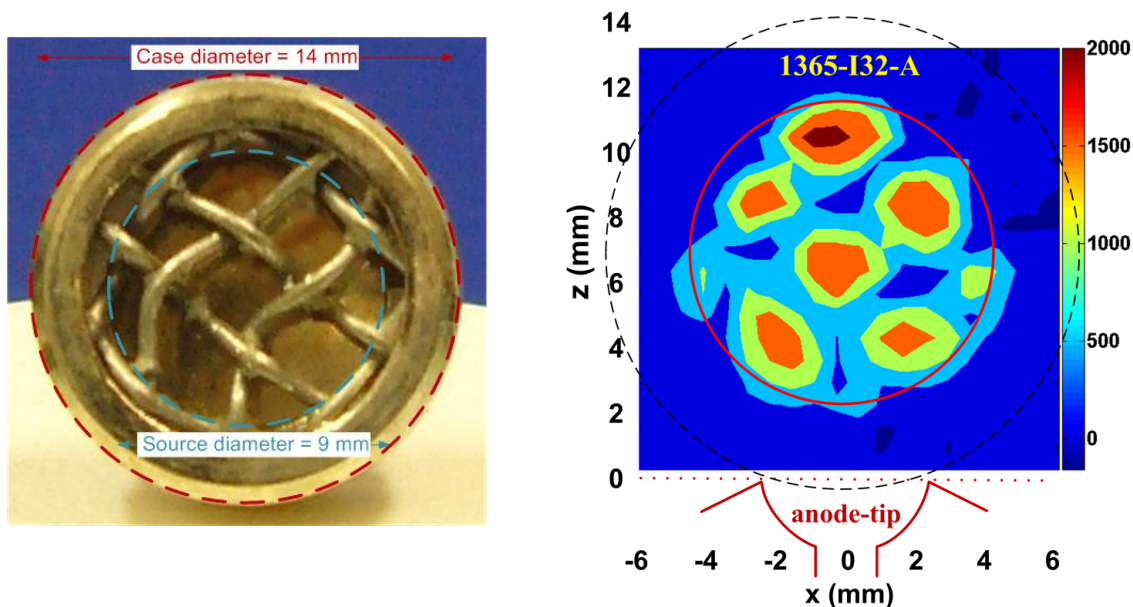


Figure 118: Am-241 α source (left) with 9 mm actual diameter and 14 mm case diameter. It is covered partially by a set of metal mesh. The reconstructed image (right) shows clearly the detail of the source and mesh.

Furthermore, as we have the adjusted z-position (the tracks' center of mass) in final log file, the above identified point can be applied for all individual reconstructed images and find the relative anode tip position in them.

To test and confirm our image registration method, another detector was exposed with a good plasma focus shot (1365-I33-A) with the neutron yield of 2.3×10^8 and then exposed with α particle for 15 min (1.66×10^8 total α particle). The position of the camera and detector were kept the same and the Kapton filter was removed for alpha exposure. The detector was then etched for 12 hours. Two different types of tracks can be distinguished very easily as seen in Figure 119. The large tracks are due to 5.5 MeV alpha particles and the small tracks are due to the ~ 2.5 MeV D-D fusion protons after they have passed through 75 μm Kapton filter.

The final reconstructed image also can be seen in Figure 120 which shows the position of each source in final image. It must be mentioned that to achieve the α source and proton source position, the tracks were split by regarding to their different area and brightness boundary. Then the de-convolution algorithm was applied for them separately.

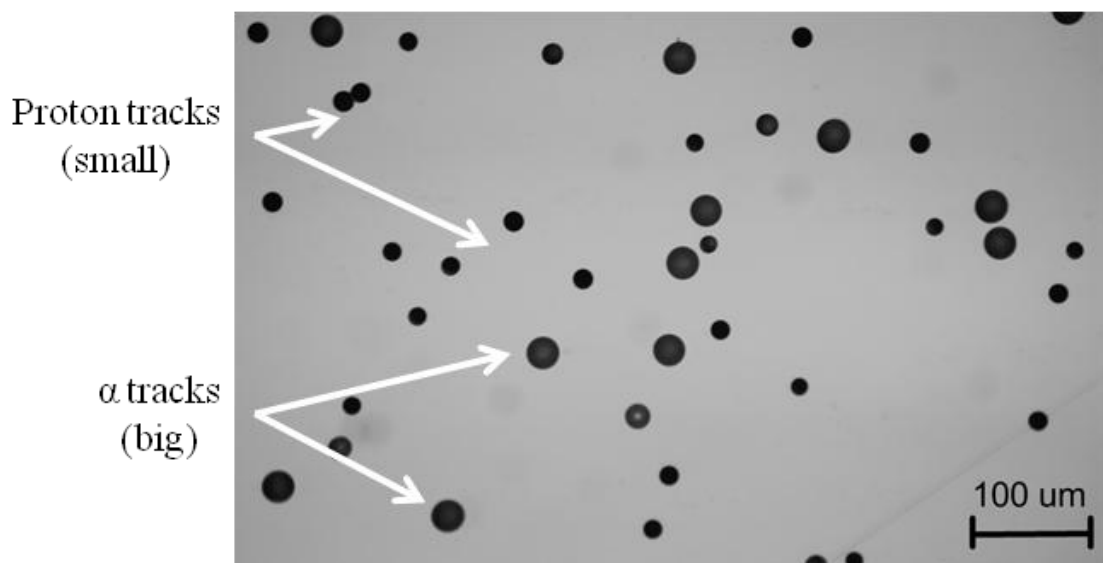


Figure 119: α tracks (upper size) and the D-D reaction proton tracks (lower size) on the detector (after 12 hours etching) can be distinguished very clearly.

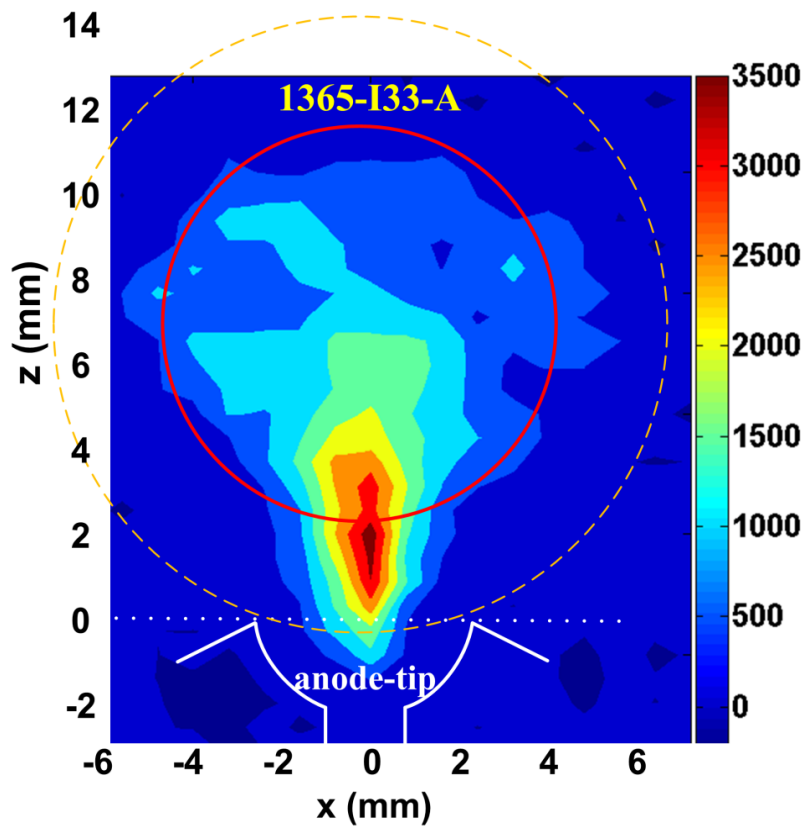


Figure 120: Reconstructed image of mixed α and proton tracks

5.3.5 Fusion imaging with pure deuterium gas

The following experiments were performed with pure deuterium as the working gas at 13 mbar and both masks were involved. The images and data are presented in chronological order.

It has to be emphasized again that the proton trajectory corrections have not applied on the reconstructed fusion source images.

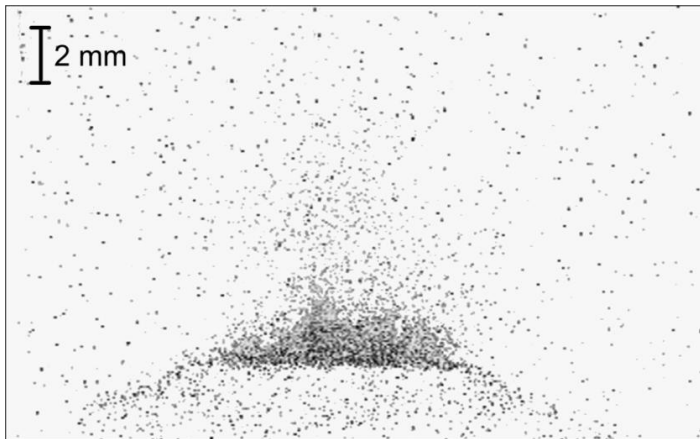
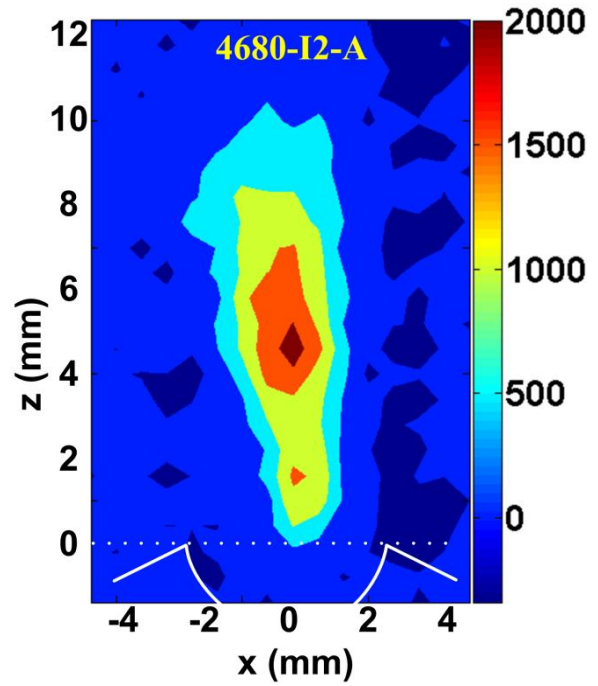
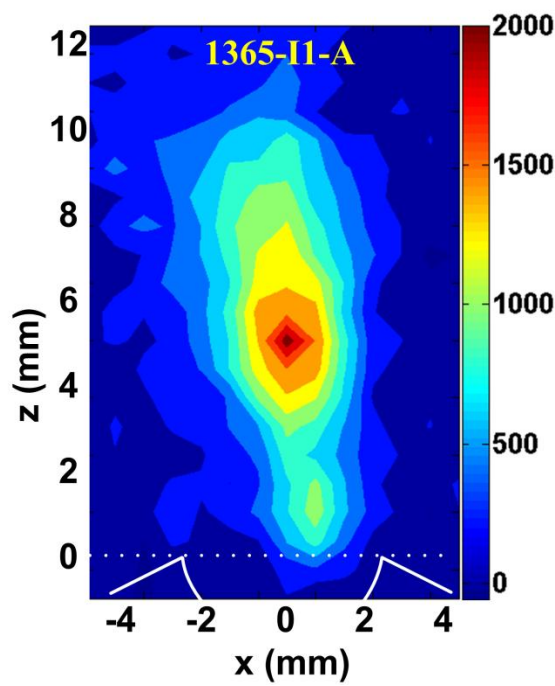
The two detectors 1365-I1-A (using the mask-1365 and detector I1, side A) and 4680-I2-A (using the mask-4680 and detector I2, side A) were exposed with a good shot ($Y_n = 1.89 \times 10^8$). Both images show an egg-shaped fusion source of ~ 3 mm width and ~ 10 mm length, tipping slightly away from the vertical. Both of them show the source more concentrated at the center (around $z = 5$ mm) and a small dense area just above anode tip (around $z = 1$ mm). Although because of different sight of view, the image 1365-I1-A looks wider which can be related to non-evenly deuterons acceleration in all direction. A slightly better SNR of mask-1365 agrees with our discussion in the previous section about open area fraction and spaced mask design. The total proton tracks for 1365-I1-A and 4680-I2-A are 102,267 and 149,880 respectively.

The x-ray image shows an intense soft x-ray emission region distributed just on top of anode and with less density above it. There is not a strong pinch column in this x-ray image which is in agreement with a wide and moderate dip current derivative signal (Full-Width-Half-Maximum (FWHM) ≈ 25 ns). Also in this image and all other x-ray images, the anode tip can be recognized by copper x-ray emission region.

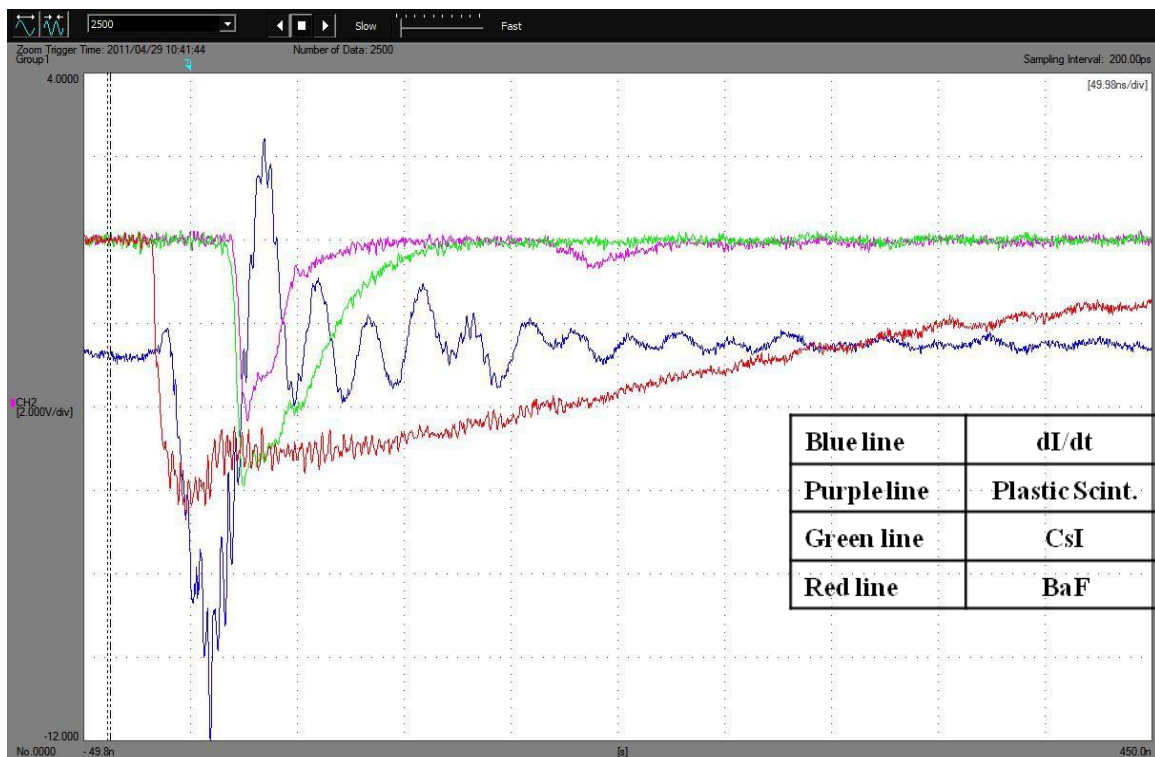
The time difference of ~ 140 ns between two peaks in the plastic scintillator signal shows neutrons time of flight with an average energy 2.45 MeV.

For this shot, there was a moderate CsI signal with integral of 241 nVs and a moderate value of neutron anisotropy (2.11).

Gas: pure D₂



Neutron Yield	1.89×10^8
Anisotropy	2.11
HX-ray Integral (nVs)	241
SNR (1365)	19.5
SNR (4680)	17.2



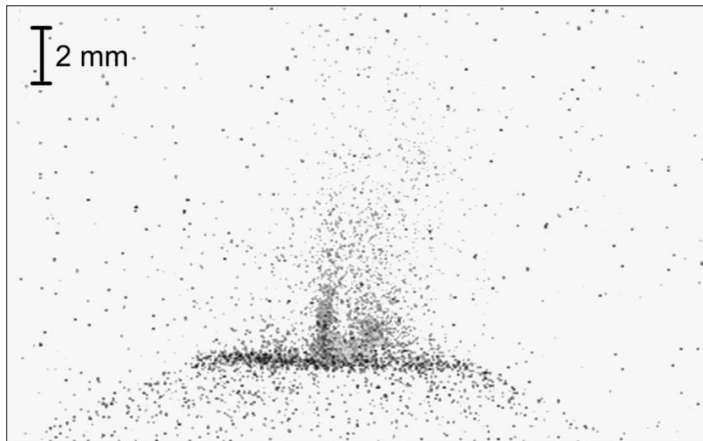
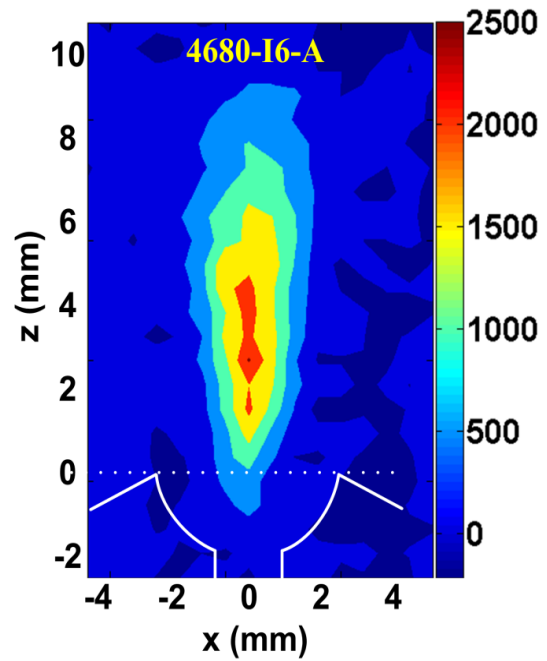
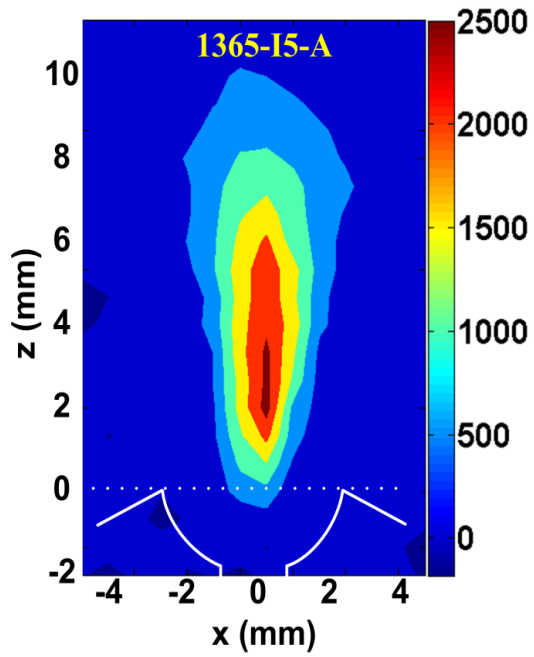
Blue line	dI/dt
Purple line	Plastic Scint.
Green line	CsI
Red line	BaF

The two detectors 1365-I5-A and 4680-I6-A were exposed with a good shot ($Y_n = 1.75 \times 10^8$). Both images show an elongated/cigar-shaped fusion source with ~ 3 mm width and ~ 8 mm length which is more concentrated at an oval shape (~ 1 mm width and ~ 4 mm length) at the center of $z = 3$ mm. The image 1365-I5-A looks somewhat wider. The SNR of mask-1365 is the better of the two. The recognized proton tracks for 1365-I5-A and 4680-I6-A are 118,471 and 154,469 respectively.

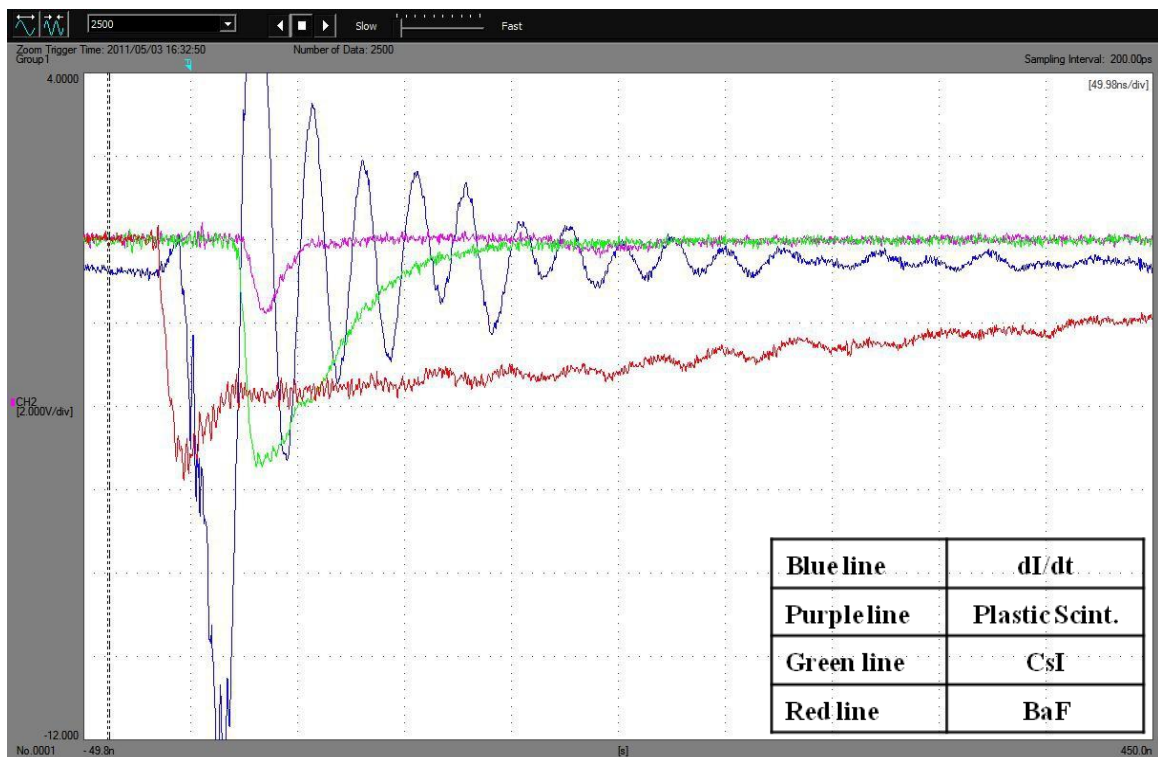
The x-ray image shows a clear dense pinch region above the center of anode (~ 4 mm length) with an extra pinch with less density just beside it. A narrow pinch shape usually indicates to a well and evenly formed fusion source.

It can be seen also that there is a narrow ($\text{FWHM} \approx 15$ ns) and strong dip in the current derivative signal. For this shot, there was a moderate CsI signal with integral of 260 nVs and a moderate value of neutron anisotropy (2.17).

Gas: pure D₂



Neutron Yield	1.75×10^8
Anisotropy	2.17
HX-ray Integral (nVs)	260
SNR (1365)	23.2
SNR (4680)	19.4

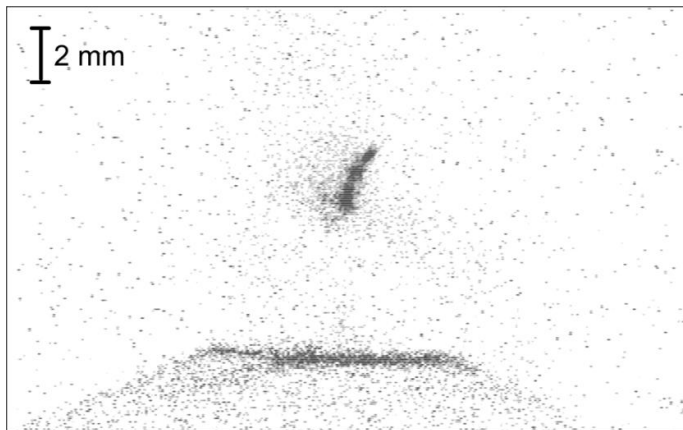
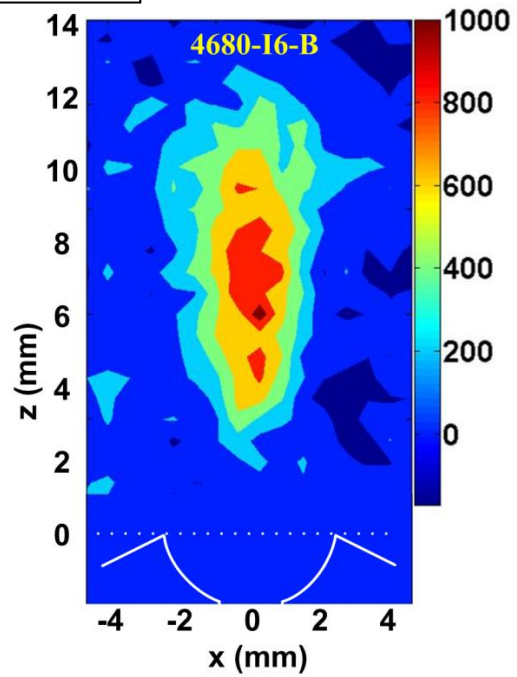
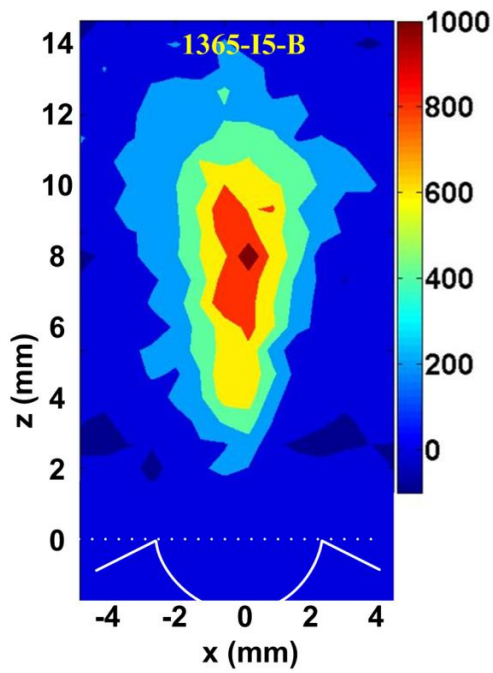


The two detectors 1365-I5-B and 4680-I6-B were exposed with a good shot ($Y_n = 1.01 \times 10^8$). Both images show an egg-shaped fusion source with ~ 4 mm width and ~ 11 mm length which is more concentrated at the center (~ 1 mm width and ~ 3 mm length). The total image size of 1365-I5-B looks wider and longer than 4680-I6-B. The SNR of mask-1365 is the better of the two, and both of them are less intense than before because of fewer proton tracks and also bigger object size. The total proton tracks for 1365-I5-B and 4680-I6-B are 71,690 and 100,889 respectively.

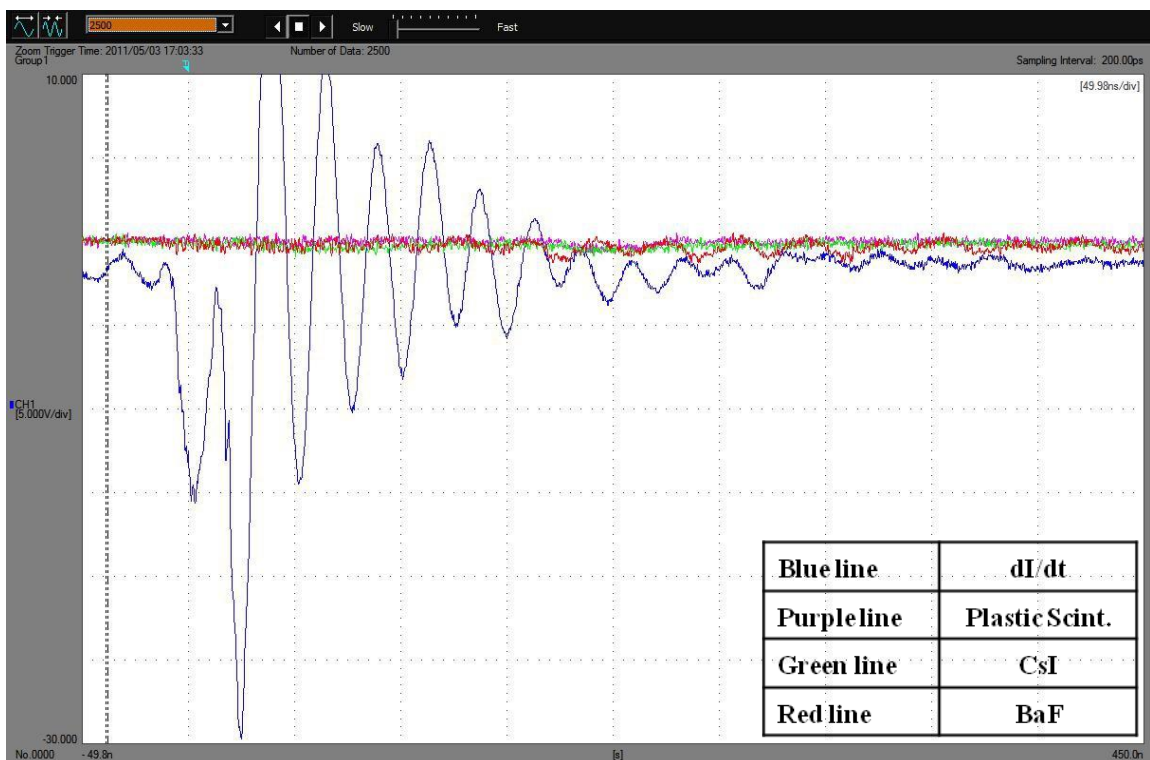
The unexpected x-ray image shows a bent ~ 2 mm length dense pinch region on ~ 5 mm above the anode, almost the same place of densest part of the fusion source. Although the fusion source image is also bent, however it is hard to conclude that there exists a correlation between fusion source and x-ray image.

Except the current derivative signal which shows a double pinch and weak dip with $\text{FWHM} \approx 12$ ns for each pinch, all other signals are almost flat. The integral of CsI hard x-ray detector is very small, only 17 nVs in contrary to a moderate neutron anisotropy 1.88.

Gas: pure D₂



Neutron Yield	1.01×10^8
Anisotropy	1.88
HX-ray Integral (nVs)	17
SNR (1365)	15.6
SNR (4680)	11.5

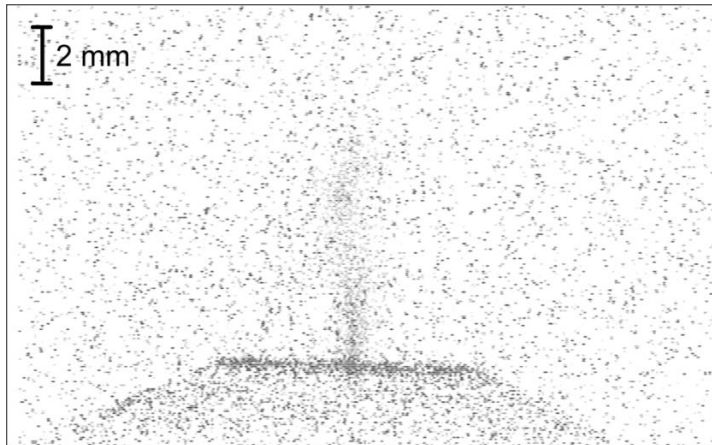
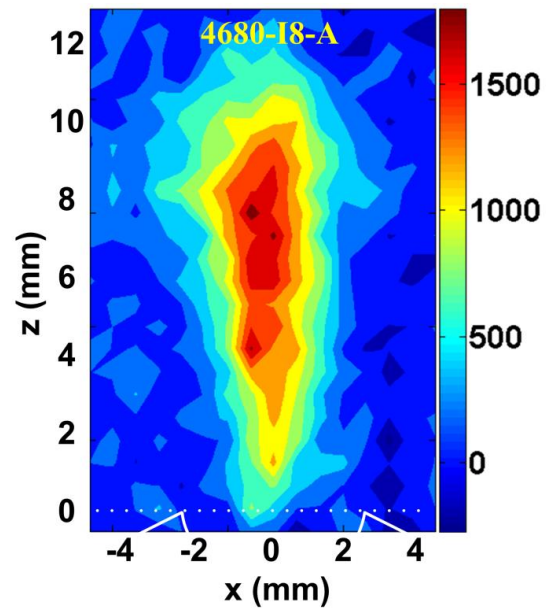
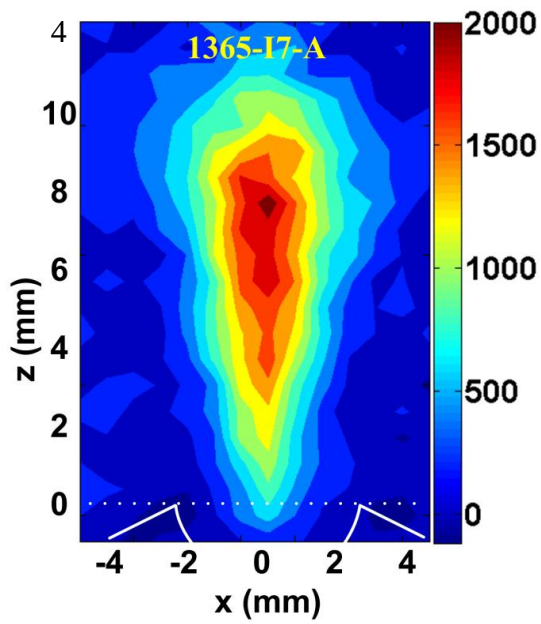


The two detectors 1365-I7-A and 4680-I8-A were exposed with a very good shot ($Y_n = 2.68 \times 10^8$). Again, both images show a cone-shaped fusion source with ~ 4 mm width and ~ 11 mm length which is more concentrated at the center (around $z = 8$ mm above the anode) with ~ 1 mm width and ~ 3 mm length. In opposite of good proton tracks, the SNR of both masks are quite small, because of the extended object size. The total proton tracks for 1365-I7-A and 4680-I8-A are 158,278 and 208,333 respectively.

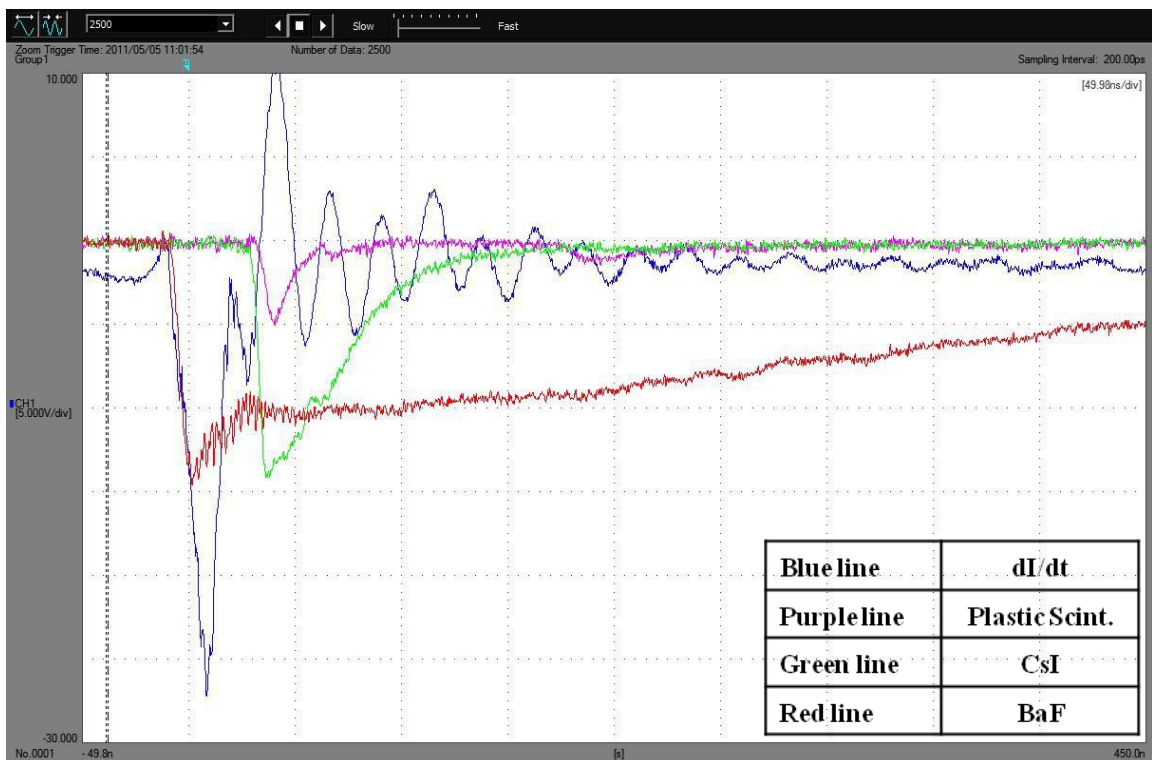
The x-ray image shows a ~ 6 mm length and narrow pinch region on top of the anode and also fairly evenly distributed along the z-axis. The high density and expanded x-ray source indicates the high probability of D-D fusion reaction. That is why such this x-ray distribution is usually associated with a large neutron yield.

Despite of current derivative signal which shows a weak dip pinch with $\text{FWHM} \approx 15$ ns, the hard x-ray signal is strong. The integral of CsI hard x-ray detector is 268 nVs in agreement with moderate neutron anisotropy 2.14.

Gas: pure D₂



Neutron Yield	2.68×10^8
Anisotropy	2.14
HX-ray Integral (nVs)	268
SNR (1365)	15.8
SNR (4680)	12.3

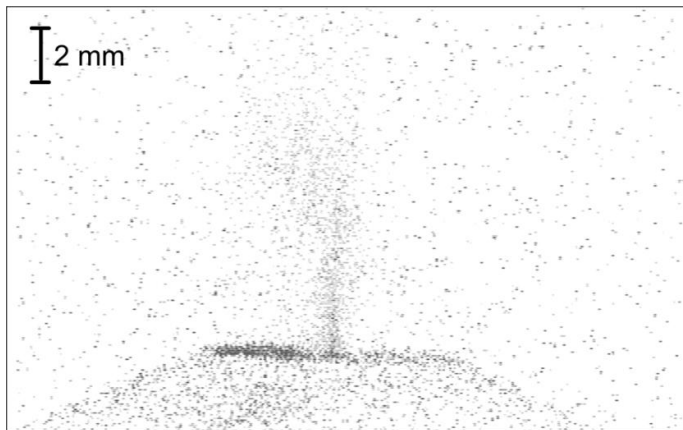
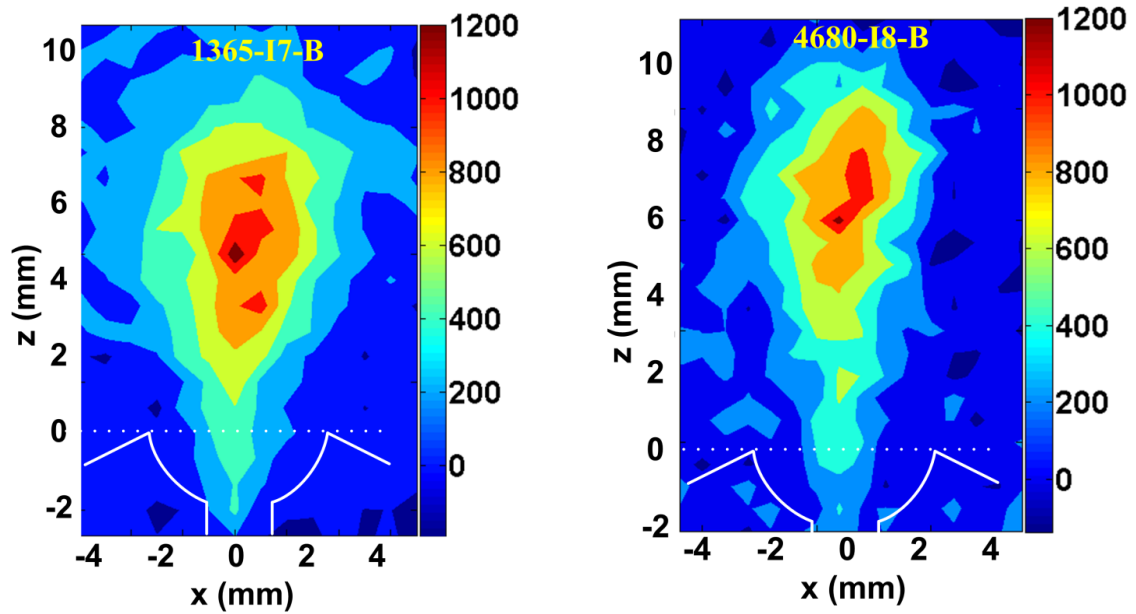


The two detectors 1365-I7-B and 4680-I8-B were exposed with a good shot ($Y_n = 1.35 \times 10^8$). Both images show a cone-shaped fusion source with ~ 5 mm width and ~ 11 mm length which is more concentrated at center around $z = 6$ mm above the anode. The same as previous shot, the SNR of both masks are quite low, because of the extended object size. The image 1365-I7-B looks wider than 4680-I8-B. The total proton tracks for 1365-I7-B and 4680-I8-B are 88,434 and 116,500 respectively.

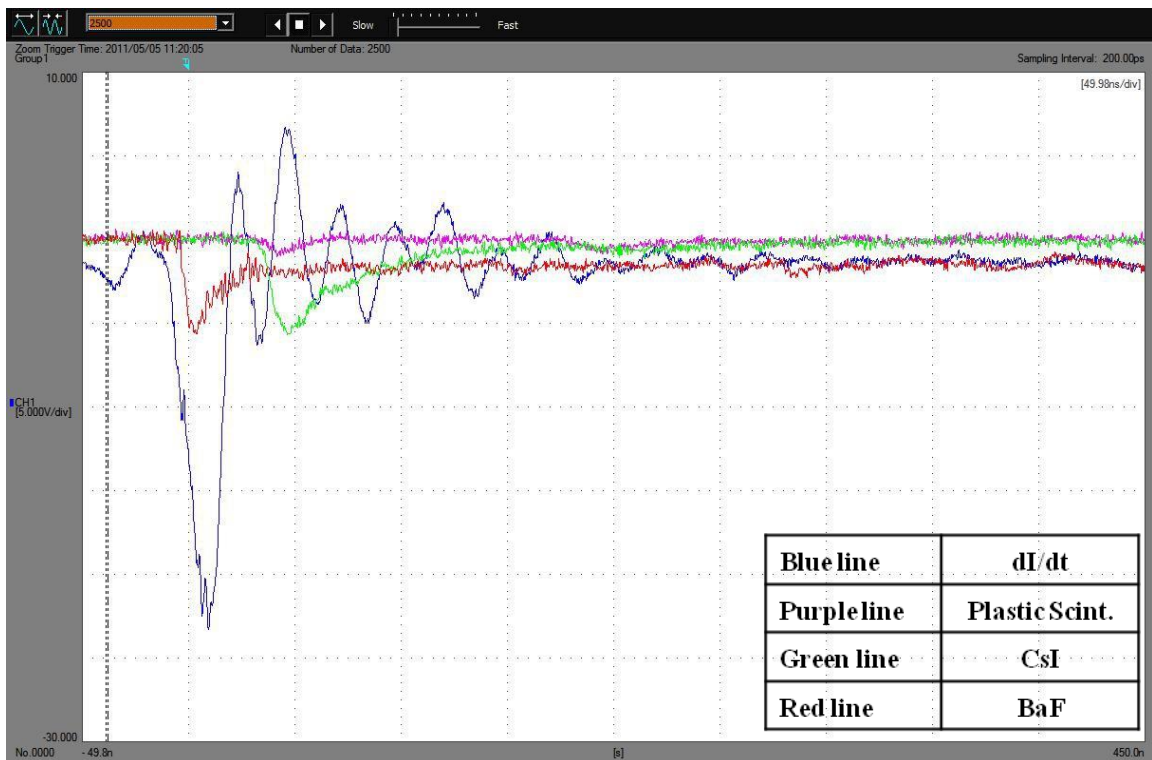
The x-ray image shows a low dense pinch region with ~ 6 mm length, narrow at the top of the anode and fairly evenly distributed along the z-axis. The expansion of hot plasma region can be correlated to the extended fusion source.

There is a relative similarity between this shot and previous shot regarding the fusion source shape and x-ray images. Both of them have very extended fusion source and x-ray images.

Current derivative signal shows a weak dip pinch with $\text{FWHM} \approx 20$ ns and the hard x-ray signal is moderate. The integral of CsI hard x-ray detector is 102 nVs in agreement with moderate neutron anisotropy of 2.04.

Gas: pure D₂

Neutron Yield	1.35×10^8
Anisotropy	2.04
HX-ray Integral (nVs)	102
SNR (1365)	14.0
SNR (4680)	11.2

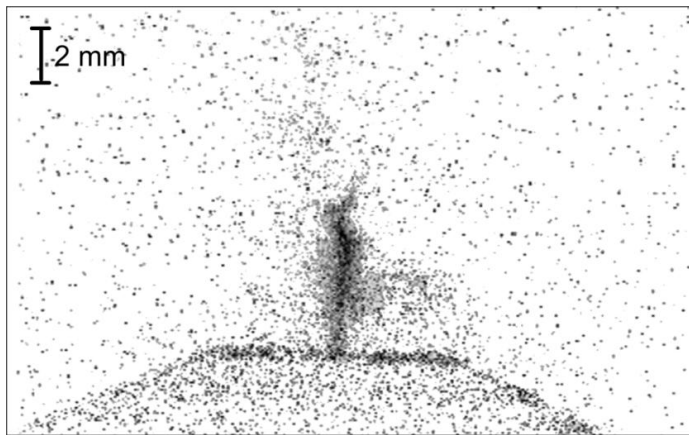
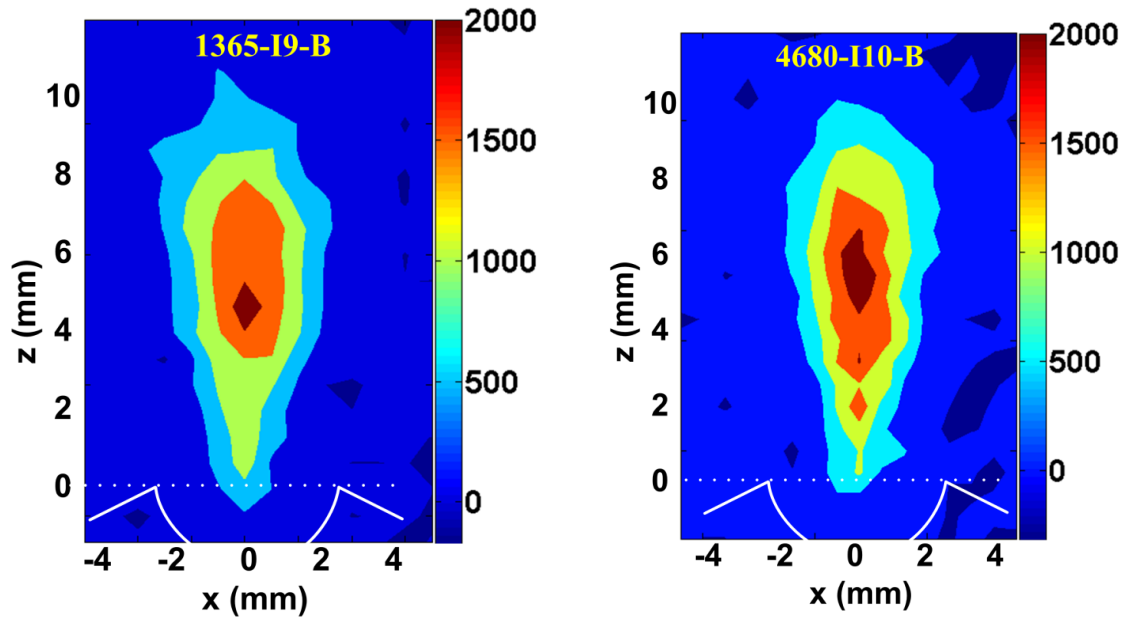


The two detectors 1365-I9-B and 4680-I10-B were exposed with a very good shot ($Y_n = 2.31 \times 10^8$). Both images show an egg-shaped fusion source with ~ 3 mm width and ~ 10 mm length. The high concentration region is in about $z = 5$ mm. The SNR of mask-1365 is, as usual, the better of the two. The total proton tracks for 1365-I9-B and 4680-I10-B are 135,071 and 193,658 respectively.

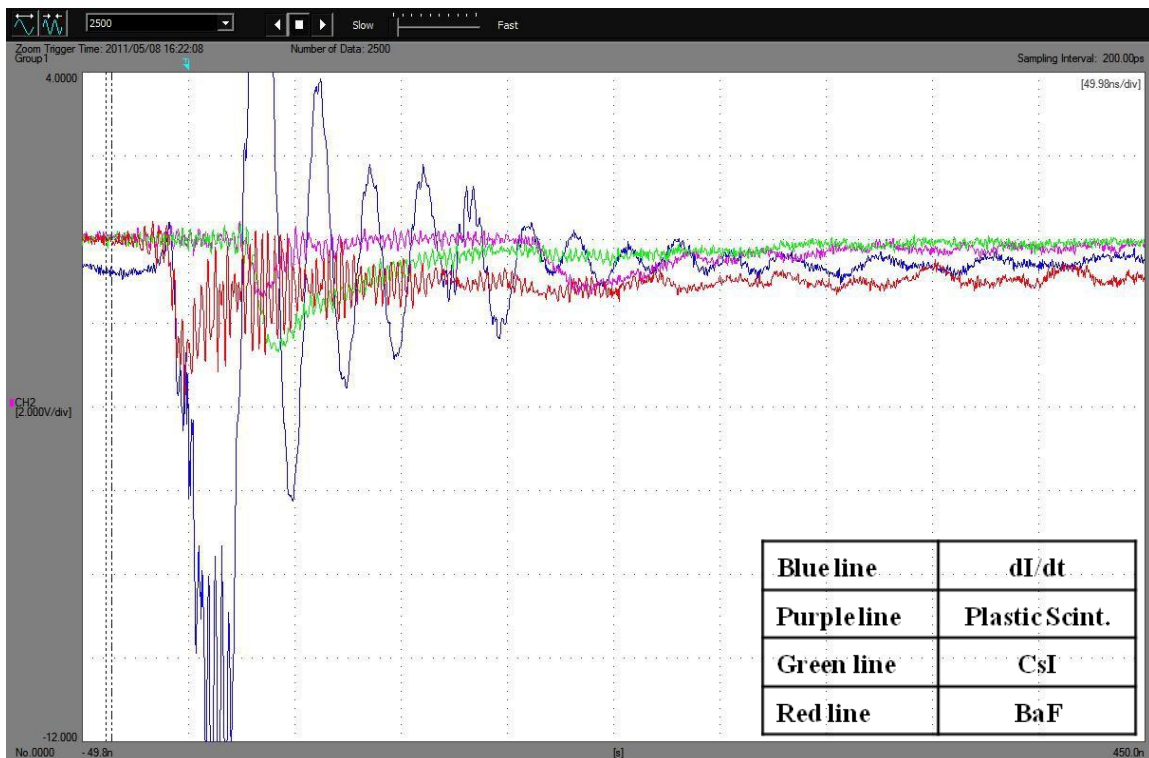
The x-ray image shows a clear dense pinch region on the top and center of anode (~ 6 mm length) and a quite evenly distributed plasma along the length of the pinch column. A narrow pinch shape indicates a nice and evenly formed fusion source.

It can be seen also a narrow ($\text{FWHM} \approx 25$ ns) and strong dip in current derivative signal. For this shot, there was a moderate CsI signal with integral of 139 nVs and a moderate value of neutron anisotropy (1.99).

Gas: pure D₂



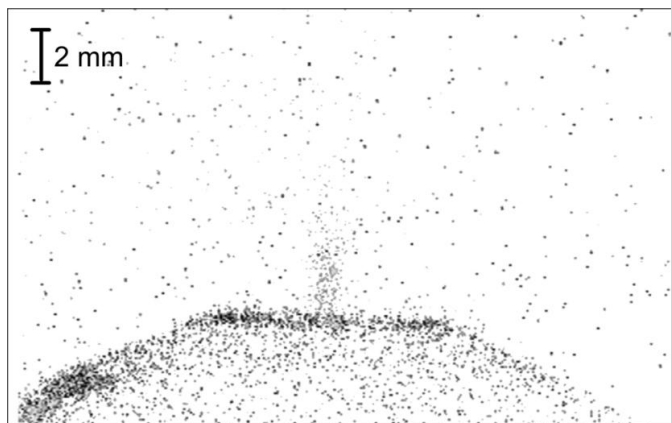
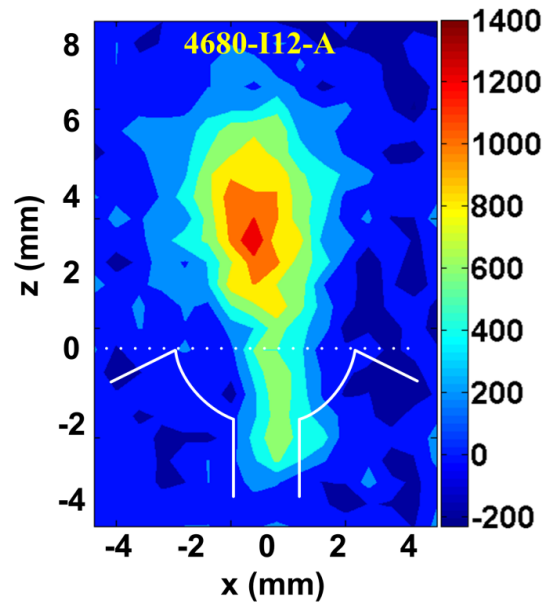
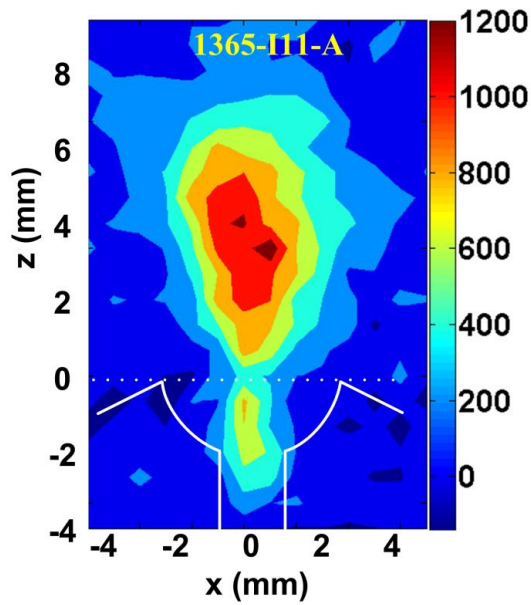
Neutron Yield	2.31×10^8
Anisotropy	1.99
HX-ray Integral (nVs)	139
SNR (1365)	19.7
SNR (4680)	16.0



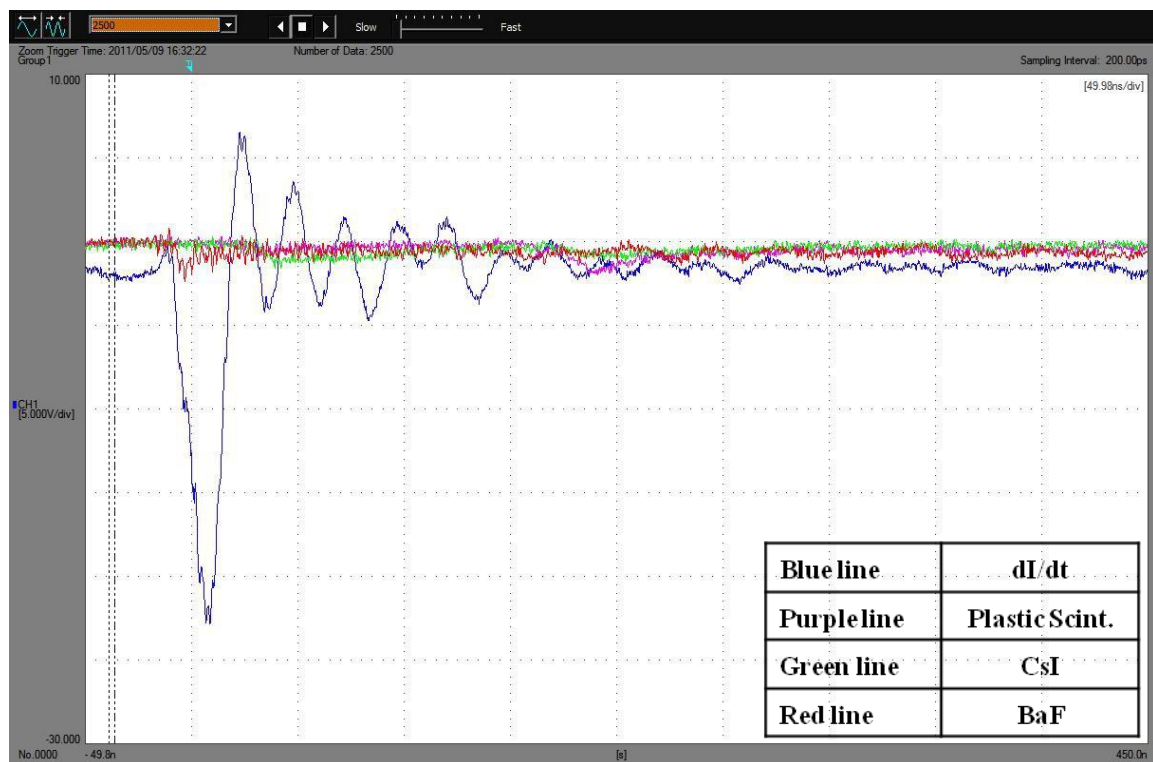
The two detectors 1365-I11-A and 4680-I12-A were exposed with a good shot ($Y_n = 1.47 \times 10^8$). As usual, both images show an egg-shaped fusion source with ~ 4 mm width and ~ 10 mm length which is more concentrated at the center around $z = 3$ mm above the anode. It is clear that the images are positioned inside the anode because of the trajectory affect. The SNR of both masks are quite low, because of the extended object size. The total proton tracks for 1365-I11-A and 4680-I12-A are 86,246 and 113,617 respectively.

The x-ray image shows a weak and narrow pinch region with ~ 3 mm length on top of the anode and a little distributed along the z-axis. The expansion of hot plasma region can be correlated to the extended fusion source.

In this shot, the current derivative signal which shows a weak dip pinch with $\text{FWHM} \approx 20$ ns. Moderate neutron anisotropy, 1.87 is in opposition of very weak hard x-ray signal as the integral of CsI hard x-ray detector is only 44 nVs.

Gas: pure D₂

Neutron Yield	1.47×10^8
Anisotropy	1.87
HX-ray Integral (nVs)	44
SNR (1365)	14.1
SNR (4680)	12.5

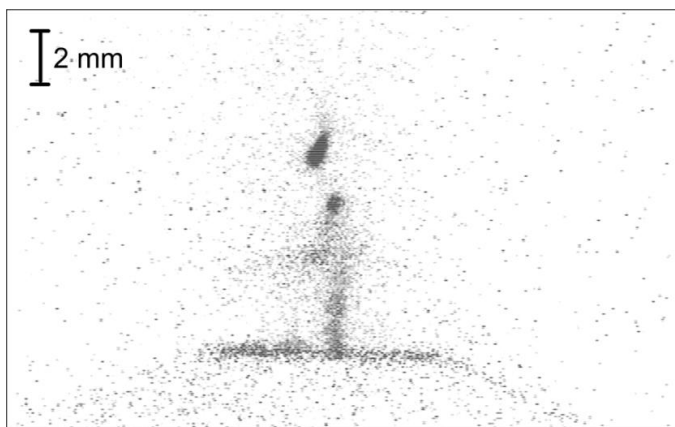
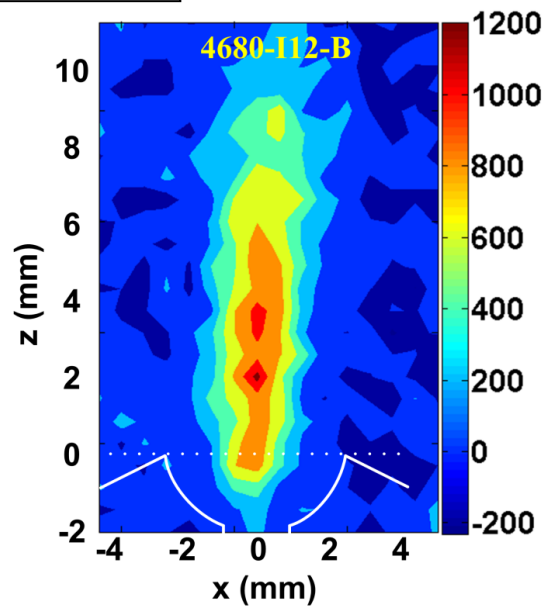
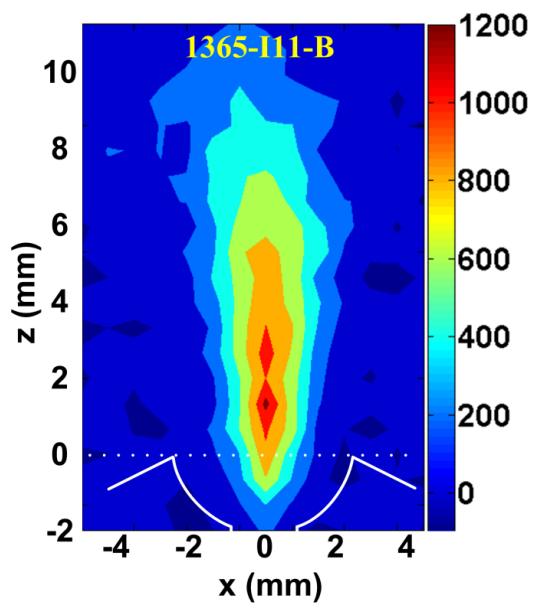


The two detectors 1365-I11-B and 4680-I12-B were exposed with a weak shot ($Y_n = 0.85 \times 10^8$). The final images are a little bit different from most of the proceeding shots, showing an elongated cigar-shape fusion source. However both of them have more or less the same size with ~ 2 mm width and ~ 12 mm length which is more concentrated at two separated small areas (each ~ 1 mm²) at $z = 2$ mm above the anode tip. Although the number of proton tracks is quite low, because of relatively smaller object size, both SNRs are reasonably high. The total proton tracks for 1365-I11-B and 4680-I12-B are 63,555 and 87,204 respectively.

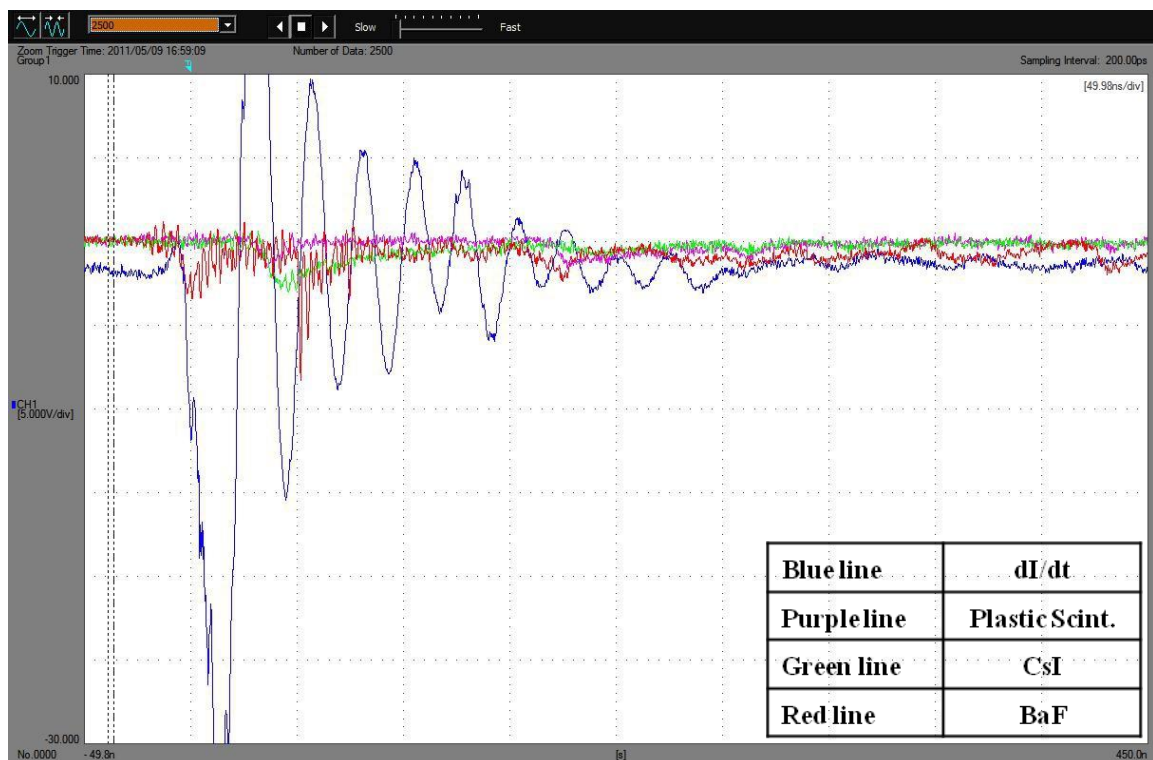
The x-ray image shows the forming of a usual shape and quite dense pinch region with ~ 5 mm length on top of anode and two high dense plasma areas on ~ 6 mm above the anode. These dense areas (may be called hot spots) can be correlated to formation of two dense fusion source regions. The different distance ($z = 4$ mm for fusion source place and $z = 6$ mm for dense plasma region) is probably related to the proton trajectory; Increasing the real pinch current (more than 300 kA as we consider it in our calculation) and so increasing the magnetic field may cause bigger trajectory angle and shifting down the final image (the real proton place on the detector shifts up). The existing of higher pinch current can be true by regarding to very dip and narrow (FWHM ≈ 20 ns) discharge current signal.

The integral of CsI hard x-ray detector is 53 nVs in agreement with moderate neutron anisotropy 1.93.

Gas: pure D₂



Neutron Yield	0.85×10^8
Anisotropy	1.93
HX-ray Integral (nVs)	53
SNR (1365)	17.7
SNR (4680)	13.9

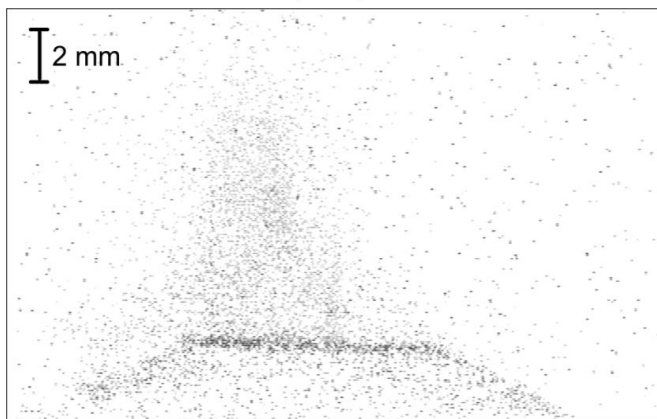
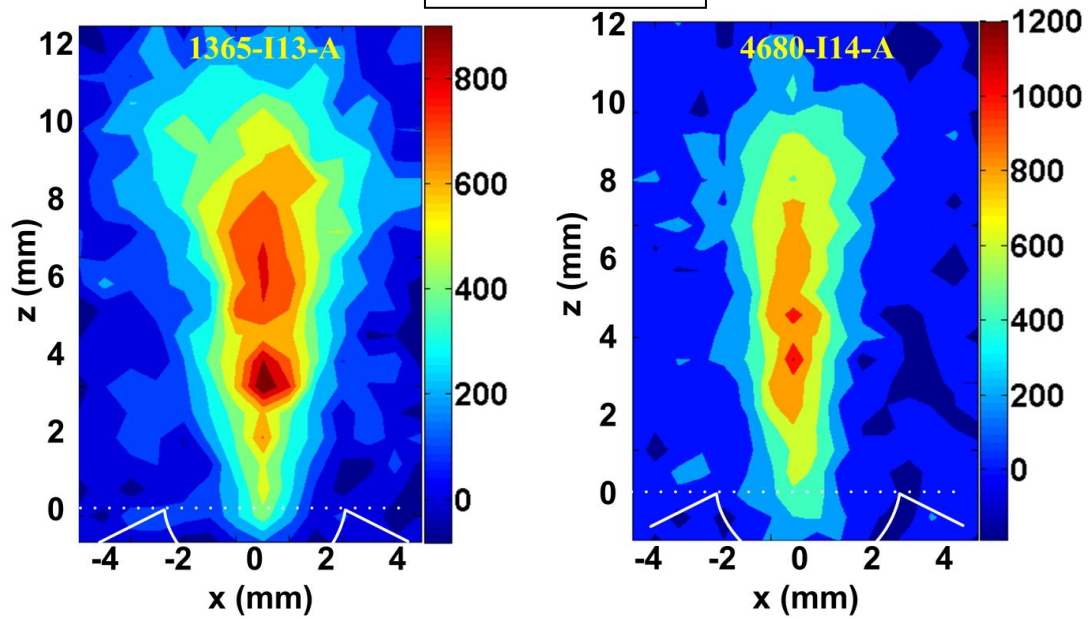


The two detectors 1365-I13-A and 4680-I14-A were exposed with a good shot ($Y_n = 1.47 \times 10^8$). The final images are a little bit different from each other. 1365-I13-A image shows an egg-shaped fusion source, much wider on top. Its width is ~ 2 mm on the top of anode (bottom of the source image) and reaches to ~ 5 mm on top end of image. However 4680-I14-A looks more ellipsoid with minimum and maximum width of ~ 2 mm and ~ 3 mm respectively. Their lengths are quite similar, being about ~ 11 mm. It must be mentioned that the colour bars arrangement are different for these images which does sometimes exaggerate the size of the fusion source in the mask-1365 image. Also both of them show two separated more dense sources around $z = 4$ mm. For this shot, the SNR of both masks are quite low but similar which is because of extended object size. The total proton tracks for 1365-I13-A and 4680-I14-A are 76,843 and 104,700 respectively.

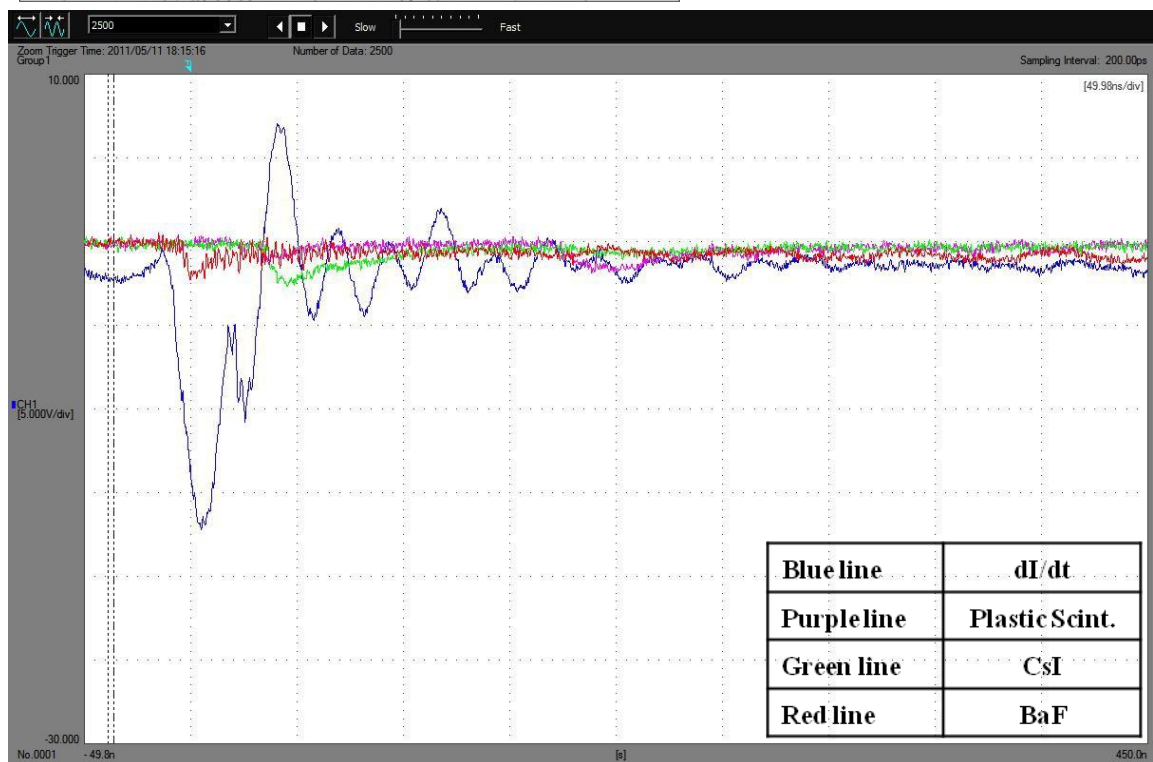
The x-ray image barely shows a low dense cylindrical pinch region with ~ 4 mm length on top and center of anode and also evenly distributed on a big area mostly on the left side of the pinch. The expansion of hot plasma region can be correlated to the extended fusion source and its shifting to one side may cause the different shape of reconstructed images (since there is a 60° difference in viewing angle).

The current derivative signal shows a very weak dip pinch with $\text{FWHM} \approx 30$ ns. The integral of CsI detector is 60 nVs which shows moderate hard x-ray signal in contrariety of very small neutron anisotropy of 1.43.

Gas: pure D₂



Neutron Yield	1.47×10^8
Anisotropy	1.43
HX-ray Integral (nVs)	60
SNR (1365)	12.5
SNR (4680)	12.0

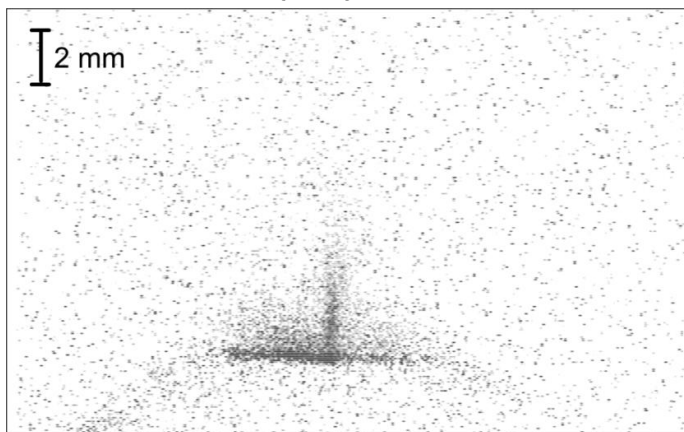
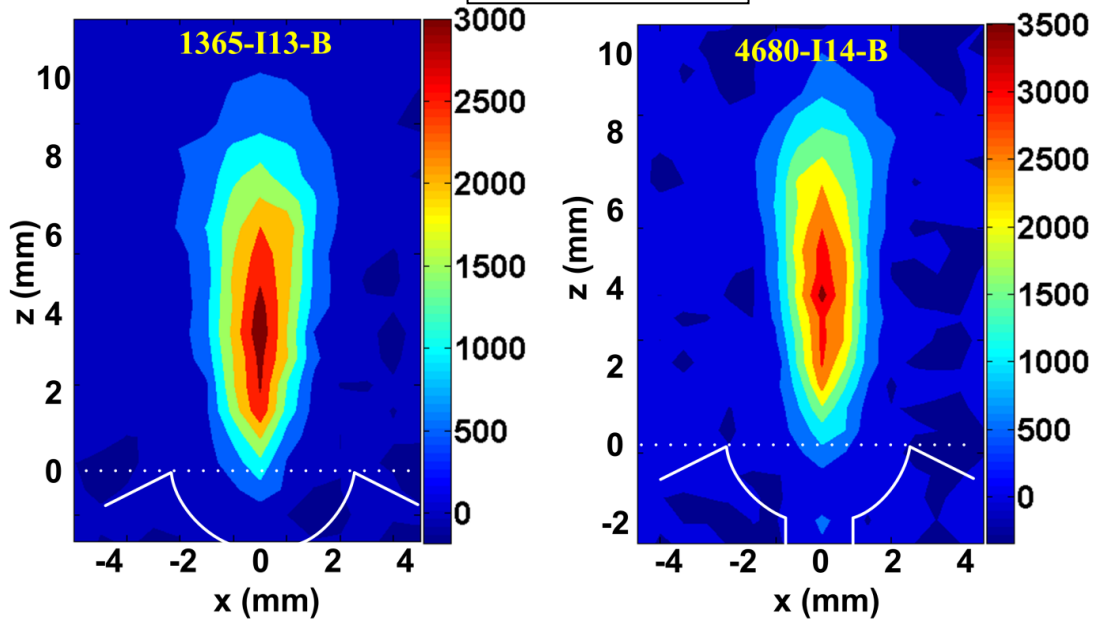


The two detectors 1365-I13-B and 4680-I14-B were exposed with a very good shot ($Y_n = 2.38 \times 10^8$). Both images show an egg-shaped (very near to ellipsoidal) fusion source with ~ 3 mm width and ~ 10 mm length. The high concentration region is at $z = 4$ mm. The SNR of both images are quite high because of high neutron (proton) yield and very compact object size. The total proton tracks for 1365-I13-B and 4680-I14-B are 141,773 and 200,024 respectively.

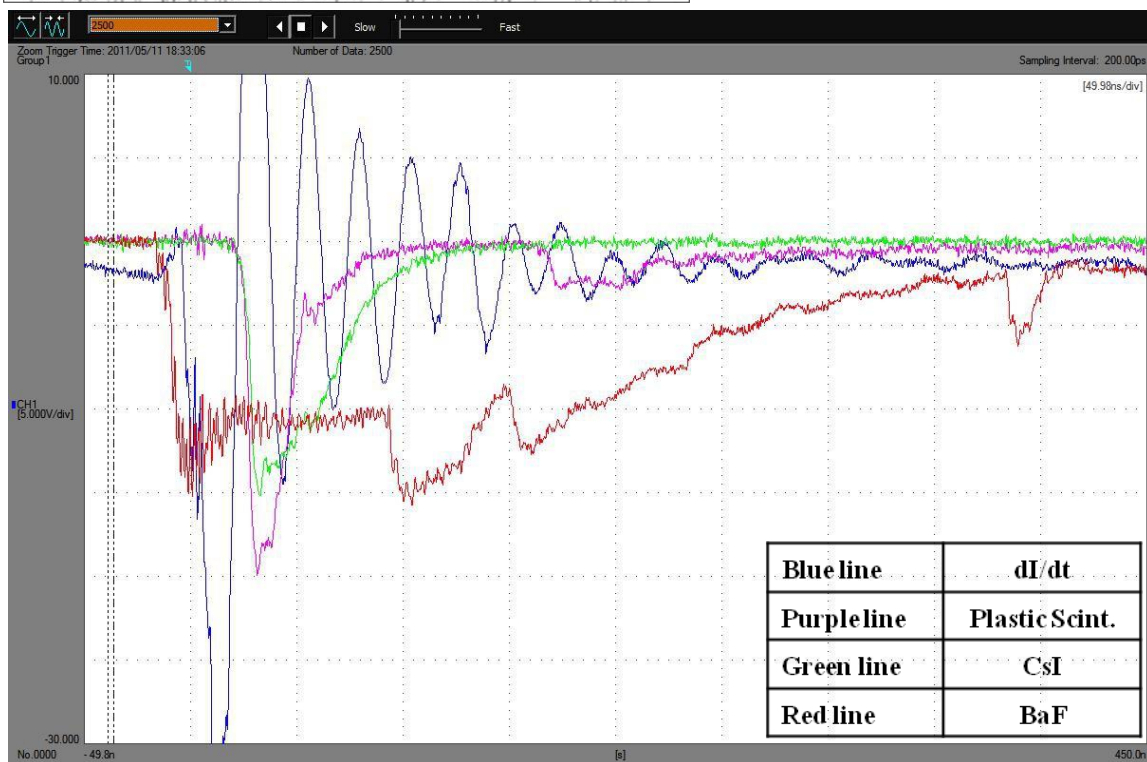
The x-ray image shows a clear dense pinch region on the top and center of anode (~ 4 mm length) and evenly distributed along the z-axis. A narrow pinch shape indicates to a nice and evenly formed fusion source.

It can be seen also a narrow (FWHM ≈ 15 ns) and strong dip in current derivative signal. The integral of CsI hard x-ray detector 257 nVs and neutron anisotropy 2.41, both are showing a strong acceleration electric field for deuterons and electrons in pinch region for this shot. Moreover the plastic scintillator signal shows a relatively wide time range of neutron emission which may be justified by longer confinement time for high energy deuterons in pinch region. The concentrated fusion source size, dense x-ray image and high neutron yield also support this idea. Need to comment on the unexpected secondary peaks in the BaF₂ scintillator trace. In view of fact, that there are no corresponding peaks in the CsI scintillator trace, these cannot presumably be genuine x-ray pulses. They can only be assumed to be the result of electronic noise.

Gas: pure D₂



Neutron Yield	2.38×10^8
Anisotropy	2.41
HX-ray Integral (nVs)	257
SNR (1365)	23.0
SNR (4680)	22.6

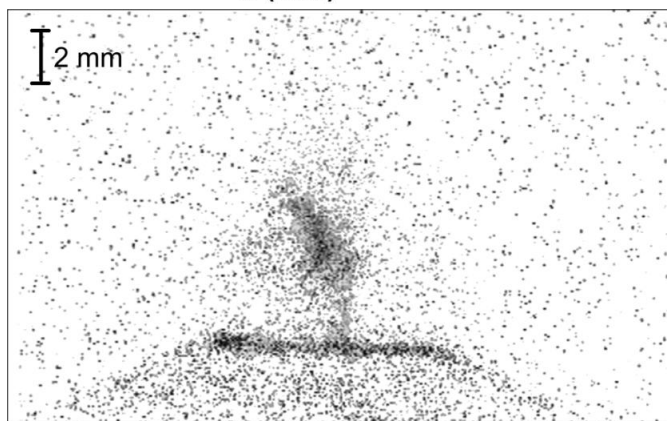
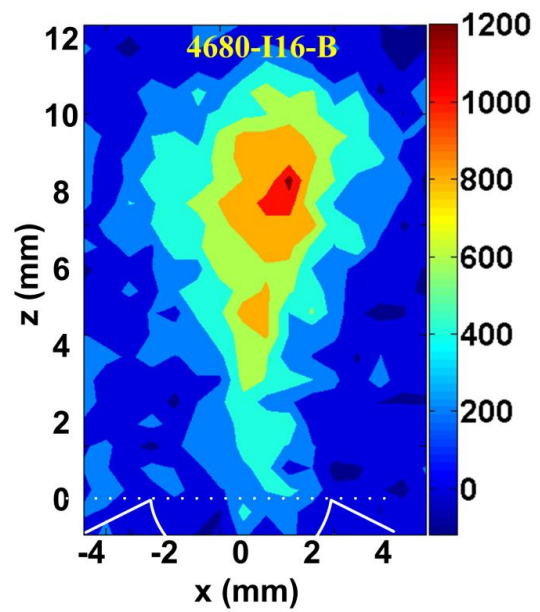
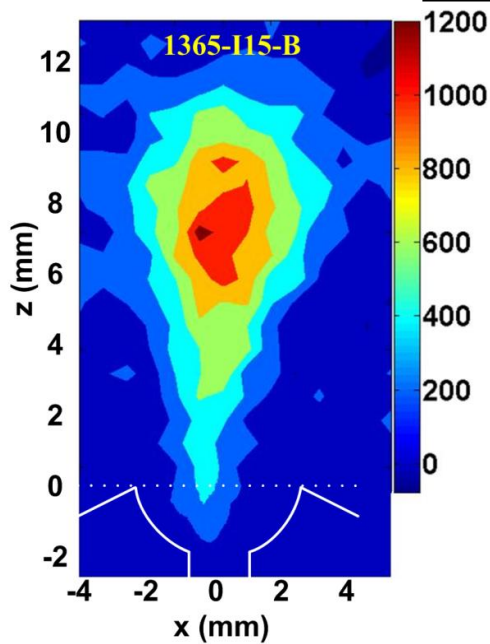


The two detectors 1365-I15-B and 4680-I16-B were exposed with a good shot ($Y_n = 2.02 \times 10^8$). Both images show an egg-shaped fusion source with 1-6 mm width (top section is much wider than bottom) and ~ 10 mm length which is more concentrated at center around $z = 7$ mm (for 1365-I15-B) and $z = 8$ mm (for 4680-I16-B) above the anode. The SNR of mask-1365 is the better of the two, but both of them are relatively low, because of extended object size. The total proton tracks for 1365-I15-B and 4680-I16-B are 104,175 and 147,463 respectively.

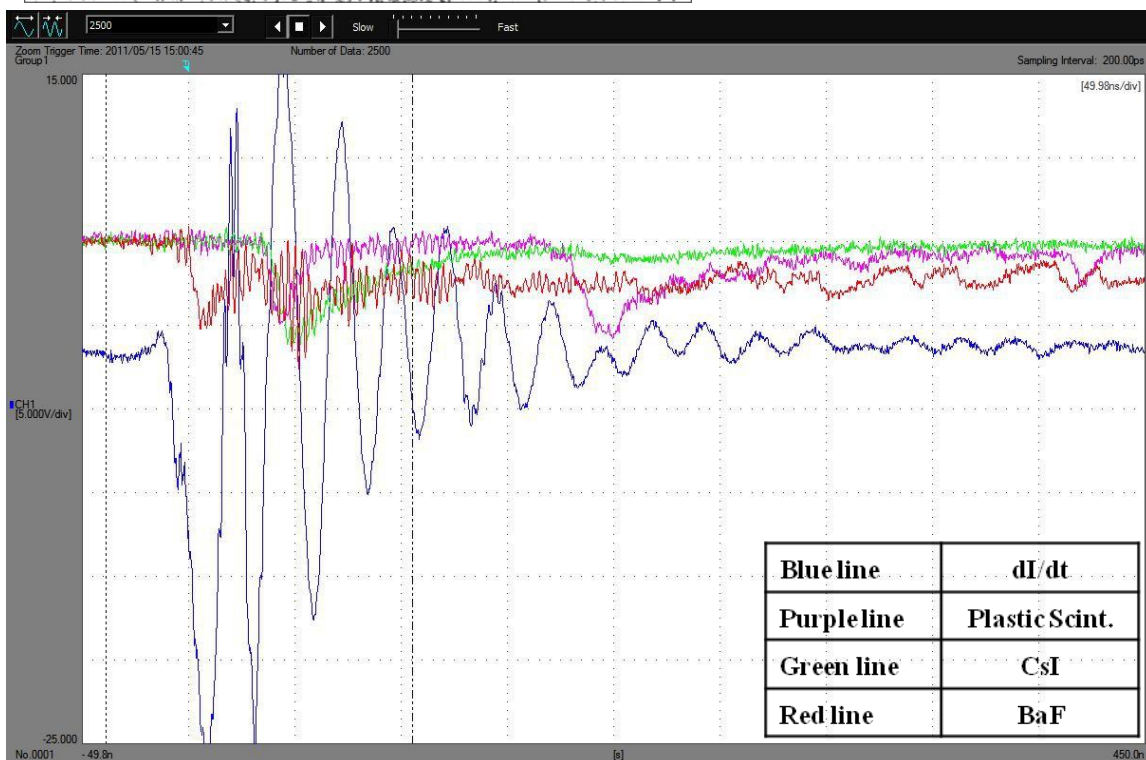
The rather unexpected x-ray image shows a low dense pinch region with ~ 2 mm length on top of the anode and exactly after that a dense plasma region is started with about 4 mm^2 area and bent to the left direction. Also low dense plasma is distributed along the z-axis. The expansion of hot plasma region can be correlated to the extended fusion source. Bending the fusion source image also is in agreement with bent dense plasma region. This unexpected shape also may cause different trajectory for D-D reaction protons and so the 1365-I15-B image is shifted more down.

The two separated pinch regions may go back to the current derivative signal which shows double dip pinch with FWHM ≈ 15 ns and 7 ns respectively. The integral of CsI hard x-ray detector is 117 nVs in spite of the low neutron anisotropy of 1.89.

Gas: pure D₂



Neutron Yield	2.02×10^8
Anisotropy	1.89
HX-ray Integral (nVs)	117
SNR (1365)	13.7
SNR (4680)	9.9



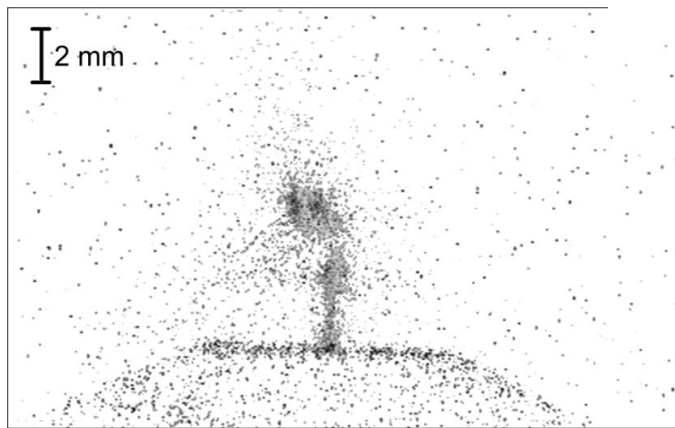
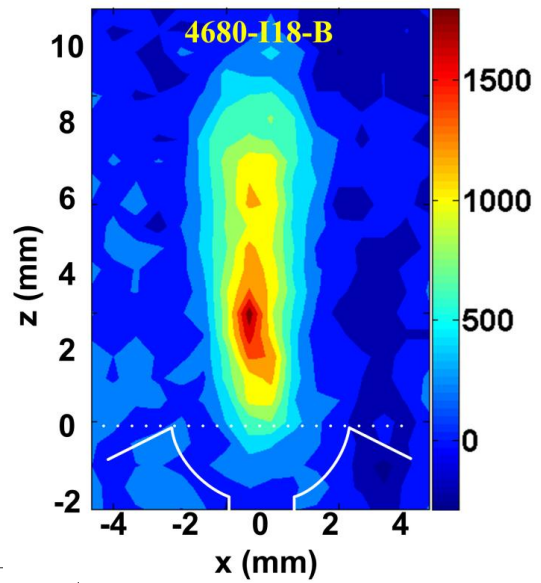
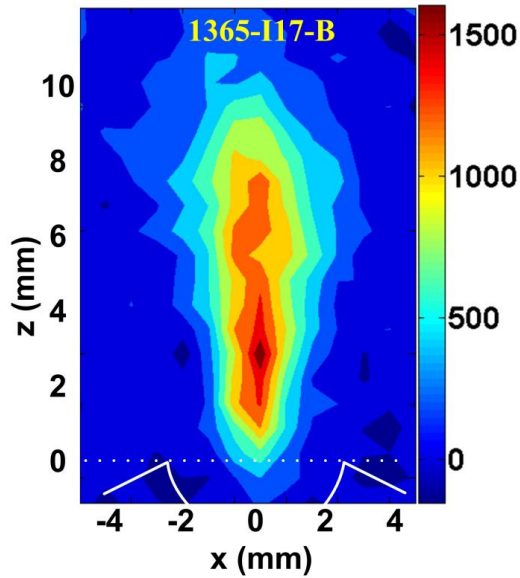
Blue line	dI/dt
Purple line	Plastic Scint.
Green line	CsI
Red line	BaF

The two detectors 1365-I17-B and 4680-I18-B were exposed with a good shot ($Y_n = 1.6 \times 10^8$). 1365-I17-B image shows an egg-shaped fusion source with ~ 2 mm width on bottom of the source and ~ 4 mm on top end. However 4680-I18-B looks more ellipsoid with ~ 2 mm width. Their length is ~ 10 mm with a concentrated fusion region at around $z = 3$ mm. The SNR of mask-1365 is the better of the two, but both of them are quite low. The total proton tracks for 1365-I17-B and 4680-I18-B are 91,657 and 118,022 respectively.

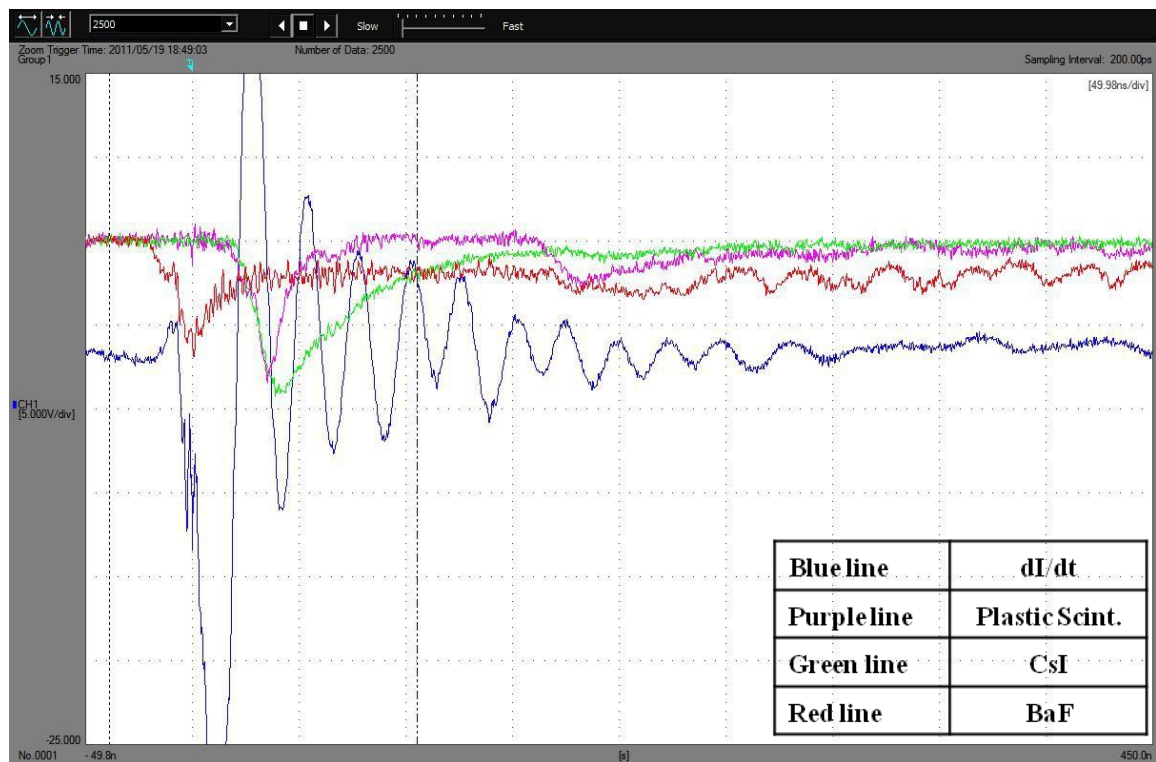
As with the previous shot, the unexpected x-ray image shows a usual dense pinch region with ~ 4 mm length on top of the anode and exactly after that a dense plasma region is apparent with about 2 mm^2 area and bent to the left direction. The same as before, this bent shape may cause different trajectory for D-D reaction protons and so the 1365-I17-B image is shifted more down by ~ 1 mm.

There is a quite strong dip in current derivative signal with $\text{FWHM} \approx 20$ ns. The integral of CsI hard x-ray detector is 196 nVs in agreement with moderate neutron anisotropy 2.02.

Gas: pure D₂



Neutron Yield	1.6×10^8
Anisotropy	2.02
HX-ray Integral (nVs)	196
SNR (1365)	17.5
SNR (4680)	12.9



The detector 1365-I19-A was exposed with a weak shot ($Y_n = 0.64 \times 10^8$). The image shows an ellipsoid fusion source with ~ 2 mm width (slightly wider at the top) and ~ 10 mm length which is more concentrated at $z = 3$ mm. The total proton tracks for 1365-I19-A is 52,250.

About detector 4680-I20-A it must be mentioned that erroneously this side was exposed two times, for second shot, I forgot to turn it back. So the image 4680-I20-A is like combination of two images: 1365-I19-A and 1365-I19-B. The total proton tracks for 4680-I20-A is 170,834.

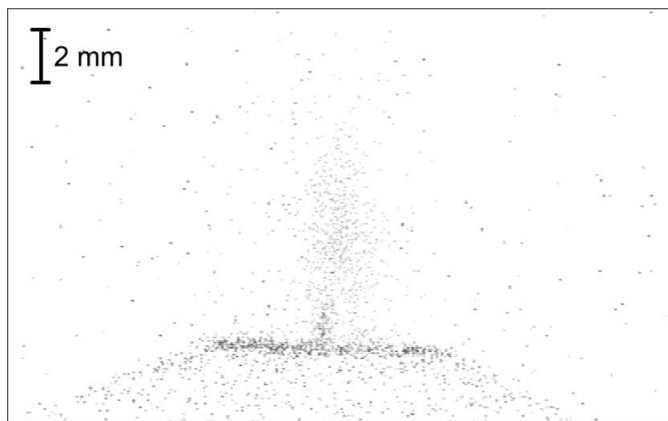
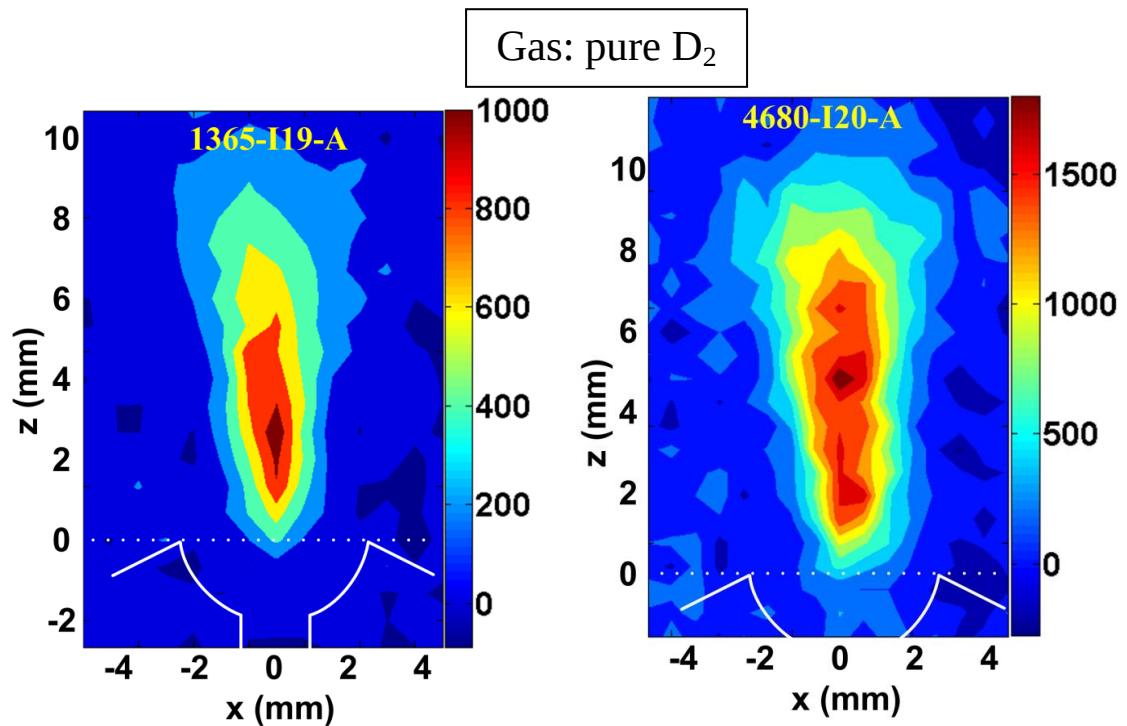
The x-ray image shows a weak pinch region with ~ 2 mm length on top of the anode and a then stretched with less density above that (~ 2 mm width and ~ 4 mm length). So the less density of fusion source image on the top section can be related to this kind of plasma distribution.

It can be seen also a weak dip in current derivative signal with $\text{FWHM} \approx 15$ ns. The all other signals are quite weak. The integral of CsI hard x-ray detector is only 35 nVs.

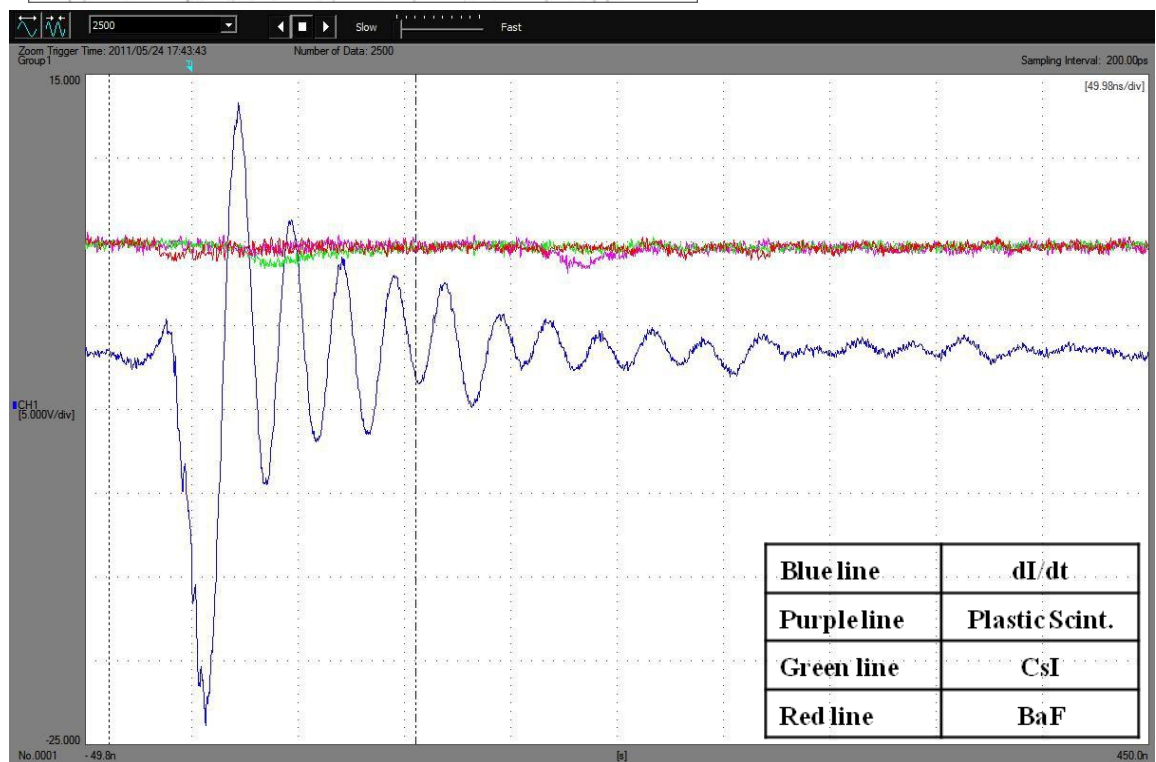
For the next shot, the detectors 1365-I19-B was exposed with a good shot ($Y_n = 1.15 \times 10^8$). Its image shows an egg-shaped fusion source with ~ 2 mm width on bottom of the source and ~ 4 mm on the top end and the total length is ~ 10 mm with a concentrated fusion region at around $z = 3$ mm. The total proton tracks for 1365-I19-B is 77,261.

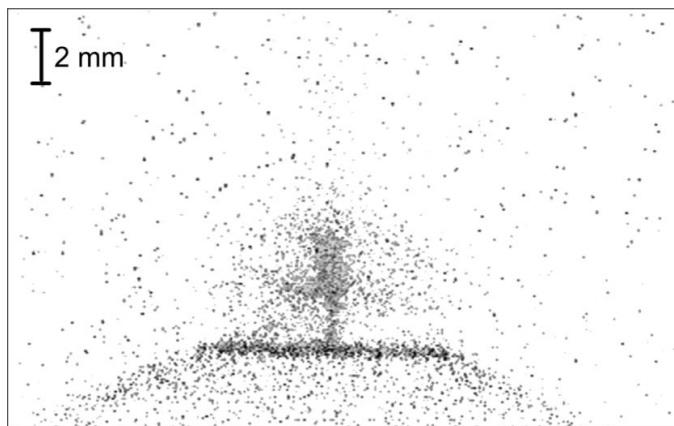
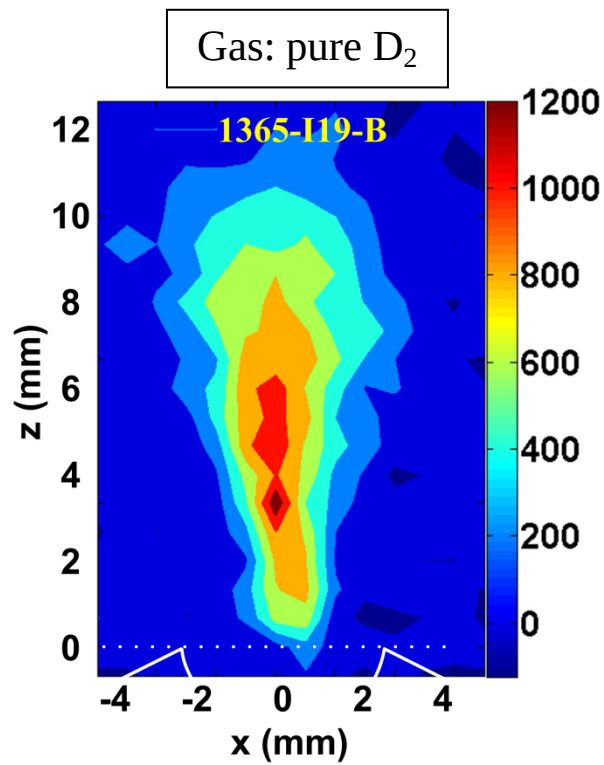
The same as previous shot, the x-ray image shows a short dense pinch region with ~ 2 mm length on top of the anode and exactly after that a dense plasma area is started with ~ 3 mm length and ~ 2 mm width which a mushroom shape less dense plasma surrounds it.

There is a relatively strong dip in current derivative signal with $\text{FWHM} \approx 15$ ns. The integral of CsI hard x-ray detector is 62 nVs in agreement with moderate neutron anisotropy 2.13.

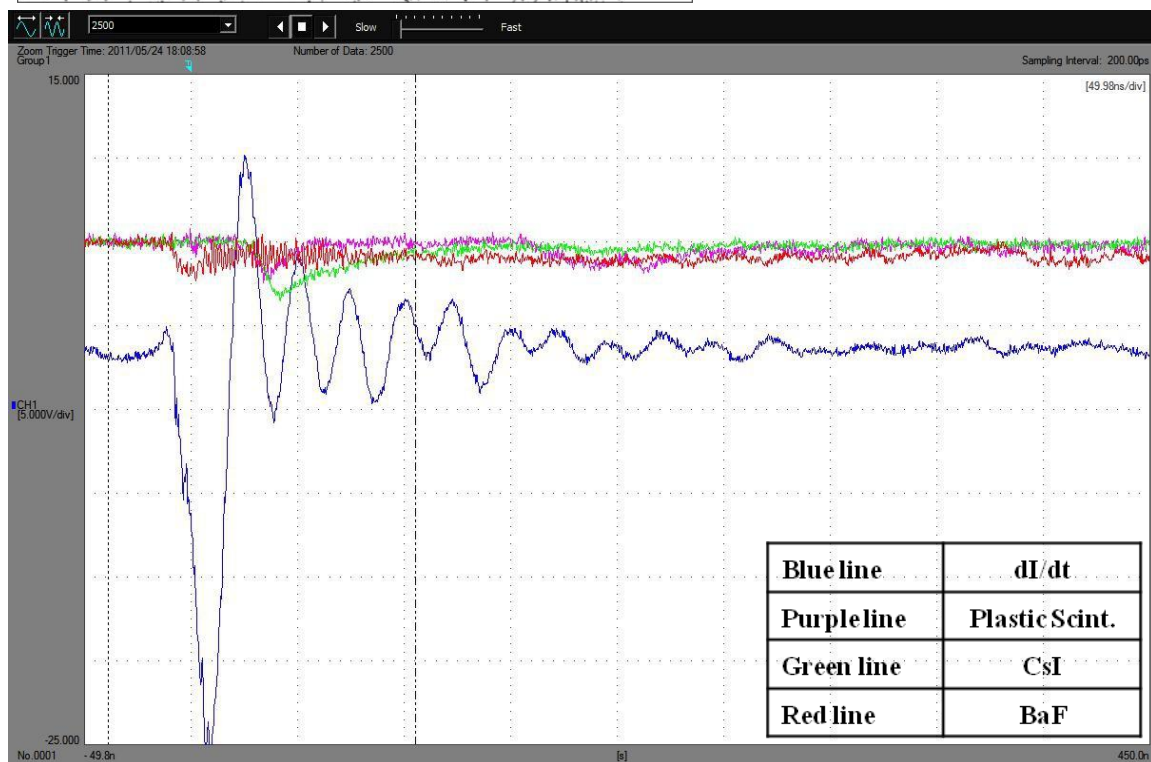


Neutron Yield	0.64×10^8
Anisotropy	1.91
HX-ray Integral (nVs)	35
SNR (1365)	18.5
SNR (4680)	14.6





Neutron Yield	1.15×10^8
Anisotropy	2.13
HX-ray Integral (nVs)	62
SNR (1365)	16.8
SNR (4680)	



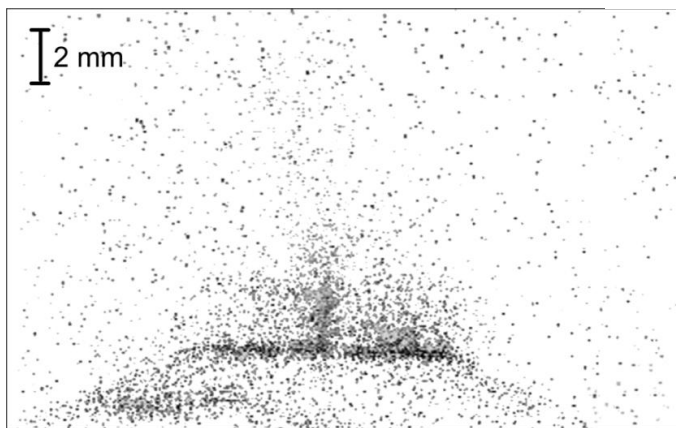
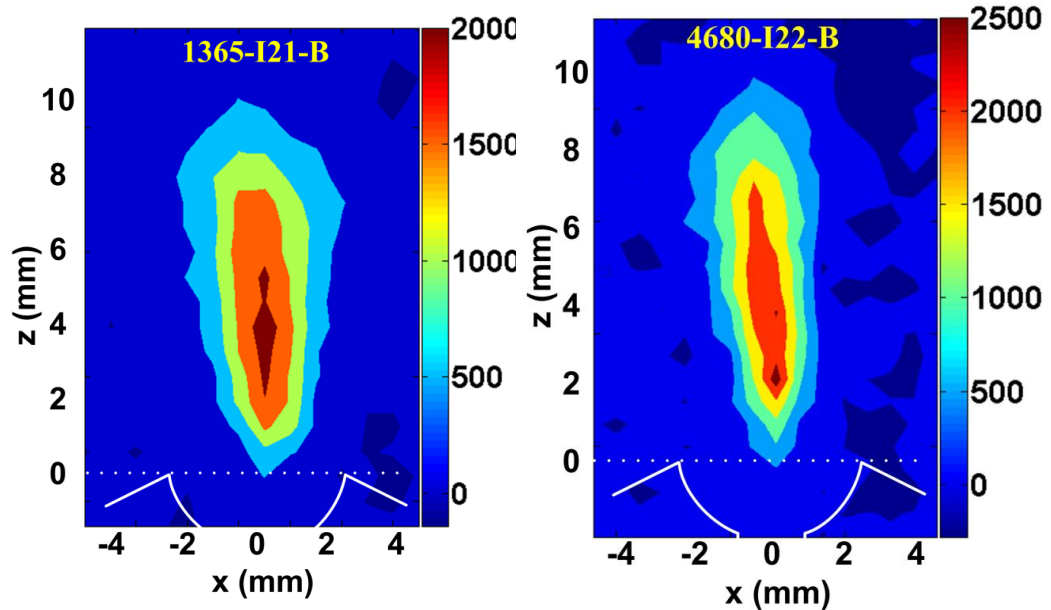
Blue line	dI/dt
Purple line	Plastic Scint.
Green line	CsI
Red line	BaF

The two detectors 1365-I21-B and 4680-I22-B were exposed with a good shot ($Y_n = 1.58 \times 10^8$). Both images show an ellipsoid fusion source with ~ 2 mm width and ~ 8 mm length which is more concentrated at an oval shape (~ 1 mm width and ~ 5 mm length) at the center of $z = 4$ mm. The image 1365-I21-B looks slightly wider. It must be mentioned again that for these images, the colour bars arrangement are different. The SNR of mask-1365 is the better of the two. The total proton tracks for 1365-I21-B and 4680-I22-B are 109,749 and 158,344 respectively.

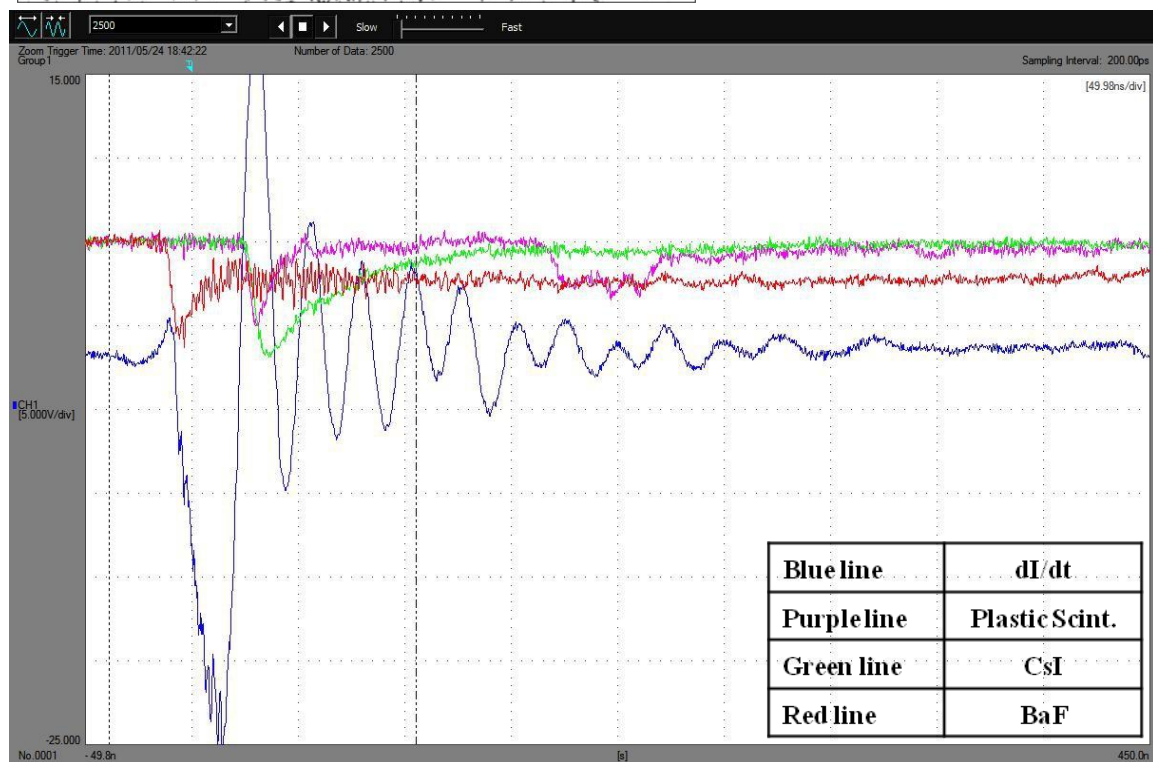
The x-ray image shows a clear dense pinch region on the top and center of anode (~ 4 mm length) with an extra dense area just beside that. A narrow pinch shape usually refers to a nice and evenly formed fusion source.

It can be seen also a wide ($\text{FWHM} \approx 55$ ns) but strong dip in current derivative signal. The integral of CsI hard x-ray detector 128 nVs is showing a relative strong x-ray, however the neutron anisotropy 2.01 is just moderate.

Gas: pure D₂



Neutron Yield	1.58×10^8
Anisotropy	2.01
HX-ray Integral (nVs)	128
SNR (1365)	19.0
SNR (4680)	16.7

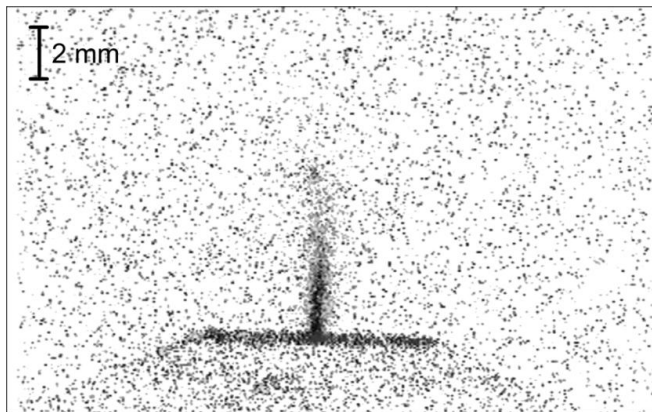
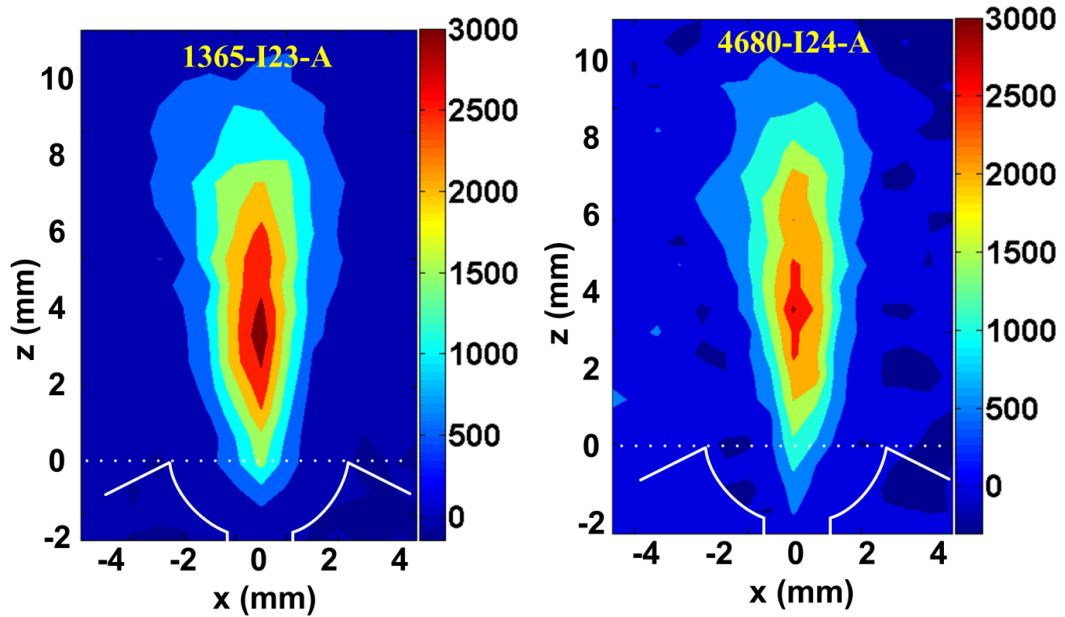


The two detectors 1365-I23-A and 4680-I24-A were exposed with a very good shot ($Y_n = 3.15 \times 10^8$). Both images show an egg-shaped (near to ellipsoidal) fusion source with ~ 3 mm width and ~ 10 mm length with high concentration region in $z = 3$ mm. The SNR of both images are quite high because of high neutron (proton) yield and concentrated object size. The total proton tracks for 1365-I23-A and 4680-I24-A are 173,675 and 220,147 respectively.

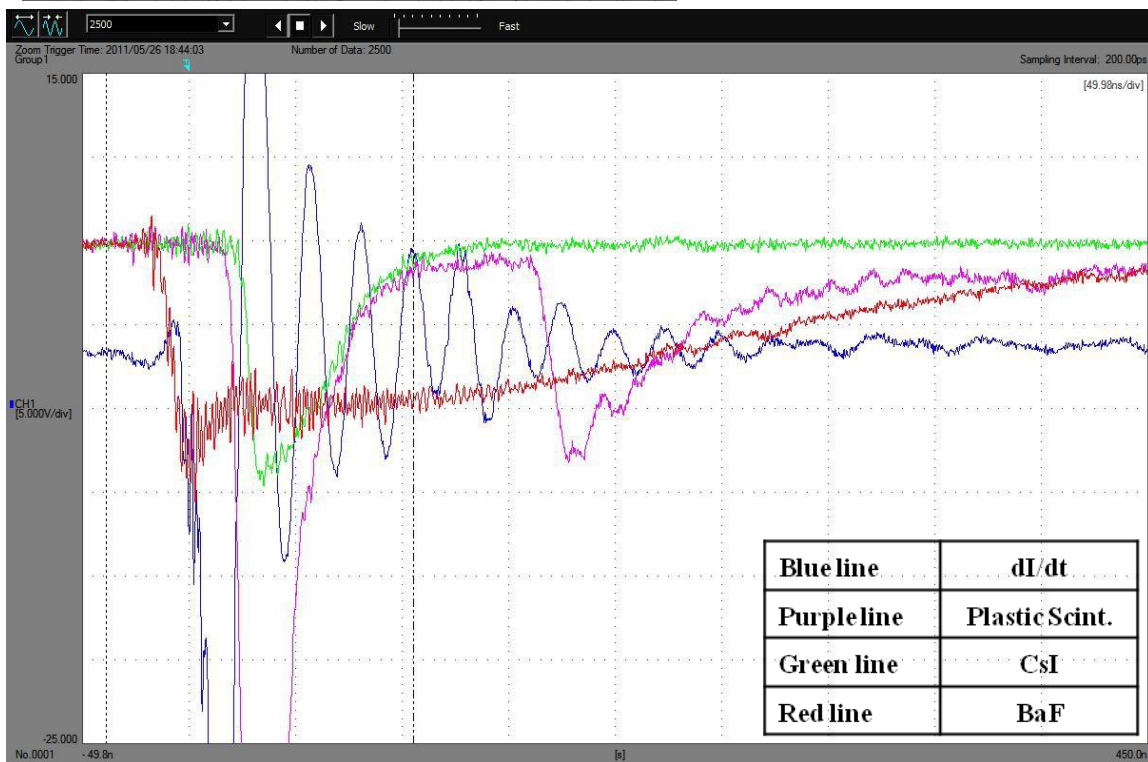
The x-ray image shows a clear dense pinch region on the top and center of anode (~ 6 mm length) and evenly distributed plasma along the z-axis. A narrow pinch shape indicates to a nice and evenly formed fusion source. Comparison with fusion source and x-ray images of 1365-I13-B and 4680-I14-B leads us to find a relation between their lengths; longer x-ray pinch, longer fusion source.

It can be seen also a narrow ($\text{FWHM} \approx 15$ ns) and very strong dip in current derivative signal which can be referred to narrow and strong pinch shape. The integral of CsI hard x-ray detector 253 nVs and neutron anisotropy 2.39, both are indicating a strong accelerating electric field for deuterons and electrons in pinch region for this shot. Moreover the plastic scintillator signal shows a relatively wide time range of neutron emission which may be justified by longer confinement time for high energy deuterons in pinch region. The concentrated fusion source size, dense x-ray image and high neutron yield also support this idea.

Gas: pure D₂



Neutron Yield	3.15×10^8
Anisotropy	2.39
HX-ray Integral (nVs)	253
SNR (1365)	21.8
SNR (4680)	17.5

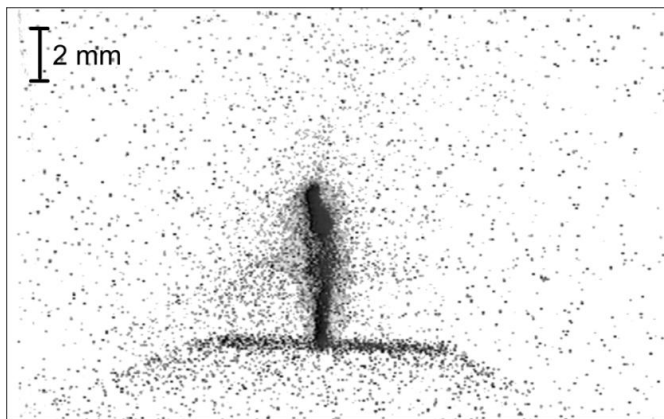
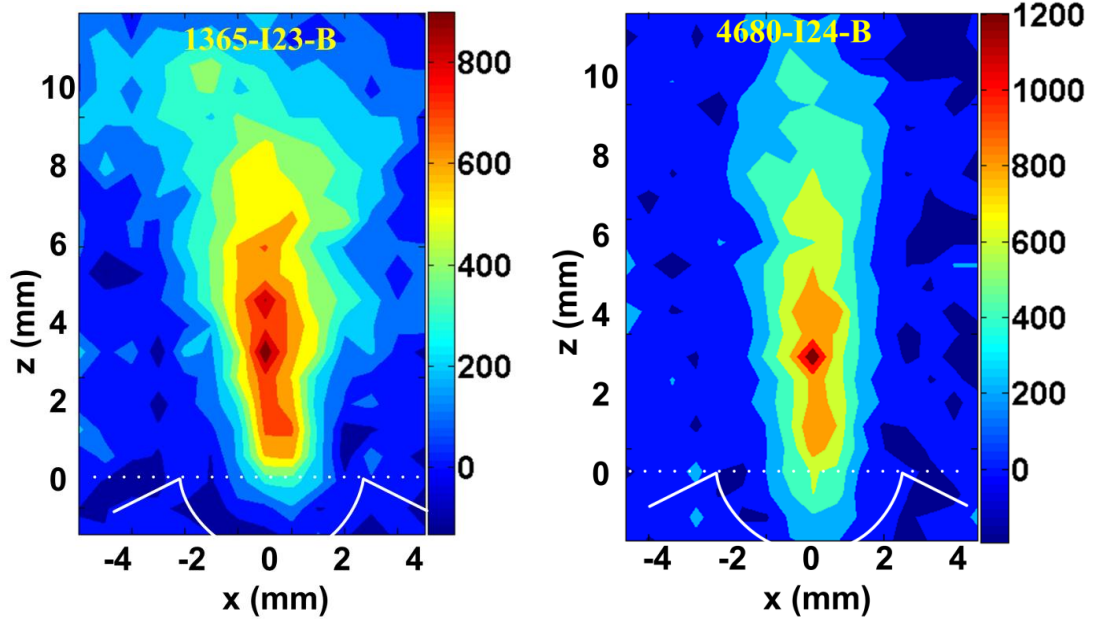


The two detectors 1365-I23-B and 4680-I24-B were exposed with a weak shot ($Y_n = 0.83 \times 10^8$). 1365-I23-B image shows an egg-shaped fusion source with ~ 2 mm width on bottom of the source and ~ 5 mm on top end. However 4680-I24-B looks more ellipsoid with ~ 2 mm width. Their length is ~ 11 mm with a concentrated fusion region at around $z = 3$ mm. It must be noted again that for these images, the colour bars arrangement are different. In spite of other shots, this time, the SNR of mask-1365 is worse than mask-4680, because of much expanded image. The total proton tracks for 1365-I23-B and 4680-I24-B are 65,800 and 88,043 respectively.

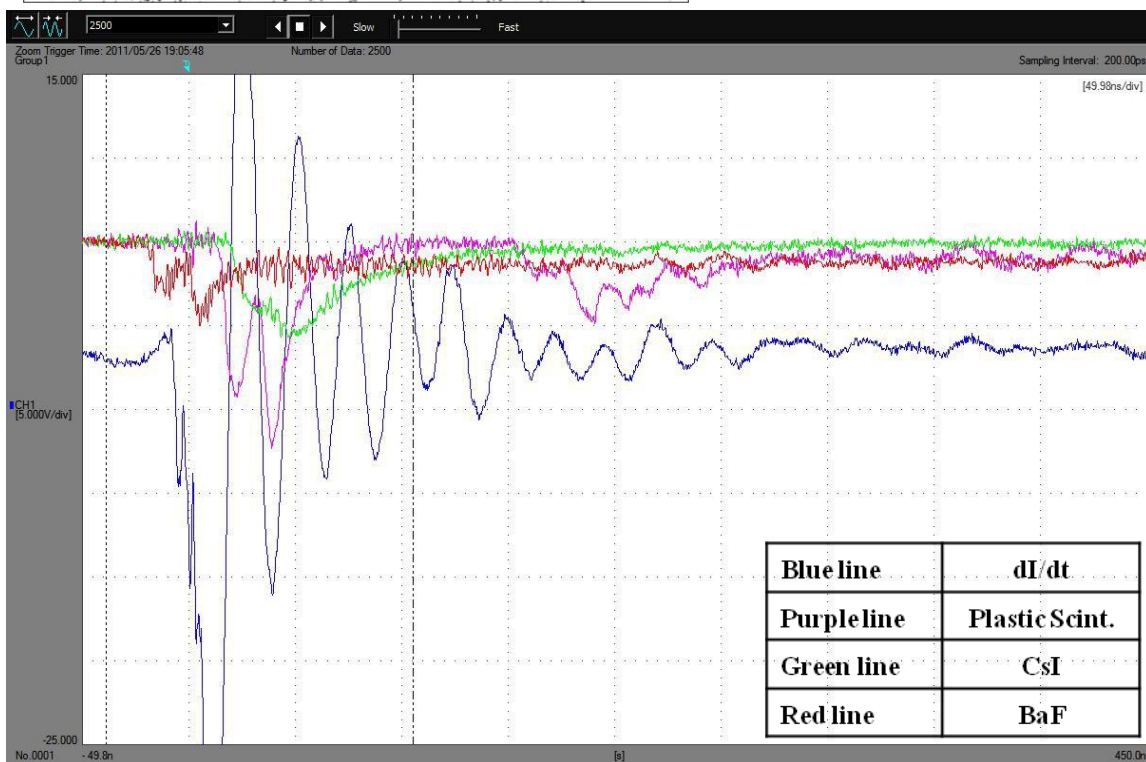
The same as previous shot, the x-ray image shows a usual dense pinch region with ~ 4 mm length on top of the anode and exactly after that a very dense plasma region is started with about 2 mm length and a bit bent to the left direction. This bent shape may cause different trajectory for D-D reaction protons and so the 1365-I23-B image is so expanded. Moreover it seems that this dense area is very close to the dense fusion source in both reconstructed images.

There is a quite strong and narrow dip in current derivative signal with $\text{FWHM} \approx 15$ ns, i.e. a strong dip current may cause a strong pinch but not necessarily high neutron yield. The integral of CsI detector, 132 nVs shows a moderate hard x-ray, in opposition of small neutron anisotropy 1.88. Furthermore, although there is only a single pinch in the current signal, double hard x-ray signal can be seen easily in all detectors signals.

Gas: pure D₂



Neutron Yield	0.83×10^8
Anisotropy	1.88
HX-ray Integral (nVs)	132
SNR (1365)	13.2
SNR (4680)	15.2



5.3.6 Fusion imaging with admixture gas

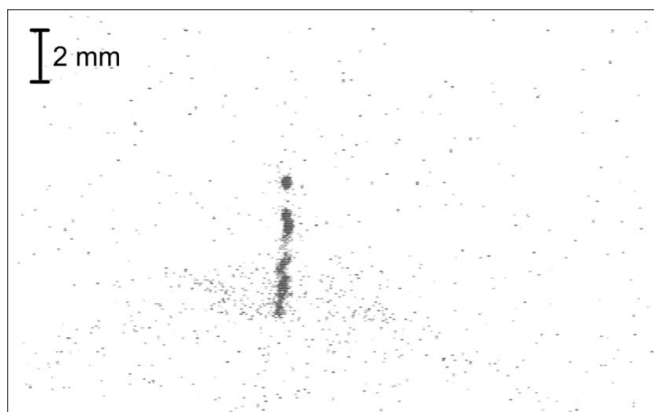
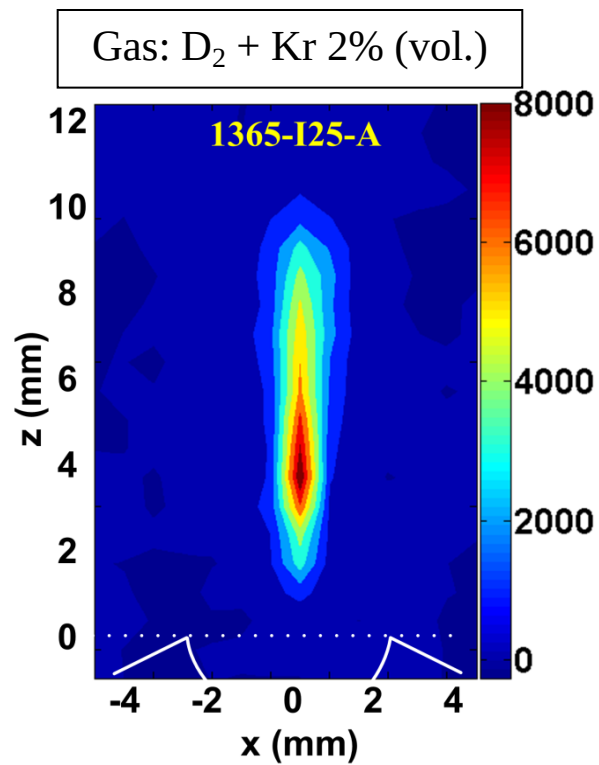
The following experiments were performed with deuterium-krypton admixture gas at 13 mbar and only mask-1365 was employed. Using only one CAI, reduces the overall processing time for each experimental plasma focus shot by approximately one day, and therefore speeds up the experimental procedure. For x-ray imaging, many test exposures were performed using 20 μm Be filter for $\text{D}_2\text{-Kr}$ admixture, and most images were quite saturated. Therefore the filter was changed to combination of 50 μm Be + 12.5 μm Ti (titanium) to avoid x-ray image saturation.

Detector 1365-I25-A was exposed with a good shot ($Y_n = 2.01 \times 10^8$). The fusion source image looks like a nice prolate ellipsoid with ~ 1 mm width and ~ 8 mm length, more concentrated in the lower half of the source at $z = 4$ mm. Generally, the fusion source in admixture gas operation is narrower than pure deuterium gas operation, causes much higher precision of final image (the highest concentration proton track in this shot is ~ 8000 units, in comparison with ~ 3000 in the best case of pure deuterium). This is clearly because of smaller object volume in admixture gas operation. A very high SNR (about twice as before) also results from the small fusion source region and even background. The total proton tracks of 1365-I25-A is 143,346.

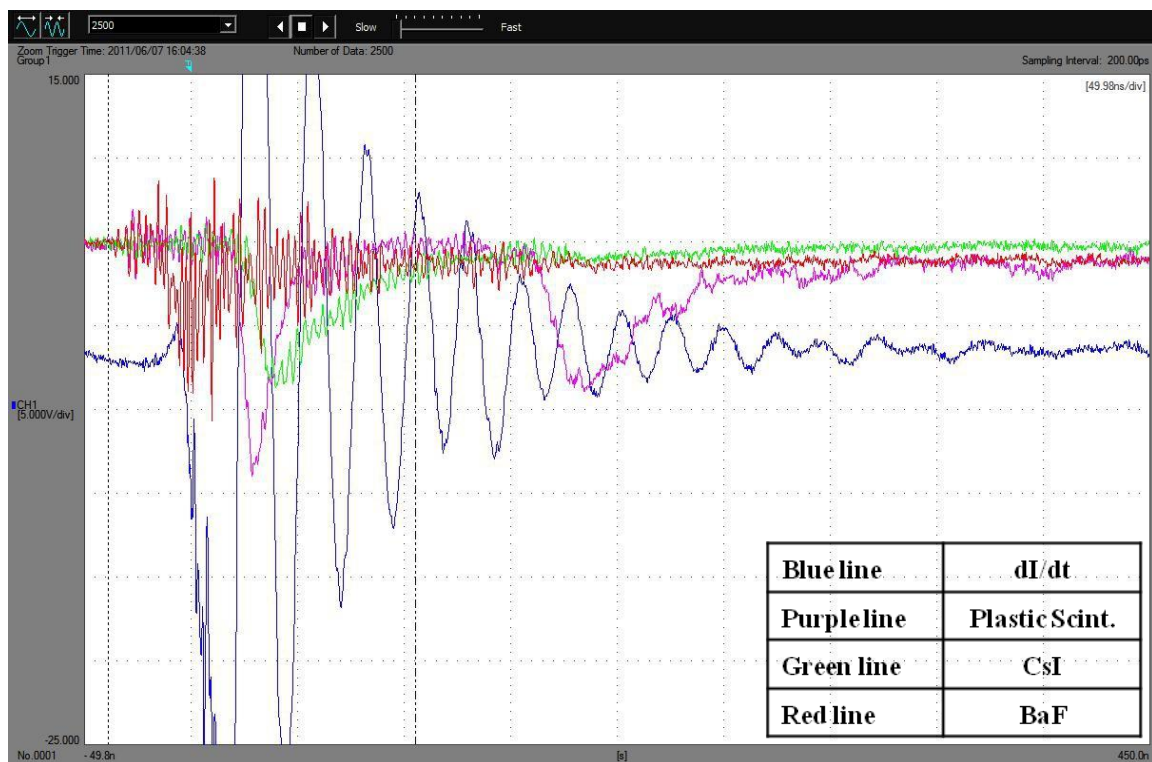
In contrary of pure deuterium, in admixture gas operation, the x-ray image clearly shows the existence of hot spots and/or $m = 0$ instability in final compression. Moreover it shows the high emission rate of x-ray which allows the plasma column becomes more compressed. The size of x-ray image (~ 6 mm length) in energy range above 1 keV is shorter than fusion source. The pinch column is started exactly on top of anode; however, the fusion is shifted up by ~ 2 mm and then elongated in downstream (further of anode). Moreover there is no evidence for existing of hot spots or $m = 0$ instability in fusion source image. So it seems that there is better correlation between fusion source and lower energy x-ray (according to our previous experiment with admixture gas, the x-ray image in range of 0.7 keV-1 keV is longer and more smooth).

The lower anisotropy indicates the lower average deuteron energy which naturally agrees with smaller fusion source size, concentrated inside the pinch region. Another reason of such a small source goes back to this reality that krypton ions are heavier than deuterons and so move behind them in the axial acceleration phase and radial compression. So that, the deuterons somehow are surrounded by krypton ions causes of existing of smaller volume for their fusion reaction. The longer neutron emission time also confirms this idea.

The current derivative signal shows a very strong dip ($\text{FWHM} \approx 20$ ns), which is same for almost all other admixture gas shots. The integral of CsI hard x-ray detector is 177 nVs. However the neutron anisotropy is 1.82, showing a less deuteron average energy in pinch region.



Neutron Yield	2.01×10^8
Anisotropy	1.82
HX-ray Integral (nVs)	177
SNR (1365)	44.3

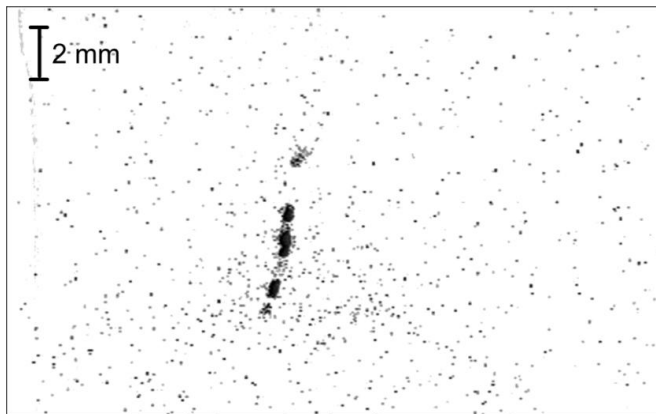
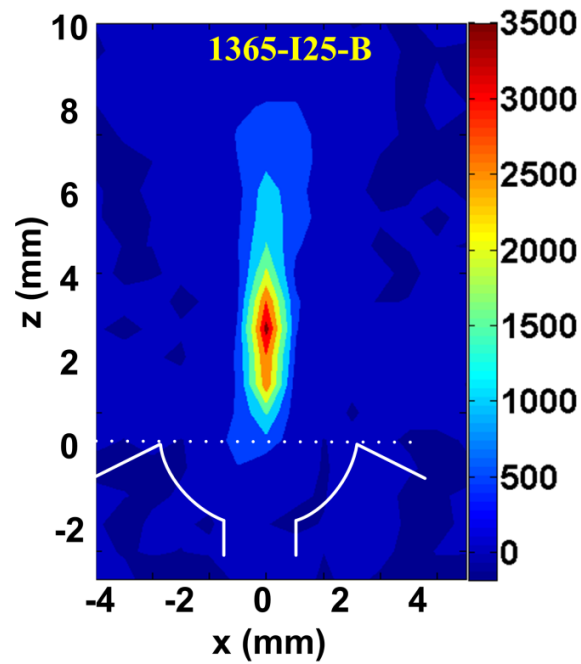


Detector 1365-I25-B was exposed with a weak shot ($Y_n = 0.68 \times 10^8$) even though, the fusion source image still shows a concentrated object, looks like a prolate ellipsoid with ~ 1 mm width and ~ 6 mm length, more concentrated in center at $z = 3$ mm. A very high SNR is just because of small object size. The total proton tracks of 1365-I25-B is 54,808.

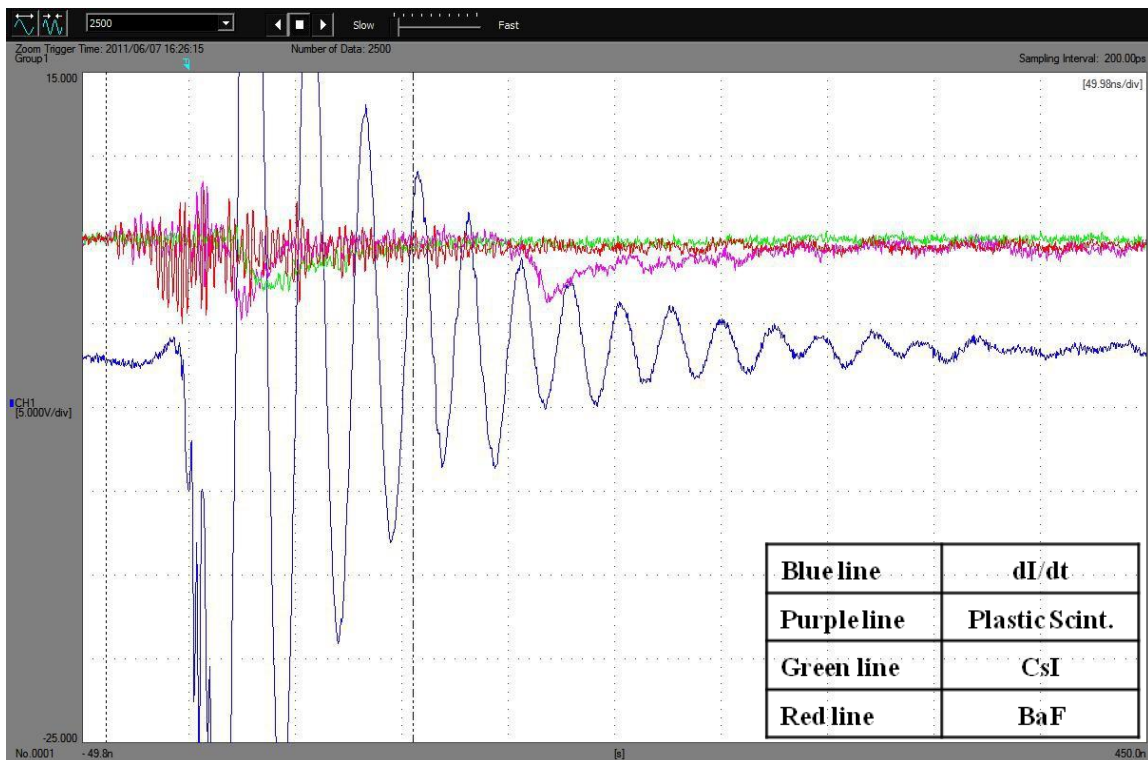
Again, the x-ray image clearly shows the existence of hot spots and/or $m = 0$ instability in final compression. The pinch column with ~ 7 mm length leans to the right.

The same as before, the lower anisotropy indicates the lower average deuteron energy. The current derivative signal shows a very strong dip ($\text{FWHM} \approx 20$ ns). For this shot, the integral of CsI hard x-ray detector, 52 nVs and the neutron anisotropy, 1.75 both are showing a low electron and deuteron average energy in pinch region.

Gas: D₂ + Kr 2% (vol.)



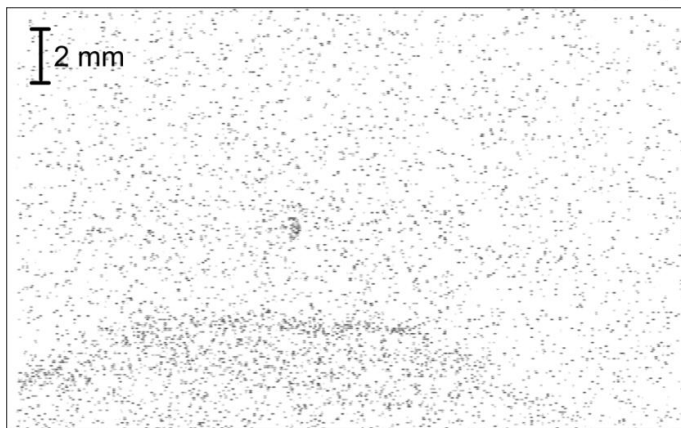
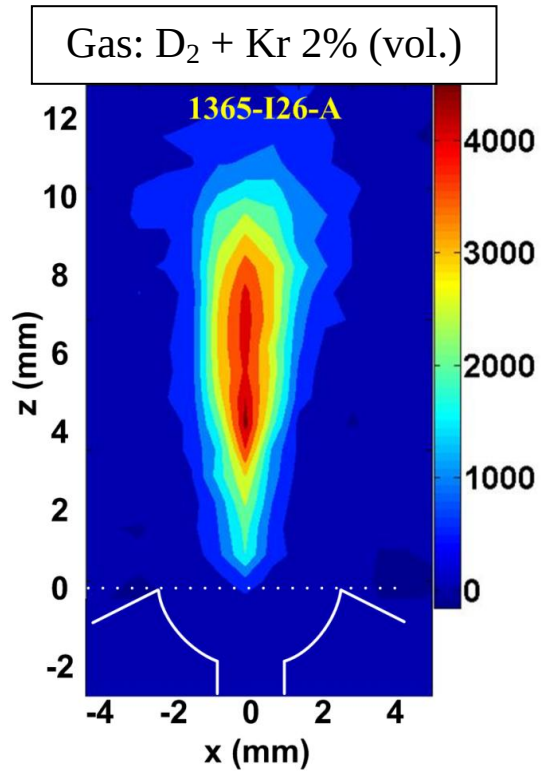
Neutron Yield	0.68×10^8
Anisotropy	1.75
HX-ray Integral (nVs)	52
SNR (1365)	43.4



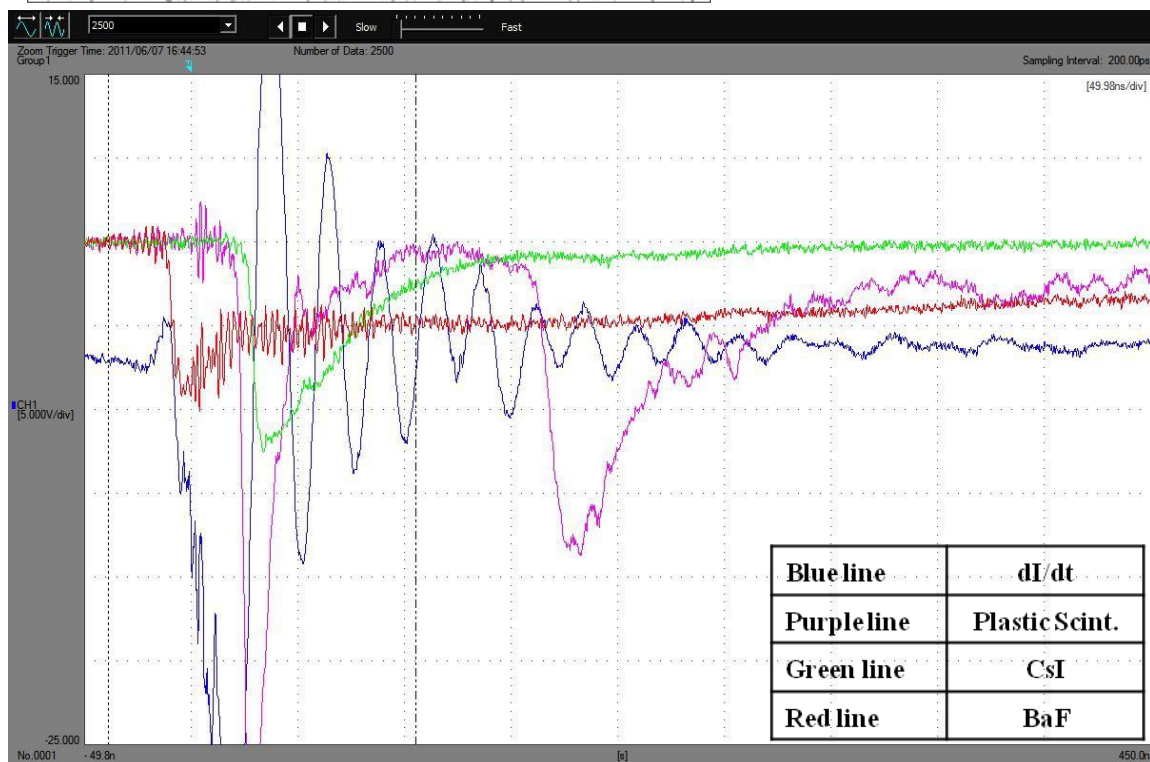
Detector 1365-I26-A was exposed with a very good shot ($Y_n = 4.17 \times 10^8$). The image looks like a prolate ellipsoid concentrated source with ~ 0.5 mm width and ~ 4 mm length in the center of an egg-shaped object with ~ 11 mm length, much wider on the top. A relatively small SNR is just because of extended object size. The total proton tracks of 1365-I26-A is 222,354.

In spite of previous x-ray images, this time there is no hot spots or even a dense plasma region (of course in higher energy range) in final compression. However, the hot plasma is distributed evenly along the z-axis which is corresponded to high neutron yield.

The same as before, the lower anisotropy indicates the lower average deuteron energy. The current derivative signal shows a very strong and wide dip ($\text{FWHM} \approx 25$ ns) which is correlated to long neutron emission time or long confinement time. The integral of CsI detector, 266 nVs shows a strong hard x-ray.



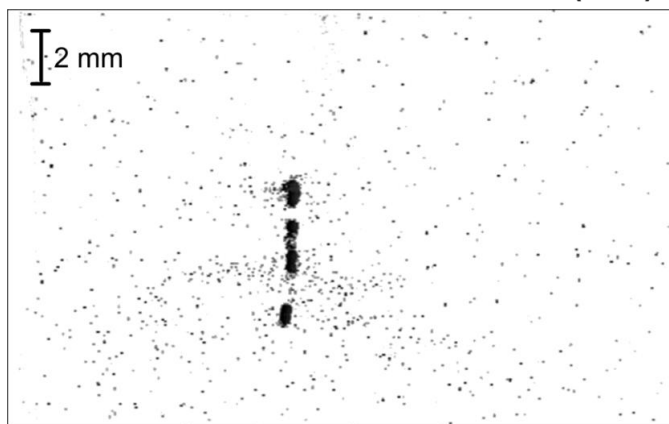
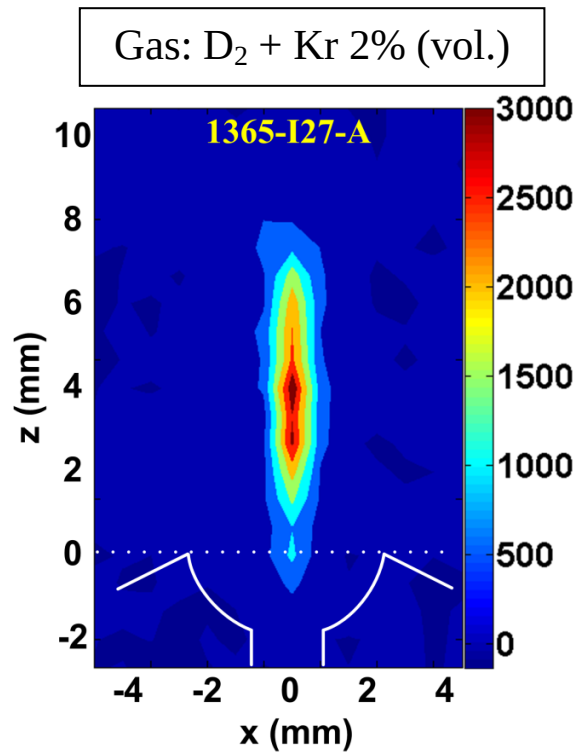
Neutron Yield	4.17×10^8
Anisotropy	1.93
HX-ray Integral (nVs)	266
SNR (1365)	23.3



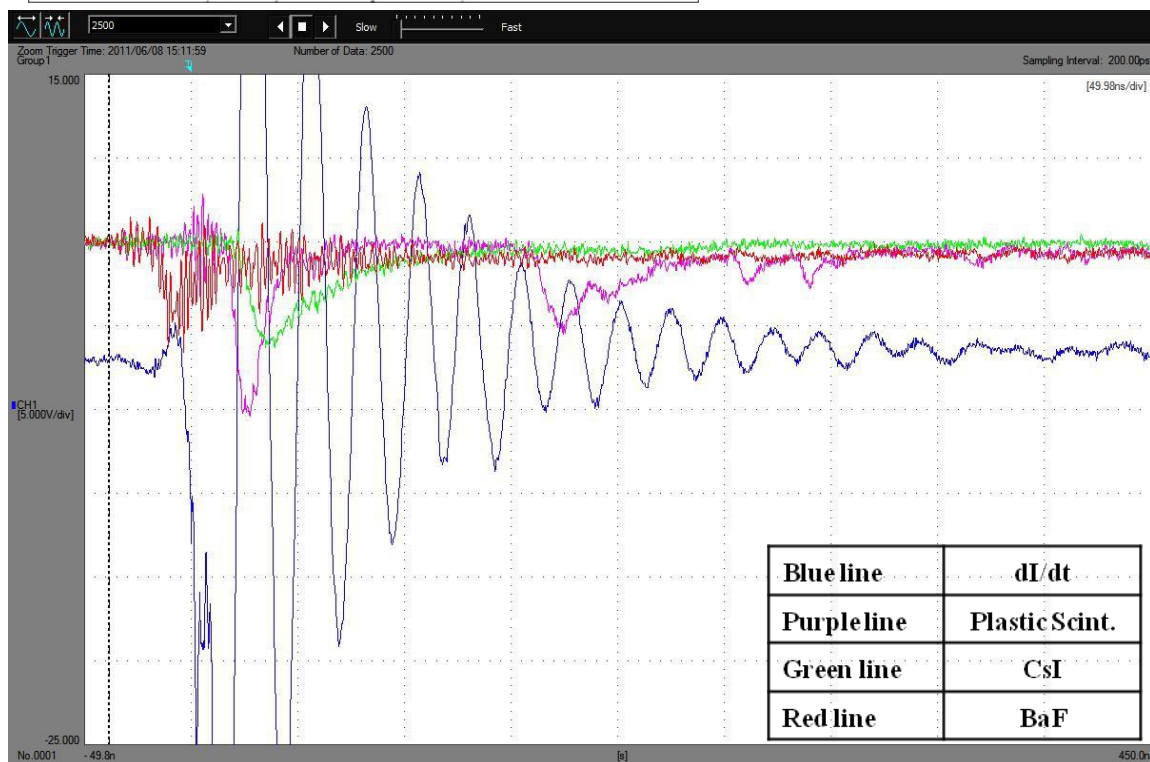
Detector 1365-I27-A was exposed with a weak shot ($Y_n = 0.98 \times 10^8$). The fusion source image shows a concentrated object, looks like a prolate ellipsoid with ~ 1 mm width and ~ 6 mm length. A very high SNR is because of small object size. The total proton tracks of 1365-I27-A is 62,248.

The x-ray image clearly shows the existing of hot spots and/or $m = 0$ instability in final compression. The pinch column length is ~ 6 mm, the same as fusion source.

The same as before, the low anisotropy, 1.75 indicates the low average deuteron energy. The current derivative signal shows a very strong dip (FWHM ≈ 20 ns). For this shot, the integral of CsI detector, 123 nVs shows a moderate hard x-ray signal.



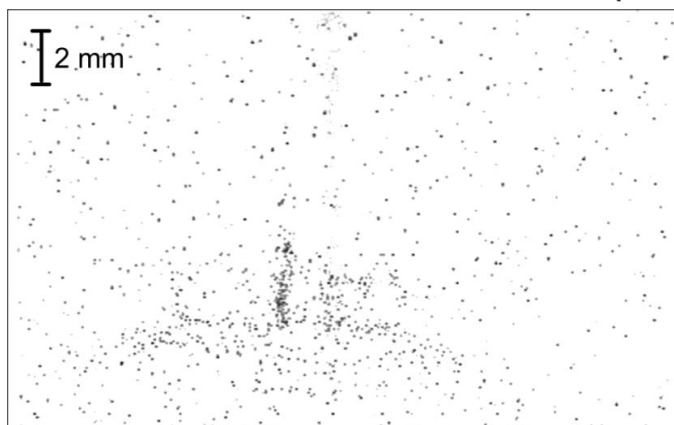
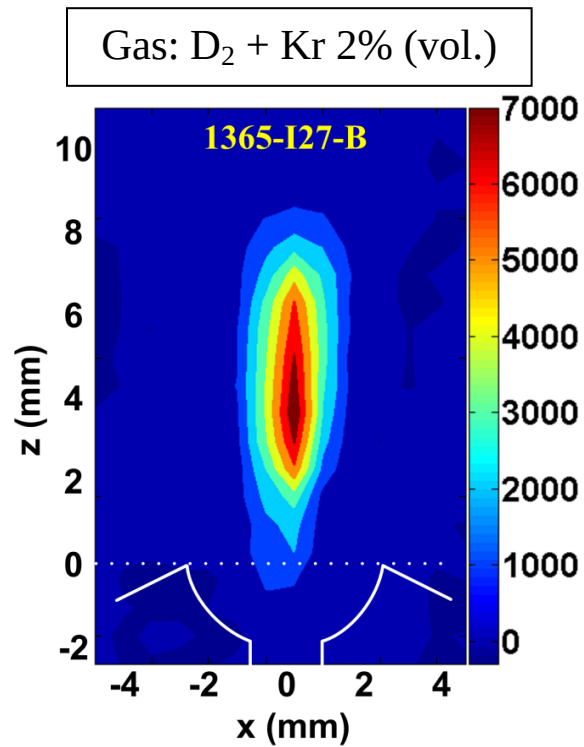
Neutron Yield	0.98×10^8
Anisotropy	1.62
HX-ray Integral (nVs)	123
SNR (1365)	35.1



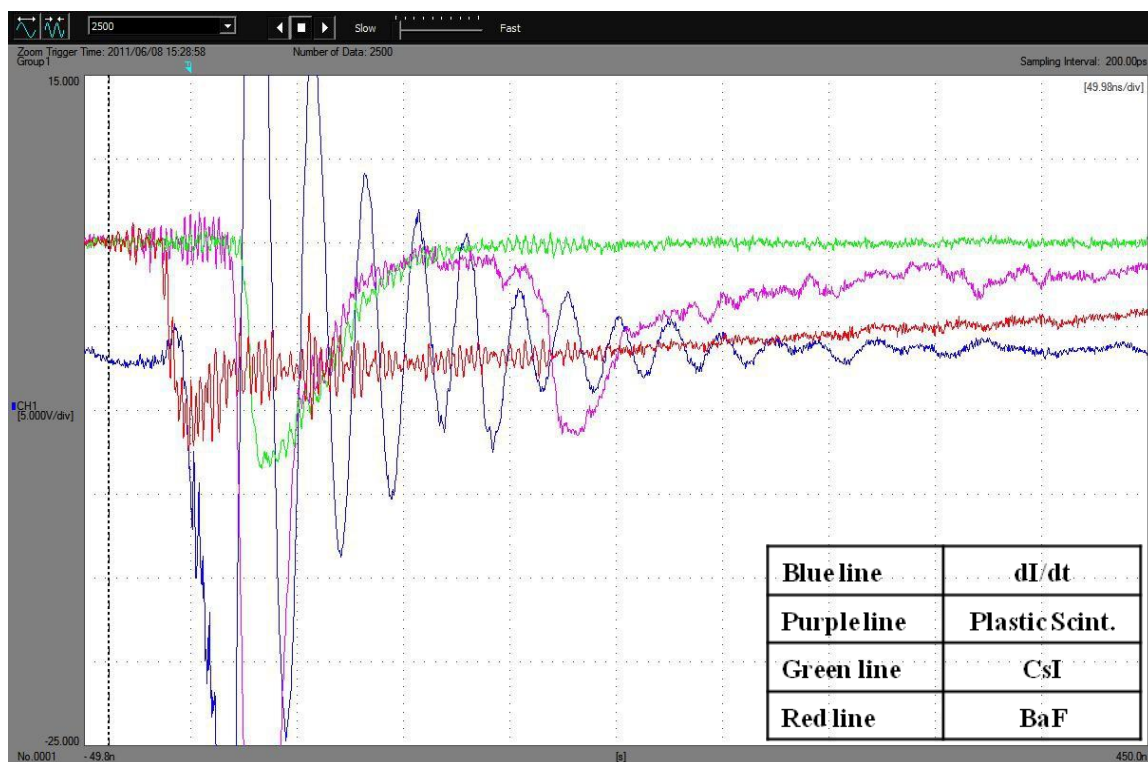
Detector 1365-I27-B was exposed with a very good shot ($Y_n = 2.94 \times 10^8$). The fusion source image looks like a nice prolate ellipsoid with ~ 2 mm width and ~ 8 mm length, more concentrated in center at $z = 4$ mm. A very high SNR is because of high neutron (proton) yield and small object size. The total proton tracks of 1365-I27-B is 190,073.

The same as detector 1365-I26-A, there is no hot spots in final compression. In x-ray image, only a low dense pinch can be seen with ~ 4 mm length. However, the hot plasma is distributed evenly in whole over the space which is corresponded to high neutron yield.

For this shot, existing relatively high neutron anisotropy (2.22) indicates high average deuteron energy in agreement with the integral of CsI detector, 235 nVs showing a strong hard x-ray. The current derivative signal shows a very strong and wide dip ($\text{FWHM} \approx 25$ ns) which is correlated to long neutron emission time or long confinement time.



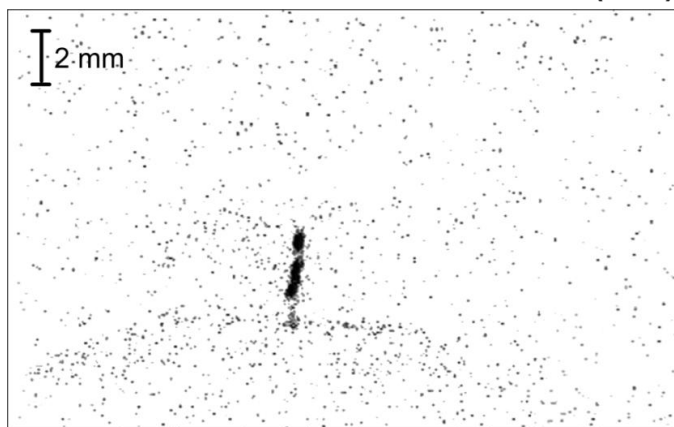
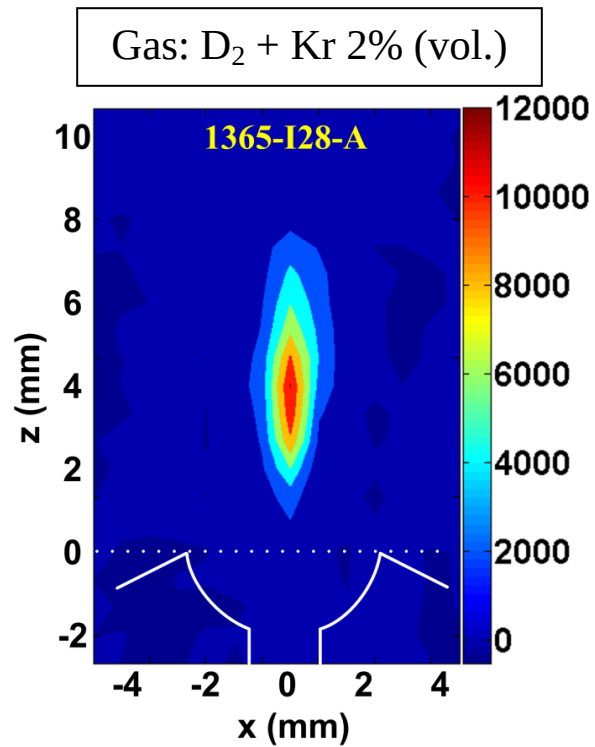
Neutron Yield	2.94×10^8
Anisotropy	2.22
HX-ray Integral (nVs)	235
SNR (1365)	37.6



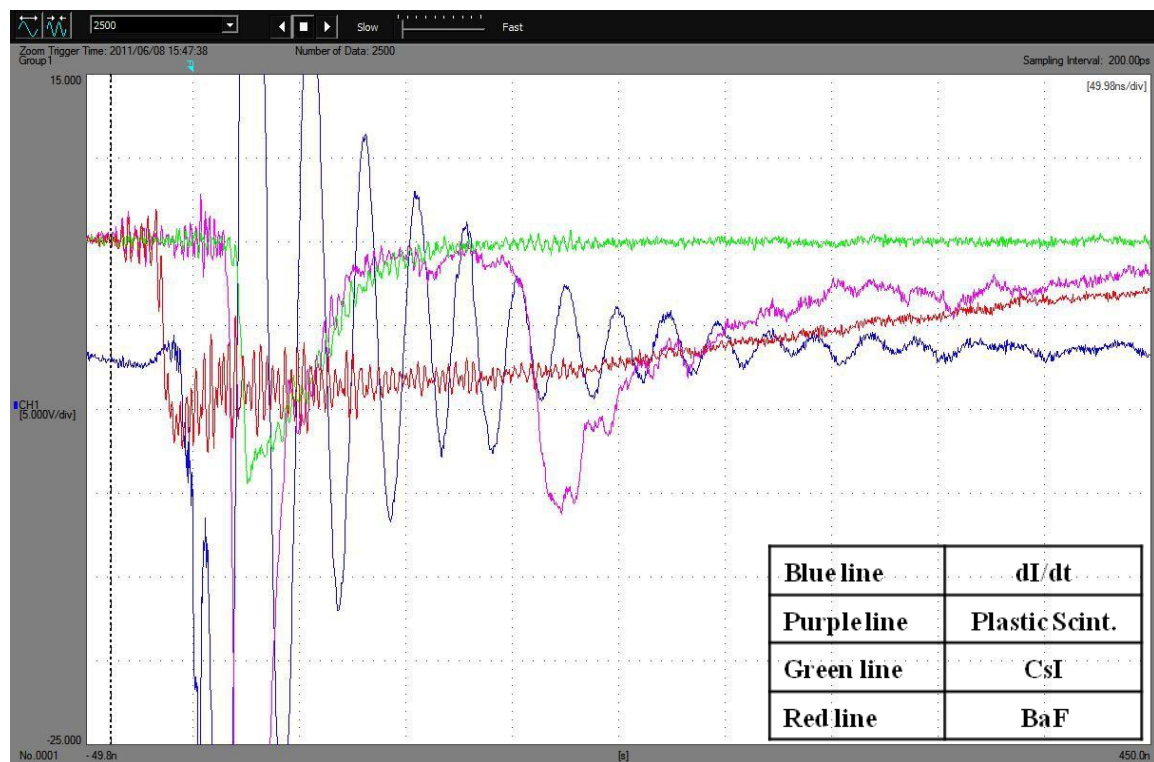
Detector 1365-I28-A was exposed with a very good shot ($Y_n = 4.05 \times 10^8$). The image looks like a small concentrated prolate ellipsoid source with ~ 1.5 mm width and ~ 6 mm length. A very large SNR is because of huge proton number and very small object size. The total proton tracks of 1365-I28-A is 187,435.

The x-ray image clearly shows the existing of high dense hot spots and/or $m = 0$ instability in final compression. The pinch column length is just ~ 3 mm. The hot plasma is also distributed evenly along the z-axis.

The same as before, the relatively high neutron anisotropy indicates the high average deuteron energy. The current derivative signal shows a very strong and wide dip (FWHM ≈ 25 ns) which is correlated to long neutron emission time or long confinement time. The integral of CsI detector, 224 nVs shows a strong hard x-ray which is correlated to high anisotropy both show the high acceleration mechanism in pinch region.



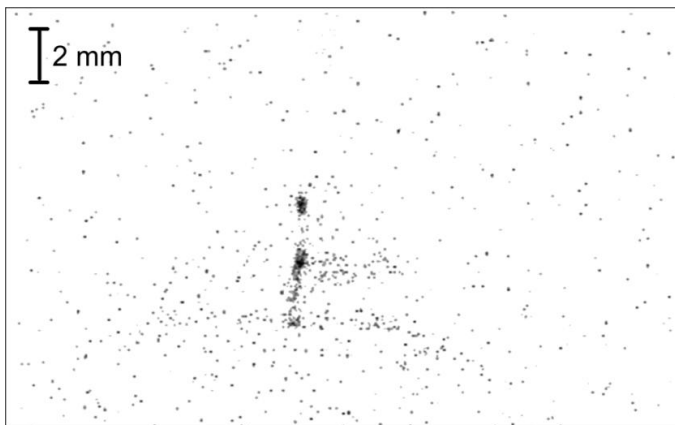
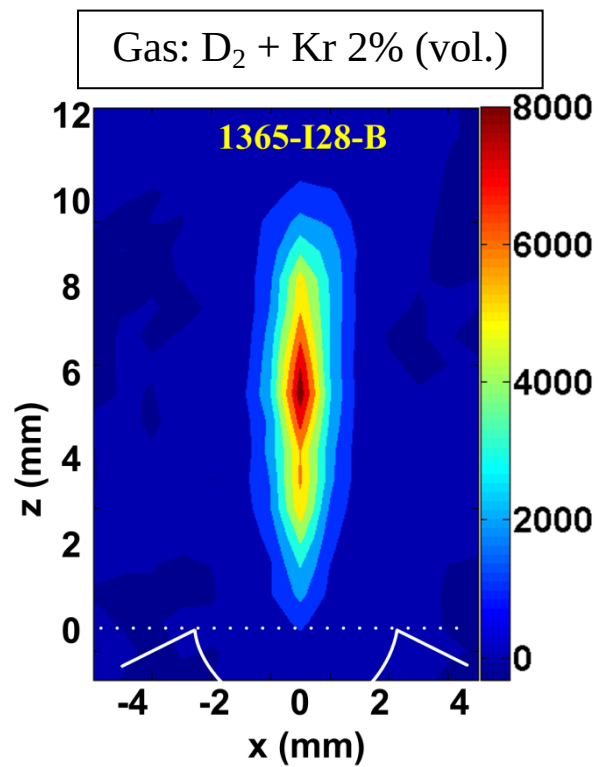
Neutron Yield	4.05×10^8
Anisotropy	2.17
HX-ray Integral (nVs)	224
SNR (1365)	50.8



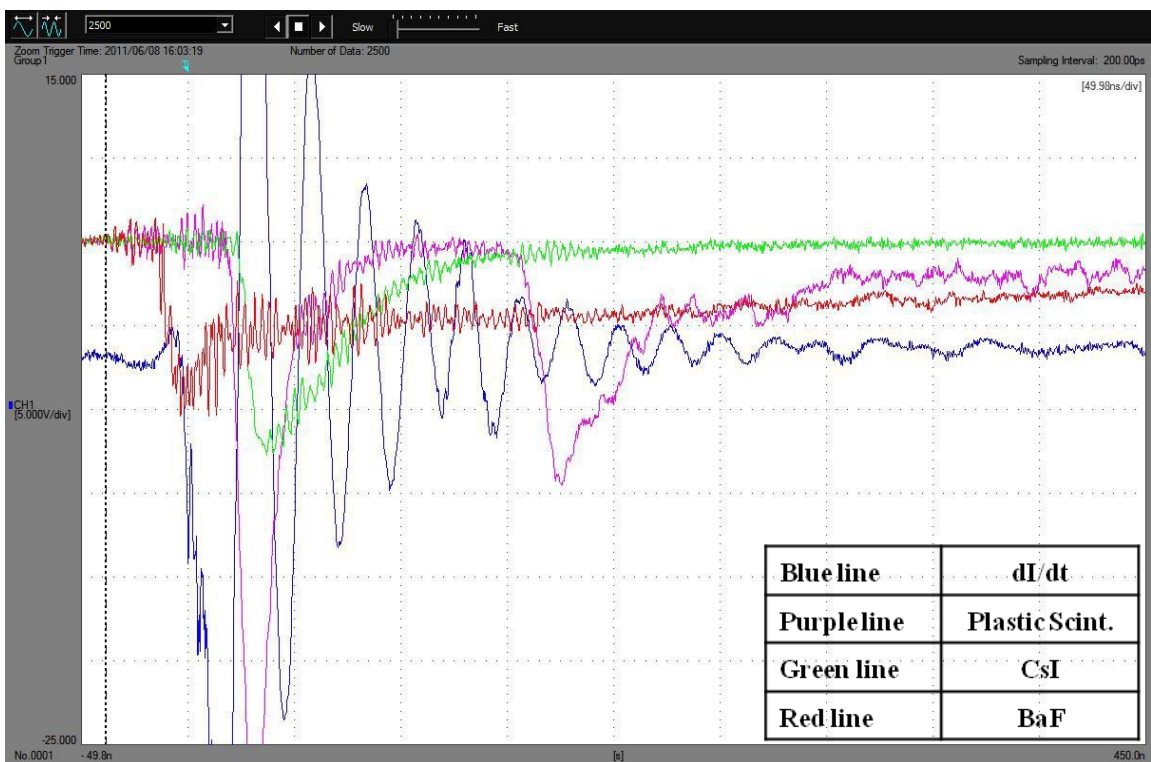
Detector 1365-I28-B was exposed with a very good shot ($Y_n = 2.92 \times 10^8$). The fusion source image looks like a nice prolate ellipsoid with ~ 2 mm width and ~ 9 mm length, more concentrated in center at $z = 6$ mm. A very high SNR is because of high neutron (proton) yield and small object size. The total proton tracks of 1365-I28-B is 183,179.

The x-ray image clearly shows the existing of dense spots and/or $m = 0$ instability in final compression. The pinch column length is ~ 5 mm.

For this shot, existing relatively high neutron anisotropy (2.14) indicates high average deuteron energy in agreement with the integral of CsI detector, 266 nVs shows a strong hard x-ray. The current derivative signal shows a very strong and wide dip (FWHM ≈ 25 ns) which is correlated to long neutron emission time or long confinement time.



Neutron Yield	2.92×10^8
Anisotropy	2.14
HX-ray Integral (nVs)	266
SNR (1365)	41.3



Blue line	dI/dt
Purple line	Plastic Scint.
Green line	CsI
Red line	BaF

Chapter 6

Discussion

6 Discussion

The main goal of this project was introducing, design and manufacturing of a new diagnostic for proton (fusion product) source imaging in small and moderate energy range of plasma focus devices, although this method can be applied for any other kind of fusion source like inertial confinement fusion (ICF). Moreover, as it is mentioned in introduction, it can be used in x-ray and γ -ray source imaging in medical and industrial applications. This new approach to decide the open area fraction by regarding to the object (source) size and its distance to mask, prepare a flexible mask parameter to reach the best resolution, precision and signal to noise ratio. In particular for medical imaging application, getting a clearness image with the minimum dosage for patient is a key point.

In plasma focus device, also we need a high resolution and high intensity fusion source image for understanding the behavior of fusion reaction and its correlation with other parameters. Since invention of plasma focus device, the fusion source imaging has been done on a few devices with pinhole imaging technique for big plasma focus machine and mostly based on neutron imaging. One of the best proton pinhole imaging experiments has been done on Poseidon (280 kJ, $Y_n \approx 10^{11}$ per shot). However it could not be done for small or medium size because of no enough protons particle to make a clear image. However, the coded aperture imaging technique solves this problem and offers a much bigger open area fraction together with a proper resolution by using a number of small pinhole (in our case, hundreds holes).

6.1 Mask design:

It is worth to emphasizing that for designing of a mask, many of parameters must be taken in account. In the following section, I will try to collect some of these parameters and conclude the best mask configuration for our application.

1. Distances: As the whole camera (mask and detector) must be put inside the vacuumed chamber, so the source to detector distance cannot be bigger than the chamber radius (in NX2, it is 9.5 cm). Quotient of source-detector distance and source-mask distance gives the camera magnification which identifies the detector pixel size and so the minimum detector physical size. So it cannot be too big, as the detector must be occupied in the chamber. Moreover, the mask should not put too close to source

which is hot plasma, firstly to avoid of damaging the mask and secondly to prevent of disturbing of plasma sheath formation between anode and cathodes. Also it should not be too small because of losing the object pixel size or resolution factor (quotient of source-detector distance and mask-detector distance). Although some other technical criteria also can affect on our decision, but in our case, mask was placed in almost in middle of source and detector. The source-mask and mask-detector distances are 4.4 cm and 4.0 cm respectively, so the magnification and resolution factor are both about 2.

2. Total open area: The total mask throughput is the most important parameter in imaging which is related to open solid angle area. According to the theoretical calculation, the minimum number of particle must be in order of 10^4 which (in our case) is a fraction of 10^{-4} out of average total 10^8 protons per shot. The first mask (mask-400, 57 square holes with 0.4 mm side) has $\sim 9 \text{ mm}^2$ open area, but two other recent masks (mask-1365, 341 square holes with 0.3 mm side and mask-4680, 585 square holes with 0.27 mm side) have ~ 30 and $\sim 42 \text{ mm}^2$ total open area respectively which predicts an order of magnitude 5 for proton tracks on each detector for a moderate shot. Furthermore by increasing the total proton tracks, signal to noise ratio (SNR) will be increased. However it cannot be selected too big because of track overlapping. A rough calculation shows the detector effective area for these two masks is $\sim 2,500 \text{ mm}^2$ and for a good shot ($Y_n = 3 \times 10^8$), the total proton tracks will be around 6×10^5 . With 15 hours etching (NaOH 6.25 n at 70°C), the average proton track diameter and area are $\sim 23 \text{ }\mu\text{m}$ and $\sim 415 \text{ }\mu\text{m}^2$ respectively and so the total tracks' area will be $\sim 250 \text{ mm}^2$ which is 10% of detector area. It means the chance of track overlapping is about 10%. By increasing the open area, the track size must be decreased by reducing the etching time. However, the suggested time for this kind of experiments is 10-20 hours to have a good discrimination between genuine particle tracks and spurious dark features.

3. Resolution: The object pixel size or resolution is an important parameter for an accurate source image investigation. Especially in plasma focus device that we are looking for a possible correlation between fusion source and bright spots and/or $m = 0$ instability. The usual pinch size in NX2 plasma focus device is a cylinder with $\sim 10 \text{ mm}$ in height and $\sim 2 \text{ mm}$ in diameter (from side-on view this will be its width). According to our x-ray imaging (especially in using of admixture gas), there are few hot spots with around 1-2 mm gap. So the resolution of imaging system must be better than 1 mm with given geometry, the mask pixel size (hole size) should not be bigger than

0.5 mm. However, the resolution and therefore hole size cannot be chosen too small, firstly because of low precision (number of passed particle) which has been discussed before and secondly to avoid the collimation affect. It needs to be explained that if the hole diameter is smaller than its thickness, it will become a collimator. In collimator, only those particles have chance to pass the hole without wall friction that come in perpendicular direction. This arrangement is completely different from pinhole imaging that usually particles are coming from a relatively big solid angle. Regarding to the mask substrate which is 50 μm HAVAR alloy or Stainless Still, the hole size must be bigger than that.

4. Open area fraction: Contrary to what was generally assumed of 50% open area fraction (ρ) for Hadamard cyclic difference set, in Singer set, ρ is flexible. The best SNR is strongly depended to choosing the optimum ρ which can be identified with the object pixel number (m). For mask-400, the open area fraction has been chosen as $\rho = 1/7 \approx 14\%$, for mask-1365, it is $\rho = 1/4 \approx 25\%$ and for mask-4680, $\rho = 1/8 \approx 12.5\%$. It must be mentioned that the availability of coded aperture with desired number of pixel and open area fraction is an important condition. For some modulo, there is very limited number of mask definition, according to lack of suitable irreducible polynomial.

5. Mask physical size: The mask size together with the object size identify the detector physical size. Regarding to the pinch plasma shape which likes a cylinder with ~ 10 mm in height and ~ 2 mm in diameter, so the optimum coded aperture shape is a spaced mask with a ratio 5:1. That is why, as it can be seen in simulation and experimental results, the mask-1365 (spaced factor $k = 6$) represents a better reconstructed image and better signal to noise ratio comparison to mask-4680 (spaced factor $k = 2$), even with smaller total open area and naturally less proton tracks. Furthermore, the physical size of detector is limited by reserved space which is identified with size of the vacuumed chamber.

So, regarding to all these parameters leads us to identify the mask definition. After the prototype mask-400, two different masks with different parameters such as prime power, modulo, irreducible polynomial, spaced mask factor and even different hole size were designed and applied for fusion imaging in plasma focus device at the same time. The final reconstructed images from both systems were compared and it was

shown that the de-convolution algorithm works well for coded aperture imaging with Singer cyclic difference set.

6.2 Simulation:

In order to check the correctness of the theoretical analysis and the decoding algorithm (image reconstruction), a Monte Carlo program was written to simulate the results to be expected for various experimental CAI arrangements. Inputs to the program include the definition of the source distribution and intensity, the coded mask pattern, and the geometry of the CAI camera. For each point in the source distribution emitted rays were then generated isotropically within a cone which fully illuminates the CAI mask. Rays impinging on the mask plane at the position of an open hole were transmitted to detector plane and the (x,y) arrival position in the detector plane was written to a file; while rays encountering opaque regions of the mask were terminated. In this way a coded image file is produced which is the equivalent of the experimental coded image file. The image decoding was then performed with the same program as for the experimental data. The number of rays emitted from the simulated source is the equivalent of anticipated experimental exposures. SNR values in these images were calculated in the same way as for the experimental images.

6.3 Detector:

Position sensitive Solid State Nuclear Track Detector (CR-39) was used to detect the proton fusion products with almost 100% efficiency. It is not sensitive at all to the visible light, UV, x-ray and γ -ray and has a very low cross-section with neutron. However all energetic charge particles (mostly deuterons in plasma focus device) can make an invisible latent track which will be miscounted as a real proton track. To avoid this, a proper filter (75 μm Kapton) was used to stop all energetic deuterons (with energy less than 3 MeV) and according to our previous experimental results, the population of deuterons with more energy is much less than the average neutron yield (and so proton yield). Also, the age of detector is a very effective parameter on its efficiency; older detector has less sensitivity with charge particles.

After exposure, the CR-39 detector is chemically etched to reveal an etch-pit or track where the energetic charged particle entered the detector. In principle, it is evident that the rate of chemical attack on the damaged trail will not be strictly constant. The

main etching parameters are: etchant chemistry, molarity, temperature and etching time. In our experiments for fusion source imaging, we have used NaOH 6.25 n solution at 70 °C for 15 hours. So for different range of proton energy (assumed ~2.6-3.4 MeV) causes a different track diameter ~20-30 μm (bigger track is corresponding to the lower proton energy).

6.4 Scanning and Analyzing:

To acquire the images of CR-39 surface, an optical microscope has been used with a $\times 20$ magnification objective lens with two different CCD cameras. For mask-400 experiments, 1.4 Mpx CCD camera was used with $528 \times 388 \mu\text{m}^2$ field of view. However, for mask-1365 and mask-4680, 4.9 Mpx CCD camera was employed with bigger field of view ($726 \times 544 \mu\text{m}^2$). It also includes a motorized stage controlled by Stage-Pro software to scan 12,000 frames for each detector. To obtain the best focusing on detector, automatic laser focusing was used by sending a laser wave to detector surface, detecting the reflected wave and finding the best focus stage. For using this option the detector surface must be highly reflective, so it needs to be coat by Indium before scanning.

An Image-Pro macro was written to automatically analyze all the stored images, and write a data file containing tracks' measured parameters (area, average brightness or gray-value, roundness and position) for each recognized dark candidate track. The tracks are counted by proper pre-criteria to avoid having a huge output file, e.g. unusual too small or big tracks or too bright tracks are considered as fake tracks and are omitted in initial analysis.

Filtering operations reduce or increase the rate of change that occurs in the intensity transitions within an image. Two of most important filters which have been used in our image analysis are Flatten and Watershed. The Flatten filter causes to even out background variations which prepare an image for a count/size operation if its objects are difficult to isolate because the background contains pixels of the same intensity as the objects of interest. Flatten reduces the intensity variations in the background pixels. The Watershed filter also causes to separate objects that are touching.

To discriminate between genuine particle tracks and spurious dark features, within-polygon criteria are applied to scatter plots of (i) brightness (gray-value) vs. area,

(i) roundness vs. area, and (iii) roundness vs. brightness. Genuine particle tracks are selected by applying the logical AND of these three criteria, and the (x,y) positions of all selected tracks (representing the coded image) are written to a new file.

Finally the coded data was then processed by our convolution-based decoding algorithm to achieve the output reconstructed file. It has been tried to correct the inaccuracy of magnification, tracks' center of mass and rotation to achieve the best Signal to Noise Ratio.

6.5 Other diagnostics:

Two sensitive beryllium activation detectors for fast neutrons (especially D-D fusion neutrons) have been constructed and then an absolute and reliable calibration has been done based on the evaluated ${}^9\text{Be}(n,\alpha){}^6\text{He}$ reaction cross-section and energy data from the ENDF database, energy calibration of the xenon-filled proportional counters using ${}^{55}\text{Fe}$ x-ray source and finally using the MCNP5 code for neutron and electron transportation. The response function of the beryllium activation detector has a maximum in ~ 3.0 MeV and for D-D thermonuclear neutrons, of 2.45 MeV energy, the response function is about 23 % below the peak value; while for energies below 1 MeV the response is negligible. This is a useful attribute of the detector as it is insensitive to the majority of room scattered neutrons. The two constructed detectors are fairly compact, enabling them to be positioned close to fusion source and its short relevant half-life of ~ 0.8 s, allows it to be used for moderate repetition rate sources (up to ~ 1 Hz). With placing them at end-on and side-on, the total neutron yield, anisotropy and average deuteron energy (involved in fusion reaction) will be calculated.

To investigate the correlation between the fusion source image with x-ray source image which illustrates the bright x-ray region (pinch), the x-ray pinhole imaging system was applied with a CCD image sensor. To prevent visible light or ultra violet radiation reaching the CCD, it is covered by $20\ \mu\text{m}$ beryllium filter (for pure deuterium operation) or $12.5\ \mu\text{m}$ titanium plus $50\ \mu\text{m}$ beryllium filter (for krypton seeding admixture operation). To synchronize the frame capturing with plasma focus discharge (exposure) time, a synchronization circuit was designed

For time resolved comparisons, three different scintillation detectors were employed simultaneously. Plastic scintillator was used for hard x-ray and neutron detection, while cesium-iodide (CsI) and barium-fluoride (BaF_2) were employed for

hard x-ray measurements. Concurrently with these, a Rogowski coil was used to measure the discharge current derivative signal. The signals were captured by a high-bandwidth digital storage oscilloscope.

6.6 Alpha source imaging:

In order to check the correctness of the theoretical analysis and the decoding algorithm and to confirm the CAI technique and its superiority over the pinhole imaging, experiments were done with different alpha source shapes with both a CAI mask and a pinhole having the same angular resolution. In this study, the mask-400 with 57 holes was fabricated and arranged within a vacuum enclosure to view a ^{226}Ra source to image 1-spot, 2-spot and ridge alpha sources. The nuclear track detector, CR-39 was used to register the ~ 5 to 8 MeV alpha particles passing through the coded mask.

It is observed from the final images that the CAI system has superior SNR and fidelity by comparison with pinhole images. The CAI system produces a much brighter and more faithful image of the source, as reflected in the considerably larger SNR for the CAI vs. pinhole image. The present measurements are sufficient to validate the CAI technique with Singer set based masks before proceeding to image fusion protons from the plasma focus device.

6.7 Fusion source imaging with mask-400:

The NX2 plasma focus device with a hollow anode was employed to investigate the spatial distribution of the fusion source with deuterium working gas. Five CAI cameras with mask-400 were used to image the source from different directions; one CAI camera is located at 0° (axial direction) and four others are at 45° “compass points”. It was anticipated that the fusion source dimension would be somewhat larger than the size of the pinch, because of gyration of medium energy deuterons (~ 20 to 50 keV) within, and around, the pinch column due to the high magnetic field. The detector (CR-3) registered the arrival positions of the ~ 3 MeV proton passing through the mask and it is covered by 75 μm Kapton film to prevent the high energy deuterons up to 3 MeV from reaching the detector.

For proton imaging it is essential to consider the curvature of proton trajectories due to the magnetic field in and around the pinch column, and its influence on the images recorded by the CAI camera. For this purpose, a program was written in

MATLAB to calculate trajectories of D-D fusion protons that impinge on the CAI mask, and to calculate the apparent source position.

After chemical etching, the positions of several 10^4 proton tracks were successfully read from the surface of each CR-39 polymer nuclear track detector by an automated scanning system. The list of track coordinates was processed by a de-convolution algorithm to produce a final image of the pinch fusion source for each CAI camera.

From the CAI images, the fusion reaction source in the NX2 device appears to be a “cigar shaped” cylindroid with a height of about 13 mm and with a maximum diameter of about 3 mm surrounding the plasma pinch column. It is interesting to observe that the size of the fusion source is larger than the pinch column itself, although the highest fusion densities are concentrated in the pinch column. It is also interesting that there is no indication of sausage or kink instabilities which are typically observed in shadographic images of PF pinches.

6.8 Fusion source imaging with spaced mask:

The most important discussion about the fusion source shape in plasma focus device goes back to its correlation with hot spots and/or $m = 0$ instability. For this investigation, the camera must be placed in 90° (side-on) to see more details of pinch column. To avoid missing some fraction of fusion source with using the hollow anode, for this series of experiments, a tapered anode was used. Because of special anode shape, the pinch column height is less than previous experiment with hollow anode, as it can be seen in x-ray images. Two different working gases (pure deuterium and $D_2+2\%Kr$ admixture) were used to consider their different shapes in fusion and x-ray sources.

6.8.1 Pure deuterium:

The NX2 plasma focus device was fired in optimum neutron emission yield (13 mbar and 11 kV/1.6 kJ). The both masks were applied in order to compare the reconstructed images from mask-1365 and mask-4680. Each detector was exposed only by one shot and the final image was compared with other diagnostics such as neutron yield detector, x-ray image, discharge current derivative, plastic scintillator, Cesium Iodide and Barium Fluoride.

As a general conclusion, it can be seen that regardless of x-ray or pinch region, neutron emission yield and other diagnostics, the fusion source in pure deuterium looks like an egg-shaped, narrower in the nearby anode tip and wider on the top, more concentrate in the center and become less where the egg-shaped source is expanding. The two most nominated instabilities of $m = 0$ (sausage instability) and $m = 1$ (kink instability) are not evident in our fusion source image or maybe they are not in vicinity of $m = 0$ places. Even, these instabilities cannot be seen in pinch x-ray region (as rule of thumb, the hot spot usually is referred to the high dense and high temperature, more than 1 keV). For this anode geometry, the size of fusion source is $\sim 2\text{-}4$ mm wide and $\sim 8\text{-}12$ mm length in comparison to x-ray (in range of 0.7 keV-1 keV) images size with $\sim 1\text{-}2$ mm wide and $\sim 4\text{-}8$ mm length. It clearly shows the gyration of high energy deuterons (~ 50 keV) inside the pinch current magnetic field and fusion inside and outside of pinch. There is higher chance for collision inside the high dense plasma and less probability for deuterons to escape the pinch and fuse outside. Moreover, it indicates the anisotropy of fusion reaction because of high energy deuteron movement along a down-stream cone which proves the beam target mechanism for fusion reaction in plasma focus device.

Furthermore, regarding to performance of 50 shots without fusion imaging system, a relative direct correlation can be seen between neutron yield, neutron anisotropy and hard x-ray signal. This can be justified by giving the same mechanism for deuteron and electron acceleration: higher energy deuterons are gyrating less inside the current magnetic field and move more in end-on direction and causes higher anisotropy; the high energy electrons also produce more hard x-ray via their movement in plasma and Bremsstrahlung emission.

6.8.2 D₂+2%Kr Admixture gas:

For operation with admixture gas, krypton ($z=84$) was added to deuterium (with Kr volumetric ratio of 2%) at the same as previous condition (13 mbar and 11 kV/1.6 kJ). It was unexpected that with increasing the total mass by $\sim 50\%$, the quarter period time ($T/4$) increases only by 5% (~ 50 ns). For these experiments, only the mask-1365 was applied for fusion source imaging. The same as it was done for pure deuterium, each detector was exposed by one shot and the final image was compared with other diagnostics.

The only change was using of different filter for x-ray image. To avoid saturating of x-ray image, 20 μm beryllium was replaced by the combination of 50 μm beryllium and 12.5 μm titanium which allows the x-ray with energy of bigger than ~ 1 keV pass through the filter. In contrary of pure deuterium, in admixture gas operation, the images clearly show the existing of hot spots and/or $m = 0$ instability in final compression. Moreover it shows the high emission rate of x-ray which allows the plasma column becomes more compressed (~ 1 mm wide and ~ 3 -6 mm length of x-ray image size in energy range of bigger than 1 keV).

The fusion source shape looks like a prolate ellipsoid, more concentrate in the center with the average size of ~ 1 -2 mm wide and ~ 3 -6 mm length, the same size as x-ray image. However, in spite of existing of $m = 0$ instability in x-ray images, there is no separated dense points in fusion source. The lower anisotropy indicates the lower average deuteron energy which naturally agrees with smaller fusion source size, concentrated inside the pinch region. Another reason of such a small source goes back to this reality that krypton ions are heavier than deuterons and so move behind them in axial acceleration phase and radial compression. So that, the deuterons somehow are surrounded by krypton ions causes of existing of smaller volume for their fusion reaction.

Moreover, regarding to performance of 50 shots without fusion imaging system, there is very weak correlation between neutron yield, neutron anisotropy and hard x-ray signal. A very high x-ray yield in admixture gas operation is because of higher probability of collision between electrons and bigger krypton nuclei and so more Bremsstrahlung emission. Also, there is a good agreement between lower average neutron yield (lower deuteron energy has lesser cross section) and lower hard x-ray emission (lower electron energy).

6.9 Final conclusion and suggested future work:

The purpose of this thesis was to introduce a new technique for fusion source imaging in low yield devices where the pinhole imaging won't give a precise image. With this sense, Coded Aperture Imaging has been applied for NX2 plasma focus device. The final reconstructed images show the ability of this technique for proton imaging. Getting the better SNR by using of Singer Cyclic Difference Set rather than the more usual Hadamard set is the strength point of this project.

So far, it can be said that the fusion source images of NX2 plasma focus device look like similar as Poseidon. Although Jager used pinhole imaging (with the size of 1.5 mm and 3 mm at 220 mm away) which did not suffers so much from displacement of proton trajectory, however in both results and images, $m = 0$ or $m = 1$ instabilities are not evident or maybe they are not in vicinity of such points. Although still it is too soon to conclude this, however, it seems that the other models which the ion acceleration are not based on $m = 0$ instability such as surfing model, plasma diode and double-layer pulse current model can be applied for our machine.

To understand of plasma fusion source production and ion acceleration in plasma focus device needs more effort and more experiment which some of them can be advised here:

1. Shadowgraphy of current sheath must be done to understand and calculate the magnetic field distribution which is needed for proton trajectory and smearing the fusion source image in z-direction.
2. The actual amount of current passing through the pinch must be measured which is not easy for small device like NX2. In big PF devices such as PF1000, a small magnetic coil has been installed just on top of the anode to measure the pinch current.
3. As a fraction of source might be inside the pinch and affected by magnetic trajectory, the program must be written carefully to calculate the real source position.
4. The operation with admixture gas can be done with different heavy gases such as Argon, Neon or Xenon and also with different volumetric ratio.
5. The same imaging can be done on other plasma focus devices or even other high energy proton (fusion) source such as z-pinch or inertial confinement fusion (ICF) systems.

Appendices

Appendix A:

Macro to analyze the proton tracks by using the filters such as flattening, watershed, etc. and record the (x,y) track position, area, mean gray density and roundness.

Language: Visual basic

Environment: Image-Pro (Media Cybernetics)

```
=====

' 7 Jan. 2011
'ali-macro-6 for new Camera (bigger frame), asks the input folder for scanning and
output file name to record the data.
'it gets mean density and put 70% of this as mean grey value to apply for manually
setting range
' Default Script
Option Explicit
Sub Mask1365_Ali()
'F8 : Shortcut key!
' This Macro opens all images of a detector and Calculate the absolute X & Y of Tracks
in each of them with some conditions.
    Dim nFile As Integer 'Number of files
    Dim DStat As Integer
    Dim strImageFile As String *500
    Dim strFileName As String *500
    Dim countFileName As String * 500
    Dim Iname As String *250
    Dim Xpos_Frame As Double 'X poistion of Frame
    Dim Ypos_Frame As Double 'Y poistion of Frame
    Dim posXdoc As IPDOCPOS, posYdoc As IPDOCPOS
    Dim Xpos_Frame_Zero As Double 'X poistion of first Frame
    Dim Ypos_Frame_Zero As Double 'Y poistion of first Frame
    Dim X_pix_um As Single 'X of pixel in microns
    Dim Y_pix_um As Single 'Y of pixel in microns
    Dim X_pix_origin As Single
    Dim Y_pix_origin As Single
    ' Close all images/documents that are open presently
    ret = IpAppCloseAll()
    ret=IpMacroPause("Enter OUTPUT filename",MS_EXCLAM,2000)
    ret=IpStGetName("Convolved Data Output file","E:\Ali-CR39
Images\ProcessedScan\","*.cnt",countFileName)
    " Debug.Print countFileName
```

```

' allow user to select direcorry containing the images
ret=IpMacroPause("Enter DIRECTORY for Images",MS_EXCLAM,2000)
ret = IpStGetName("DIRECTORY for Images ", "E:\Ali-CR39
Images\","*.TIF",strFileName )
    Debug.Print strFileName
' turn OFF graphic updating
' ret=IpAppUpdateDoc(DOCSEL_NONE)
' load the first image
nFile=0
DStat = IpStSearchDir(strFileName,"*.TIF",nFile,strImageFile)
' while images are being succesfully loaded => loop through each image in
directory
Do While DStat = 1
    ret = IpWsLoad(strImageFile, "TIF")
    " Debug.Print strImageFile
    ' flatten this image (before performing count/measurement)
    Flatten_This()
' Get the (x,y) Offset for this Frame
' Get the (x,y) Offset for this Frame
    ret = IpDocGetPosition( DOCSEL_ACTIVE, INF_XPOSITION, 0, posXdoc)
    ret = IpDocGetPosition( DOCSEL_ACTIVE, INF_YPOSITION, 0, posYdoc)
    Xpos_Frame=posXdoc.Position
    Ypos_Frame=posYdoc.Position
' Store the original/beginning (x,y) offset
    If nFile=0 Then
        Xpos_Frame_Zero = Xpos_Frame
        Ypos_Frame_Zero = Ypos_Frame
        ' id_scal=IpSCalSetLong(0,SCAL_APPLY_RESOLUTION,0)
        IpSCalGetSng(SCAL_CURRENT_CAL,SCAL_SCALE_X,X_pix_um)
        IpSCalGetSng(SCAL_CURRENT_CAL,SCAL_SCALE_Y,Y_pix_um)
    End If
' Calculate the reduced (x,y)offset for current frame (units are um)
    Xpos_Frame = Xpos_Frame - Xpos_Frame_Zero
    Ypos_Frame = Ypos_Frame - Ypos_Frame_Zero
' Calculate Stage-Scan origin for this frame
    X_pix_origin = -X_pix_um * Xpos_Frame
    Y_pix_origin = -Y_pix_um * Ypos_Frame
' And Set the ORIGIN
    ret = IpSCalSetOrigin(X_pix_origin, Y_pix_origin)
    ret = IpSCalSetLong(SCAL_CURRENT_CAL, SCAL_APPLY, 0)
' Process the current Image
    Areas_Centroids()
' Write extracted measurement info to output file
    ret = IpBlbSaveData(countFileName, S_APPEND)
' now close current image/document
    ret = IpDocClose()
    Rem load the next (n'th) file as the current image/document
    nFile =nFile + 1
'DStat = IpStSearchDir("E:\Ali-CR39 Images\Test","*.TIF",nFile,strImageFile)
    DStat = IpStSearchDir(strFileName,"*.TIF",nFile,strImageFile)

```

```

Loop
Debug.Print nFile
' turn ON graphic updating
ret=IpAppUpdateDoc(DOCSEL_ALL)
End Sub
Sub Areas_Centroids()
    Dim stats(10) As Single
    Dim darkobj_gv As Single
    Dim micro_area As Single
    Flatten_This() '** Flatten
    ret = IpBlbEnableMeas(BLBM_ALL, 0)
    ret = IpBlbEnableMeas(BLBM_AREA, 1)          'for activating of Area filter!
    'ret = IpBlbSetFilterRange(BLBM_AREA, 124.38, 10*124.38)
    '10<Area<100      micro_area=12.43742285
    ret = IpBlbSetFilterRange(BLBM_AREA, 50*micro_area, 1000*micro_area)
    '50<Area<1000      !!!!! must be changed
    ret = IpBlbEnableMeas(BLBM_CENTRX, 1)
    ret = IpBlbEnableMeas(BLBM_CENTRY, 1)
    ret = IpBlbEnableMeas(BLBM_DENSITY, 1)
    ret = IpBlbSetAttr(BLOB_BRIGHTOBJ, 0)
    ret = IpBlbSetAttr(BLOB_AUTORANGE, 0)      'set the range of Gray value
manually.
    ret=IpHstGet(GETSTATS, 0, stats(0))        ' get the histogram statistics
    darkobj_gv=stats(0)*0.7                    ' 70% of mean grey value for whole
image
    ret = IpSegSetRange(0, 0, darkobj_gv)
    ret = IpBlbEnableMeas(BLBM_ROUNDNESS, 1)    'for activating of Roundness
filter!
    'ret = IpBlbSetFilterRange(BLBM_ROUNDNESS, 1.0, 1.2)
    '1<Roundness<1.2
    ret = IpBlbSetAttr(BLOB_FILLHOLES,1) 'fill holes in option
    ret = IpBlbSetAttr(BLOB_DISPLAY,1)
    ret = IpBlbSetAttr(BLOB_FILTEROBJECTS, 0)  ' measurement filters OFF
    ret = IpBlbCount()                        ' count the dark objects
    ret = IpBlbSetAttr(BLOB_FILTEROBJECTS, 1)  ' measurement filters ON
    ret = IpBlbSplitObjects(1)                'watershed split must be put after Count
!!!!
End Sub
' routine to flatten the current image by grey-scale dilation and smoothing
Sub Flatten_This()
    Dim id_First As Integer
    Dim id_Smoothed As Integer
    Dim stats(10) As Single
    Dim mean_gv As Single
    ret = IpDocGet(GETACTDOC, 0, id_First) ' get current image ID
    id_Smoothed = IpWsDuplicate()          ' get working-image ID
    ret = IpFltMedian(3, 1) ' median filter eliminates the one bright pixel for Scion
camera
    ret = IpFltDilate(MORPHO_11x11OCTAGON, 5) ' hefty dilation
    ret = IpLSFltApply(LF_LOPASS,51,51,1,1) ' large filter smoothing

```

```
ret = IpHstGet(GETSTATS, 0, stats(0))
mean_gv=stats(0) ' obtain mean grey value for image histogram
ret = IpAppSelectDoc(id_First) ' select original image
ret = IpOpImageArithmetics(id_Smoothed, mean_gv, OPA_DIV, 0) 'flatten the
image
Rem Delete the working-image
ret = IpAppSelectDoc(id_Smoothed)
ret = IpDocClose()
ret = IpAppSelectDoc(id_First) ' select original image (just in case there are
others)
End Sub
```


Appendix B:

Program to discriminate between genuine particle tracks and spurious dark features, within polygon criteria which are applied to scatter plots, allow the user to select the desired region

Language: MATLAB Macro Language

Environment: MATLAB (From Mathworks Inc.)

```
=====

% Track Data Scatter Plots
% Script to read in a CR-39 track data file, selected by the user, and
% generates scatter plots of:
% 1) grey-value vs. area
% 2) circularity vs. grey-value
% 3) circularity vs. area
% The user can then adjust the size of axes etc. before running another
% script which prompts him to select a polygon on each of the scatter
% plots for the selection of genuine tracks
% Date: 5th August 2010
% clear the Command Window, clear all variable, and close all figures
clc
clear all
close all
SecondTime=0;
% *****
% * Read in Track Data File *
% *****
% save the current working directory (so that it can be reset)
CurrWorkDir=cd;
% change it to where the data files usually are
cd 'C:\Ali-CR39 Images\ProcessedScan';
in_filename = uigetfile('*..*', 'Select Track Data File to Open...');
% load all the track data as a 2D array of name 'trackdat'
trackdata=load(in_filename);
cd(CurrWorkDir);
% *****
% * Plot three standard Scatter Plots *
% *****
% display them and get their handles
% scatterplot: Area vs. Grey-value
scatter(trackdata(:,1),trackdata(:,4),1)
hScatPlot1=gcf;
xlabel('Area')
ylabel('Grey-value')
```

```
figure;
% scatterplot: Roundness vs. Grey-value
scatter (trackdata(:,1),trackdata(:,5),1)
hScatPlot2=gcf;
ylim([0.9 2.0])
xlabel('Area')
ylabel('Roundness')
figure;
scatter (trackdata(:,4),trackdata(:,5),1)
hScatPlot3=gcf;
ylim([0.9 2.0])
xlabel('Grey-value')
ylabel('Roundness')
%subplot (2,2,1)=gcf;
disp('Scatter Plots are now displayed.')
disp(' ')
disp('You can modify the ranges of the plot axes, and then')
disp('run the Polygon_Selection routine... Step_2')
```

Appendix C:

Program to select genuine particle tracks within rectangular criteria on boundaries of each parameter (area, average brightness and roundness)

Language: Visual Basic

Environment: Microsoft Visual studio 2010 Express

=====

```
Option Explicit On
Imports System
Imports System.IO
Imports System.Collections
Module Module1
    ' Track Selection from Image_pro output file
    ' last modification date
    ' 10 January 2011
    ' Filenames ... as Public variable
    Public Track_Dir As String = "C:\Ali\Project\Coded Aperture Imaging\Monte-
Carlo\VBASIC\Data\NX2-Mask(1365-4680)\Raw-data\" ' Raw data file directory
    Public Selected_Dir As String = "C:\Ali\Project\Coded Aperture Imaging\Monte-
Carlo\VBASIC\Data\NX2-Mask(1365-4680)\Sele-----cted-data\" ' Selected Tracks file
directory
    Public CAIData_Dir As String = "C:\CAI_Process\Track_Data\" ' CAI data
directory for deconvolution
    Public Criteria_Dir As String = "C:\Ali\Project\Coded Aperture Imaging\Monte-
Carlo\VBASIC\Data\NX2-Mask(1365-4680)\Criteria\" ' criteria definition file directory
    Public Track_Input_FileName As String = Track_Dir & "Dummy1" ' Track
DATA read from CR_39 filename
    Public Selected_Extens As String = ".slct" ' Extention of Selected file
(output file)
    Public Selected_FileName As String = "" ' Selected Tracks output
filename
    Public CAIData_Extens As String = ".caid" ' Extention of CAI data
file (input for Deconvolution prog)
    Public CAIData_FileName As String = "" ' CAI data filename
    Public Criteria_Extens As String = ".crt" ' Extention of Criteria file
    Public Criteria_FileName As String = "" ' Criteria filename
    ' some variable
    Public minArea, maxArea, minBright, maxBright, minRound, maxRound As Single '
Criteria read from file
    Sub Main()
        ' display the program banner
        banner()
        ' ask the Track Input file-name for reduction
        get_filenames()
```

```

' read Criteria file
read_Crietra()
' read the Track Input-file and write the reduced file
Select_track()
' let user press a key to terminate the program
wait_user()
End Sub
'***** Subroutine to display the program banner *****
Private Sub banner()
    Console.WriteLine(" =====", vbNewLine)
    Console.WriteLine(" |      Tracks Selection      |", vbNewLine)
    Console.WriteLine(" |      from Image_pro file   |", vbNewLine)
    Console.WriteLine(" |                               |", vbNewLine)
    Console.WriteLine(" =====", vbNewLine)
    Console.WriteLine(vbNewLine, vbNewLine)
End Sub
'***** Subroutine to get the needed filenames *****
Private Sub get_filenames()
    Dim Track_FileName As String = ""
    Console.WriteLine(vbNewLine & "Input file-name (without extension): ")
    Track_FileName = Console.ReadLine() ' Track DATA read from CR_39
filename
    Track_Input_FileName = Track_Dir & Track_FileName
    Selected_FileName = Selected_Dir & Track_FileName & Selected_Extens
    CAIData_FileName = CAIData_Dir & Track_FileName & CAIData_Extens
    Criteria_FileName = Criteria_Dir & Track_FileName & Criteria_Extens
    ' does the Track Input file exist?
    If Not File.Exists(Track_Input_FileName) Then
        Console.WriteLine("{0} does not exist!", Track_Input_FileName)
        wait_user()
        End
        Return
    End If
    ' does the Criteria file exist?
    If Not File.Exists(Criteria_FileName) Then
        Console.WriteLine("{0} does not exist!", Criteria_FileName)
        wait_user()
        End
        Return
    End If
    Console.WriteLine(vbNewLine & "===== " & vbNewLine)
    Console.WriteLine("TRACK DATA Input File-name: " & Track_Input_FileName
& vbNewLine)
End Sub
'***** Subroutine to read the Criteria parameters/definition
*****
Private Sub read_Crietra()
    Dim comment As String = ""
    ' open the Criteria Defn File, read in and set values of Conditions
    FileOpen(1, Criteria_FileName, OpenMode.Input)

```

```

' read and ignore a couple of comment/line-header lines
Input(1, comment)
Input(1, comment)
' read the maximum and minimum of Area
Input(1, minArea)
Input(1, maxArea)
' read and ignore a line-header line
Input(1, comment)
' read the maximum and minimum of Brightness (Mean Density)
Input(1, minBright)
Input(1, maxBright)
" read and ignore a line-header line
Input(1, comment)
' read the maximum and minimum of Roundness
Input(1, minRound)
Input(1, maxRound)
FileClose(1)
End Sub
'***** Subroutine to Select the Tracks with Critria from Image_Pro file *****
Private Sub Select_track()
' variables
Dim Number_of_Tracks As Single = 0
Dim Selected_Tracks As Single = 0
Dim Area, xTrack, yTrack, MeanDensity, Roundness As Single
' open the Input Track and check their parameter,
FileOpen(2, Track_Input_FileName, OpenMode.Input)
FileOpen(3, Selected_FileName, OpenMode.Output)
FileOpen(4, CAIData_FileName, OpenMode.Output)
Do Until EOF(2)
    Input(2, Area)
    Input(2, xTrack)
    Input(2, yTrack)
    Input(2, MeanDensity)
    Input(2, Roundness)
    If (Area > minArea And Area < maxArea And MeanDensity > minBright And
MeanDensity < maxBright And Roundness > minRound And Roundness < maxRound)
Then
        ' the Track is accepted, so print it in Selected file
        Print(3, Area, TAB(), xTrack, TAB(), yTrack, TAB(), MeanDensity, TAB(),
Roundness)
        Print(3, vbNewLine)
        ' the Track is accepted, so print x & y in CAI Data file
        ' as the detector (x,y) is Scanning (y,x) so we have exchanged them !!!!
        Print(4, yTrack, TAB(), xTrack)
        Print(4, vbNewLine)
        Selected_Tracks += 1
    End If
    Number_of_Tracks += 1
Loop
FileClose(2)

```

```

FileClose(3)
FileClose(4)
' write summary data to the console
Console.WriteLine("Number of Tracks in Data File= " & Number_of_Tracks,
vbNewLine)
Console.WriteLine("Number of Selected Tracks= " & Selected_Tracks,
vbNewLine)
Console.WriteLine(vbNewLine)
End Sub
'***** Subroutine to wait press a key *****
Private Sub wait_user()
Dim cki As ConsoleKeyInfo
Console.WriteLine(vbNewLine & vbNewLine & "Press ANY key to terminate" &
vbNewLine)
cki = Console.ReadKey()
End Sub
End Module

```

Appendix D:

Program to reduce number of tracks to be opened in Excel file

Language: Visual Basic

Environment: Microsoft Visual studio 2010 Express

```

=====

Option Explicit On
Imports System
Imports System.IO
Imports System.Collections
Module Module1
    ' =====
    ' | Track Number Reducing from Image_pro output file |
    ' =====
    ' last modification date
    ' 10 January 2011
    ' Filenames ... as Public variable
    Public Track_Dir As String = "C:\Ali\Project\Coded Aperture Imaging\Monte-
Carlo\VBasic\Data\NX2-Mask(1365-4680)\Raw-data\" ' Raw data file directory
    Public Selected_Dir As String = "C:\Ali\Project\Coded Aperture Imaging\Monte-
Carlo\VBasic\Data\NX2-Mask(1365-4680)\Selected-data\" ' Selected Tracks file
    directory
    Public CAIData_Dir As String = "C:\CAI_Process\Track_Data\" ' CAI data
    directory for deconvolution
    Public Criteria_Dir As String = "C:\Ali\Project\Coded Aperture Imaging\Monte-
Carlo\VBasic\Data\NX2-Mask(1365-4680)\Criteria\" ' criteria definition file directory
    Public Track_Input_FileName As String = Track_Dir & "Dummy1" ' Track
    DATA read from CR_39 filename
    Public Selected_Extens As String = ".slct" ' Extention of Selected file
    (output file)
    Public Selected_FileName As String = "" ' Selected Tracks output
    filename
    Public CAIData_Extens As String = ".caid" ' Extention of CAI data
    file (input for Deconvolution prog)
    Public CAIData_FileName As String = "" ' CAI data filename
    Public Criteria_Extens As String = ".crt" ' Extention of Criteria file
    Public Criteria_FileName As String = "" ' Criteria filename
    ' some variable
    Public minArea, maxArea, minBright, maxBright, minRound, maxRound As Single '
    Criteria read from file
    Sub Main()
        ' display the program banner
        banner()
        ' ask the Track Input file-name for reduction
        get_filenames()

```

```

' read Criteria file
read_Crietra()
' read the Track Input-file and write the reduced file
Select_track()
' let user press a key to terminate the program
wait_user()
End Sub
'***** Subroutine to display the program banner *****
Private Sub banner()
    Console.WriteLine(" =====", vbNewLine)
    Console.WriteLine(" |      Tracks Selection      |", vbNewLine)
    Console.WriteLine(" |      from Image_pro file   |", vbNewLine)
    Console.WriteLine(" |                               |", vbNewLine)
    Console.WriteLine(" =====", vbNewLine)
    Console.WriteLine(vbNewLine, vbNewLine)
End Sub
'***** Subroutine to get the needed filenames *****
Private Sub get_filenames()
    Dim Track_FileName As String = ""
    Console.WriteLine(vbNewLine & "Input file-name (without extension): ")
    Track_FileName = Console.ReadLine() ' Track DATA read from CR_39
filename
    Track_Input_FileName = Track_Dir & Track_FileName
    Selected_FileName = Selected_Dir & Track_FileName & Selected_Extens
    CAIData_FileName = CAIData_Dir & Track_FileName & CAIData_Extens
    Criteria_FileName = Criteria_Dir & Track_FileName & Criteria_Extens
    ' does the Track Input file exist?
    If Not File.Exists(Track_Input_FileName) Then
        Console.WriteLine("{0} does not exist!", Track_Input_FileName)
        wait_user()
        End
        Return
    End If
    ' does the Criteria file exist?
    If Not File.Exists(Criteria_FileName) Then
        Console.WriteLine("{0} does not exist!", Criteria_FileName)
        wait_user()
        End
        Return
    End If
    Console.WriteLine(vbNewLine & "===== " & vbNewLine)
    Console.WriteLine("TRACK DATA Input File-name: " & Track_Input_FileName
& vbNewLine)
End Sub
'***** Subroutine to read the Criteria parameters/definition
*****
Private Sub read_Crietra()
    Dim comment As String = ""
    ' open the Criteria Defn File, read in and set values of Conditions
    FileOpen(1, Criteria_FileName, OpenMode.Input)

```



```

' read and ignore a couple of comment/line-header lines
Input(1, comment)
Input(1, comment)
' read the maximum and minimum of Area
Input(1, minArea)
Input(1, maxArea)
' read and ignore a line-header line
Input(1, comment)
' read the maximum and minimum of Brightness (Mean Density)
Input(1, minBright)
Input(1, maxBright)
" read and ignore a line-header line
Input(1, comment)
' read the maximum and minimum of Roundness
Input(1, minRound)
Input(1, maxRound)
FileClose(1)
End Sub
'***** Subroutine to Select the Tracks with Critria from Image_Pro file
*****

Private Sub Select_track()
' variables
Dim Number_of_Tracks As Single = 0
Dim Selected_Tracks As Single = 0
Dim Area, xTrack, yTrack, MeanDensity, Roundness As Single
' open the Input Track and check their parameter,
FileOpen(2, Track_Input_FileName, OpenMode.Input)
FileOpen(3, Selected_FileName, OpenMode.Output)
FileOpen(4, CAIData_FileName, OpenMode.Output)
Do Until EOF(2)
    Input(2, Area)
    Input(2, xTrack)
    Input(2, yTrack)
    Input(2, MeanDensity)
    Input(2, Roundness)
    If (Area > minArea And Area < maxArea And MeanDensity > minBright And
MeanDensity < maxBright And Roundness > minRound And Roundness < maxRound)
Then
        ' the Track is accepted, so print it in Selected file
        Print(3, Area, TAB(), xTrack, TAB(), yTrack, TAB(), MeanDensity, TAB(),
Roundness)
        Print(3, vbNewLine)
        ' the Track is accepted, so print x & y in CAI Data file
        ' as the detector (x,y) is Scanning (y,x) so we have exchanged them !!!!
        Print(4, yTrack, TAB(), xTrack)
        Print(4, vbNewLine)
        Selected_Tracks += 1
    End If
    Number_of_Tracks += 1
Loop

```

```

FileClose(2)
FileClose(3)
FileClose(4)
' write summary data to the console
Console.WriteLine("Number of Tracks in Data File= " & Number_of_Tracks,
vbNewLine)
Console.WriteLine("Number of Selected Tracks= " & Selected_Tracks,
vbNewLine)
Console.WriteLine(vbNewLine)
End Sub

'***** Subroutine to wait press a key *****
Private Sub wait_user()
Dim cki As ConsoleKeyInfo
Console.WriteLine(vbNewLine & vbNewLine & "Press ANY key to terminate" &
vbNewLine)
cki = Console.ReadKey()
End Sub
End Module

```

Appendix E:

Program to reconstruct the image (de-convolution) with correction of rotation, magnification and center of mass to achieve the best Signal to Noise Ratio (SNR)

Language: Visual Basic

Environment: Microsoft Visual studio 2010 Express

```

=====

' =====
' |   Coded Aperture Deconvolution   |
' |       with Optimization of       |
' |   Signal-to-Noise Ratio (SNR)   |
' =====
' Version 18
' Last modification date & time
' 1st June 2011
' Filenames ... as Public variable
Public CAI_Dir As String = "C:\CAI_Process\"
Public Experiment_Dir As String = CAI_Dir & "Experiment\"    ' EXPERIMENT
definition directory
Public Coded_Mask_Dir As String = CAI_Dir & "Masks\"
Public Track_Data_Dir As String = CAI_Dir & "Track_Data\"    ' Track DATA
read from CR_39 directory
Public Convoluted_Dir As String = CAI_Dir & "Conv_Data\"
Public Deconvoluted_Dir As String = CAI_Dir & "Dcnv_Data\"
Public Logs_Dir As String = CAI_Dir & "Log_Files\"
Public Params_Dir As String = CAI_Dir & "Parameters\"
Public Experiment_Extens As String = ".caix"    ' EXPERIMENT definition
directory
Public Coded_Mask_Extens As String = ".cam"
Public Track_Data_Extens As String = ".caid"    ' Track DATA read from CR_39
directory
Public Convoluted_Extens As String = ".conv"
Public Deconvoluted_Extens As String = ".dat"
Public Logs_Extens As String = ".calg"
Public Params_Extens As String = ".par"
Public Previous_Filename As String = CAI_Dir & "Previous.txt"
Public Experiment_FileName As String = Experiment_Dir & "Dummy1" &
Experiment_Extens    ' EXPERIMENT definition filename
Public Coded_Mask_FileName As String = CAI_Dir & "Dummy2" &
Coded_Mask_Extens
Public Track_Data_FileName As String = Track_Data_Dir & "Dummy3" &
Track_Data_Extens    ' Track DATA read from CR_39 filename

```

```

Public Convoluted_Output_FileName As String = Convoluted_Dir & "Dummy4" &
Convoluted_Extens ' output CONVOLUTED/coded array filename
Public Deconvoluted_Output_FileName As String = Deconvoluted_Dir &
"Dummy5" & Deconvoluted_Extens ' output DE-CONVOLUTED array/image
filename
Public Logs_Output_FileName As String = Logs_Dir & "Dummy6" & Logs_Extens
Public Params_Output_FileName As String = Params_Dir & "Dummy7" &
Params_Extens
Public expfnam As String = ""
Public trkfnam As String = ""
' cyclic difference set (CDS) parameters
Public cds_Pixels As Integer
Public cds_Holes As Integer
Public cds_SideLobe As Integer
Public cds_Pix_Truncated As Integer = 0
Public scale_factor As Double, lobe_factor As Double
' mask spacing
Public K As Integer = 1
Public Kminus1 As Integer = K - 1
' pixel dimensions of mask (zero-based) i.e. the notional default below is a 10x10
mask
Public NX As Integer = 9
Public NY As Integer = 9
Public NXplus1 As Integer = NX + 1
Public NYplus1 As Integer = NY + 1
Public NPIX As Integer = NXplus1 * NYplus1 * K
Public NPIXminus1 As Integer = NPIX - 1
' pixel limits for folded-out mask
Public NXLim As Integer = NX * 2
Public NYLim As Integer = NY * 2
' buch of variable definitions...
Public Mpixel_size, Dpixel_size As Single
Public SrcToMask, DetToMask As Single
Public Mtransform As Integer
Public x_mask_offset, y_mask_offset As Single
Public Mask(NX, NY) As Boolean ' Mathematical Pattern for 2D Coded Aperture
Mask
Public Detector(NX, NY, Kminus1) As Integer ' Array to hold number of tracks per
pixel (on detector plane)
Public Deconv(NX, NY, Kminus1) As Integer ' Deconvoluted plane
Public Number_of_Tracks As Integer = 0
Public Number_of_Holes As Integer = 0
Public Trunc_Deconv(Kminus1) As Integer ' Deconvoluted plane for truncated
pixels
Public Trunc_Pix As Boolean = False ' to hold the value of the one pixel truncated
from the CDS
Public Num_Tracks_in_Kplane(2) As Integer
Public xData(1000), yData(1000) As Single
Public numData As Integer
Dim ReRun As Boolean

```

```

Dim ZoomUnity As Boolean
Sub Main()
    Dim xCenter, yCenter As Single
    Dim Theta As Single = 0.0
    Dim Zoom As Single = 1.0
    ' display the program banner
    banner()
    get_filenames()
    Ask_Options()
    ' read experimental definition file
    read_expt()
    ' call subroutine to open and read the mask definition file
    read_mask()
    ' perform any necessary transformation on the mask pattern
    transform_mask()
    ' read the coded image from the CR-39 detector
    read_coded_img(xCenter, yCenter)
    If ReRun Then
        Read_Stored_Params(xCenter, yCenter, Theta, Zoom)
    Else
        ' try to locate the centre of the exposure on the CR-39 more accurately
        iterate_for_centre(xCenter, yCenter)
        ' adjust these 4 parameters to maximize the SNR using simplex method of non-
linear optimization
        optimize_for_snr(xCenter, yCenter, Theta, Zoom)
    End If
    ' now accumulate coded_image with parameters set for Optimized (maximum)
SNR image
    accum_coded_img(xCenter, yCenter, Theta, Zoom)
    write_coded_img()
    ' obtain decoded image by convoluting the mask pattern with the coded-image
array
    decode_image()
    ' normalise the image so that it is Integer, but sums exactly to correct number of
tracks
    normalize_image()
    ' write out the final image to the deconvoluted-image file
    output_image()
    ' save the filenames in the "previous" file for the next time this program is run
    save_filenames()
    ' let user press a key to terminate the program
    wait_user()
End Sub
***** Subroutine to display the program banner *****
Private Sub banner()
    Console.WriteLine(" =====", vbNewLine)
    Console.WriteLine(" |", vbNewLine)
    Console.WriteLine(" | Coded Aperture Deconvolution |", vbNewLine)
    Console.WriteLine(" |", vbNewLine)
    Console.WriteLine(" | with Optimization of |", vbNewLine)

```

```

Console.WriteLine(" |", vbNewLine)
Console.WriteLine(" | Signal-to-Noise Ratio (SNR) |", vbNewLine)
Console.WriteLine(" |", vbNewLine)
Console.WriteLine(" =====", vbNewLine)
Console.WriteLine(vbNewLine, vbNewLine)
Console.WriteLine(" CAI_Optimisation: Version 18", vbNewLine, vbNewLine)
Console.WriteLine(" (Modified on 1st June 2011)", vbNewLine)
Console.WriteLine(vbNewLine, vbNewLine)
End Sub
'***** Subroutine to get the needed filenames *****
Private Sub get_filenames()
' open the Mask Defn File, read in and set values of mask pixels
FileOpen(1, Previous_Filename, OpenMode.Input)
' read in the main part of file-name for Experimental file
Input(1, expfnam)
Console.WriteLine("Previously used EXPERIMENTAL File: " & expfnam &
vbNewLine)
expfnam = Prompt_name(expfnam)
Experiment_FileName = Experiment_Dir & expfnam & Experiment_Extens
' read in the main part of file-name for Track Data file
Input(1, trkfnam)
Console.WriteLine("Previously used TRACK DATA File: " & trkfnam &
vbNewLine)
trkfnam = Prompt_name(trkfnam)
Track_Data_FileName = Track_Data_Dir & trkfnam & Track_Data_Extens
Track DATA read from CR_39 filename
Convoluted_Output_FileName = Convoluted_Dir & trkfnam &
Convoluted_Extens ' output CONVOLUTED/coded array filename
Deconvoluted_Output_FileName = Deconvoluted_Dir & trkfnam &
Deconvoluted_Extens
Logs_Output_FileName = Logs_Dir & trkfnam & Logs_Extens
Params_Output_FileName = Params_Dir & trkfnam & Params_Extens
FileClose(1)
Console.WriteLine(vbNewLine & "===== " & vbNewLine)
Console.WriteLine("EXPERIMENTAL File: " & Experiment_FileName &
vbNewLine & vbNewLine)
Console.WriteLine("TRACK DATA File: " & Track_Data_FileName &
vbNewLine & vbNewLine)
' write stuff to the log file
FileOpen(8, Logs_Output_FileName, OpenMode.Output)
Print(8, " CAI_Optimisation: Version 18" & vbNewLine & vbNewLine)
Print(8, " (Modified on 1st June 2011)" & vbNewLine)
Print(8, vbNewLine & vbNewLine)
Print(8, "EXPERIMENTAL File: " & Experiment_FileName & vbNewLine &
vbNewLine)
Print(8, "TRACK DATA File: " & Track_Data_FileName & vbNewLine &
vbNewLine)
End Sub
'***** Subroutine to prompt the user for a filename *****
Private Function Prompt_name(ByVal inName As String) As String

```

```

Dim cki As ConsoleKeyInfo
Dim key As Char
Dim yes, no, gotit As Boolean
Console.WriteLine(vbNewLine & "Use this file again?" & vbNewLine)
Do
    cki = Console.ReadKey()
    key = cki.Key.ToString()
    Console.WriteLine(vbNewLine)
    yes = (key = "Y")
    no = (key = "N")
    gotit = (yes Or no)
    If Not gotit Then Console.WriteLine("Type 'Y' or 'N'" & vbNewLine)
Loop Until gotit
If no Then
    Console.WriteLine(vbNewLine & "Input new file-name (without extension): ")
    Prompt_name = Console.ReadLine()
Else
    Prompt_name = inName
End If
End Function
'***** Subroutine to ask the user if the program should use the
Stored Parameters file *****
Private Sub Ask_Options()
    Dim cki As ConsoleKeyInfo
    Dim key As Char
    Dim yes, no, gotit As Boolean
    ' Ask user if this is a Re-Run with stored parameters
    Console.WriteLine(vbNewLine & "RE-RUN using values from the STORED
PARAMETERS file?" & vbNewLine)
    Do
        cki = Console.ReadKey()
        key = cki.Key.ToString()
        Console.WriteLine(vbNewLine)
        yes = (key = "Y")
        no = (key = "N")
        gotit = (yes Or no)
        If Not gotit Then Console.WriteLine("Type 'Y' or 'N'" & vbNewLine)
    Loop Until gotit
    ReRun = yes
    ' If not a Re-Run then ask if Zoom Factor should be held at unity
    If no Then
        Console.WriteLine(vbNewLine & "Do you want to keep ZOOM equal to
UNITY (i.e. 1.000) ?" & vbNewLine)
        Do
            cki = Console.ReadKey()
            key = cki.Key.ToString()
            Console.WriteLine(vbNewLine)
            yes = (key = "Y")
            no = (key = "N")
            gotit = (yes Or no)

```

```

        If Not gotit Then Console.WriteLine("Type 'Y' or 'N'" & vbNewLine)
        Loop Until gotit
        ZoomUnity = yes
    End If
End Sub
'***** Subroutine to save the filenames for the next run of program
*****
Private Sub save_filenames()
    ' open the Previous.txt file, and write the main filenames used in this run
    FileOpen(6, Previous_Filename, OpenMode.Output)
    Print(6, expfnam, vbNewLine)
    Print(6, trkfnam, vbNewLine)
    FileClose(6)
End Sub
Private Sub wait_user()
    Dim cki As ConsoleKeyInfo
    ' close the log file
    FileClose(8)
    Console.WriteLine(vbNewLine & vbNewLine & "Press ANY key to terminate" &
vbNewLine)
    cki = Console.ReadKey()
End Sub
'***** Subroutine to read the experimental parameters/definition
*****
Private Sub read_expt()
    ' does the Mask Definiton file exist?
    If Not File.Exists(Experiment_FileName) Then
        Console.WriteLine("{0} does not exist.", Experiment_FileName)
        Return
    End If
    Dim comment As String = ""
    ' open the Mask Defn File, read in and set values of mask pixels
    FileOpen(1, Experiment_FileName, OpenMode.Input)
    ' read and ignore a couple of comment/line-header lines
    Input(1, comment)
    Input(1, comment)
    ' read the name of the mask definition file
    Input(1, Coded_Mask_FileName)
    ' read and ignore a line-header line
    Input(1, comment)
    ' read the mask transform value
    Input(1, Mtransform)
    ' read the mask pixel size (in microns)
    Input(1, Mpixel_size)
    ' read and ignore a line-header line
    Input(1, comment)
    ' read the SrcToMask and DetToMask distances (any units, since only ratio
actually matters)
    Input(1, SrcToMask)
    Input(1, DetToMask)

```



```

FileClose(1)
' calculate the pixel-size on the detector (they're magnified by comparison with
mask pixel-size
Dpixel_size = Mpixel_size * (SrcToMask + DetToMask) / SrcToMask
End Sub
'***** Subroutine to open and read the NEW FORMAT mask definition file
*****
Private Sub read_mask()
' does the Mask Definiton file exist?
If Not File.Exists(Coded_Mask_FileName) Then
    Console.WriteLine("{0} does not exist.", Coded_Mask_FileName)
    Return
End If
Dim comment As String = ""
Dim col, row As Integer
Dim colSum, rowSum As Integer
' open the Mask Defn File, read in and set values of mask pixels
FileOpen(1, Coded_Mask_FileName, OpenMode.Input)
' read and ignore the two comment lines and a line-header
Input(1, comment)
Input(1, comment)
Input(1, comment)
' read the cyclic difference set parameters
Input(1, cds_Pixels)
Input(1, cds_Holes)
Input(1, cds_SideLobe)
' read the mask dimensions
Input(1, NXplus1)
Input(1, NYplus1)
' read the mask spacing (presently unused)
Input(1, K)
Kminus1 = K - 1
' set the factors required for subtraction of sidelobe and re-scaling of image
scale_factor = cds_Holes / (cds_Holes - cds_SideLobe)
lobe_factor = cds_SideLobe / cds_Holes
cds_Pix_Truncated = cds_Pixels - NXplus1 * NYplus1
' check that the number of pixels is consistent
If (cds_Pix_Truncated <> 0) Then
    If (cds_Pix_Truncated < 0) Then
        Console.WriteLine("**** ERROR *** Number of PIXELS in CDS & Mask
Definition are not consistent")
        Print(8, "**** ERROR *** Number of PIXELS in CDS & Mask Definition are
not consistent" & vbNewLine)
        cds_Pix_Truncated = 0
        Return
    End If
End If
' read and ignore the "Columns Rows" line
Input(1, comment)
' set various pixel dimensions for mask (zero-based)

```

```

NX = NXplus1 - 1
NY = NYplus1 - 1
NPIX = NXplus1 * NYplus1 * K
NPIXminus1 = NPIX - 1
' pixel limits for folded-out mask
NXLim = NX * 2
NYLim = NY * 2
' re-dimension the arrays
ReDim Mask(NX, NY) ' Boolean mask definition (TRUE => Open Pixel (i.e. a
hole))
ReDim Detector(NX, NY, Kminus1) ' Array to hold number of tracks per pixel
(on detector plane)
ReDim Deconv(NX, NY, Kminus1) ' Deconvoluted plane
ReDim Num_Tracks_in_Kplane(Kminus1)
ReDim Trunc_Deconv(Kminus1) ' Deconvoluted plane for truncated pixels
' initialize the mask array to FALSE (i.e. no holes)
For row = 0 To NY
    For col = 0 To NX
        Mask(col, row) = False
    Next col
Next row
' zero these variables
Number_of_Holes = 0
colSum = 0
rowSum = 0
' now read in the positions of the holes
Do Until EOF(1)
    Input(1, col)
    Input(1, row)
    Mask(col, row) = True
    Number_of_Holes = Number_of_Holes + 1
    colSum += col
    rowSum += row
Loop
FileClose(1)
' check if number of holes is consistent
If (Number_of_Holes <> cds_Holes) Then
    Console.WriteLine("**** ERROR *** Number of HOLES in CDS & Mask
Definition are not consistent")
    Return
End If
x_mask_offset = ((colSum / Number_of_Holes) - 0.5 * NX) * Dpixel_size
y_mask_offset = ((rowSum / Number_of_Holes) - 0.5 * NY) * Dpixel_size * K
Dim rho_recip As Integer
rho_recip = Int(cds_Pixels / cds_Holes + 0.5)
' output CDS info to the console
Console.WriteLine("Parameters of Cyclic Difference Set (CDS):", vbNewLine,
vbNewLine)
Console.WriteLine("Pixels   = {0}", cds_Pixels, vbNewLine)
Console.WriteLine("Holes    = {0}", cds_Holes, vbNewLine)

```

```

Console.WriteLine("Sidelobe = {0}", cds_SideLobe, vbNewLine)
Console.WriteLine("Rho    = 1/{0}", rho_recip, vbNewLine)
Console.WriteLine(vbNewLine, vbNewLine)
Print(8, "Parameters of Cyclic Difference Set (CDS):", vbNewLine, vbNewLine)
Print(8, "Pixels  = ", cds_Pixels, vbNewLine)
Print(8, "Holes   = ", cds_Holes, vbNewLine)
Print(8, "Sidelobe = ", cds_SideLobe, vbNewLine)
Print(8, "Rho    = 1/", rho_recip, vbNewLine)
Print(8, vbNewLine, vbNewLine)
' output mask dimensions info to the console
Console.WriteLine("Dimensions of Coded Aperture Mask:", vbNewLine,
vbNewLine)
Console.WriteLine("Pixels in X direction = {0}", NXplus1, vbNewLine)
Console.WriteLine("Pixels in Y direction = {0}", NYplus1, vbNewLine)
Console.WriteLine("Row-Spacing factor   = {0}", K, vbNewLine)
Console.WriteLine("Pixel Size on Mask = {0}", Mpixel_size, vbNewLine)
Console.WriteLine("Pixel Size on CR-39 = {0}", Dpixel_size, vbNewLine)
Console.WriteLine(vbNewLine, vbNewLine)
Print(8, "Dimensions of Coded Aperture Mask:", vbNewLine, vbNewLine)
Print(8, "Pixels in X direction = ", NXplus1, vbNewLine)
Print(8, "Pixels in Y direction = ", NYplus1, vbNewLine)
Print(8, "Row-Spacing factor   = ", K, vbNewLine)
Print(8, "Pixel Size on Mask = ", Mpixel_size, vbNewLine)
Print(8, "Pixel Size on CR-39 = ", Dpixel_size, vbNewLine)
Print(8, vbNewLine, vbNewLine)
End Sub
'***** Subroutine to perform any necessary transformation of the mask pattern
*****
Private Sub transform_mask()
    Select Case Mtransform
        Case 0
            Console.WriteLine("No Transformation performed on Mask Pattern",
vbNewLine)
            Console.WriteLine(vbNewLine)
            Print(8, "No Transformation performed on Mask Pattern", vbNewLine)
            Print(8, vbNewLine)
        Case 1
            reflect_mask_vert()
        Case Else
            Console.WriteLine("Mask Transformation {0} not coded yet!", vbNewLine)
            Console.WriteLine("No Transformation performed on Mask Pattern",
Mtransform, vbNewLine)
            Console.WriteLine(vbNewLine)
    End Select
End Sub
'***** Subroutine to reflect the mask pattern through the vertical axis
*****
Private Sub reflect_mask_vert()
    Dim col, row As Integer
    Dim btemp As Boolean

```

```

For row = 0 To NY
    ' only allow col index to run for half the row, otherwise the swap is performed
    twice & therefore nullified
    For col = 0 To Int(NX / 2)
        btemp = Mask(col, row)
        Mask(col, row) = Mask(NX - col, row)
        Mask(NX - col, row) = btemp
    Next col
Next row
' since mask is reflected in vertical axis, then x_mask_offset changes sign
x_mask_offset = -x_mask_offset
' output a message about reflection to the console
Console.WriteLine("Mask Pattern Transformation: REFLECTED in the Vertical
Plane", vbNewLine)
Console.WriteLine(vbNewLine)
Print(8, "Mask Pattern Transformation: REFLECTED in the Vertical Plane",
vbNewLine)
Print(8, vbNewLine)
End Sub
'***** Subroutine to read the coded image (from experiment or simulation)
*****
Private Sub read_coded_img(ByRef xc As Single, ByRef yc As Single)
    ' does the Track Data file exist?
    If Not File.Exists(Track_Data_FileName) Then
        Console.WriteLine("{0} does not exist.", Track_Data_FileName)
        Return
    End If
    ' variables to hold (x,y) posn of the next track from the CR-39 detector
    Dim xDet, yDet As Single
    Dim xSum_ALL As Double = 0.0
    Dim ySum_ALL As Double = 0.0
    numData = 0
    ' open the Mask Defn File, and count the number of tracks
    FileOpen(2, Track_Data_FileName, OpenMode.Input)
    Do Until EOF(2)
        Input(2, xDet)
        Input(2, yDet)
        ' count the number of (x,y) data pairs
        numData += 1
    Loop
    FileClose(2)
    'redimension the arrays to hold all the track data read from CR-39
    ReDim xData(numData), yData(numData)
    Dim id As Integer
    ' open the Mask Defn File again... this time to read in
    ' and store the (x,y) positions in the arrays
    FileOpen(2, Track_Data_FileName, OpenMode.Input)
    For id = 0 To numData - 1
        Input(2, xDet)
        Input(2, yDet)

```

```

    xData(id) = xDet
    yData(id) = yDet
    xSum_ALL += xDet
    ySum_ALL += yDet
Next id
FileClose(2)
' calc initial centre position, to be passed back to the
' main subroutine
xc = xSum_ALL / numData
yc = ySum_ALL / numData
' write summary data to the console
Console.WriteLine("Number of Tracks in Data File = {0}", numData, vbNewLine)
Console.WriteLine("Centre-of-Mass: X = {0}", xc, vbNewLine)
Console.WriteLine("Centre-of-Mass: Y = {0}", yc, vbNewLine)
Console.WriteLine(vbNewLine)
Print(8, "Number of Tracks in Data File = ", numData, vbNewLine)
Print(8, "Centre-of-Mass: X = ", xc, vbNewLine)
Print(8, "Centre-of-Mass: Y = ", yc, vbNewLine)
Print(8, vbNewLine)
End Sub

Private Sub Read_Stored_Params(ByRef rxc As Single, ByRef ryc As Single, ByRef
rtheta As Single, ByRef rzoom As Single)
    Dim comment As String = ""
    rtheta = 0.0
    rzoom = 1.0
    FileOpen(11, Params_Output_FileName, OpenMode.Input)
    ' read and ignore a comment lines in parameters file
    Input(11, comment)
    Input(11, rtheta) ' stored rotation angle (radians)
    Input(11, comment)
    Input(11, rxc) ' stored x-centre
    Input(11, comment)
    Input(11, ryc) ' stored y-centre
    Input(11, comment)
    Input(11, rzoom) ' stored magnification zoom factor
    FileClose(11)
    ' alert user that this is a re-run using stored paperameters and
    ' write out the parameters which have now been read in
    Print(8, vbNewLine)
    Print(8, "**** RE-RUN using Stored Parameters ****" & vbNewLine)
    Print(8, vbNewLine)
    Print(8, "Values of parameters which have been read in..." & vbNewLine)
    Print(8, vbNewLine)
    Print(8, "Angle of rotation (deg) = " & rtheta * 180 / Math.PI & vbNewLine)
    Print(8, "X-Center = " & rxc & vbNewLine)
    Print(8, "Y-Center = " & ryc & vbNewLine)
    Print(8, "Zoom Factor = " & rzoom & vbNewLine)
    Print(8, vbNewLine)
    Console.WriteLine(vbNewLine)
    Console.WriteLine("**** RE-RUN using Stored Parameters ****" & vbNewLine)

```

```

Console.WriteLine(vbNewLine)
Console.WriteLine("values of parameters which have been read in..." &
vbNewLine)
Console.WriteLine(vbNewLine)
Console.WriteLine("Angle of rotation (deg) = " & rtheta * 180 / Math.PI &
vbNewLine)
Console.WriteLine("X-Center = " & rxc & vbNewLine)
Console.WriteLine("Y-Center = " & ryc & vbNewLine)
Console.WriteLine("Zoom Factor = " & rzoom & vbNewLine)
Console.WriteLine(vbNewLine)
End Sub
'***** Subroutine to pixelize the detector plane *****
Private Sub accum_coded_img(ByVal rxc As Single, ByVal ryc As Single, ByVal
rtheta As Single, ByVal rzoom As Single)
'initialization: ensure that the array Detector and Num_Tracks_in_Kplane are filled
with zeros
Dim kplane, col, row As Integer
For kplane = 0 To Kminus1
    For row = 0 To NY
        For col = 0 To NX
            Detector(col, row, kplane) = 0
        Next col
    Next row
    Num_Tracks_in_Kplane(kplane) = 0
Next kplane
Dim xCorner, yCorner As Single
xCorner = rxc - NXplus1 * Dpixel_size
yCorner = ryc - NYplus1 * Dpixel_size * K
' variables to hold (x,y) posn of the next track from the CR-39 detector
Dim xDet, yDet As Single
Dim xdelta, ydelta As Single
Dim ix, iy As Integer
Dim costheta, sintheta As Single
costheta = Math.Cos(rtheta)
sintheta = Math.Sin(rtheta)
Number_of_Tracks = 0
For id = 0 To numData - 1
    ' rotate the data around the centre
    xdelta = (xData(id) - rxc) * rzoom
    ydelta = (yData(id) - ryc) * rzoom
    xDet = rxc + xdelta * costheta - ydelta * sintheta
    yDet = ryc + xdelta * sintheta + ydelta * costheta
    ix = Int((xDet - xCorner) / Dpixel_size)
    iy = Int((yDet - yCorner) / Dpixel_size)
    kplane = iy Mod K
    iy = Int(iy / K)
    ' check that the track position lies within the folded-out mask area
    ' if its outside then just skip this track
    If (ix < 0 Or ix > NXLim) Then Continue For
    If (iy < 0 Or iy > NYLim) Then Continue For

```

```

' OK it is within the folded-out mask area
' if it is in the right fold-out area then shift it upwards (in y-dir) by one pixel,
' and then left in x-dir by NX pixels
If (ix > NX) Then
    iy += 1
    ix -= NXplus1
End If
' if it is (now) in the top fold-out area then shift it downwards (in y-dir) by NY
pixels
If (iy > NY) Then iy -= NYplus1
' by this point the track-position indices (ix,iy) have been shifted and folded into
the main area
' count the number of tracks
Number_of_Tracks += 1
' count the number of tracks in each Kplane
Num_Tracks_in_Kplane(kplane) += 1
' and increment the count stored in the the relevant pixel
Detector(ix, iy, kplane) += 1
Next id
End Sub
'***** Subroutine to locate the centre of the track distribution
*****
Private Sub iterate_for_centre(ByRef xc As Single, ByRef yc As Single)

    Dim xSum_ALL As Double = 0.0
    Dim ySum_ALL As Double = 0.0
    ' variables to hold (x,y) posn of the next track from the CR-39 detector
    Dim xDet, yDet As Single
    Dim i, id As Integer
    Dim xCorner_WIDE_BL, yCorner_WIDE_BL, xCorner_WIDE_TR,
yCorner_WIDE_TR As Single
    Dim xSum_WIDE, ySum_WIDE As Single
    Dim Count_WIDE As Integer
    Dim x_WIDE_centre, y_WIDE_centre As Single
    Dim factor As Single
    ' initial centre position
    x_WIDE_centre = xc
    y_WIDE_centre = yc
    ' five iterations should be sufficient
    For i = 5 To 0 Step -1
        ' decrease this factor from 0.6 to 0.5, to narrow in on the exposed area
        factor = 0.5 + i * 0.02
        ' zero some stuff
        xSum_WIDE = 0.0
        ySum_WIDE = 0.0
        Count_WIDE = 0
        ' and 10% wider limits
        ' Bottom-Left (BL) corner
        xCorner_WIDE_BL = x_WIDE_centre - factor * NXplus1 * Dpixel_size
        yCorner_WIDE_BL = y_WIDE_centre - factor * NYplus1 * Dpixel_size * K

```

```

' Top-Right (TR) corner
xCorner_WIDE_TR = x_WIDE_centre + factor * NXplus1 * Dpixel_size
yCorner_WIDE_TR = y_WIDE_centre + factor * NYplus1 * Dpixel_size * K
For id = 0 To numData - 1
    xDet = xData(id)
    yDet = yData(id)
    If xDet >= xCorner_WIDE_BL And xDet <= xCorner_WIDE_TR And yDet
    >= yCorner_WIDE_BL And yDet <= yCorner_WIDE_TR Then
        xSum_WIDE += xDet
        ySum_WIDE += yDet
        Count_WIDE += 1
    End If
Next id
x_WIDE_centre = xSum_WIDE / Count_WIDE - x_mask_offset
y_WIDE_centre = ySum_WIDE / Count_WIDE - y_mask_offset
Next i
' output message to console about the re-centering
Console.WriteLine("Central Position for CAI Exposure estimated to be:",
vbNewLine)
Print(8, "Central Position for CAI Exposure estimated to be:", vbNewLine)
' re-centre the CR-39 image, and pass values back to
'the Main subroutine
xc = x_WIDE_centre
yc = y_WIDE_centre
Console.WriteLine("Adjusted X-centre = {0}", xc, vbNewLine)
Console.WriteLine("Adjusted Y-centre = {0}", yc, vbNewLine)
Console.WriteLine(vbNewLine, vbNewLine)
Print(8, "Adjusted X-centre = ", xc, vbNewLine)
Print(8, "Adjusted Y-centre = ", yc, vbNewLine)
Print(8, vbNewLine, vbNewLine)
End Sub
'***** Subroutine to write out the decoconvoluted array/iamge *****
Private Sub decode_image()
    Dim kplane, col, row As Integer
    For row = 0 To NY
        For col = 0 To NX
            For kplane = 0 To Kminus1
                Deconv(col, row, kplane) = convarr(kplane)
            Next kplane
            Circ_Rotate_Mask_Array()
        Next
    Next row
    ' if there are a truncated pixels then calculate deconvolution values for them
    ' and rotate the mask once for each truncated pixel
    If (cde_Pix_Truncated > 0) Then
        For kplane = 0 To Kminus1
            Trunc_Deconv(kplane) = convarr(kplane)
        Next kplane
        Circ_Rotate_Mask_Array()
    End If

```



```

End Sub
'***** Function to caculate the track-sum for this mask cyclic-position
*****
Private Function convarr(ByVal kp As Integer) As Integer
    Dim col, row As Integer
    Dim s As Integer = 0
    For row = 0 To NY
        For col = 0 To NX
            If Mask(col, row) Then s += Detector(col, row, kp)
        Next col
    Next row
    Return s
End Function
'***** Subroutine to perform the cyclic-rotation of the mask pattern
*****
Private Sub Circ_Rotate_Mask_Array()
    Dim Temp_Column(NY) As Boolean
    Dim TempBool As Boolean
    Dim row_2, row_1 As Integer
    ' this next bit of code copies the last column of the mask into a temporary array,
while also
    ' performing a ROTATE-UP on its contents
    TempBool = Mask(NX, NY)
    row_2 = NY
    For row_1 = NY - 1 To 0 Step -1
        Temp_Column(row_2) = Mask(NX, row_1)
        row_2 = row_1
    Next row_1
    Temp_Column(0) = TempBool
    ' now SHIFT-RIGHT all the columns of mask array (except the last)
    Dim row As Integer
    Dim col_2, col_1 As Integer
    col_2 = NX
    For col_1 = NX - 1 To 0 Step -1
        For row = 0 To NY
            Mask(col_2, row) = Mask(col_1, row)
        Next row
        col_2 = col_1
    Next col_1
    ' and lastly, copy the temporary (column) array into column zero of the mask array
    For row = 0 To NY
        Mask(0, row) = Temp_Column(row)
    Next
    If (cds_Pix_Truncated = 1) Then
        Dim btemp As Boolean
        btemp = Mask(0, 0)
        Mask(0, 0) = Trunc_Pix
        Trunc_Pix = btemp
    End If
End Sub

```

```

Private Sub optimize_for_snr(ByRef rxc As Single, ByRef ryc As Single, ByRef
rtheta As Single, ByRef rzoom As Single)
    'now score & optimize the image over a range of simplexs
    Dim xCv, yCv As Single
    Dim snr As Single
    Dim xy_away As Single
    Dim Theta_away As Single = 2.5 * Math.PI / 180 ' step simplex 2.5 degrees away
from starting point
    Dim Zoom_away As Single = 0.05 ' step simplex 5% in magnification away from
starting point
    xy_away = 0.333 * Dpixel_size ' step simplex 1/3 of a detector pixel away from
starting point
    xCv = rxc
    yCv = ryc
    'setup the simplex
    Dim iNrDims
    If ZoomUnity Then
        iNrDims = 3 ' 3 dimensions (x,y,theta)
    Else
        iNrDims = 4 ' 4 dimensions (x,y,theta,magnification)
    End If
    Dim iNrVertices As Integer = iNrDims + 1 ' 5 vertices
    Dim sSimplex(iNrVertices - 1, iNrDims - 1) As Single ' the simplex
    Dim sSimplexVal(iNrVertices - 1) As Single ' image_score values at
simplex
    Dim sCoef(iNrDims - 1) As Single
    Dim away(iNrDims - 1) As Single
    If ZoomUnity Then
        away = {xy_away, xy_away, Theta_away}
    Else
        away = {xy_away, xy_away, Theta_away, Zoom_away}
    End If
    Dim i, j As Integer
    For i = 0 To iNrVertices - 1
        If ZoomUnity Then
            sCoef = {xCv, yCv, rtheta}
        Else
            sCoef = {xCv, yCv, rtheta, rzoom}
        End If
        If i > 0 Then
            sCoef(j) += away(j)
        End If
        'Put in simplex and compute function value
        Algebra.SetMatrixRow(sSimplex, sCoef, i)
        sSimplexVal(i) = image_score(sCoef)
        j = i
    Next i
    ' solve for the best signal-to-noise-ratio
    Algebra.SolveNonLinear(sSimplex, sSimplexVal, 5000, AddressOf image_score)
    Console.WriteLine("Error for current transform: " & sSimplexVal(0))

```

```

Print(8, "Error for current transform: " & sSimplexVal(0))
'Copy result back to transform
Algebra.GetMatrixRow(sSimplex, sCoef, 0)
' write out the parameters for the solution (by simplex method) giving maximum
signal-to-noise-ratio
    Console.WriteLine(vbNewLine)
    Console.WriteLine("Percentage change in X-Center (% of Det Pixel) = " & 100.0 *
(sCoef(0) - rxc) / Dpixel_size)
    Console.WriteLine("Percentage change in Y-Center (% of Det Pixel) = " & 100.0 *
(sCoef(1) - ryc) / Dpixel_size)
    If ZoomUnity Then
        Console.WriteLine("Zoom Factor set to Unity")
    Else
        Console.WriteLine("Percentage change in Track Posn. Magnification = " &
100.0 * (sCoef(3) - 1.0))
    End If
    Console.WriteLine(vbNewLine)
    Print(8, vbNewLine)
    Print(8, "Percentage change in X-Center (% of Det Pixel) = " & 100.0 * (sCoef(0) -
rxc) / Dpixel_size)
    Print(8, vbNewLine)
    Print(8, "Percentage change in Y-Center (% of Det Pixel) = " & 100.0 * (sCoef(1) -
ryc) / Dpixel_size)
    Print(8, vbNewLine)
    If ZoomUnity Then
        Print(8, "Zoom Factor set to Unity")
    Else
        Print(8, "Percentage change in Track Posn. Magnification = " & 100.0 *
(sCoef(3) - 1.0))
    End If
    Print(8, vbNewLine & vbNewLine)
' set centre position, rotation angle, and zoom for Optimized (maximum) SNR
image
    rxc = sCoef(0)
    ryc = sCoef(1)
    rtheta = sCoef(2)
    If Not ZoomUnity Then
        rzoom = sCoef(3)
    End If
    snr = 1.0 / sSimplexVal(0)
' write out the parameters for the solution (by simplex method) giving maximum
signal-to-noise-ratio
    Console.WriteLine("Optimized angle of rotation (deg) = " & rtheta * 180 /
Math.PI)
    Console.WriteLine("Optimized value of X-Center    = " & rxc)
    Console.WriteLine("Optimized value of Y-Center    = " & ryc)
    Console.WriteLine("Optimized Pixel Size on CR-39   = " & Dpixel_size / rzoom)
    Console.WriteLine("Optimized Signal-to-Noise Ratio = " & snr)
    Console.WriteLine(vbNewLine)
    Print(8, "Optimized angle of rotation (deg) = " & rtheta * 180 / Math.PI)

```

```

Print(8, vbNewLine)
Print(8, "Optimized value of X-Center    = " & rxc)
Print(8, vbNewLine)
Print(8, "Optimized value of Y-Center    = " & ryc)
Print(8, vbNewLine)
Print(8, "Optimized Pixel Size on CR-39   = " & Dpixel_size / rzoom)
Print(8, vbNewLine)
Print(8, "Optimized Signal-to-Noise Ratio = " & snr)
Print(8, vbNewLine & vbNewLine)
' write out the parameters for the solution to the Parameters file which
' the user can then modify and use in the non-optimizing deconvolution program
FileOpen(10, Params_Output_FileName, OpenMode.Output)
Print(10, "Rotation angle (radians)" & vbNewLine)
Print(10, rtheta & vbNewLine)
Print(10, "X-Center (microns)" & vbNewLine)
Print(10, rxc & vbNewLine)
Print(10, "Y-Center (microns)" & vbNewLine)
Print(10, ryc & vbNewLine)
Print(10, "Zoom factor (adjustment of magnification)" & vbNewLine)
Print(10, rzoom & vbNewLine)
FileClose(10)
End Sub
'***** Subroutine to subtract the sidelobe and rescale the image
*****
Public Function recipSNR() As Single
' subtract off the side-lobe level and then
' re-scale counts according to the cyclic difference set (CDS) parameters
' that the mask pattern is based on
Dim kplane, col, row As Integer
Dim icntr, jcntr, kcntr As Integer
Dim sidelobe As Double
Dim rPixVal(NPIXminus1) As Double
Dim rPixSum As Double = 0.0
Dim r_Trunc_PixVal(Kminus1) As Double
Dim NXp1NYp1 As Integer
NXp1NYp1 = NXplus1 * NYplus1
' calculate the re-scaled deconvoluted (real value) image
Dim iElement As Integer
For kplane = 0 To Kminus1
    sidelobe = Num_Tracks_in_Kplane(kplane) * lobe_factor
    For row = 0 To NY
        For col = 0 To NX
            iElement = col + row * NXplus1 + kplane * NXp1NYp1
            rPixVal(iElement) = (Deconv(col, row, kplane) - sidelobe) * scale_factor
            rPixSum += rPixVal(iElement)
        Next col
    Next row
    If (cds_Pix_Truncated > 0) Then
        r_Trunc_PixVal(kplane) = (Trunc_Deconv(kplane) - sidelobe) * scale_factor
        rPixSum += r_Trunc_PixVal(kplane)
    End If
Next kplane

```

```

    End If
Next kplane
' DEBUG: output the sum for the rPix array
'Console.WriteLine("Sum of rPixVal array = " & rPixSum)
'Console.WriteLine(vbNewLine)
Dim val_high As Double
Dim high_cntr As Integer
Dim val_total As Double = 0.0
Dim swap As Boolean
' sort the first (Number_of_Tracks)-worth of the pixel value array
For icntr = 0 To NPIXminus1 - 1
    val_high = rPixVal(icntr)
    swap = False
    For jcntr = icntr + 1 To NPIXminus1
        If rPixVal(jcntr) > val_high Then
            swap = True
            high_cntr = jcntr
            val_high = rPixVal(jcntr)
        End If
    Next jcntr
    If swap Then
        rPixVal(high_cntr) = rPixVal(icntr)
        rPixVal(icntr) = val_high
    End If
    val_total += val_high
    If val_total >= Number_of_Tracks Then Exit For
Next icntr
rPixVal(icntr) = (val_total - Number_of_Tracks)
' DEBUG: check that the remainder of the image (i.e. background) sums to zero
Dim rsum As Double = 0.0
For jcntr = icntr To NPIXminus1
    rsum += rPixVal(jcntr)
Next jcntr
For kplane = 0 To Kminus1
    rsum += r_Trunc_PixVal(kplane)
Next
' only for debugging
'Console.WriteLine("Sum of background pixels = " & rsum)
'Console.WriteLine(vbNewLine)
' now square everything beyond the first (Number_of_Tracks)-worth
For jcntr = icntr To NPIXminus1
    rPixVal(jcntr) *= rPixVal(jcntr)
Next jcntr
For kplane = 0 To Kminus1
    r_Trunc_PixVal(kplane) *= r_Trunc_PixVal(kplane)
Next
' and sort them in descending order
For kcctr = icntr To NPIXminus1 - 1
    val_high = rPixVal(kcctr)
    swap = False

```

```

For jcntr = kcntr + 1 To NPIXminus1
  If rPixVal(jcntr) > val_high Then
    swap = True
    high_cntr = jcntr
    val_high = rPixVal(jcntr)
  End If
Next jcntr
If swap Then
  rPixVal(high_cntr) = rPixVal(kcntr)
  rPixVal(kcntr) = val_high
End If
Next kcntr
' sum these squares from lowest to highest value, to preserve numerical
significance
Dim SquareSum As Double = 0.0
For kcntr = NPIXminus1 To icntr Step -1
  SquareSum += rPixVal(kcntr)
Next
For kplane = 0 To Kminus1
  SquareSum += r_Trunc_PixVal(kplane)
Next
recipSNR = Math.Sqrt(SquareSum / (NPIX - icntr)) / rPixVal(0)
End Function
Private Function image_score(ByVal sCoef() As Single) As Single
  'Private Function image_score(ByVal xc As Single, ByVal yc As Single, ByVal th
As Single, ByVal zm As Single) As Single
  Dim xc, yc, th, zm As Single
  xc = sCoef(0)
  yc = sCoef(1)
  th = sCoef(2)
  If ZoomUnity Then
    zm = 1.0
  Else
    zm = sCoef(3)
  End If
  ' call subroutine to process the track positions and accumulate tracks-counts for
each pixel in the coded-image
  ' and the perform the deconvolution
  accum_coded_img(xc, yc, th, zm)
  ' obtain decoded image by convoluting the mask pattern with the coded-image
array
  decode_image()
  ' subtract the sidelobe level and rescale, so that number of tracks in image is
preserved
  image_score = recipSNR()
End Function
'***** Subroutine to subtract the sidelobe and rescale the image
*****
Private Sub normalize_image()
  ' subtract off the side-lobe level and then

```

```

' re-scale counts according to the cyclic difference set (CDS) parameters
' that the mask pattern is based on
Dim col, row, kplane As Integer
Dim sidelobe As Double
Dim rDcnv(NX, NY, Kminus1) As Double
Dim rfp(NPIXminus1) As Double ' (real) fractional part
Dim icntr As Integer = 0
Dim sum_intfp As Integer = 0
Dim intfp As Integer
' calculate the re-scaled deconvoluted (real value) image
' also obtain a list the fractional (0<rfp<1) of these values
For kplane = 0 To Kminus1
    sidelobe = Num_Tracks_in_Kplane(kplane) * lobe_factor
    For row = 0 To NY
        For col = 0 To NX
            ' add on a small random number in order to separate the values which
            ' are otherwise exactly equal (due to the integer-ness of Deconv)
            rDcnv(col, row, kplane) = (Deconv(col, row, kplane) - sidelobe) * scale_factor +
0.001 * Rnd()
            intfp = Int(rDcnv(col, row, kplane))
            ' subtract off the integer part
            rfp(icntr) = rDcnv(col, row, kplane) - intfp
            icntr += 1 'increment counter
            sum_intfp += intfp ' sum integer parts of deconvoluted array
        Next col
    Next row
Next kplane
Dim rfp_high As Double
Dim jcntr, high_cntr As Integer
Dim swap As Boolean
' sort the fractional-part array in descending order
For icntr = 0 To NPIXminus1 - 1
    swap = False
    rfp_high = rfp(icntr)
    For jcntr = icntr + 1 To NPIXminus1
        If rfp(jcntr) > rfp_high Then
            swap = True
            high_cntr = jcntr
            rfp_high = rfp(jcntr)
        End If
    Next
    If swap Then
        rfp(high_cntr) = rfp(icntr)
        rfp(icntr) = rfp_high
    End If
Next
' find the point where the cut must be made so as to integerise the rDcnv array
' such that the sum of the integer-array is exactly equal to the number of tracks
Dim rCutVal As Double
jcntr = Number_of_Tracks - sum_intfp

```

```

icntr = jcntr - 1
rCutVal = 1.0 - 0.5 * (rfp(icntr) + rfp(jcntr))
' integerise the rDcnv array, so that the deconvoluted image is an integer array
For kplane = 0 To Kminus1
    For row = 0 To NY
        For col = 0 To NX
            Deconv(col, row, kplane) = Int(rCutVal + rDcnv(col, row, kplane))
        Next col
    Next row
Next kplane
Console.WriteLine("Number of Tracks in CAI Image = {0}", Number_of_Tracks,
vbNewLine)
Console.WriteLine(vbNewLine)
Print(8, "Number of Tracks in CAI Image = ", Number_of_Tracks, vbNewLine)
Print(8, vbNewLine)
End Sub
'***** Subroutine to write out the decovoluted array/iamge
*****
Private Sub ouput_image()
    FileOpen(4, Deconvoluted_Output_FileName, OpenMode.Output)
    Print(4, "Deconvoluted Image Output", vbNewLine)
    ' for the time-being just write out the value in the 10th column of each row
    Dim col, row, kplane As Integer
    For row = 0 To NY
        For kplane = 0 To Kminus1
            For col = 0 To NX
                Print(4, vbTab & Deconv(col, row, kplane))
            Next col
            Print(4, vbNewLine)
        Next kplane
    Next row
    FileClose(4)
    Console.WriteLine("Deconvoluted Image written to file: ")
    Console.WriteLine(vbTab & Deconvoluted_Output_FileName)
    Console.WriteLine(vbNewLine, vbNewLine)
    Console.WriteLine(" --- Program Completed ---")
    Print(8, "Deconvoluted Image written to file: ")
    Print(8, vbTab & Deconvoluted_Output_FileName)
    Print(8, vbNewLine, vbNewLine)
    Print(8, " --- Program Completed ---")
End Sub
'***** Subroutine to write out the mask pattern to a file
*****
Private Sub write_mask()
    Dim Mask_Output_FileName As String = "C:\CAI_Process\conv_data\mask.dat"
    FileOpen(3, Mask_Output_FileName, OpenMode.Output)
    Print(3, "Output of As-Read Mask Image ", vbNewLine)
    Dim col, row As Integer
    For row = 0 To NY
        For col = 0 To NX

```



```

        If Mask(col, row) Then
            Print(3, vbTab & 1)
        Else
            Print(3, vbTab & 0)
        End If
    Next col
    Print(3, vbNewLine)
Next row
FileClose(3)
End Sub
'***** Subroutine to write out the coded image to a file *****
Private Sub write_coded_img()
    FileOpen(3, Convolutd_Output_FileName, OpenMode.Output)
    Print(3, "Coded Image Output", vbNewLine)
    Dim col, row, kplane As Integer
    ' corrected order of loops on 24/2/11
    For row = 0 To NY
        For kplane = 0 To Kminus1
            For col = 0 To NX
                Print(3, vbTab & Detector(col, row, kplane))
            Next col
            Print(3, vbNewLine)
        Next kplane
    Next row
    FileClose(3)
End Sub
End Module

```

Appendix F:

Program, based on Monte Carlo Simulation generates numerical data which is equivalent to the coded image data extracted by scanning CR-39 detectors in the experimental situation to check the correctness of theoretical analysis and decoding algorithm.

Language: Visual Basic

Environment: Microsoft Visual studio 2010 Express

```

Option Explicit On
Imports System
Imports System.IO
Imports System.Collections
Module Module1
    ' =====
    ' |                               |
    ' |   Coded Aperture Simulation   |
    ' |                               |
    ' | =====
    ' last modification date
    ' 22 December 2010
    ' Filenames ... as Public variable
    Public CAI_Dir As String = "C:\CAI_Simulate\"
    Public Experiment_Dir As String = CAI_Dir & "Experiment\"      ' EXPERIMENT
definition directory
    Public Coded_Mask_Dir As String = CAI_Dir & "Masks\"
    Public Track_Data_Dir As String = CAI_Dir & "Track_Data\"      ' Track DATA
read from CR_39 directory
    Public Convoluted_Dir As String = CAI_Dir & "Conv_Data\"
    Public Logs_Dir As String = CAI_Dir & "Log_Files\"
    Public Experiment_Extens As String = ".caix"      ' EXPERIMENT definition
directory
    Public Coded_Mask_Extens As String = ".cam"
    Public Track_Data_Extens As String = ".caid"      ' Track DATA read from CR_39
directory
    Public Convoluted_Extens As String = ".conv"
    Public Logs_Extens As String = ".calg"
    Public Previous_Filename As String = CAI_Dir & "Previous.txt"
    Public Experiment_FileName As String = Experiment_Dir & "Dummy1" &
Experiment_Extens      ' EXPERIMENT definition filename
    Public Coded_Mask_FileName As String = CAI_Dir & "Dummy2" &
Coded_Mask_Extens
    Public Track_Data_FileName As String = Track_Data_Dir & "Dummy3" &
Track_Data_Extens      ' Track DATA read from CR_39 filename
    Public Convoluted_Output_FileName As String = Convoluted_Dir & "Dummy4" &
Convoluted_Extens      ' output CONVOLUTED/coded array filename

```

```

Public Logs_Output_FileName As String = Logs_Dir & "Dummy6" & Logs_Extens
Public expfnam As String = ""
Public trkfnam As String = ""
' cyclic difference set (CDS) parameters
Public cds_Pixels As Integer
Public cds_Holes As Integer
Public cds_SideLobe As Integer
' pixel dimensions of mask (zero-based) i.e. the notional default below is a 10x10
mask
Public NX As Integer = 9
Public NY As Integer = 9
Public NXplus1 As Integer = NX + 1
Public NYplus1 As Integer = NY + 1
Public NPIX As Integer = NXplus1 * NYplus1
Public NPIXminus1 As Integer = NPIX - 1
Public halfNX As Integer
Public halfNY As Integer
' pixel limits for folded-out mask
Public NXLim As Integer = NX * 2
Public NYLim As Integer = NY * 2
' mask spacing
Public K As Integer = 1
' buch of variable definitions...
Public Mpixel_size, Dpixel_size As Single
Public SrcToMask, DetToMask As Single
Public Mtransform As Integer
Public x_mask_offset, y_mask_offset As Single
Public Mask(NX, NY) As Boolean ' Mathematical Pattern for 2D Coded Aperture
Mask
Public nDet(NX, NY) As Integer ' Array to hold number of tracks per pixel (on
detector plane)
Public Deconv(NX, NY) As Integer ' Deconvoluted plane
Public Number_of_Tracks As Integer = 0
Public Number_of_Holes As Integer = 0
Public xData(1000), yData(1000) As Single
'*****
' Source parameters...
Public np As Integer 'number of particle
Public th_max As Single 'maximum emission angle
Public Sreq, Styp As Integer 'required Source: 1=required , -1=unrequired. type
of Source: 1=Circular , 2=Rectangular , 3=Point Source
Public rSrc, xCenter, yCenter As Single 'Circular Source Radius and Center
Public xLftTop, yLftTop, xRgtBot, yRgtBot As Single 'Rectangular Source corners
Public xSrcPt(25), ySrcPt(25) As Single 'number of point sources and their
coordinates (upto 10 Point)
Public nSrcPt As Integer
' Declare a PRNG and initialize it randomly based on the
' system clock (True is required but ignored)
Dim NewRandom As New MTRandom(True)
Sub Main()

```

```

' display the program banner
banner()
get_filenames()
' read experimental definition file
read_expt()
' call subroutine to open and read the mask definition file
read_mask()
' save the filenames in the "previous" file for the next time this program is run
save_filenames()
' write the random track to the file like the CR-39 detector
write_random_track()
' write out the coded image to a file
write_coded_img()
' let user press a key to terminate the program
wait_user()
End Sub
***** Subroutine to display the program banner *****
Private Sub banner()
Console.WriteLine(" =====", vbNewLine)
Console.WriteLine(" | Coded Aperture Simulation |", vbNewLine)
Console.WriteLine(" =====", vbNewLine)
Console.WriteLine(vbNewLine, vbNewLine)
End Sub
***** Subroutine to get the needed filenames *****
Private Sub get_filenames()
' open the Mask Defn File, read in and set values of mask pixels
FileOpen(1, Previous_Filename, OpenMode.Input)
' read in the main part of file-name for Experimental file
Input(1, expfnam)
Console.WriteLine("Previously used EXPERIMENTAL File: " & expfnam &
vbNewLine)
expfnam = Prompt_name(expfnam)
Experiment_FileName = Experiment_Dir & expfnam & Experiment_Extens
' read in the main part of file-name for Track Data file
Input(1, trkfnam)
Console.WriteLine("Previously used TRACK DATA File: " & trkfnam &
vbNewLine)
trkfnam = Prompt_name(trkfnam)
Track_Data_FileName = Track_Data_Dir & trkfnam & Track_Data_Extens
Track DATA read from CR_39 filename
Convoluted_Output_FileName = Convoluted_Dir & trkfnam &
Convoluted_Extens ' output CONVOLUTED/coded array filename
Logs_Output_FileName = Logs_Dir & trkfnam & Logs_Extens
FileClose(1)
Console.WriteLine(vbNewLine & "===== " & vbNewLine)
Console.WriteLine("EXPERIMENTAL File: " & Experiment_FileName &
vbNewLine & vbNewLine)
Console.WriteLine("TRACK DATA File: " & Track_Data_FileName &
vbNewLine & vbNewLine)
' write stuff to the log file

```

```

    FileOpen(8, Logs_Output_FileName, OpenMode.Output)
    Print(8, "EXPERIMENTAL File: " & Experiment_FileName & vbNewLine &
vbNewLine)
    Print(8, "TRACK DATA File: " & Track_Data_FileName & vbNewLine &
vbNewLine)
End Sub
Private Function Prompt_name(ByVal inName As String) As String
    Dim cki As ConsoleKeyInfo
    Dim key As Char
    Dim yes, no, gotit As Boolean
    Console.WriteLine(vbNewLine & "Use this file again?" & vbNewLine)
    Do
        cki = Console.ReadKey()
        key = cki.Key.ToString()
        Console.WriteLine(vbNewLine)
        yes = (key = "Y")
        no = (key = "N")
        gotit = (yes Or no)
        If Not gotit Then Console.WriteLine("Type 'Y' or 'N'" & vbNewLine)
    Loop Until gotit
    If no Then
        Console.WriteLine(vbNewLine & "Input new file-name (without extension): ")
        Prompt_name = Console.ReadLine()
    Else
        Prompt_name = inName
    End If
End Function
'***** Subroutine to save the filenames for the next run of program
*****
Private Sub save_filenames()
    ' open the Previous.txt file, and write the main filenames used in this run
    FileOpen(6, Previous_Filename, OpenMode.Output)
    Print(6, expfnam, vbNewLine)
    Print(6, trkfnam, vbNewLine)
    FileClose(6)
End Sub
Private Sub wait_user()
    Dim cki As ConsoleKeyInfo
    ' close the log file
    FileClose(8)
    Console.WriteLine(vbNewLine & vbNewLine & "Press ANY key to terminate" &
vbNewLine)
    cki = Console.ReadKey()
End Sub
'***** Subroutine to read the experimental parameters/definition
*****
Private Sub read_expt()
    ' does the Mask Definiton file exist?
    If Not File.Exists(Experiment_FileName) Then
        Console.WriteLine("{0} does not exist.", Experiment_FileName)

```

```

    Return
End If
Dim comment As String = ""
Dim i As Integer
' open the Mask Defn File, read in and set values of mask pixels
FileOpen(1, Experiment_FileName, OpenMode.Input)
' read and ignore a couple of comment/line-header lines
Input(1, comment)
Input(1, comment)
' read the name of the mask definition file
Input(1, Coded_Mask_FileName)
' read and ignore a line-header line
Input(1, comment)
' read the mask transform value
Input(1, Mtransform)
' read the mask pixel size (in microns)
Input(1, Mpixel_size)
' read and ignore a line-header line
Input(1, comment)
' read the SrcToMask and DetToMask distances (any units, since only ratio
actually matters)
Input(1, SrcToMask)
Input(1, DetToMask)
SrcToMask = SrcToMask * 1000 ' change unit mm to um
DetToMask = DetToMask * 1000 ' change unit mm to um
' read and ignore a line-header line
Input(1, comment)
' read the number of particles (np)
Input(1, np)
' read and ignore a line-header line
Input(1, comment)
' read the maximum emission angle (th_max)
Input(1, th_max)
' read and ignore a line-header line
Input(1, comment)
' read and ignore a line-header line
Input(1, comment)
' Circular Source: read the radius , xCenter , yCenter
Input(1, Sreq)
If (Sreq = 1) Then Styp = 1
Input(1, rSrc)
Input(1, xCenter)
Input(1, yCenter)
' read and ignore a line-header line
Input(1, comment)
' Rectangular Source: read the xLeftTop , yLeftTop , xRightBottom ,
yRightBottom
Input(1, Sreq)
If (Sreq = 1) Then Styp = 2
Input(1, xLeftTop)

```

```

Input(1, yLftTop)
Input(1, xRgtBot)
Input(1, yRgtBot)
' read and ignore a line-header line
Input(1, comment)
' Point Sources: read the xSrcPt , ySrcPt (upto 10)
Input(1, Sreq)
If (Sreq = 1) Then Styp = 3
Input(1, nSrcPt)
For i = 1 To nSrcPt
    Input(1, xSrcPt(i))
    Input(1, ySrcPt(i))
Next i
FileClose(1)
' calculate the pixel-size on the detector (they're magnified by comparison with mask
pixel-size
Dpixel_size = Mpixel_size * (SrcToMask + DetToMask) / SrcToMask
End Sub
'***** Subroutine to open and read the NEW FORMAT mask definition file
*****
Private Sub read_mask()
' does the Mask Definiton file exist?
If Not File.Exists(Coded_Mask_FileName) Then
    Console.WriteLine("{0} does not exist.", Coded_Mask_FileName)
    Return
End If
Dim comment As String = ""
Dim col, row As Integer
Dim colSum, rowSum As Integer
' open the Mask Defn File, read in and set values of mask pixels
FileOpen(1, Coded_Mask_FileName, OpenMode.Input)
' read and ignore the two comment lines and a line-header
Input(1, comment)
Input(1, comment)
Input(1, comment)
' read the cyclic difference set parameters
Input(1, cds_Pixels)
Input(1, cds_Holes)
Input(1, cds_SideLobe)
' read the mask dimensions
Input(1, NXplus1)
Input(1, NYplus1)
' read the mask spacing
Input(1, K)
' check that the number of pixels is consistent
If (NXplus1 * NYplus1 <> cds_Pixels) Then
    Console.WriteLine("**** ERROR *** Number of PIXELS in CDS & Mask
Definition are not consistent")
    Return
End If

```

```

' read and ignore the "Columns  Rows" line
Input(1, comment)
' set various pixel dimensions for mask (zero-based)
NX = NXplus1 - 1
NY = NYplus1 - 1
NPIX = NXplus1 * NYplus1
NPIXminus1 = NPIX - 1
halfNX = Int(NXplus1 / 2)
halfNY = Int(NYplus1 * K / 2)
' pixel limits for folded-out mask
NXLim = NXplus1 * 2 - 1
NYLim = NYplus1 * K * 2 - 1
' re-dimension the arrays
ReDim Mask(NX, NY) ' Boolean mask definition (TRUE => Open Pixel (i.e. a
hole))
ReDim nDet(NX, NY) ' Array to hold number of tracks per pixel (on detector
plane)
ReDim Deconv(NX, NY) ' Deconvoluted plane
' initialize the mask array to FALSE (i.e. no holes)
For row = 0 To NY
    For col = 0 To NX
        Mask(col, row) = False
    Next col
Next row
' zero these variables
Number_of_Holes = 0
colSum = 0
rowSum = 0
' now read in the positions of the holes
Do Until EOF(1)
    Input(1, col)
    Input(1, row)
    Mask(col, row) = True
    Number_of_Holes = Number_of_Holes + 1
    colSum += col
    rowSum += row
Loop
FileClose(1)
' check if number of holes is consistent
If (Number_of_Holes <> cds_Holes) Then
    Console.WriteLine("**** ERROR **** Number of HOLES in CDS & Mask
Definition are not consistent")
    Return
End If
x_mask_offset = ((colSum / Number_of_Holes) - 0.5 * NX) * Dpixel_size
y_mask_offset = ((rowSum / Number_of_Holes) - 0.5 * NY) * Dpixel_size
Dim rho_recip As Integer
rho_recip = Int(cds_Pixels / cds_Holes + 0.5)
' output CDS info to the console

```



```

Console.WriteLine("Parameters of Cyclic Difference Set (CDS):", vbNewLine,
vbNewLine)
Console.WriteLine("Pixels = {0}", cds_Pixels, vbNewLine)
Console.WriteLine("Holes = {0}", cds_Holes, vbNewLine)
Console.WriteLine("Sidelobe = {0}", cds_SideLobe, vbNewLine)
Console.WriteLine("Rho = 1/{0}", rho_recip, vbNewLine)
Console.WriteLine(vbNewLine, vbNewLine)
Print(8, "Parameters of Cyclic Difference Set (CDS):", vbNewLine, vbNewLine)
Print(8, "Pixels = ", cds_Pixels, vbNewLine)
Print(8, "Holes = ", cds_Holes, vbNewLine)
Print(8, "Sidelobe = ", cds_SideLobe, vbNewLine)
Print(8, "Rho = 1/", rho_recip, vbNewLine)
Print(8, vbNewLine, vbNewLine)
' output mask dimensions info to the console
Console.WriteLine("Dimensions of Coded Aperture Mask:", vbNewLine,
vbNewLine)
Console.WriteLine("Pixels in X direction = {0}", NXplus1, vbNewLine)
Console.WriteLine("Pixels in Y direction = {0}", NYplus1, vbNewLine)
Console.WriteLine("Pixel Size on Mask = {0}", Mpixel_size, vbNewLine)
Console.WriteLine("Pixel Size on CR-39 = {0}", Dpixel_size, vbNewLine)
Console.WriteLine(vbNewLine, vbNewLine)
Print(8, "Dimensions of Coded Aperture Mask:", vbNewLine, vbNewLine)
Print(8, "Pixels in X direction = ", NXplus1, vbNewLine)
Print(8, "Pixels in Y direction = ", NYplus1, vbNewLine)
Print(8, "Pixel Size on Mask = ", Mpixel_size, vbNewLine)
Print(8, "Pixel Size on CR-39 = ", Dpixel_size, vbNewLine)
Print(8, vbNewLine, vbNewLine)
End Sub
'***** Subroutine to pixelize the detector plane *****
Private Sub write_random_track()
' variables
Dim n As Integer
Dim xSrc, ySrc As Single
Dim mSrcPt As Integer = 1 ' point source counter
Dim mrSrc, alpha As Single ' variables for circular source
Dim rMask, xMask, yMask, rDet, xDet, yDet As Single
Dim im, jm As Integer
Dim ths, phi As Single
Dim xLength, yLength As Single ' half Length of Mask in x & y direction
Dim xLengthRgt As Single ' remained length of Mask (total length - half
Length)
Dim yTopBnd, yBotBnd As Single ' y_Top_Boundary and
y_Bottom_Boundary
' open the Mask Defn File,
FileOpen(2, Track_Data_FileName, OpenMode.Output)
Number_of_Tracks = 0
'*****
Randomize()
For n = 1 To np
' identify the source type and its coordinate

```

```

' if it is Circular Source:
If (Styp = 1) Then
    mrSrc = rSrc * Math.Sqrt(NewRandom.genrand_real2())
    alpha = 2 * Math.PI * NewRandom.genrand_real2()
    xSrc = mrSrc * Math.Cos(alpha) + xCenter
    ySrc = mrSrc * Math.Sin(alpha) + yCenter
' if it is Rectangular Source:
ElseIf (Styp = 2) Then
    xSrc = xLftTop + (xRgtBot - xLftTop) * NewRandom.genrand_real2()
    ySrc = yRgtBot + (yLftTop - yRgtBot) * NewRandom.genrand_real2()
' if it is Point Source:
Else
    xSrc = xSrcPt(mSrcPt)
    ySrc = ySrcPt(mSrcPt)
    mSrcPt += 1 ' next point source
    If (mSrcPt > nSrcPt) Then mSrcPt = 1 ' return to first point source
End If
phi = 2 * Math.PI * NewRandom.genrand_real2()
ths = Math.Acos(1 - (1 - Math.Cos(Math.PI * th_max / 180.0)) *
NewRandom.genrand_real2()) ' ths: theta of source
rMask = SrcToMask * Math.Tan(ths)
' xSrc & ySrc are particle coordinates on Source & xMask & yMask are particle
coordinates on Mask
xMask = rMask * Math.Cos(phi) + xSrc
yMask = rMask * Math.Sin(phi) + ySrc
xLength = halfNX * Mpixel_size
xLengthRgt = NXplus1 * Mpixel_size - xLength ' the Mask is not Symetric...
'yLength = halfNY * Mpixel_size
'yLengthTop = NYplus1 * K * Mpixel_size - yLength
' check that the particle has passed through the whole mask...
If xMask < -xLength Or xMask > xLengthRgt Then Continue For ' check the x
boundary....
jm = Int((xMask + xLength) / Mpixel_size)
If jm = NXplus1 Then jm -= 1
' check the y boundary which includes NYplus1 sections....
For l = -Int(NY / 2) To Int(NYplus1 / 2)
    im = -1
    yTopBnd = l * K * Mpixel_size
    yBotBnd = (l * K - 1) * Mpixel_size
    If yMask > yBotBnd And yMask < yTopBnd Then
        im = l + Int(NY / 2)
    Exit For
End If
Next
If (im = -1) Then Continue For ' particle is in none of the y boundaries....
' So particle passes through the mask (whole area is open!) and now must be
check with hole defenition...
If Mask(jm, im) = False Then Continue For
'so it has hitted to the pixel which has hole and must be counted...
rDet = (SrcToMask + DetToMask) * Math.Tan(ths)

```

```

    xDet = rDet * Math.Cos(phi) + xSrc
    yDet = rDet * Math.Sin(phi) + ySrc
    Print(2, xDet, yDet)
    Print(2, vbNewLine)
    '*****
    ' count the number of (x,y) data pairs
    Number_of_Tracks += 1
Next
FileClose(2)
' write summary data to the console
Console.WriteLine("Number of Tracks in Data File = {0}", Number_of_Tracks,
vbNewLine)
Console.WriteLine(vbNewLine)
Print(8, "Number of Tracks in Data file = ", Number_of_Tracks, vbNewLine)
End Sub
'***** Subroutine to write out the coded image to a file *****
Private Sub write_coded_img()
    FileOpen(3, Convolved_Output_FileName, OpenMode.Output)
    Print(3, "Coded Image Output", vbNewLine)
    For row = 0 To NY
        For col = 0 To NX
            Print(3, vbTab & nDet(col, row))
        Next col
        Print(3, vbNewLine)
    Next row
    FileClose(3)
End Sub
End Module

```

References

1. J.W. Mather, Phys. of Fluids, 1960. **3**: p. 134.
2. N.V. Filippov, T.I. Filippova, and V.P. Vinogradov, Nucl. Fus. Suppl., 1962. **2**: p. 577.
3. S. Lee, *Scaling of the Plasma Focus, Plasma Dynamic Viewpoint*, in *International Workshop on Plasma Focus Research (PF98)*1998: Kudova, Poland.
4. K. Steinmetz, 1980, Ruprecht-Karls: Heidelberg.
5. P.A. Chatterton, in *Electrical Breakdown in Gases*, J.M. Meek and J.D. Craggs, Editors. 1978, Wiley New. York.
6. A. Donges, et al., Physics Review Letter 1980. **76A**: p. 391.
7. H. Krompholz, et al., Physics Review Letter 1980. **77A**: p. 246.
8. J. Nomikowski and Z. Jankowski, in *9th European Conf. on Controlled Fusion and Plasma Phys.*1979: Oxford. p. 542.
9. H. Bruzzone, in *2nd Latin American Workshop on Plasma Phys. and Controlled Thermonuclear Fusion*1985: Medellin. p. 313.
10. A. Jerzykiewicz and S. Brandt, in *12th Int. Conf. on Plasma Phys. and Contr. Nuclear Fusion Res.*1988: Nice. p. 737.
11. W.H. Bostick, L. Grunberger, and W. Prior, in *3rd European Conf. on Controlled Fusion and Plasma Phys.*1969: Utrecht. p. 120.
12. J.W. Mather, *Methods of Experimental Physics*, H. Griem and R. Lovberg, Editors. 1971, Academic Press: New York. p. 187.
13. S.P. Chow, S. Lee, and B.C. Tan, Journal Plasma Phys, 1970. **4**: p. 729.
14. K.H. Kwek, T.Y. You, and S. Lee, IEEE Trans. On Plasma Sci., 1990. **18**(826).
15. S. Lee and A. Serban, *Dimension and Lifetime of Plasma Focus Pinch*. IEEE Trans. on plasma Science, 1996. **24**(3): p. 1101-1105.
16. G.X. Zhang, et al., in *Fourth International Conference on High Density Z-Pinch*1997: Vancouver, Canada. p. 423-426.

17. N. Qi, et al., IEEE Trans. Plasma Sci., 1998. **26**(4): p. 1127.
18. A.J. Toepfer, D.R. Smith, and E.H. Beckner, Phys. Fluids, 1971(14): p. 52.
19. A. Bernard, et al., Phys. Fluids, 1975. **18**: p. 180.
20. N.J. Peacock, M. G. Hobby, and P.D. Morgan, in *5th Int. Conf. on Plasma Phys. and Contr. Nucl. Fusion Res. IAEA-CN-28/D-31972*: Tokyo.
21. Y.H. Chen and S. Lee, Int. J. Elect., 1973. **35**: p. 341.
22. M.J. Bernstein, D.A. Meskan, and H.L.L. Passen, Phys. Fluids, 1969. **12**: p. 2193.
23. H. Schmidt, in *2nd Latin American Workshop on Plasma Phys. and Contr. Thermonuclear Fusion (CIF Series)*1987: Medellin.
24. A. Serban, 1995, Nanyang Technological University: Singapore.
25. A. Jeffery and T. Taniuti, eds. *Magnetohydrodynamic Stability and Thermonuclear Confinement*. 1966, Academic Press: New York.
26. V.V. Vikhrev, V.V. Ivanov, and G.A. Rozanova, Nuclear Fusion, 1993. **33**: p. 311.
27. R. Garwin, A. Rosenbluth, and M. Rosenbluth, 1954, Los Alamos Laboratory
28. L.C. Burkhardt and R.H. Lovberg, Phys. of Fluids, 1962. **5**: p. 341.
29. S. Lee. *Theoretical Basis: Plasma Focus Model (Radiative)*. Available from: <http://www.intimal.edu.my/school/fas/UFLF>.
30. R.E. Brown and N. Jarmie, Phys. Rev. C, 1990. **41** p. 1391.
31. R.F. Post, Rev. Mod. Phys., 1956 **28** p. 338.
32. J. H. Lee, L. P. Shomo, and K.H. Kim, *Anisotropy of the Neutron Fluence from a Plasma Focus*. Phys. Fluids, 1972. **15**: p. 2433.
33. Bernard, A., et al., in *Plasma Physics Controlled Nuclear Fusion Research (1978)*1979: Innsbruck. p. 159.
34. U. Jager and H. Herold, *Fast Ion Kinetics and Fusion Reaction Mechanism in the Plasma Focus*. Nuclear Fusion, 1987. **27**: p. 407

35. M.J. Bernstein and G.G. Comisar, *Neutron Energy and Flux Distributions from a Crossed-Field Acceleration Model of Plasma Focus and Z-Pinch Discharges*. Phys. Fluids,, 1972. **15**: p. 700-707.
36. Shyke A. Goldstein and R. Lee, *Ion-Induced Pinch and the Enhancement of Ion Current by Pinched Electron Flow in Relativistic Diodes*. Physics Review Letter, 1975. **35**(16): p. 1079-1082.
37. Heinrich Hora, Reinhard Hopfl, and G.H. Miley, *Double-Layer puls current model for the plasma focus*. Plasma Phys. Contr. Fusion, 1994. **36**: p. 1075-1079.
38. Yusuke Mizuguchi, Jun-Ichi Saka, and H.R. Yousefi, *Simulation of high-energy proton production by fast magnetosonic shock waves in pinched plasma discharges*. Physics of Plasmas, 2007. **14**.
39. Mikhail A. Dorf, I.D.K., Edward A. Startsev, and R.C. Davidson, *Collective focusing of intense ion beam pulses for high-energy density physics application*. Physics of Plasmas, 2011. **18**.
40. R. Deutsch and W. Kies, Plasma Phys. Contr. Fusion, 1988. **30**: p. 263
41. M.V. Roshan and et al., *High Energy Deuteron Emission in NX2 Plasma Focus*. J. Plasma Fusion Res. Series, 2008. **8**.
42. M.V. Roshan and et al., *Nuclear activation measurements of High energy deuterons from a small plasma focus*. Physics Letter A 2009. **373**: p. 851-855.
43. M.V. Roshan and et al., *Magnetic spectroscopy of high energy deuteron beams from pulsed plasma system*. Plasma Phys. Control Fusion 2010. **52**: p. 17.
44. M.V. Roshan and et al., *Backward high energy ion beam from plasma focus*. Physics of Plasmas, 2009. **16**.
45. H. Ing, R. A. Noulty, and T. D. McLean, Radiation Measurements, 1997. **27**: p. 1.
46. S. Mahmud and et al., Rev. Sci. Instrum., 2006. **77**: p. 713-722.
47. S.V. Springham and et al., *Ferrofluidic masking of solid state nuclear track detectors during etching*. Radiation Measurements, 2009. **44**: p. 173-175.

48. S.V. Springham, S. Lee, and S.P. Moo, *Deuterium Plasma Focus Measurements using Solid State Nuclear Track Detector*. Brazilian Journal of Physics, 2002. **32** p. 172-178.
49. E. Caroli, et al., *Coded Aperture Imaging in X and Gamma Ray Astronomy*. Space Science Review, 1987. **45**: p. 349-403.
50. G.K. Skinner, *Coded-mask imaging in gamma-ray astronomy - separating the real and imaginary parts of a complex subject*, in *22nd Moriond Astrophysics Meeting 2003*, arXiv:astro: Vietnam.
51. F. Garibaldi and et al., *Small Animal Imaging by Single Photon Emission Using Pinhole and Coded Aperture Collimation*. IEEE Trans. Nuclear Science, 2005. **52**: p. 573-579.
52. L. Zhang and R.C. Lanza, *CAFNA, coded aperture fast neutron analysis for contraband detection: Preliminary results*. IEEE Trans. Nuclear Science, 1999. **46**: p. 1913-1915.
53. R. Accorsi and R.C. Lanza, *Coded Aperture Fast Neutron Analysis: Latest Design Advances*, in *16th Intl. Conf. Appl. of Accel. Res. & Indust.* 2001. p. 491-494.
54. Yen-Wei Chen and et al., *Measurement of D-D burn region using proton penumbral coded aperture imaging*. Optics Communications, 1989. **73**: p. 337-341.
55. U. Jager and et al., *Energy spectra and space resolved measurement of fusion reaction proton from plasma focus devices*. Rev. Sci. Instrum., 1985. **56**: p. 77-79.
56. C.A. Barrera and E.C. Morse, *Image reconstruction algorithms for inertial confinement fusion neutron imaging*. Rev. Sci. Instr., 2006. **77**: p. 716-725.
57. S.V. Springham, et al., *Imaging of Fusion Protons from a 3kJ Deuterium Plasma Focus*. Japanese Journal of Applied Physics, 2005. **44**: p. 4177-4121.
58. R. Accorsi, 2001, Massachusetts Institute of Technology

59. G. Groh, G.S. Hayat, and G.W. Stroke, *X-Ray and γ -Ray Imaging with Multiple-Pinhole Cameras Using a posteriori Image Synthesis*. Appl. Optics, 1972. **11**: p. 931.
60. C. Brown, *Multiplex imaging with multiple-pinhole cameras*. J. Appl. Phys., 1974. **45**: p. 1806-1811.
61. L. Mertz and N.O. Young, *Fresnel transformation of Images*, in *International Conference on Optical Instrumentation and Techniques* 1961, Chapman and Hall: London. p. 305.
62. H. Barrett, *Fresnel zone plate imaging in nuclear medicine*. Journal of Nuclear Medicine, 1972. **13**(6): p. 382-385.
63. R.H. Dicke, *Scatter-hole camera for X-rays and gamma rays*. The Astrophysical Journal, 1968. **153**(2): p. 101-106.
64. J.G. Ables, *Fourier transform photography: a new method for X-ray astronomy*, in *Astronomic Society of Australia* 1968. p. 172-173.
65. M.J.E. Golay, *Point Arrays Having Compact, Non-redundant Autocorrelation*. Journal of the Optical Society of America, 1971. **61**: p. 272.
66. E.E. Fenimore and T.M. Cannon, *Coded aperture imaging: many holes make light work*. Optical Engineering, 1980. **19**(3): p. 283-289.
67. E.E. Fenimore and T.M. Cannon, *Coded Aperture Imaging with uniformly redundant arrays*. Applied Optics, 1978. **17**: p. 337-347.
68. J.K. Skinner, *Imaging with Coded-Aperture Masks*. Nuclear Instruments and Methods in physics Research, 1984. **221**: p. 33-40.
69. E.D. Nelson and M. L. Fredman, *Hadamard Spectroscopy*. Journal of the Optical Society of America, 1970. **60**(12): p. 1664-1669.
70. C. Calabro and J. K. Wolf, *On the Synthesis of Two-Dimensional Arrays with Desirable Correlation properties*. Information and Control, 1968. **11**: p. 537-560.
71. S.R. Gottesman and E.E. Fenimore, *New family of binary arrays for coded aperture imaging*. Applied Optics, 1989. **28**: p. 4344-4352.

72. J. Singer, *A theorem in finite projective geometry and some applications to number theory*. Trans. Amer. Math. Soc., 1938. **43**: p. 377-385.
73. L. D. Baumert, *Cyclic Difference Sets*. Lecture Notes in Mathematics. Vol. 182. 1971, New York: Springer.
74. Hall, M., *Combinatorial Theory*, ed. G. Co.1967, London: Blaisdell.
75. L.D. Baumert and M. Hall, *A new construction for Hadamard matrices*. Bull. Amer. Math. Soc., 1965. **71**: p. 169-170.
76. Fenimore, E.E., *Coded aperture imaging, predicted performance of uniformly redundant arrays*. Applied Optics, 1978. **17**(22): p. 3562-3570.
77. J.J.M. In 't Zand, J. Heise, and R. Jager, *The optimum open fraction of coded apertures. With an application to the wide field X-ray cameras of SAX*. Astron. Astrophys., 1994. **288**: p. 665-674.
78. R. Accorsi, F. Gasparini, and R.C. Lanza, *Optimal coded aperture patterns for improved SNR in nuclear medicine imaging*. Nucl. Instrum. Methods Phys. Res. A, 2001. **474**: p. 273-284.
79. Fleischer, R.L., *Radon: Overview of Properties, Origin, and Transport in Radon measurements by Etched Track Detectors* ed. S.A. Durrani and R. Ilic1997a, Singapore: World Scientific Publishing.
80. Haur, S.T., 2004, NIE-NTU: Singapore.
81. D. Nikezic and K.N. Yu, *Computer program TRACK_TEST for calculating parameters and plotting profiles for etch pits in nuclear track materials*. . Computer Physics Communications 2006. **174**(2): p. 160-165.
82. D. Nikezic and K.N. Yu, *Formation and Growth of Tracks in Nuclear Track Materials*. Material Science and Engineering, 2004. **46**(3-5): p. 51-123.
83. Makoto Matsumoto and T. Nishimura, *Mersenne twister: a 623-dimensionally equidistributed uniform pseudo-random number generator*. ACM Transactions on Modeling and Computer Simulation (TOMACS)-Special issue on uniform random number generation, 1998. **8**(1).
84. D. R. Slaughter and W. L. Pickles, Nucl. Instrum. Methods Phys. Res. A, 1979. **160**: p. 87-92.

85. C. A. Ekdahl, *Rev. Sci. Instr.*, 1979. **50**: p. 941-948.
86. G. F. Knoll, *Radiation detection and measurement*. 3 ed1999: John Wiley & Sons.
87. C. L. Ruiz, R. J. Leeper, and et al., *Rev. Sci. Instr.*, 1992. **63**: p. 4889 - 4891.
88. L. Bertalot and M. Damiani et al., *Rev. Sci. Instrum.*, 1997. **68**: p. 528-531.
89. E. L. Jacobs, S. D. Bonaparte, and P. D. Thatcher, *Nucl. Instrum. Methods*, 1983. **213**: p. 387-392.
90. M.V. Roshan, S.V. Springham, and et al., *Absolute measurements of fast neutrons using yttrium*. *Rev. Sci. Instrum.*, 2010. **81**.
91. G. W. Cooper and C. L. Ruiz, *NIF total neutron yield diagnostic* *Rev. Sci. Instr.*, 2001. **72**: p. 814-817.
92. C. W. Barnes, et al., *Neutron activation for ITER*. *Rev. Sci. Instr.*, 1995. **66**: p. 888-890.
93. T. Nishitani, et al., *Neutron activation system using water flow for ITER*. *rev. Sci. Instr.*, 2003. **74**: p. 1735-1738.
94. Yu. A. Kaschuck, et al., *Gas-Jet Activation Method for a Fusion Power Measurement on ITER-FEAT*. *Fusion Science and Technology*, 2003. **43**(2): p. 176-183.
95. M.S. Rowland and J.C. Robertson, *Nucl. Instrum. Methods*,, 1984. **224**: p. 322-323.
96. A. Wolf and R. Moreh, *Utilization of Teflon-Covered Ge(Li) Diodes for Fast Neutron Detection*. *Nucl. Instrum. Methods*, 1978. **148**: p. 195-197.
97. L. Ruby and J. B. Rechen, *Nucl. Instrum. Methods*, 1984. **224**: p. 322-323.
98. B. T. Meehan, et al., *Praseodymium activation detector for measuring bursts of 14 MeV neutrons*. *nucl. Instrum. Methods*, 2010. **620**: p. 397-400.
99. V.E. Lewis and T.B. Ryves, *Measurement of d+T Neutron Pulses*. *Nucl. Instrum. Methods*, 1987. **257**: p. 462-466.
100. *National Nuclear Data Center (NNDC)*. Available from: <http://www-nds.iaea.org/exfor/endl.htm>.

101. P. H. Stelson and E. C. Campbell, *Cross Section for the $Be^9(n, \alpha)He^6$ Reaction*. Physical Review, 1957. **106** (6): p. 1252-1255.
102. J.H. Lee, Physics of Fluids, 1971. **14**(10).
103. H. Kelly and A. Marquez, Plasma Phys. Control Fusion, 1996. **38**: p. 1931-1942.
104. K. Hübner, H. Bruhns, and K. Steinmetz, Physics Letters A, 1978. **69**(4, 25): p. 269-272.
105. A. Cebanu and et al., in *8th Int. Plasma Physics and Controlled Nuclear Fusion Research* 1980 IAEA: Brussels. p. 197.
106. A. Mozer, et al., J. Appl. Phys., 1977. **13**: p. 2929.
107. R.L. Gullickson and H.L. Sahlin, Appl. Phys., 1978. **49**: p. 1099.
108. H. Krompholz and et al, Appl. Phys., 1977. **13**(29).
109. Rishi Verma and et al., Plasma Physics Control Fusion, 2009. **51**: p. 16.
110. I. Tiseanu, N. Mandache, and V. Zambreanu, Plasma Phys. Control. Fusion, 1994. **36** p. 417-432.
111. J. Koh, Plasma Source Sci. Technol., 2005. **14**: p. 12-18.
112. M Zakaullah and et al., Plasma Sources Sci. Technol., 2003. **12**: p. 443-448.
113. M. Scholz and et al., Plasma Science, IEEE Transactions,, 2002. **30**(2): p. 476-481.
114. J.H. Lee, Physics of Fluids, 1972. **15**(12).
115. *National Institute of Standard and Technology*. Available from:
<http://physics.nist.gov/PhysRefData/Star/Text/ESTAR.html>.
116. *Natinal Nuclear Data Center*. Available from:
<http://www.nndc.bnl.gov/chart/decaysearchdirect.jsp?nuc=55Fe>
<http://www.nndc.bnl.gov/chart/decaysearchdirect.jsp?nuc=109Cd>.
117. M.S. Rowland and J.C. Robertson, Radiation Effects, 1986. **96**: p. 21-24.
118. *IAEA Technical Report*, in *Handbook on Nuclear Activation Data* 1987. p. 111-115.

119. Ronald E. Brown and N. Jarmie, *Differential cross section at low energies for $^2\text{H}(d,p)^3\text{H}$ and $^2\text{H}(d,n)^3\text{He}$* . Physics Review C, 1989. **41**(4): p. 1391-1396.
120. *X-Ray Interaction With Matter*. Available from:
http://henke.lbl.gov/optical_constants/.
121. Lerner, E.J. and A.L. Peratt, *Final Report Jet Propulsion Laboratory contract 959962*, 1995.
122. Lisa Gottesfeld Brown, *A survey of image registration techniques* ACM Computing Surveys (CSUR) archive, 1992. **24**(4): p. 325-376.